

EEEE Codes

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
825-7691	1	EEEE FOR (MLB, 639-94441, JP, ULTRA)	EEEE_J08R	CRITICAL	EEEE_ULTIMATE
825-7691	1	EEEE FOR (MLB, 639-93377, JP, EXTREME)	EEEE_HNV8	CRITICAL	EEEE_EXTREME
825-7691	1	EEEE FOR (MLB, 639-94442, ROW, ULTRA)	EEEE_J08X	CRITICAL	EEEE_ULTIMATE_ROW
825-7691	1	EEEE FOR (MLB, 639-93389, ROW, EXTREME)	EEEE_HNVF	CRITICAL	EEEE_EXTREME_ROW

D21x Specific

D21x Specific

D21x Specific

D21x Specific

SOC

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
339S00416	1	SoC, H, 3GB 23nm DDR, B1	U1000	CRITICAL	COMMON

D21x Specific

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
339S00415	339S00416	BOM_TABLE_ALTS	U1000	SoC, M, 3GB 20nm DDR, B1
339S00417	339S00416	BOM_TABLE_ALTS	U1000	SoC, S, 3GB 20nm DDR, B1
339S00418	339S00416	BOM_TABLE_ALTS	U1000	SoC, S, 3GB 1xnm DDR, B1
339S00429	339S00416	BOM_TABLE_ALTS	U1000	SoC, H, 3GB 20nm DDR, B1
339S00430	339S00416	BOM_TABLE_ALTS	U1000	SoC, M, 3GB 20nm DDR, B1
339S00431	339S00416	BOM_TABLE_ALTS	U1000	SoC, S, 3GB 20nm DDR, B1
339S00432	339S00416	BOM_TABLE_ALTS	U1000	SoC, S, 3GB 18nm DDR, B1

SIPs

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
339M00011	339M00010	BOM_TABLE_ALTS	M5500	NIKI, STATS SIP
339M00013	339M00012	BOM_TABLE_ALTS	M4100	HUNT, STATS SIP

D21x Specific

SOFT-TERM CAP SUB BOMS

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
685-00152	685-00151	BOM_TABLE_ALTS	SUBBOM_CAP	SUBBOM, MLB, CAP, SOFT, D21
138S00049	138S0031	TYPICAL_CAP	ALL	CAP, R0201, 2.2UF, 6.3V, KYOCERA

To enable alt in SUBBOM

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
685-00151	1	SUBBOM, MLB, SOFT-TERM, D21	SUBBOM_CAP		
ADGES OUTPUT	138S00159	9	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C2060, C2061, C2063, C2065, C2067, C2068, C2063	SOFT_CAP
	138S0031	9	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C2060, C2061, C2063, C2065, C2067, C2068, C2063	TYPICAL_CAP
ADGES INPUT	138S00159	15	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C2051, C2052, C2054, C2055, C2052, C2060, C2060	SOFT_CAP
	138S0031	15	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C2051, C2052, C2054, C2060, C2062, C2066, C2066	TYPICAL_CAP
	138S00159	10	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C2079, C2075, C2074, C2076, C2076, C2071, C2060, C2061	SOFT_CAP
	138S0031	10	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C2079, C2075, C2074, C2076, C2076, C2071, C2060, C2061	TYPICAL_CAP
12	138S00159	2	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C3351, C3352	SOFT_CAP
	138S0031	2	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C3351, C3352	TYPICAL_CAP
SENSEORS	138S00159	3	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C3602, C3611, C3622	SOFT_CAP
	138S0031	3	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C3602, C3611, C3622	TYPICAL_CAP
ANGEL INPUT	138S00159	5	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C3791, C3795, C3798	SOFT_CAP
	138S0031	5	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C3791, C3795, C3798	TYPICAL_CAP
RCAR B203	138S00159	3	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C3909, C3925, C3926	SOFT_CAP
	138S0031	3	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C3909, C3925, C3926	TYPICAL_CAP
FOREHEAD B203	138S00159	2	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C4250, C4261	SOFT_CAP
	138S0031	2	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C4250, C4261	TYPICAL_CAP
STORAGE B203	138S00159	1	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C4303	SOFT_CAP
	138S0031	1	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C4303	TYPICAL_CAP
AUDIO	138S00159	5	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C4805, C4809, C4914, C4926, C4929	SOFT_CAP
	138S0031	5	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C4805, C4809, C4914, C4926, C4929	TYPICAL_CAP
	138S00159	5	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C5016, C5018, C5134, C5029, C5136	SOFT_CAP
	138S0031	5	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C5016, C5018, C5134, C5029, C5136	TYPICAL_CAP
NOJAVE	138S00159	1	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C5611	SOFT_CAP
	138S0031	1	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C5611	TYPICAL_CAP
DISPLAY B203	138S00159	4	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C5700, C5705, C5716, C5721	SOFT_CAP
	138S0031	4	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C5700, C5705, C5716, C5721	TYPICAL_CAP
DRS/MESA B203	138S00159	5	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C5802, C5804, C5811, C5808	SOFT_CAP
	138S0031	5	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C5802, C5804, C5811, C5808	TYPICAL_CAP
RAMBA L00	138S00159	1	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C5890	SOFT_CAP
	138S0031	1	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C5890	TYPICAL_CAP
ACC. BACK OUTPUT	138S00159	5	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C6320, C6321, C6322, C6323, C6324	SOFT_CAP
	138S0031	5	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C6320, C6321, C6322, C6323, C6324	TYPICAL_CAP
	138S00160	4	CAP, SOFT-TERM, 10UF, 10V, 0402, MURATA	C4907, C4903, C4904, C4931	SOFT_CAP
	138S0979	4	CAP, TYPICAL, 10UF, 10V, 0402, MUR/KYO	C4907, C4903, C4904, C4931	TYPICAL_CAP
CHESTNUT L00	138S00160	5	CAP, SOFT-TERM, 10UF, 10V, 0402, MURATA	C5600, C5602, C5603, C5702, C5703	SOFT_CAP
	138S0979	5	CAP, TYPICAL, 10UF, 10V, 0402, MUR/KYO	C5600, C5602, C5603, C5702, C5703	TYPICAL_CAP
RCAR (TELE)	138S00159	2	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C4825, C4826	SOFT_CAP
	138S0031	2	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C4825, C4826	TYPICAL_CAP

Default is typical

D21x Specific

D21x Specific

IKTARA DIODES SUB BOMS

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
685-00172	685-00171	BOM_TABLE_ALTS	SUBBOM_DS	SUBBOM, MLB, SCHOTTKY DIODES, ONSEMI, D21

#31526972: Update Iktara Diodes APNs

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
685-00171	1	SUBBOM, MLB, SCHOTTKY DIODES, D100ES, D21	SUBBOM_DS		
371S00133	4	DIODES, SHOTTKY DIODE, 30V, 2A, 0603	D3400, D3401, D3402, D3403		DIODES_DS
371S00132	4	ONSEMI, SHOTTKY DIODE, 30V, 2A, 0603	D3400, D3401, D3402, D3403		ONSEMI_DS

Footprint is ONSEMI

Primary is D100ES

D

NAND BOM Options

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
335S00287	1	NAND_H, ULTRA, TLC	U2600	CRITICAL	ULTIMATE
335S00240	1	NAND_H, EXTREME, TLC	U2600	CRITICAL	EXTREME

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
335S00284	335S00287	BOM_TABLE_ALTS	U2600	T, 1z, ULTIMATE
335S00285	335S00287	BOM_TABLE_ALTS	U2600	T, B1CS3, ULTIMATE
335S00288	335S00287	BOM_TABLE_ALTS	U2600	S, B1CS3, ULTIMATE
335S00286	335S00287	BOM_TABLE_ALTS	U2600	SS, 30v4, ULTIMATE
335S00228	335S00240	BOM_TABLE_ALTS	U2600	T, B1CS3, EXTREME
335S00247	335S00240	BOM_TABLE_ALTS	U2600	S, B1CS3, EXTREME
335S00276	335S00240	BOM_TABLE_ALTS	U2600	SS, 30v4, EXTREME

C

Isolator Switch

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
311S00126	311S00114	BOM_TABLE_ALTS	ALL	SPDT, TI

Mamba LDO

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
353S00932	353S00976	BOM_TABLE_ALTS	U5890	STMICRO LDO

Load Switch

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
353S00999	353S01039	BOM_TABLE_ALTS	U6100	TI, LOAD SW, TPS22915C
353S01007	353S01039	BOM_TABLE_ALTS	U6100	ONSEMI, LOAD SW, MCP333

Crystal Alts

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
197S0012	197S00118	BOM_TABLE_ALTS	Y1000	NDK, XTAL, 24MHz, 1.6x1.2
197S00120	197S00118	BOM_TABLE_ALTS	Y1000	EPSON, XTAL, 24MHz, 1.6x1.2

CRITICAL PART#	COMMENT
197S00118	XTAL, 24M, 1612, TDK
197S0012	XTAL, 24M, 1612, NDK
197S00120	XTAL, 24M, 1612, EPSON

PMID1 Zener Diode

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
371S00121	371S00030	BOM_TABLE_ALTS	DZ3300	DIODE, ZENER, 20V

A

ZRB Caps

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
138S0083	1	1uF, 25V, 0402, X5R, 10%	C3301	CRITICAL	COMMON

CRITICAL PART#	COMMENT
138S00185	MUR, CAP, ZRB, 2.2UF, 25V, 0402
138S00186	MUR, CAP, ZRB, 4.7UF, 25V, 0603

Global R/C Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
118S0764	118S0717	BOM_TABLE_ALTS	ALL	RES, 3.92K, 0.1%, 0201
138S0648	138S0652	BOM_TABLE_ALTS	ALL	CAP, X5R, 4.7UF, 6.3V, 0.60MM, 0402, TAIYO
138S00024	138S0986	BOM_TABLE_ALTS	ALL	CAP, CER, 3.3UF, 1.5V, 20%, 0.40, 0402, TAIYO/TK
138S0706	138S0739	BOM_TABLE_ALTS	ALL	CAP, CER, 10P, 20V, 20%, 10V, X5R, 0301, NIDOKA
138S0945	138S0739	BOM_TABLE_ALTS	ALL	CAP, CER, 10P, 20V, 20%, 10V, X5R, 0301, KYOCERA
138S0739	138S0706	BOM_TABLE_ALTS	ALL	CAP, CER, X5R, 0.220F, 20%, 6.3V, 20%
132S0436	132S0400	BOM_TABLE_ALTS	ALL	CAP, X5R, X5R, 0.220F, 20%, 6.3V, 0305
138S00133	138S00128	BOM_TABLE_ALTS	ALL	CAP, CER, X5R, 0.47UF, 20%, 6.3V, 01005
138S00148	138S00150	BOM_TABLE_ALTS	ALL	CAP, 0402, 15UF, 4V, KYOCERA
138S00149	138S00150	BOM_TABLE_ALTS	ALL	CAP, 0402, 14UF, 4V, MURATA
138S00151	138S00150	BOM_TABLE_ALTS	ALL	CAP, 0402, 15UF, 4V, TAIYO
138S00144	138S00143	BOM_TABLE_ALTS	ALL	CAP, 0402, 20UF, 4V, MURATA
138S00143	138S00144	BOM_TABLE_ALTS	ALL	CAP, 0402, 22UF, 4V, KYOCERA
138S00139	138S00138	BOM_TABLE_ALTS	ALL	CAP, 0201, 4UF, 4V, MURATA
138S00138	138S00139	BOM_TABLE_ALTS	ALL	CAP, 0201, 4.7UF, 4V, KYOCERA
138S00145	138S00146	BOM_TABLE_ALTS	ALL	CAP, 0402, 20UF, 6.3V, KYOCERA
138S00141	138S00140	BOM_TABLE_ALTS	ALL	CAP, 0201, 4UF, 6.3V, MURATA
138S00142	138S00140	BOM_TABLE_ALTS	ALL	CAP, 0201, 3UF, 6.3V, SAMSUNG
138S00142	138S00141	BOM_TABLE_ALTS	ALL	CAP, 0201, 3UF, 6.3V, SAMSUNG
138S00049	138S0031	BOM_TABLE_ALTS	ALL	CAP, 0201, 2.2UF, 6.3V, KYOCERA
138S00163	138S00143	BOM_TABLE_ALTS	ALL	CAP, 0402, 22UF, 4V, TY
138S00164	138S00138	BOM_TABLE_ALTS	ALL	CAP, 0201, 4UF, 4V, TY
138S00165	138S00146	BOM_TABLE_ALTS	ALL	CAP, 0402, 20UF, 6.3V, TY
138S00166	138S00140	BOM_TABLE_ALTS	ALL	CAP, 0201, 4UF, 6.3V, TY
138S00140	138S00141	BOM_TABLE_ALTS	ALL	CAP, 0201, 3UF, 6.3V, KYO
138S00048	138S00003	BOM_TABLE_ALTS	ALL	CAP, X5R, 150F, 6.3V, 0.60MM, 0402, KYO
132S0639	132S00088	BOM_TABLE_ALTS	ALL	CAP, X5R, 0.47UF, 25V, 0.30MM, 0301, MUR

CRITICAL PART#	COMMENT
118S0717	RES, 3.92K, 0.1%, 0201
138S0652	CAP, X5R, 4.7UF, 6.3V, 0.65MM, 0402
138S0706	CAP, CER, X5R, 1UF, 20%, 6.3V, 0201
132S0400	CAP, CER, X5R, 0.22UF, 20%, 6.3V, 01005
138S0031	CAP, CER, X5R, 2.2UF, 20%, 6.3V, 0201
138S00128	CAP, CER, X5R, 0.47UF, 20%, 6.3V, 01005

CRITICAL PART#	COMMENT
138S00149	0402-3T, 10.5uF@1V

CRITICAL PART#	COMMENT
138S00144	0402, 16uF@1V

CRITICAL PART#	COMMENT
138S00139	0201, 3uF@1V

CRITICAL PART#	COMMENT
138S00146	0402, 5.1uF@3V

CRITICAL PART#	COMMENT
138S00141	0201, 1.1uF@3V

CRITICAL PART#	COMMENT
155S0010	FERR BD, 1500HM, 01005

CRITICAL PART#	COMMENT
152S00617	IND, MLD, 0.1UH, 20%, 6.1A, 29MOHM, H=-.65, 1608
152S00620	IND, MLD, 0.1UH, 20%, 7.2A, 17MOHM, H=B-.8, 2012
152S00621	IND, MLD, 0.47UH, 20%, 3.5A, 53MO, H=-.65, 2012
152S00622	IND, MLD, 1.0UH, 20%, 2.1A, 100MO, H=-.65, 2012
152S00626	IND, MLD, 1.5UH, 20%, 1.1A, 160MO, H=-.65, 2012
152S00631	IND, MLD, 1.0UH, 20%, 2.5A, 78MO, H=B-.8, 2012
152S00632	IND, MLD, 1.0UH, 20%, 3.2A, 60MO, H=B-.8, 2016
152S00640	IND, MLD, 0.47UH, 3.6A, 55MO, H=B-.65MM, 2012
152S00641	IND, MLD, 0.47UH, 4A, 48MO, H=B-.8MM, 2012
152S00623	IND, MLD, 1UH, 3.6A, 60MO, H=B-.8MM, 2016
152S00651	IND, 1.2UH, 3A, 2016, 0.65Z
152S00650	IND, 0.47UH, 6.6A, 3225, 0.8Z

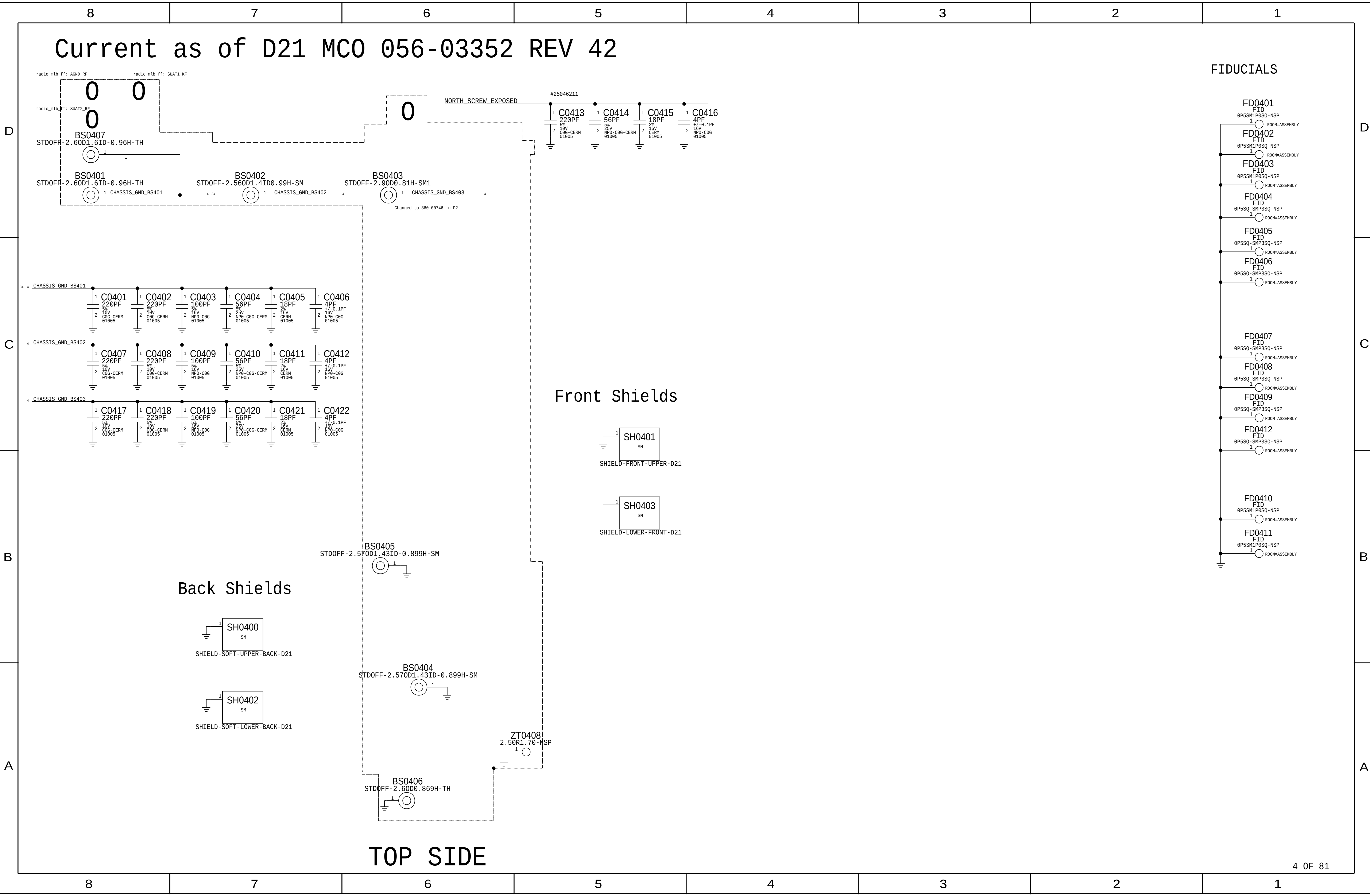
D

C

B

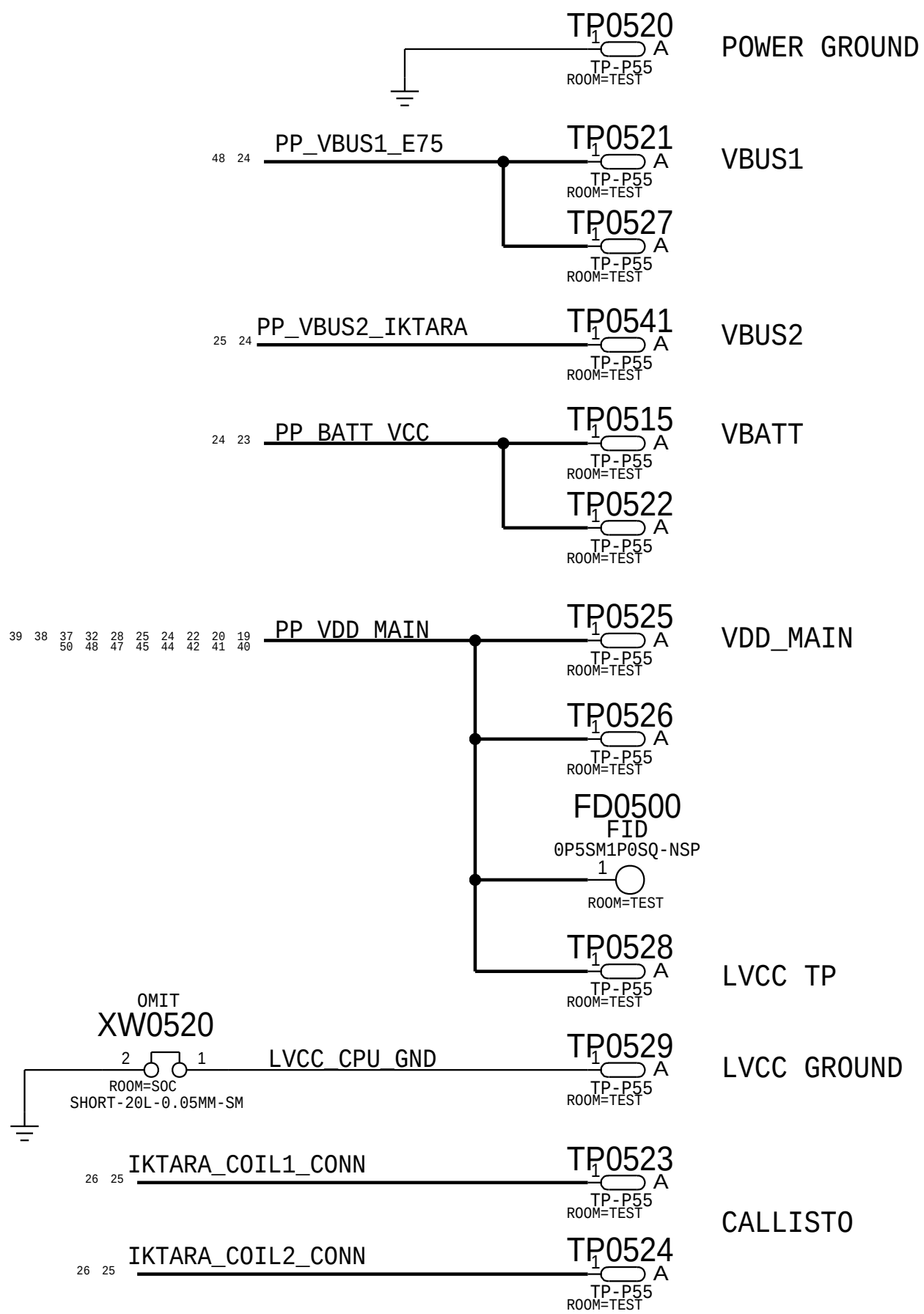
A

Current as of D21 MC0 056-03352 REV 42

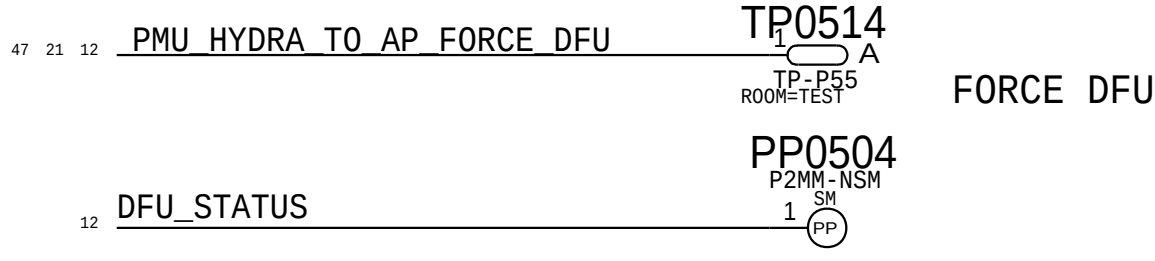


Test Points

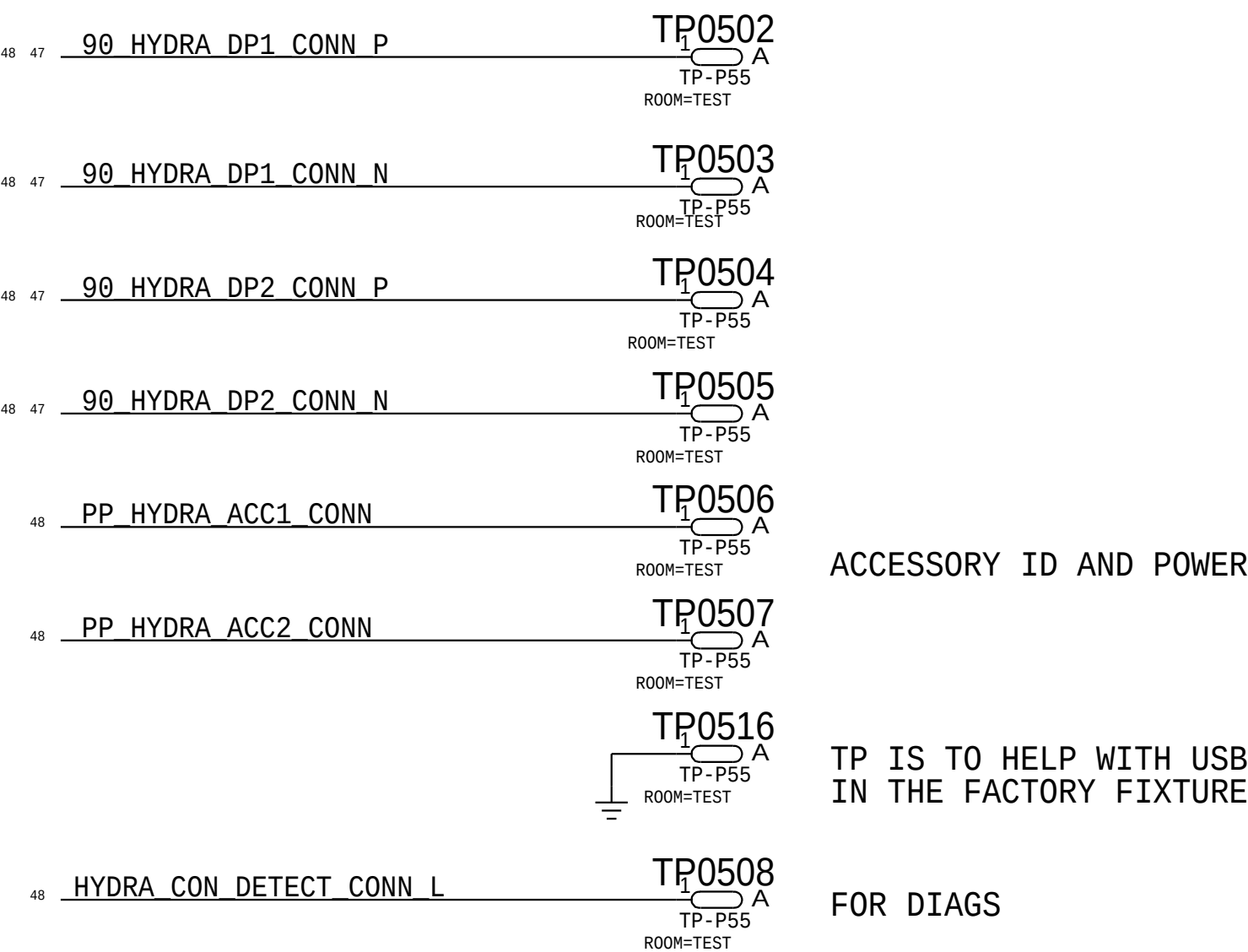
POWER



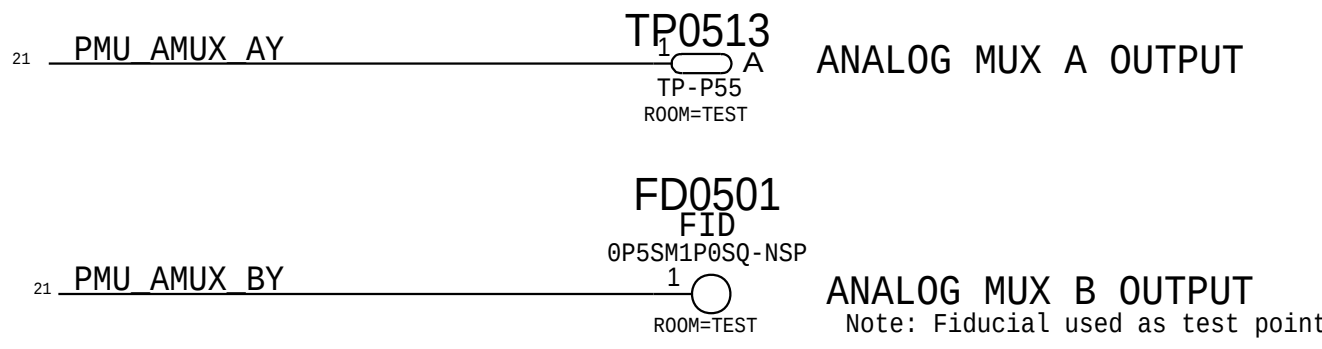
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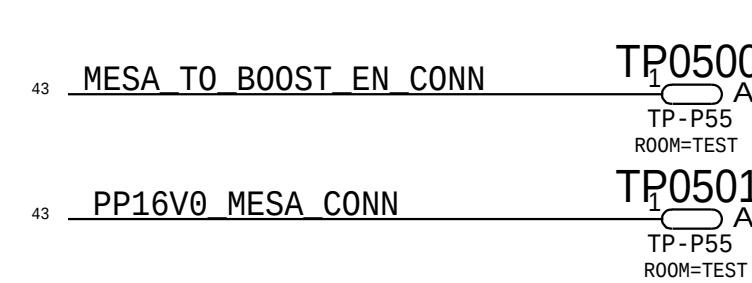
E75



AMUX



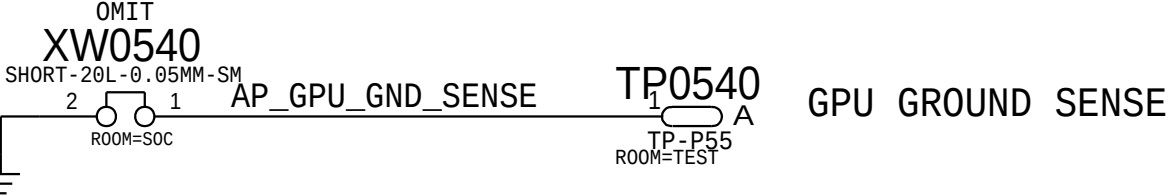
MOJAVE



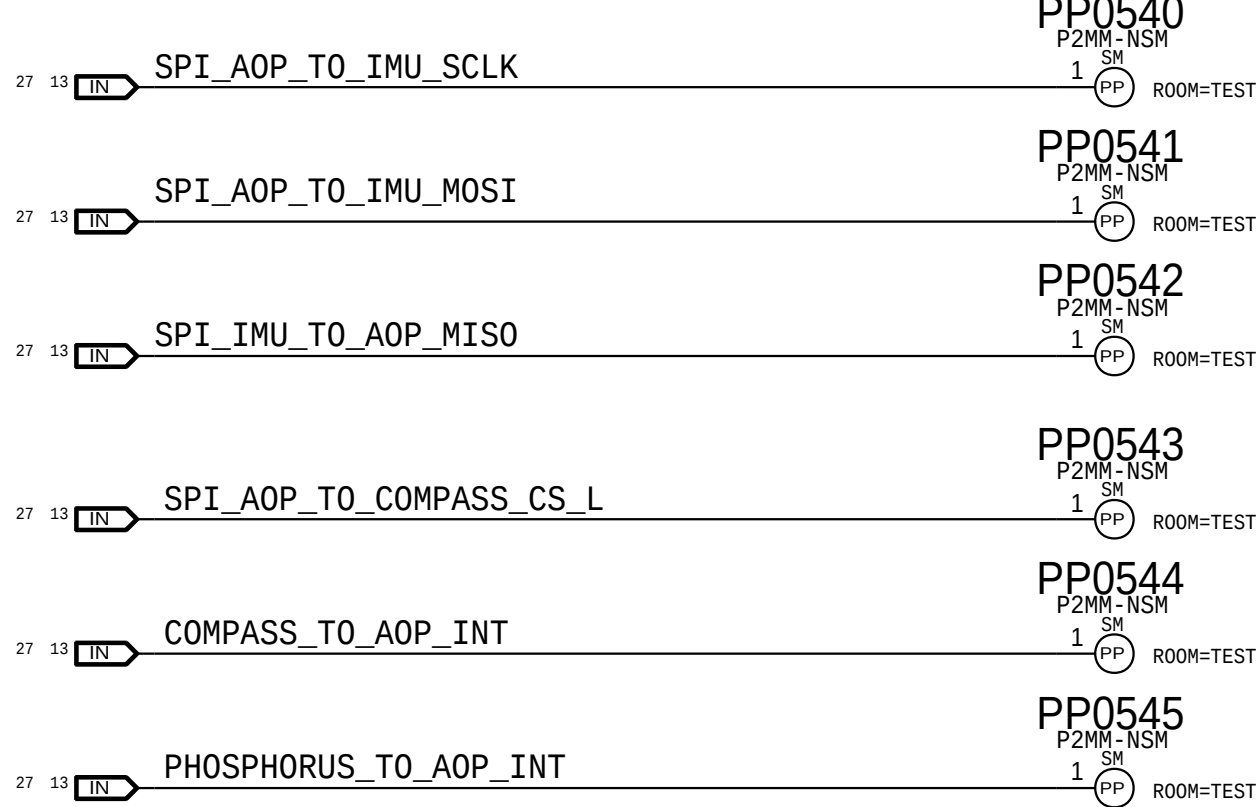
LCM



SENSE TP

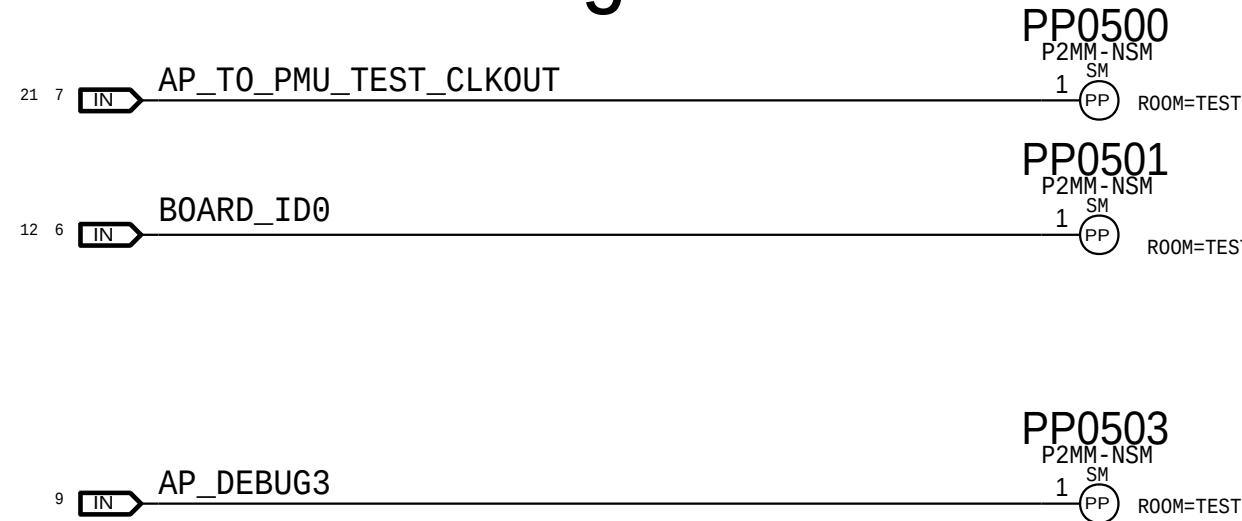


Sensor SPI

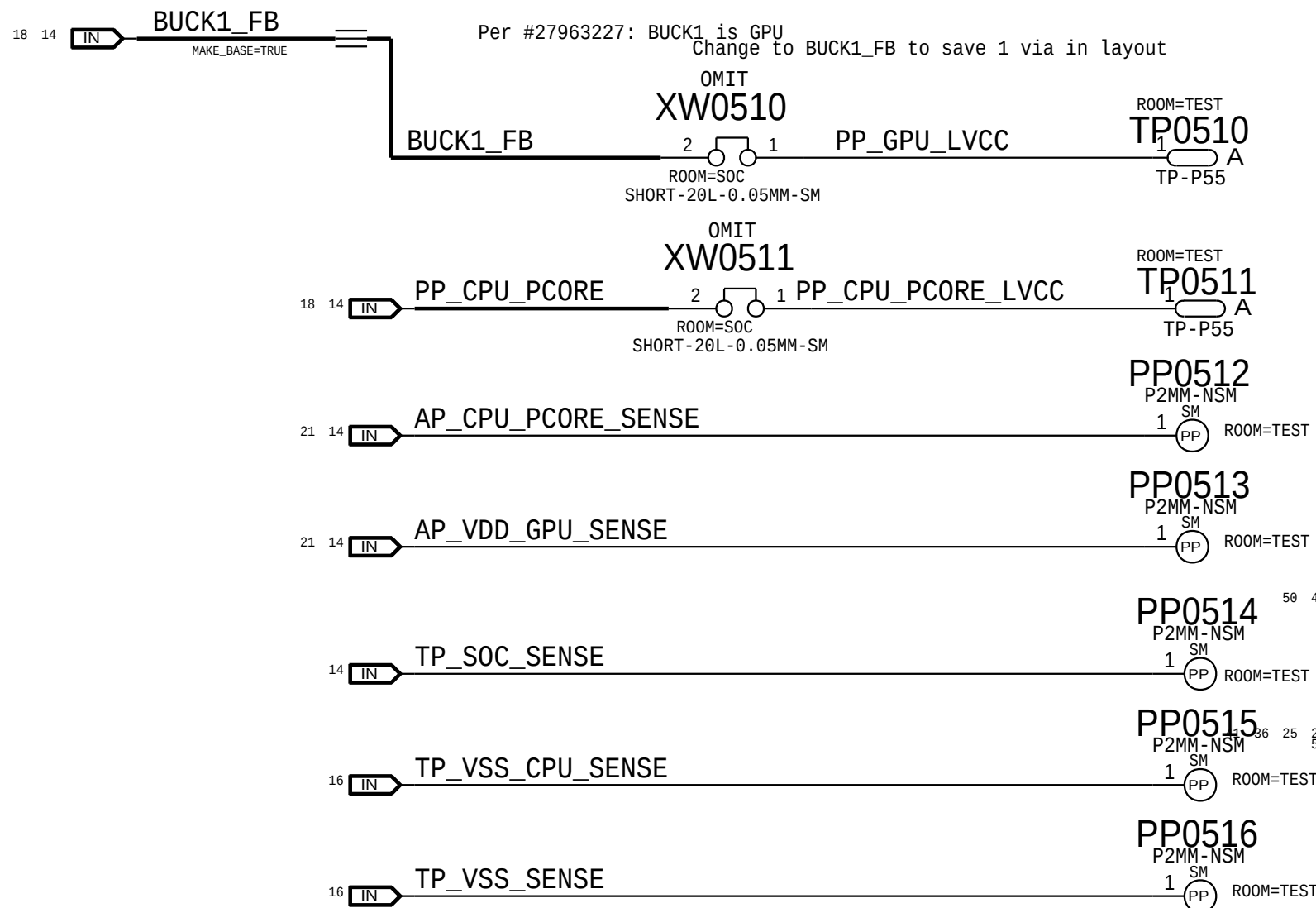


Probe Points

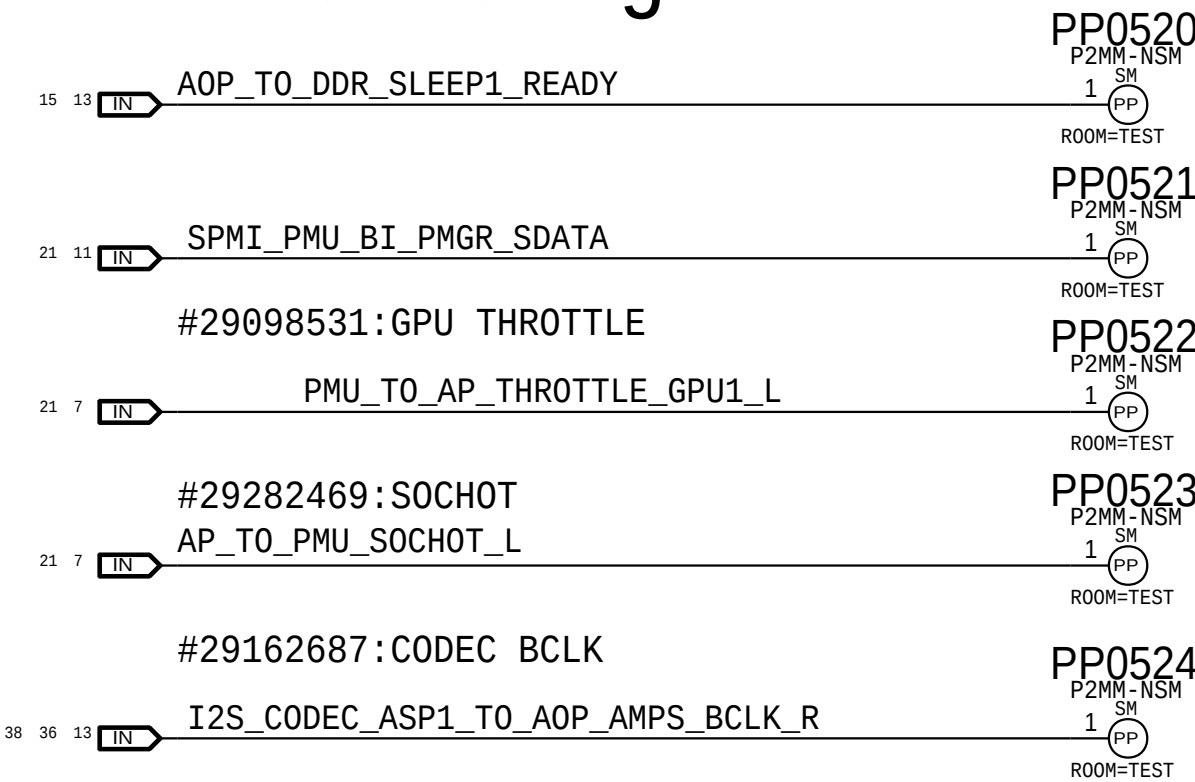
SOC Debug



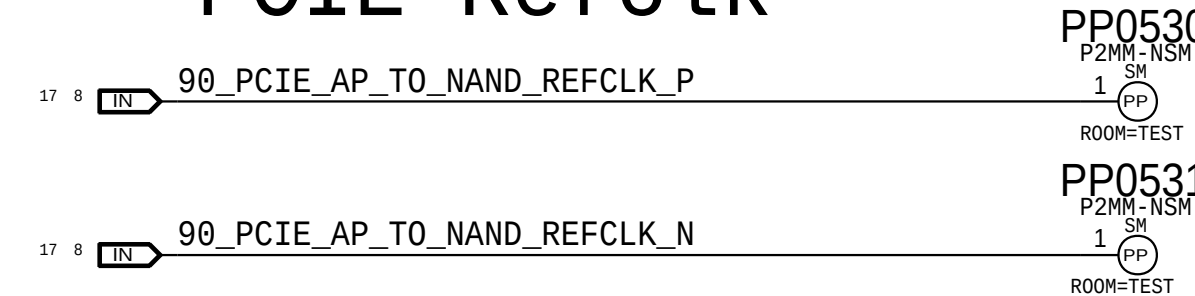
SOC CPU/GPU



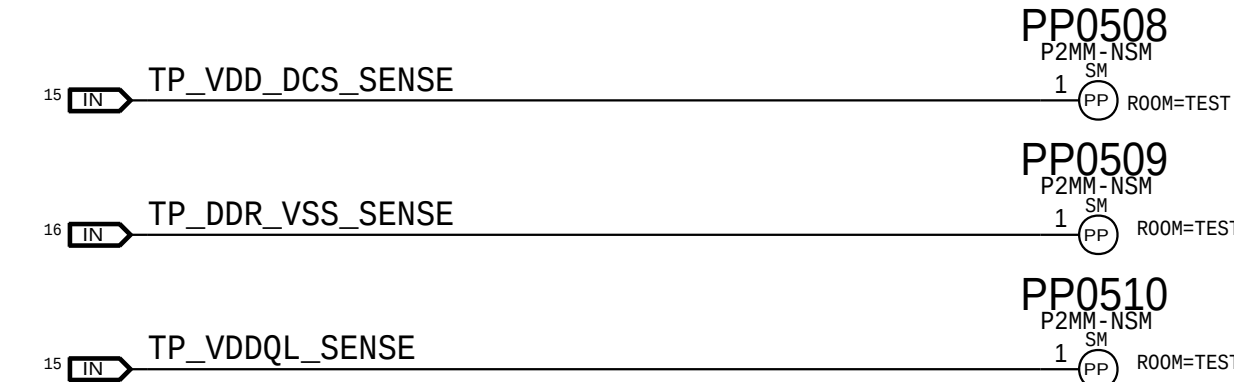
PMU Debug



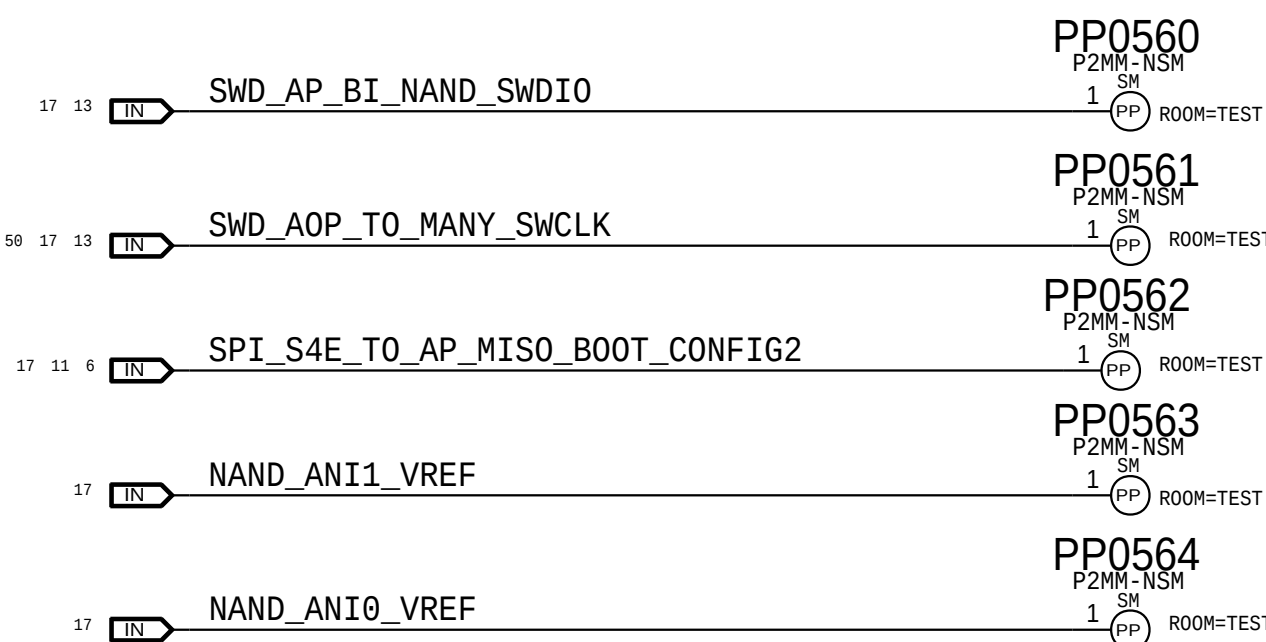
PCIE Refclk



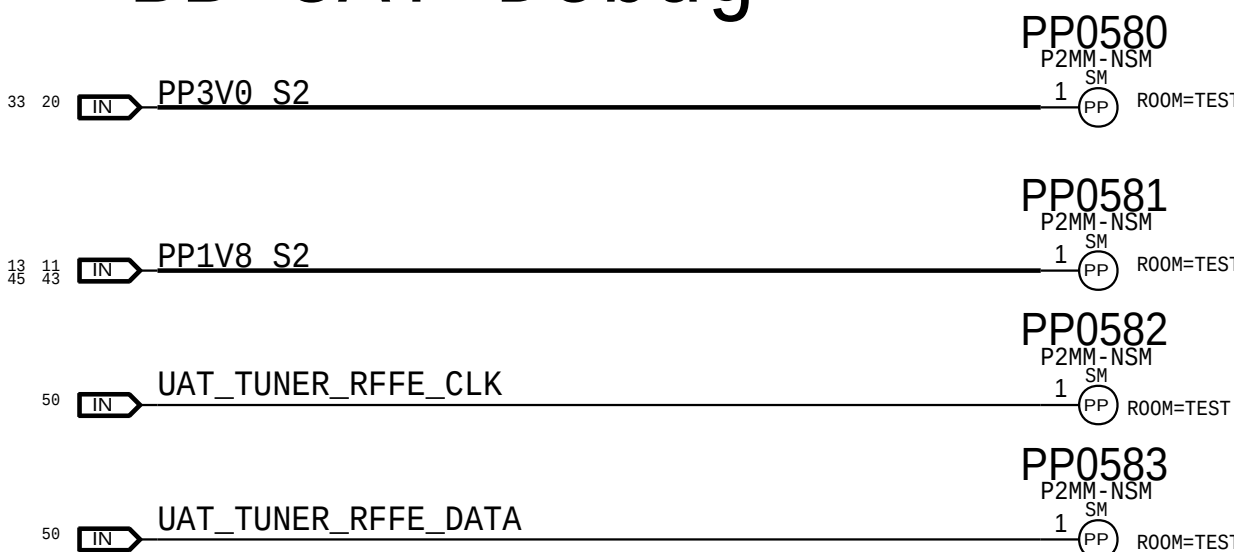
SOC DDR



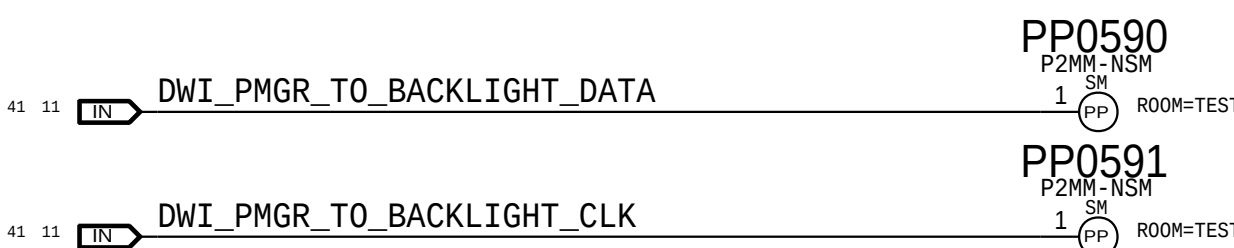
NAND Debug



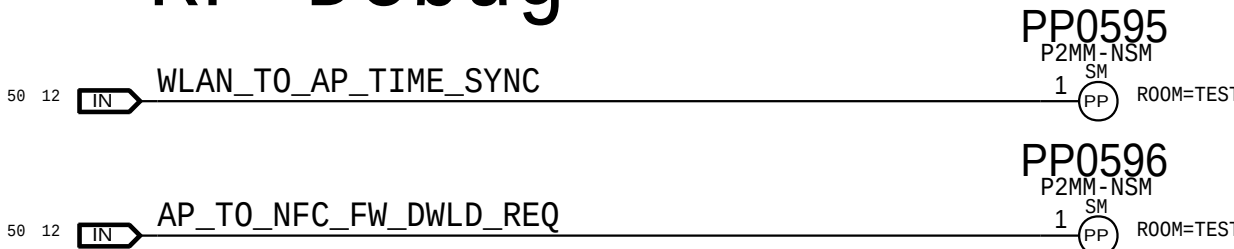
BB UAT Debug

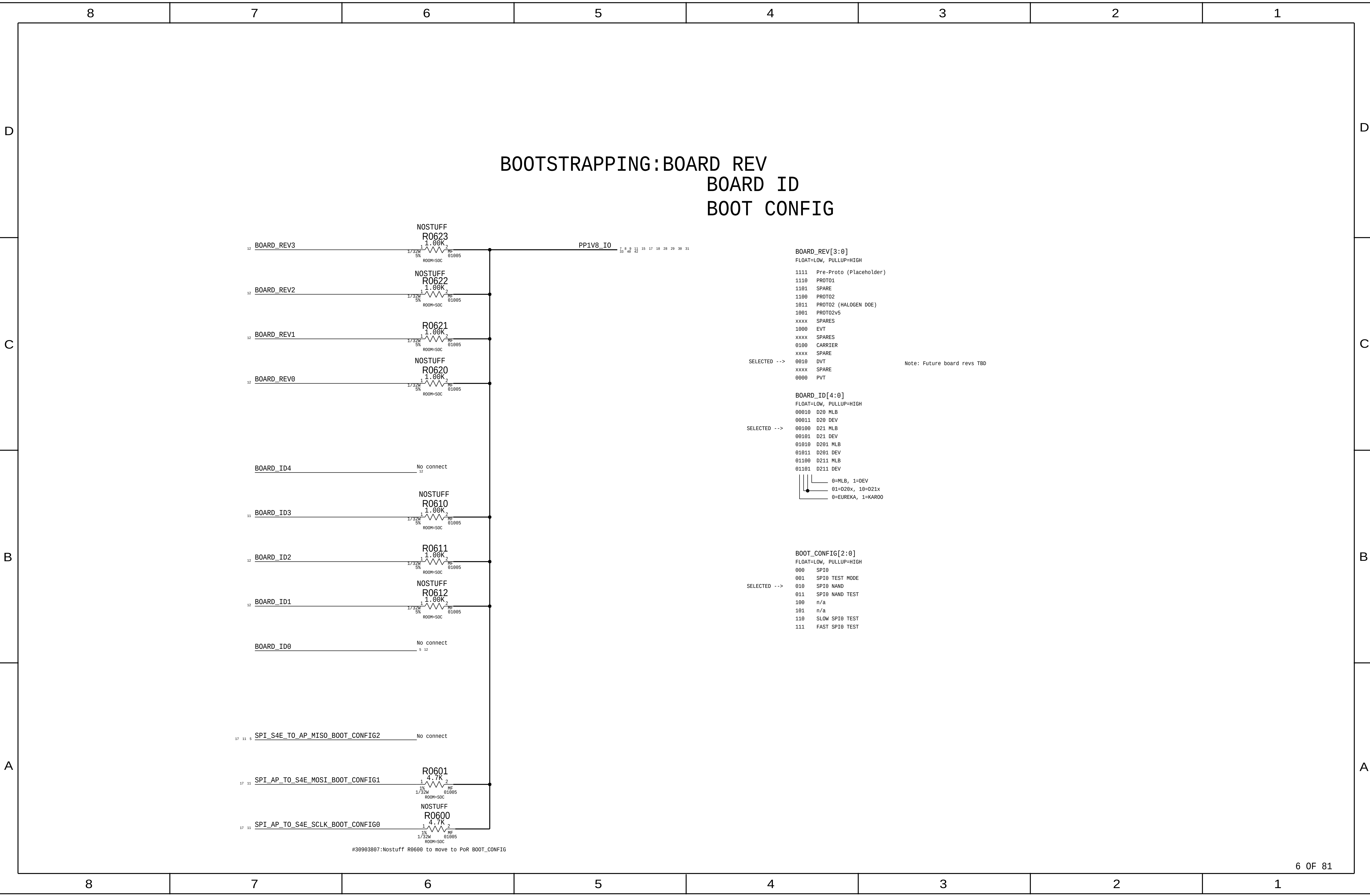


BACKLIGHT Debug

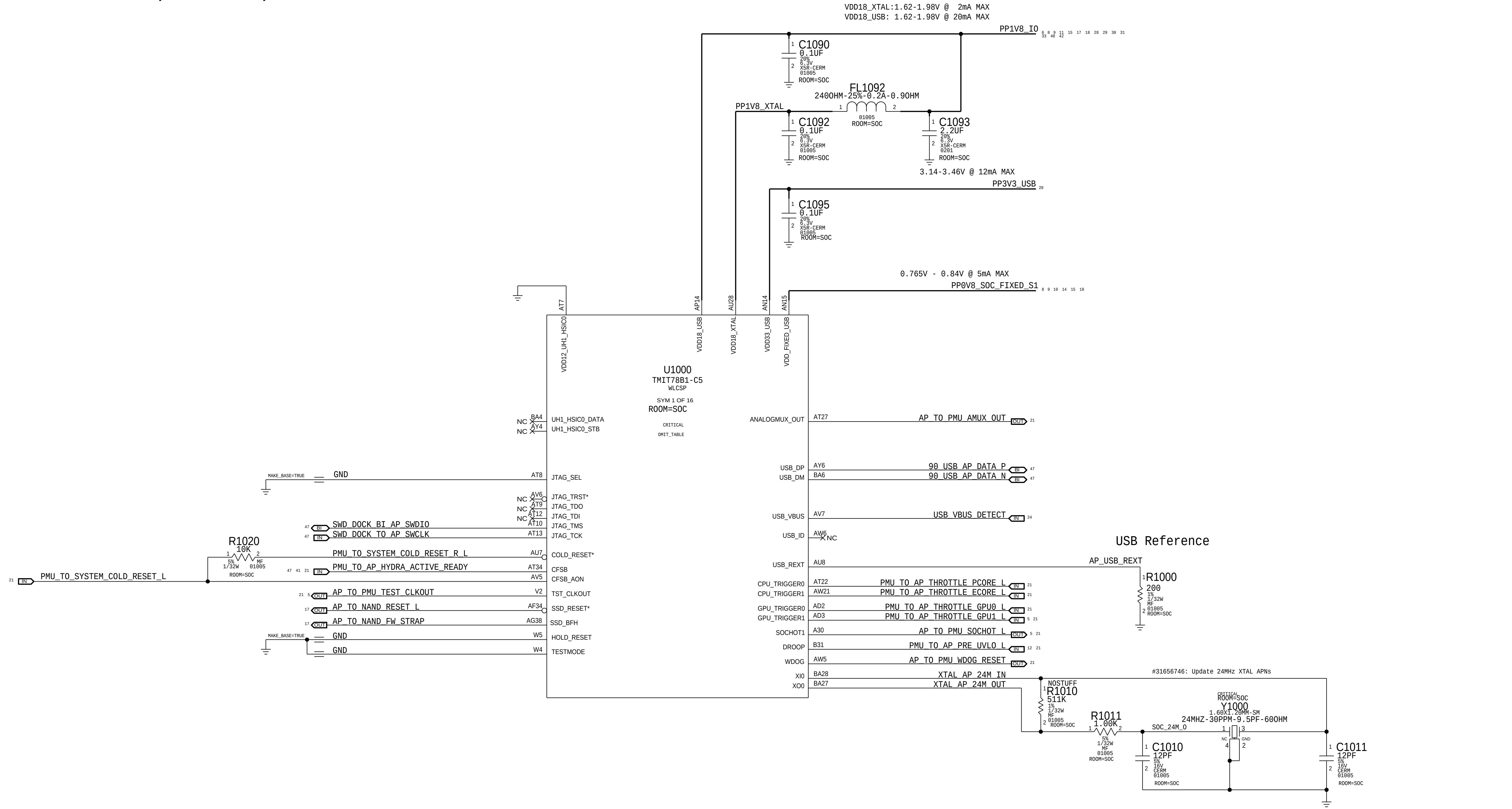


RF Debug





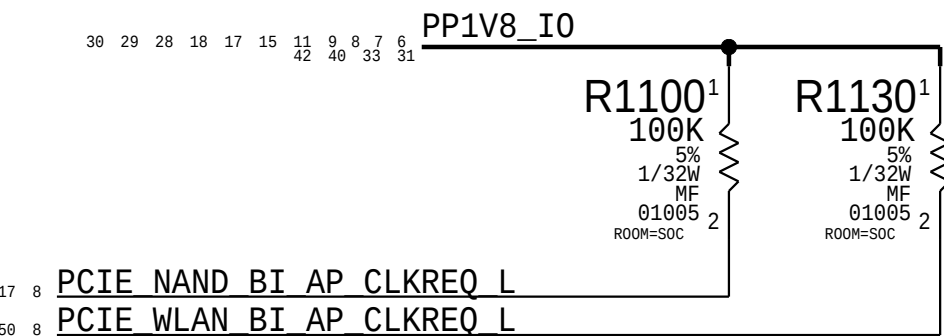
SOC - USB, JTAG, XTAL



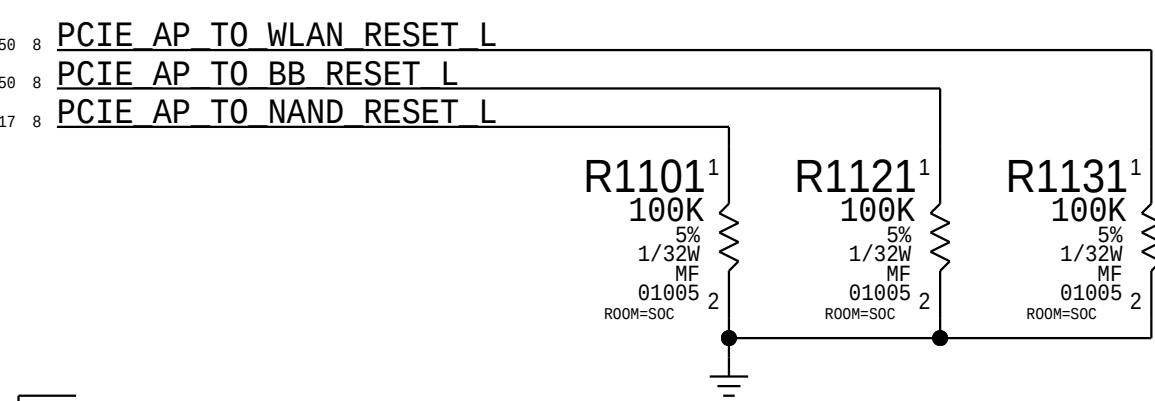
SOC - PCIE INTERFACES

D

PCie Clock Request Pull-Ups



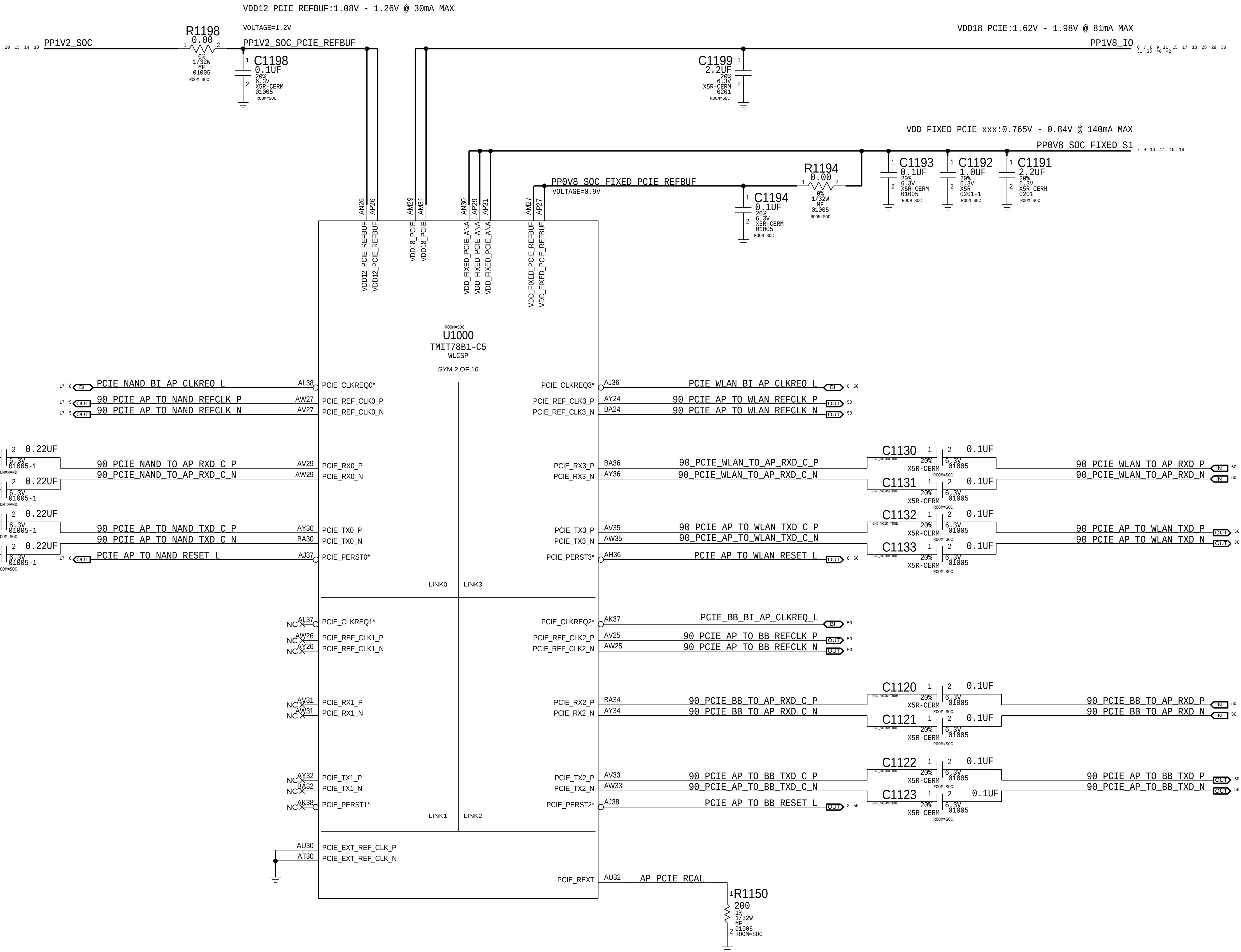
PCie Reset Pull-Downs



PCIE LINK 0

B

A



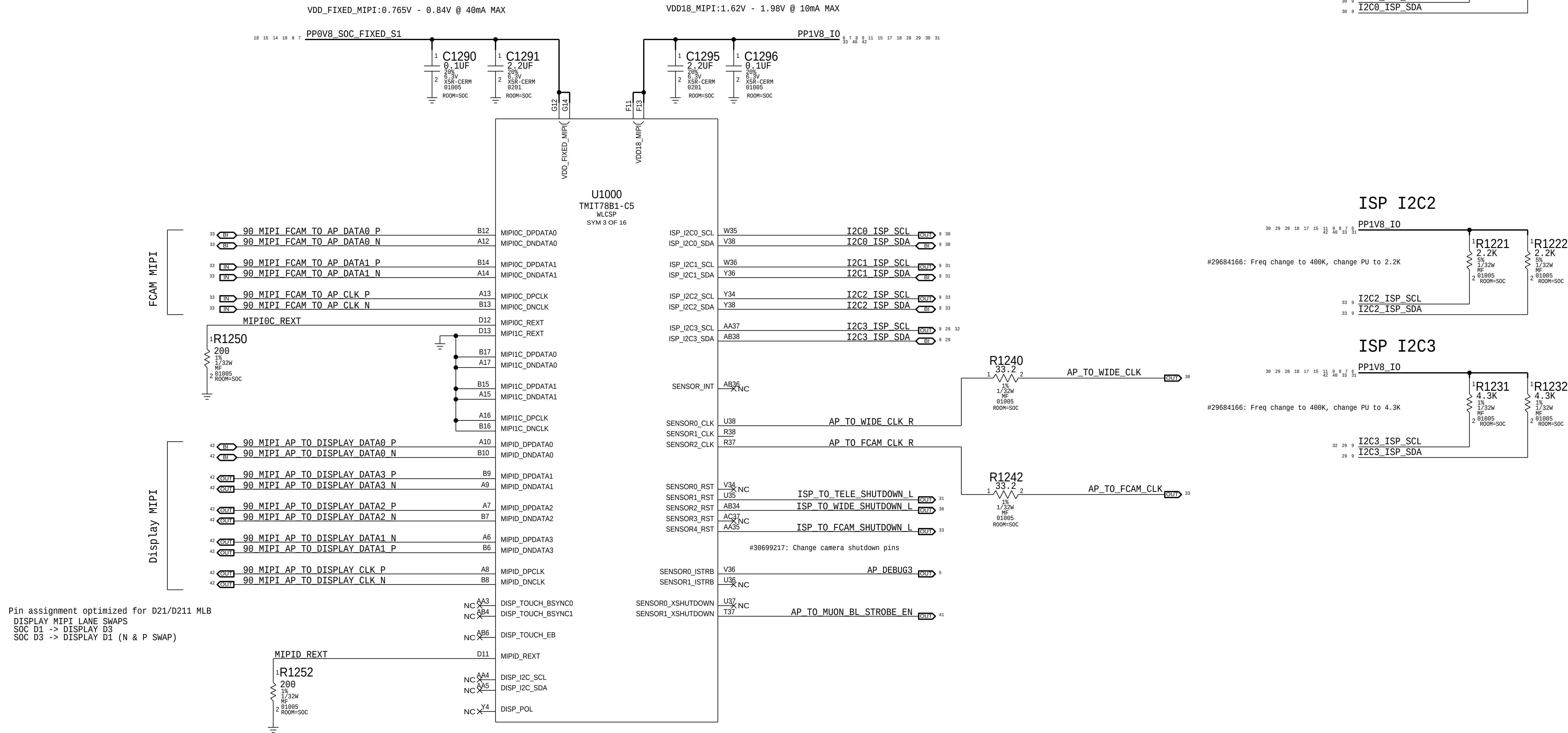
D

C

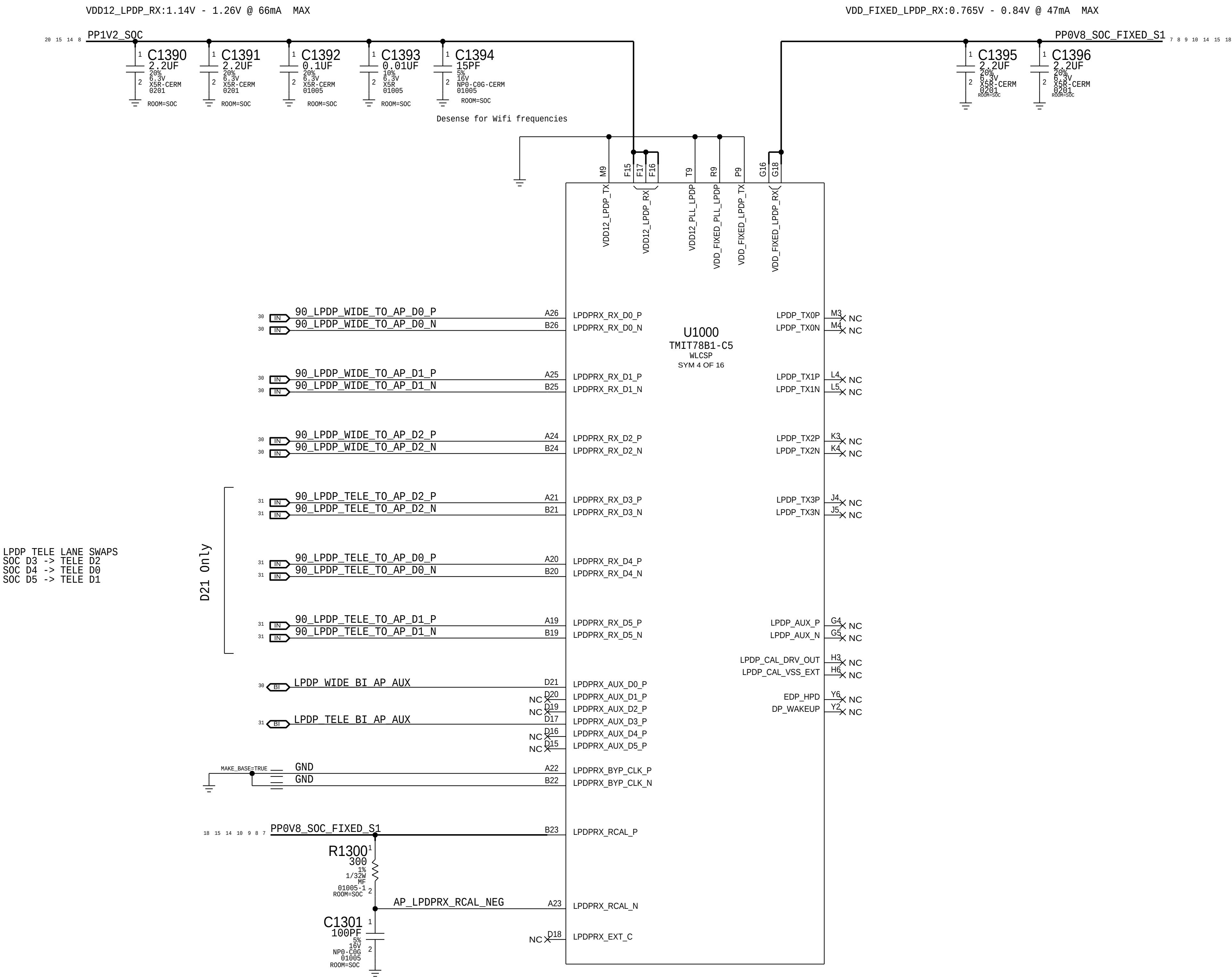
B

A

SOC - MIPI & ISP INTERFACES



SOC - LPDP INTERFACES



SOC - SERIAL INTERFACES

D

C

B

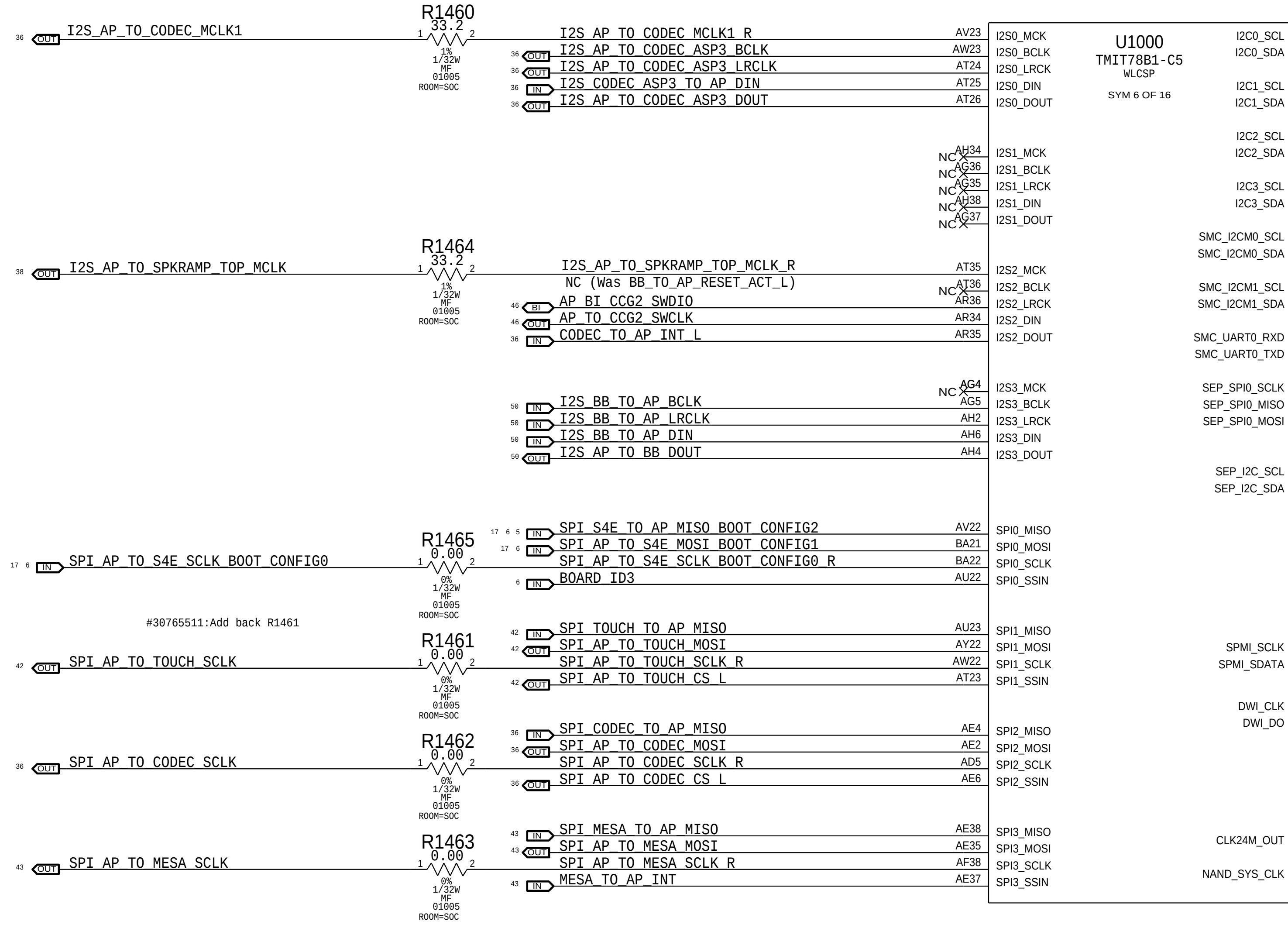
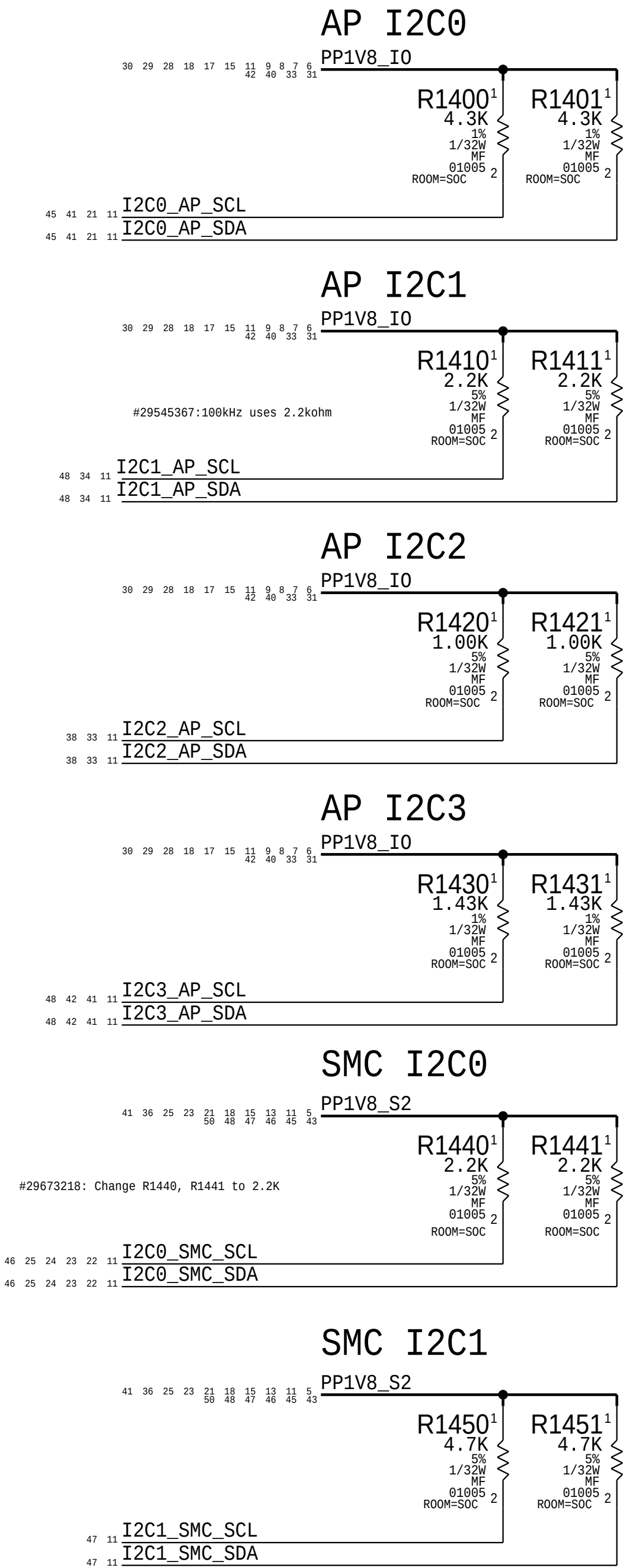
A

D

C

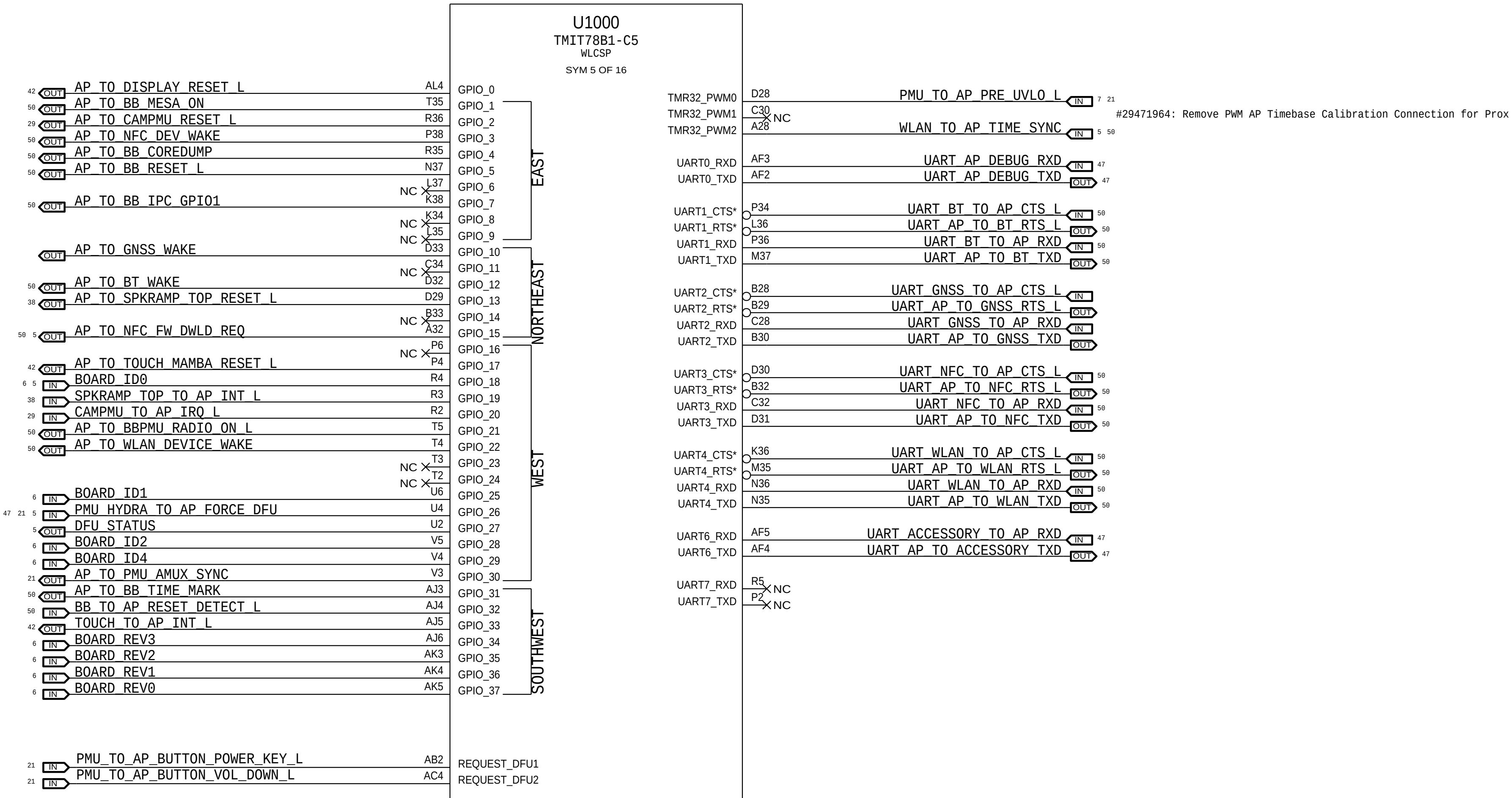
B

A



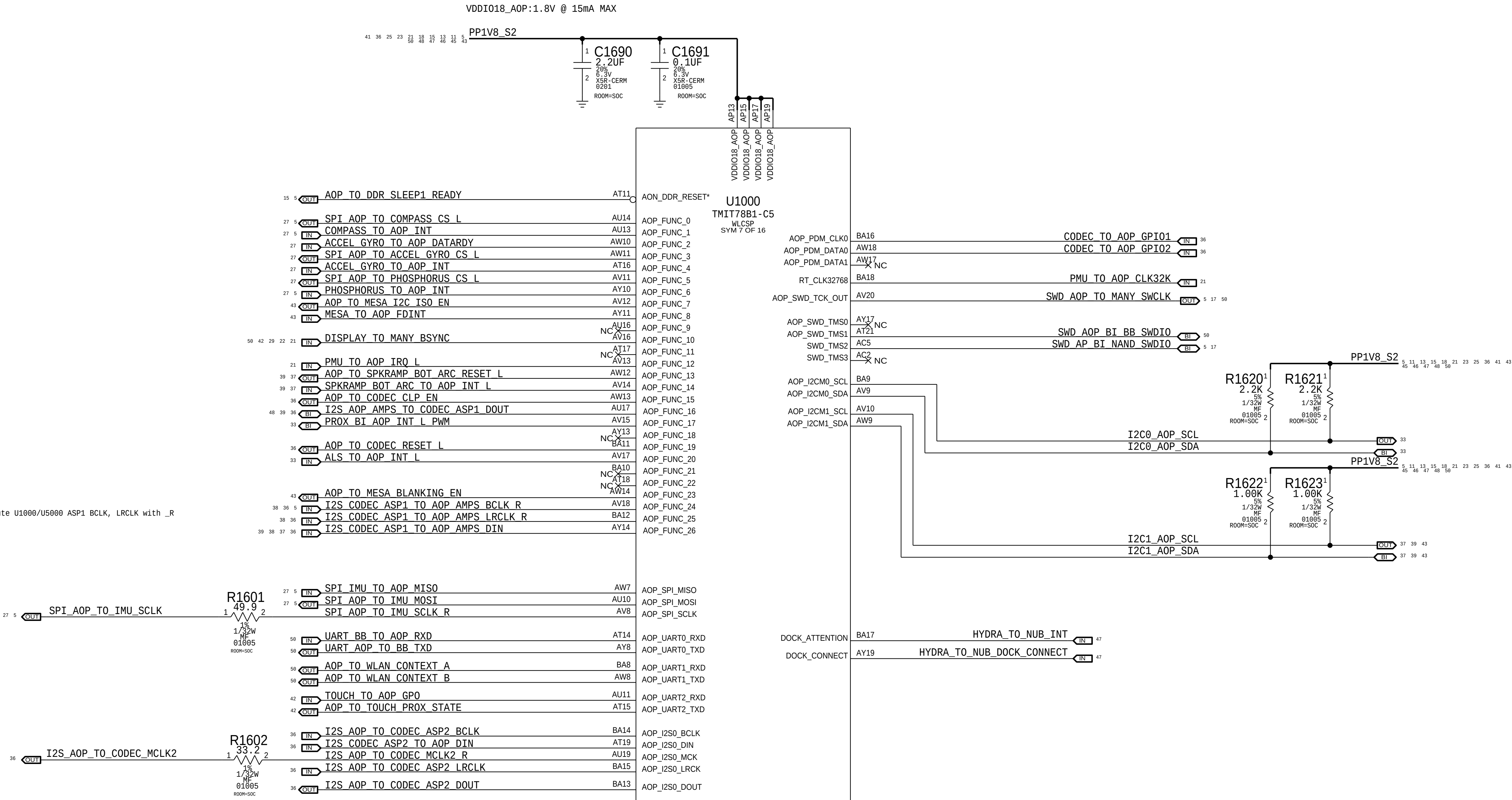
SPI: Route as Daisy-Chain (No T's Allowed)

SOC - GPIO INTERFACES



SOC - AOP

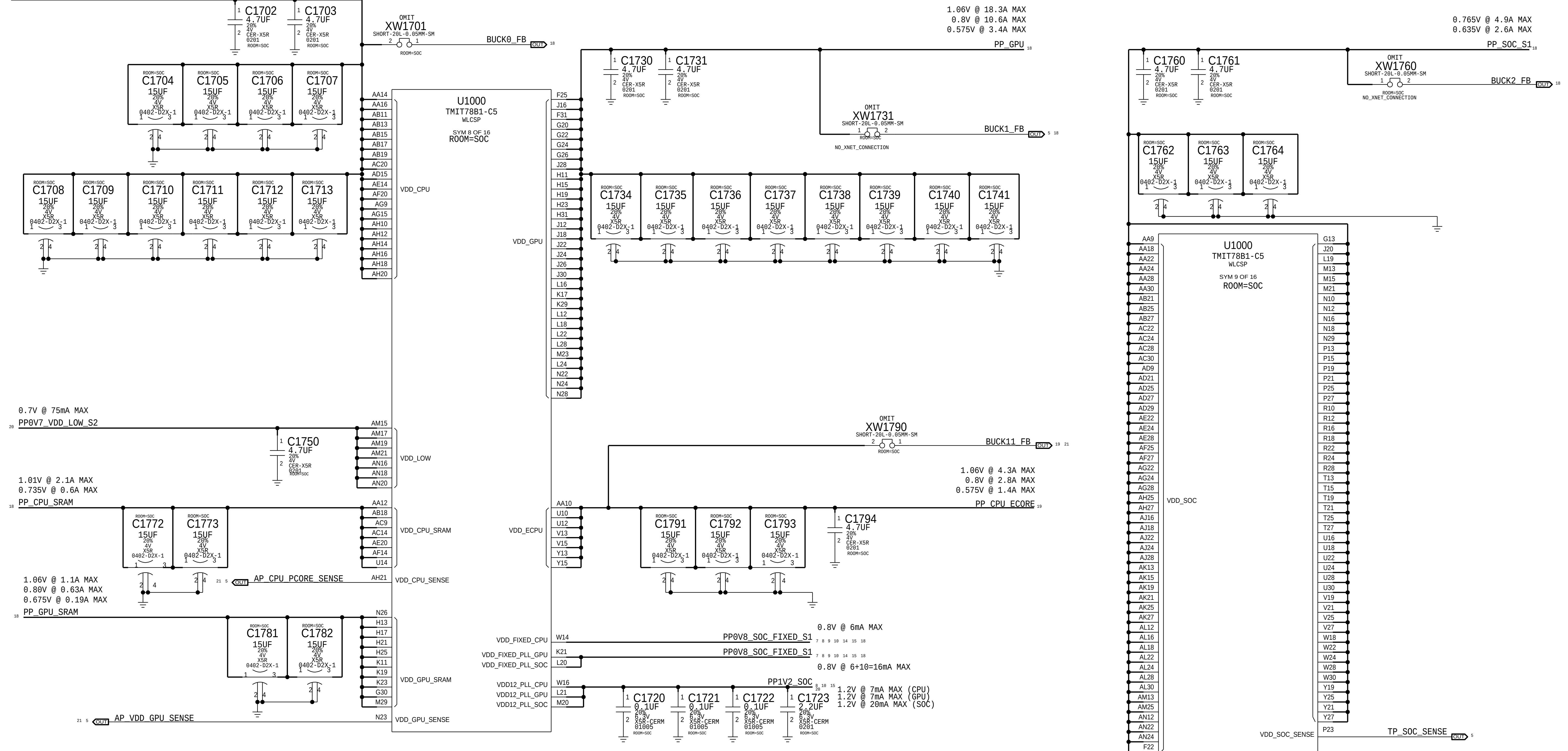
#38638490: Route U1000/U5000 ASP1 BCLK, LRCLK with _R



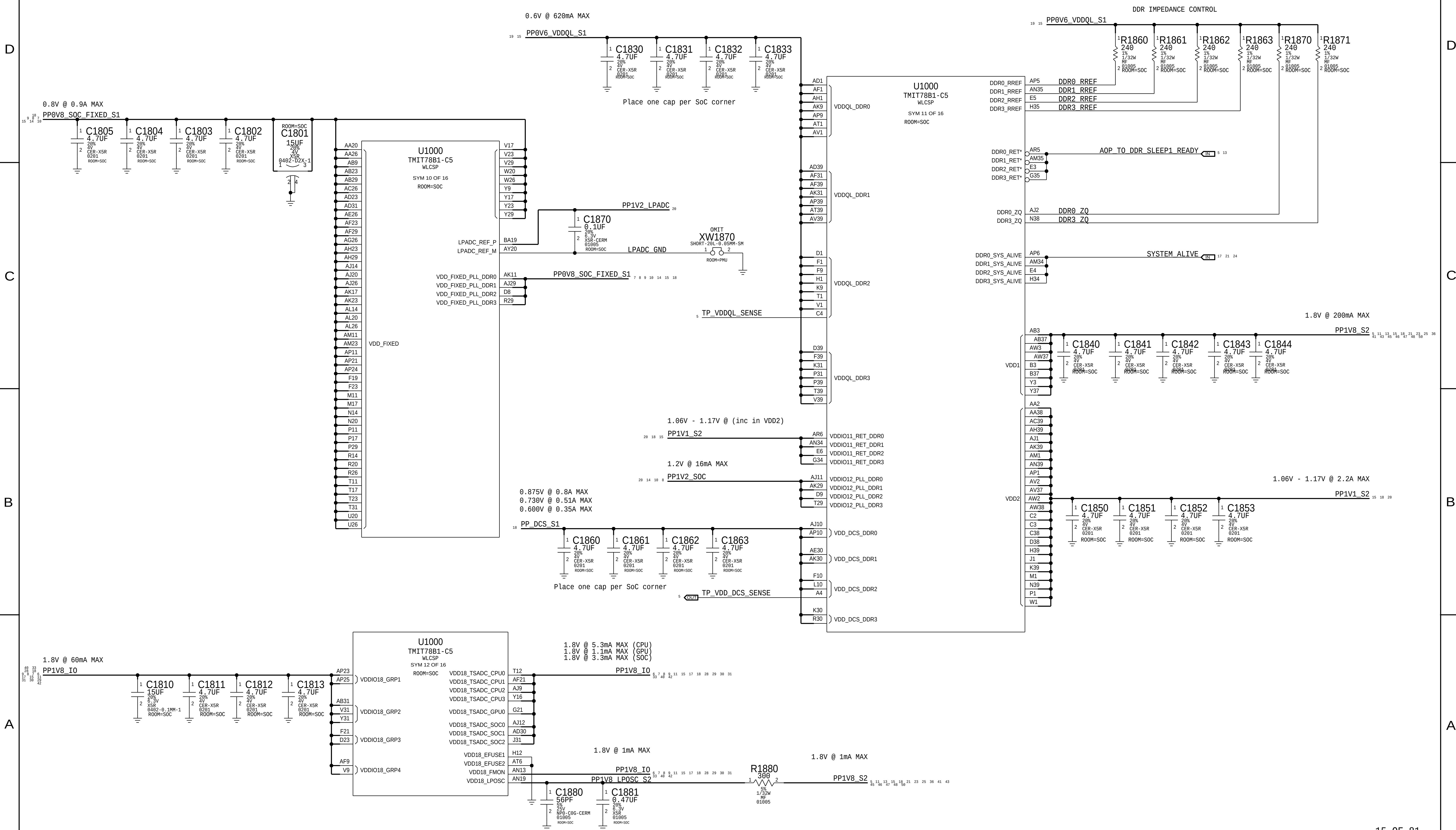
SOC - CPU, GPU & SOC RAILS

1.06V @ 11.0A MAX
0.8V @ 6A MAX
0.575V @ 2.7A MAX

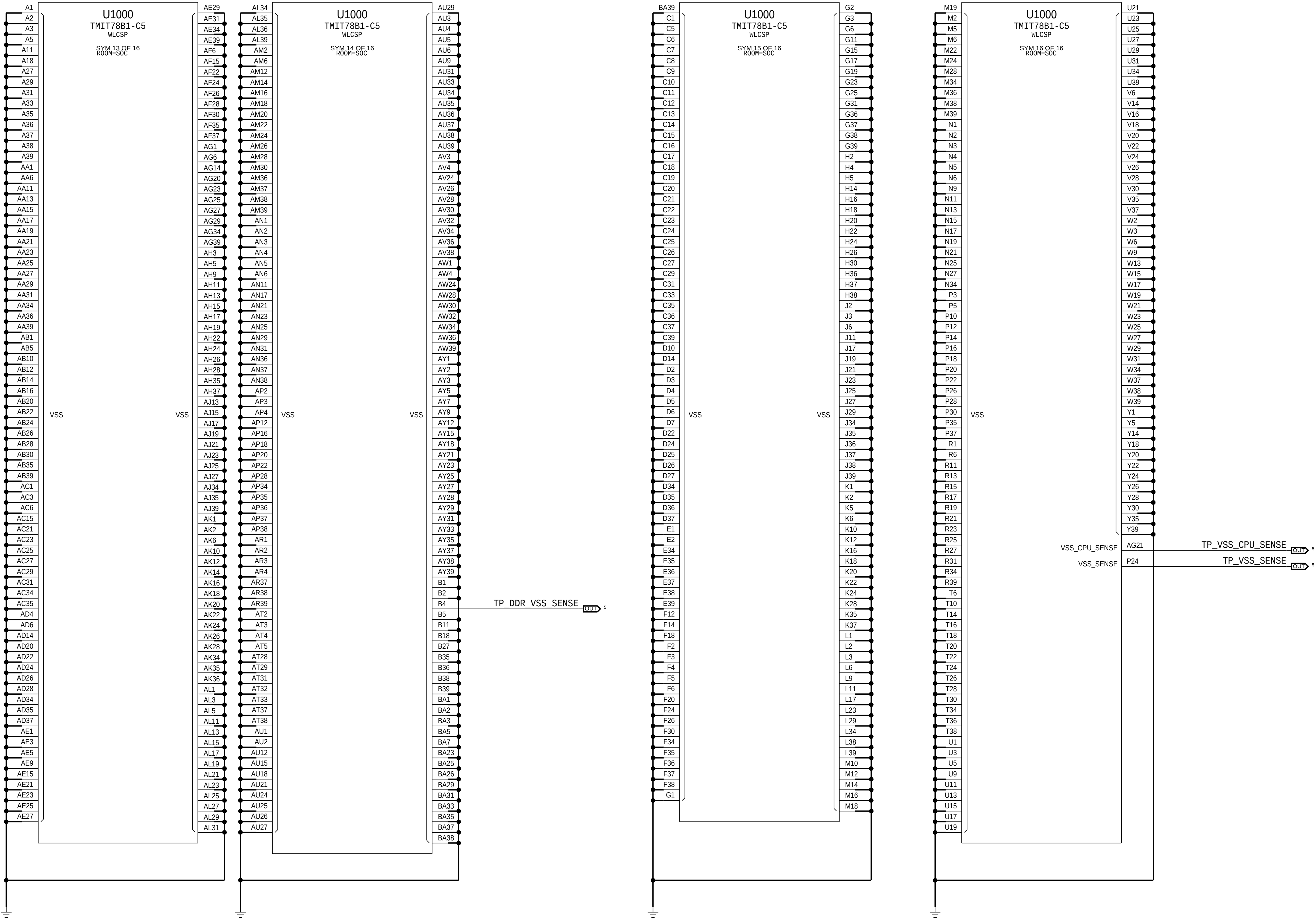
```
18 5 PP_CPU_PCORE
```

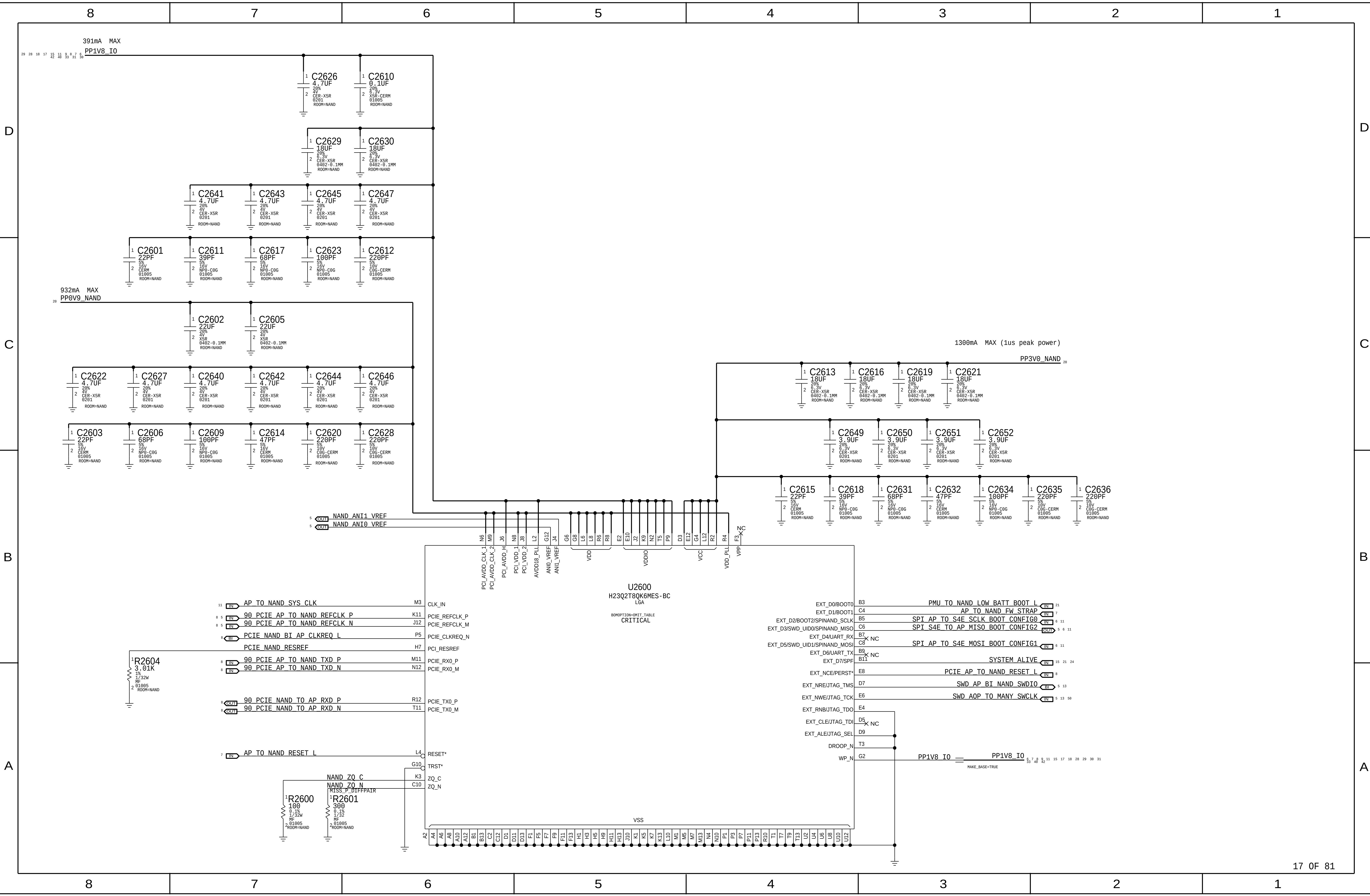


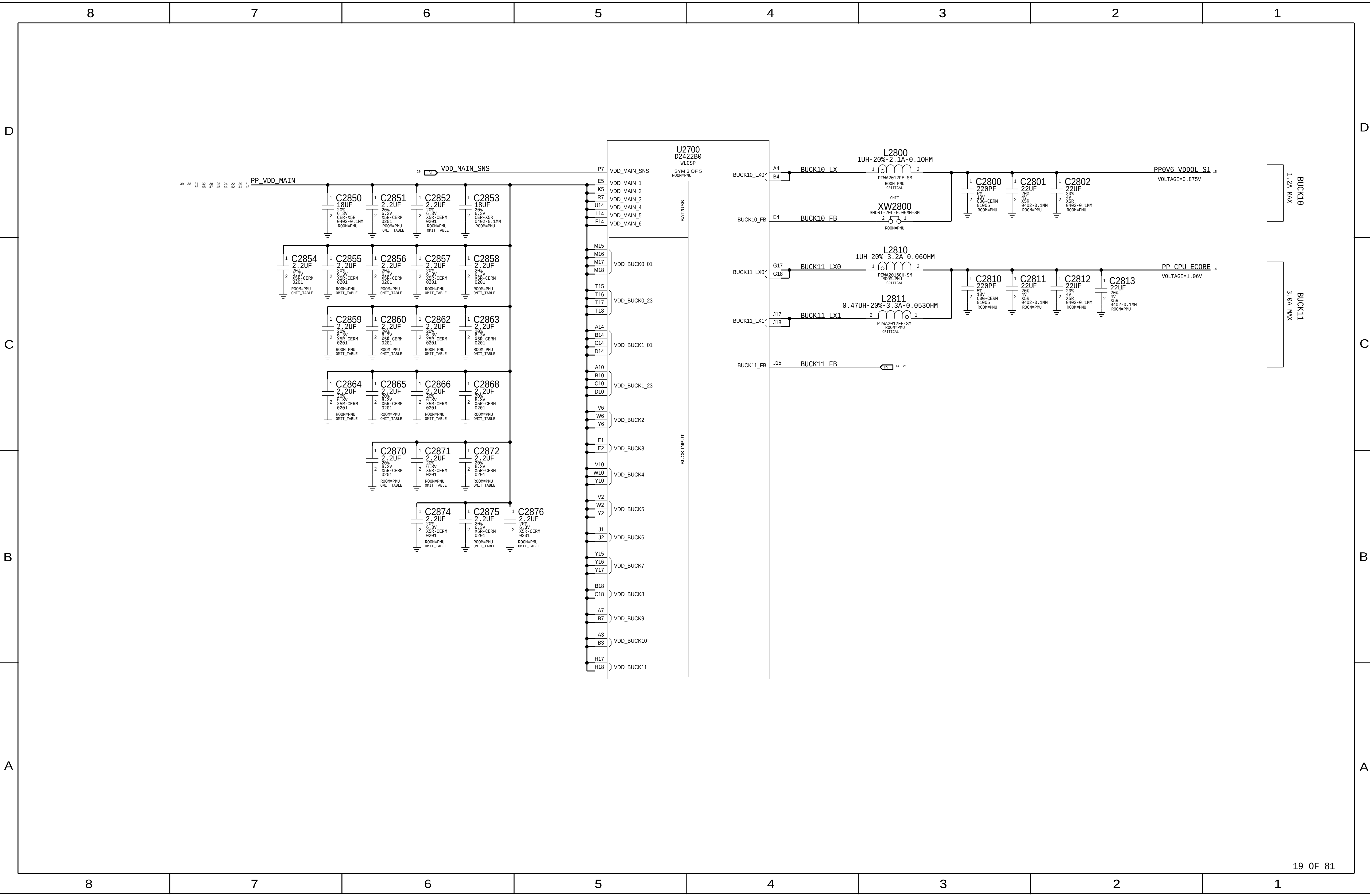
SOC - POWER SUPPLIES

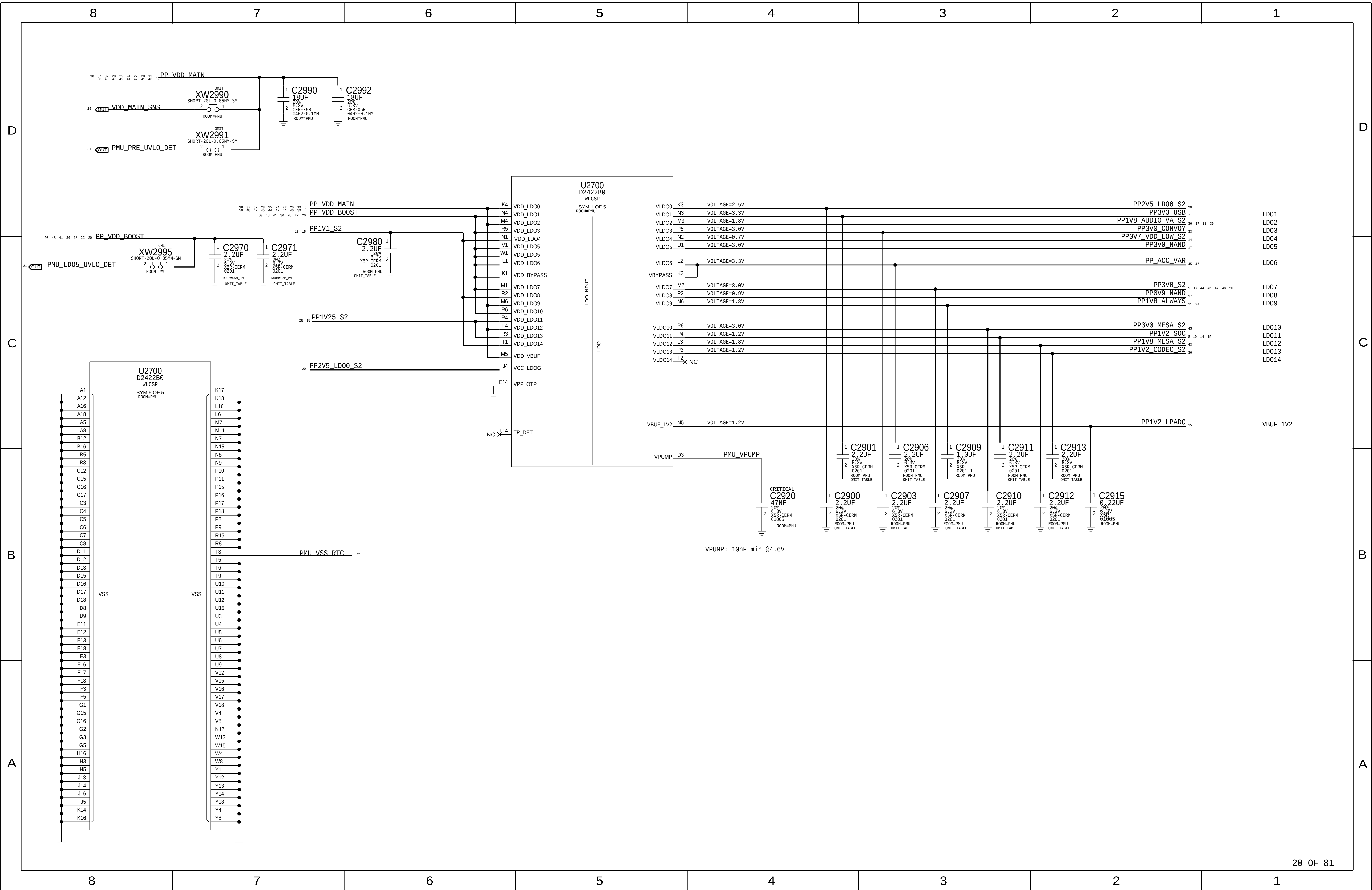


SOC - POWER SUPPLIES

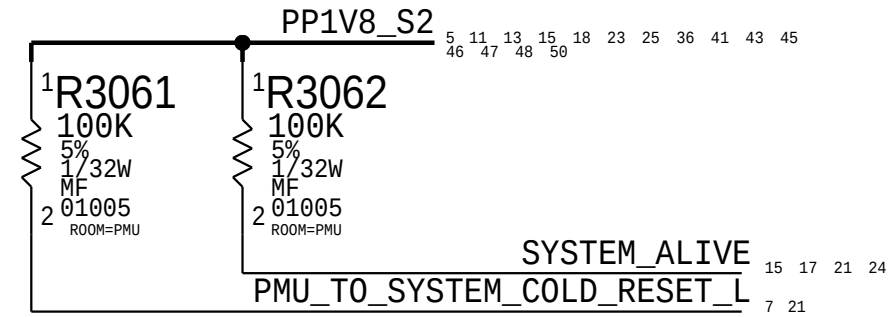




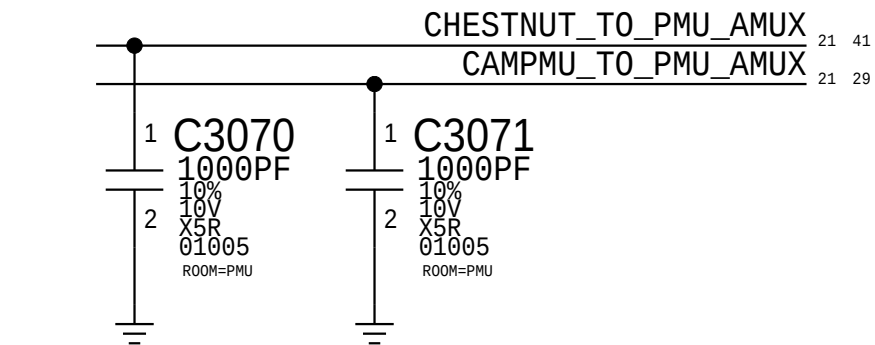




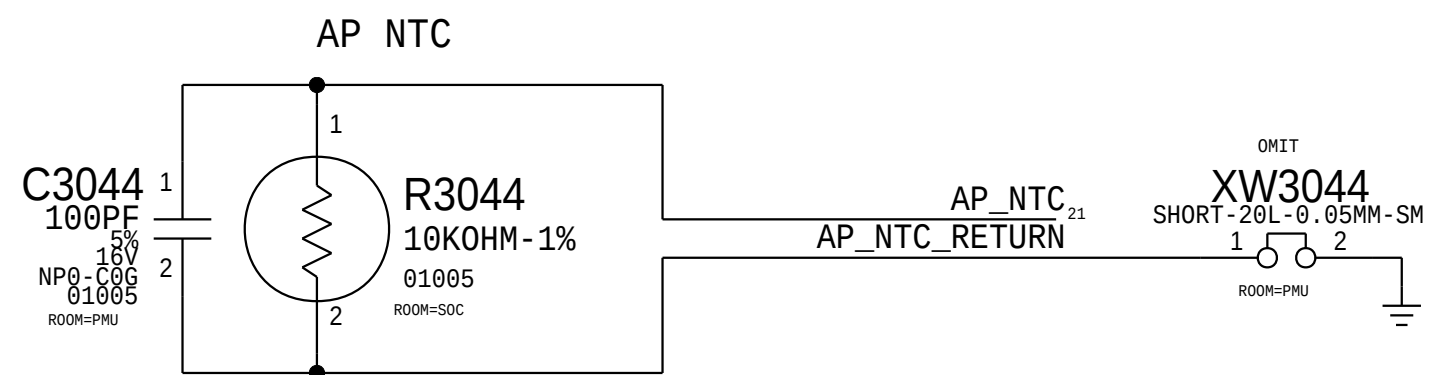
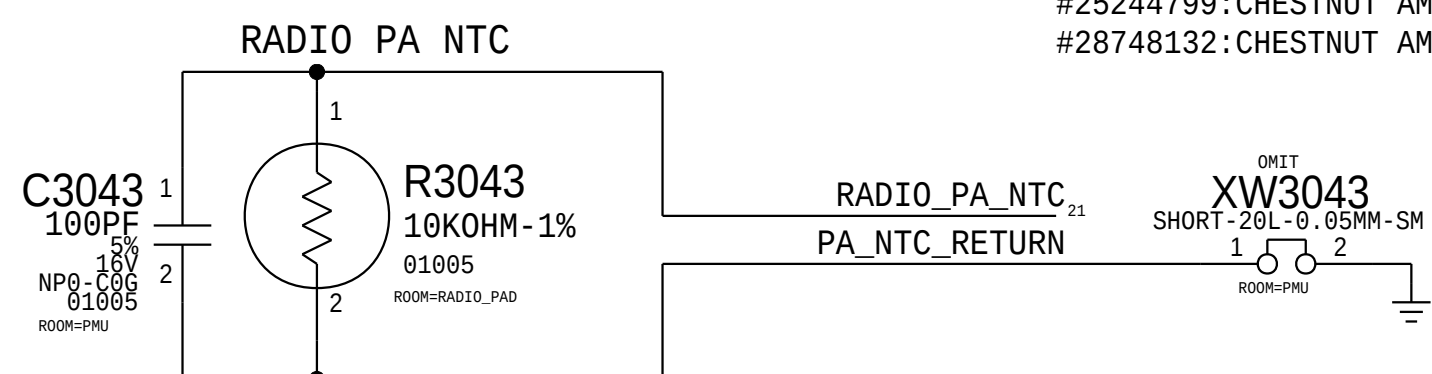
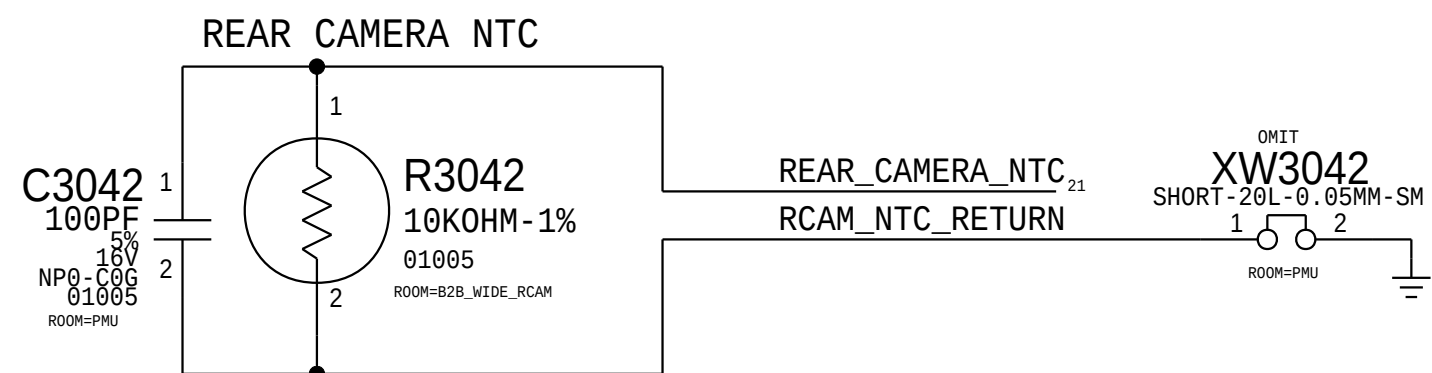
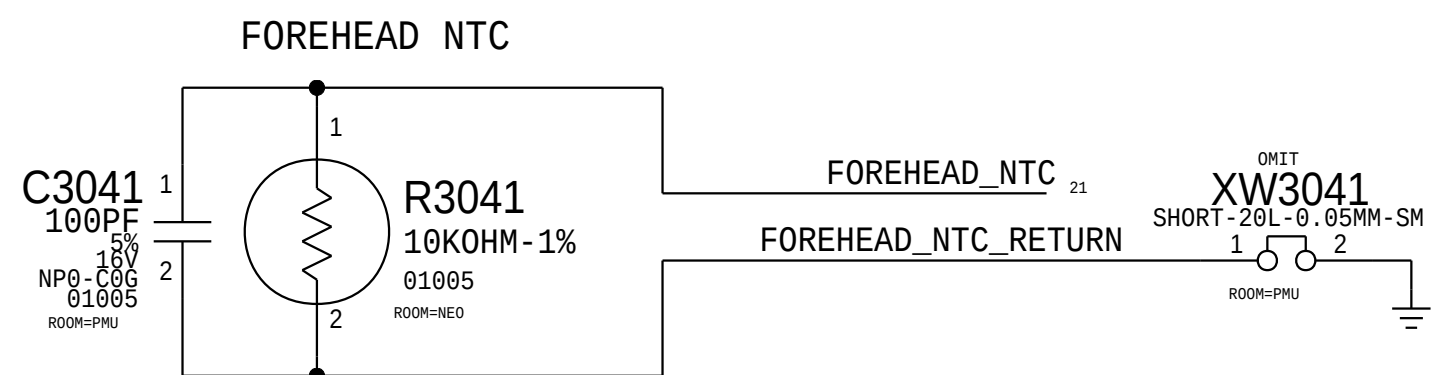
COLD_RESET & SYSTEM_ALIVE



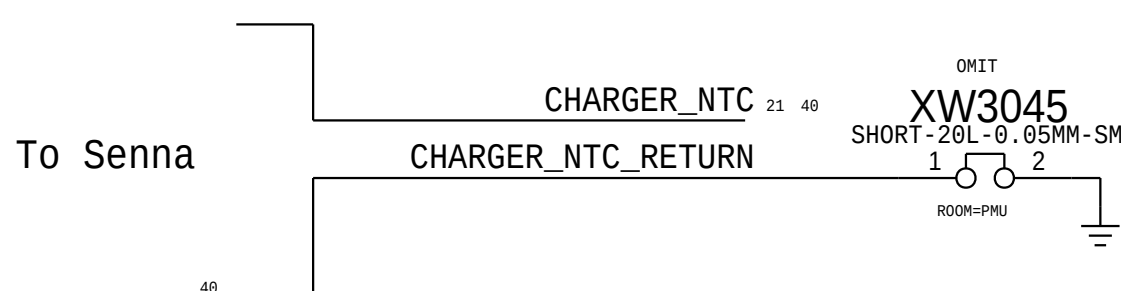
Chestnut / Ansel AMUX



NTCs



CHARGER NTC (IN SIP)



NOTE:100PF CAPS ARE THE SAMPLING CAPS FOR PMU ADC

#29404057:Damping R for ACTIVE_RDY

#29300440:Connect PMU crash to SoC

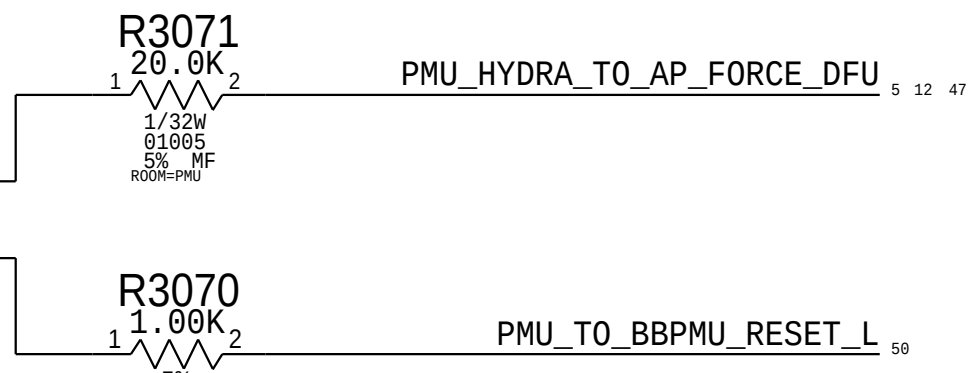
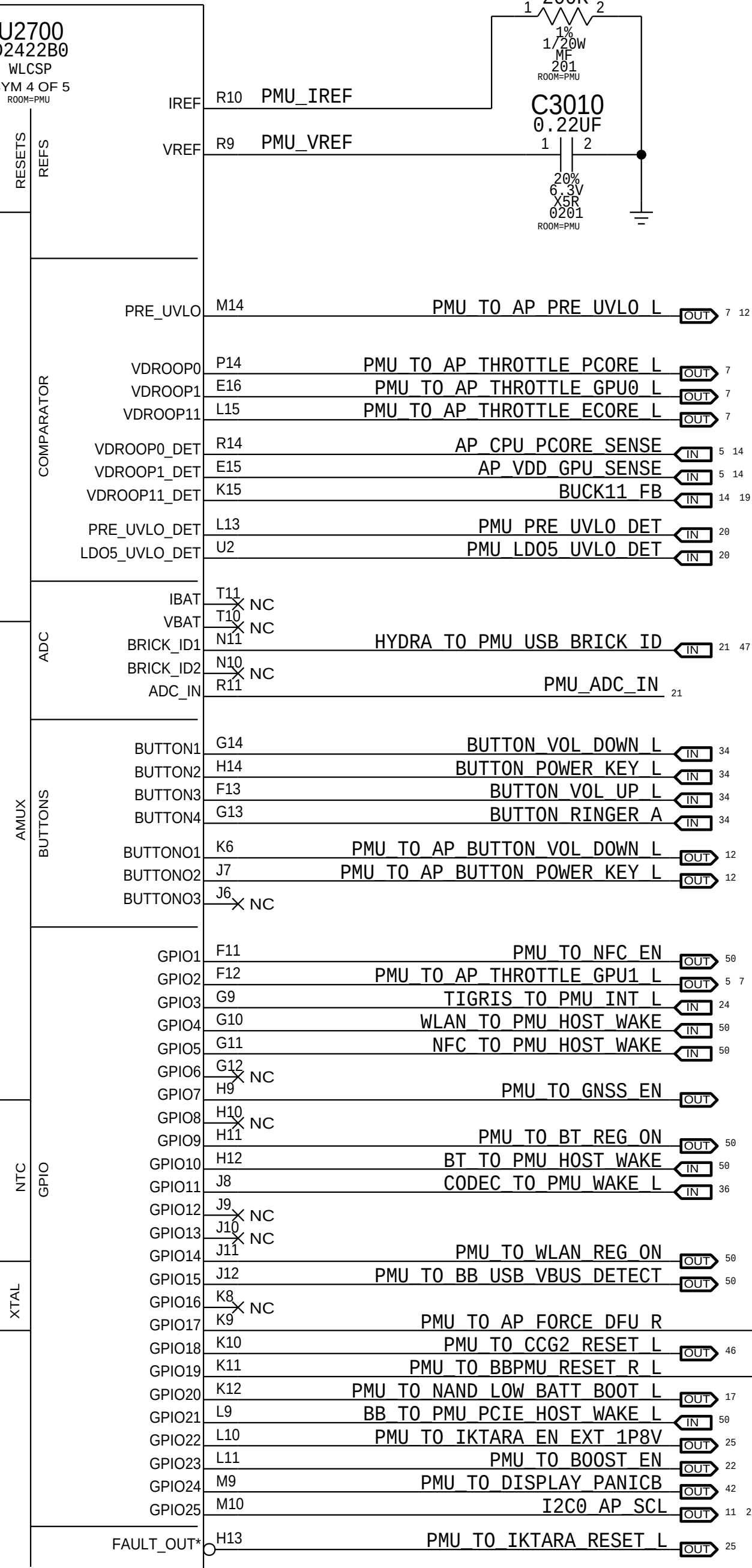
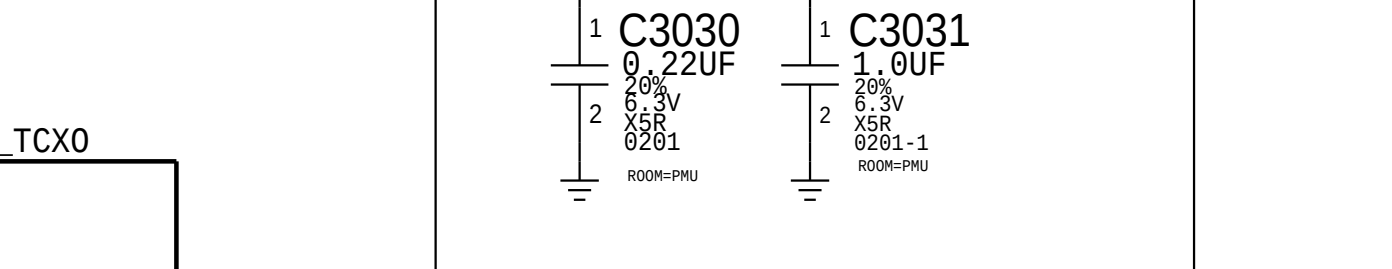
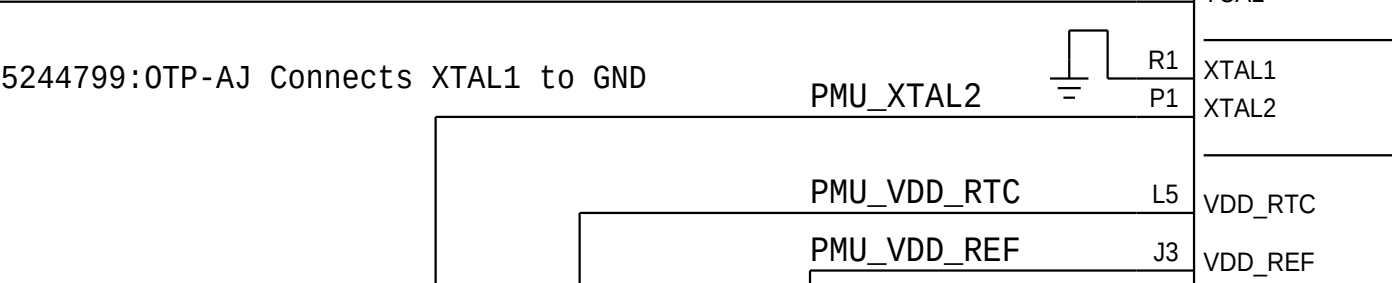
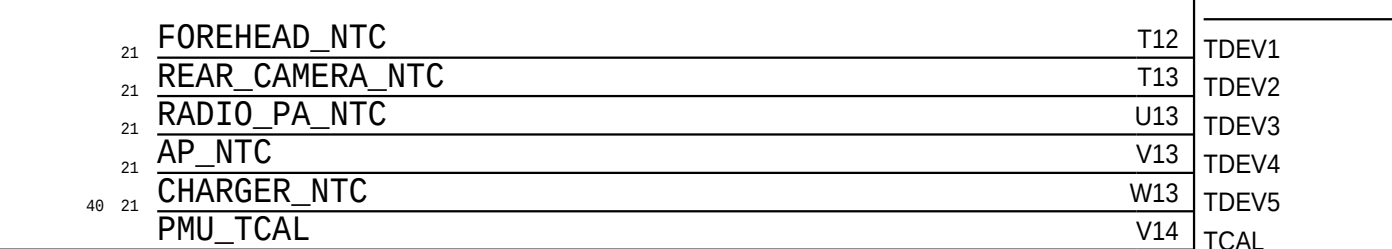
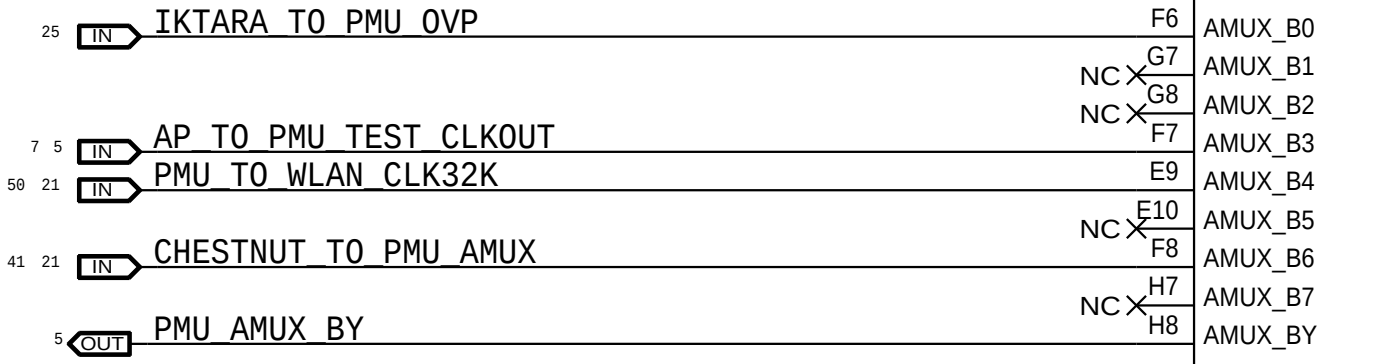
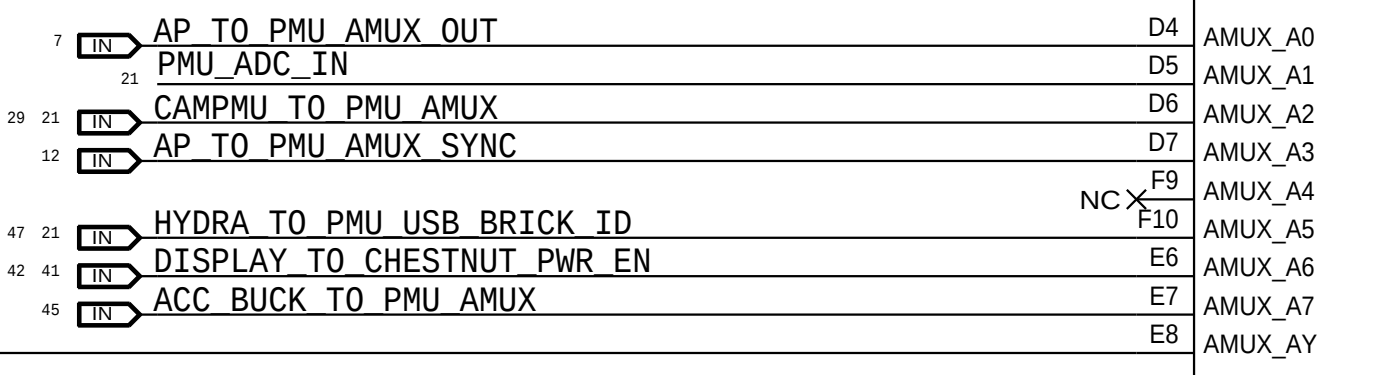
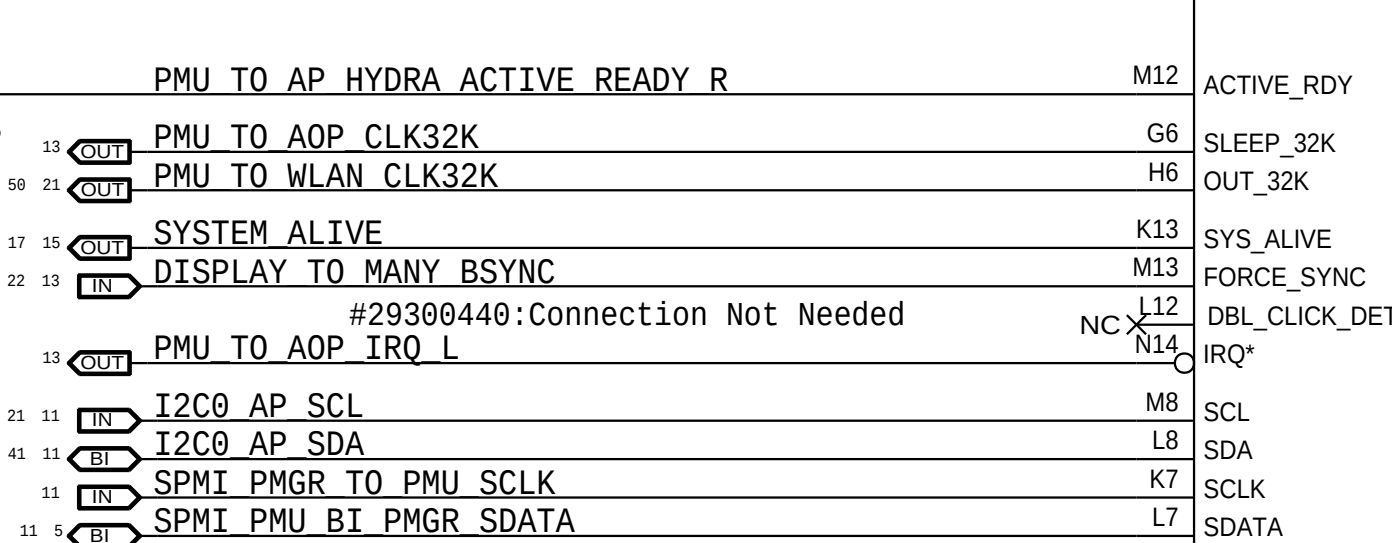
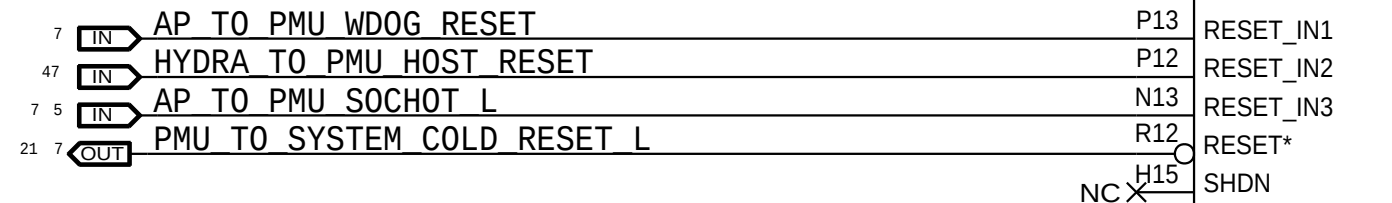
#27852679:No SDA R, PoR is SPMI

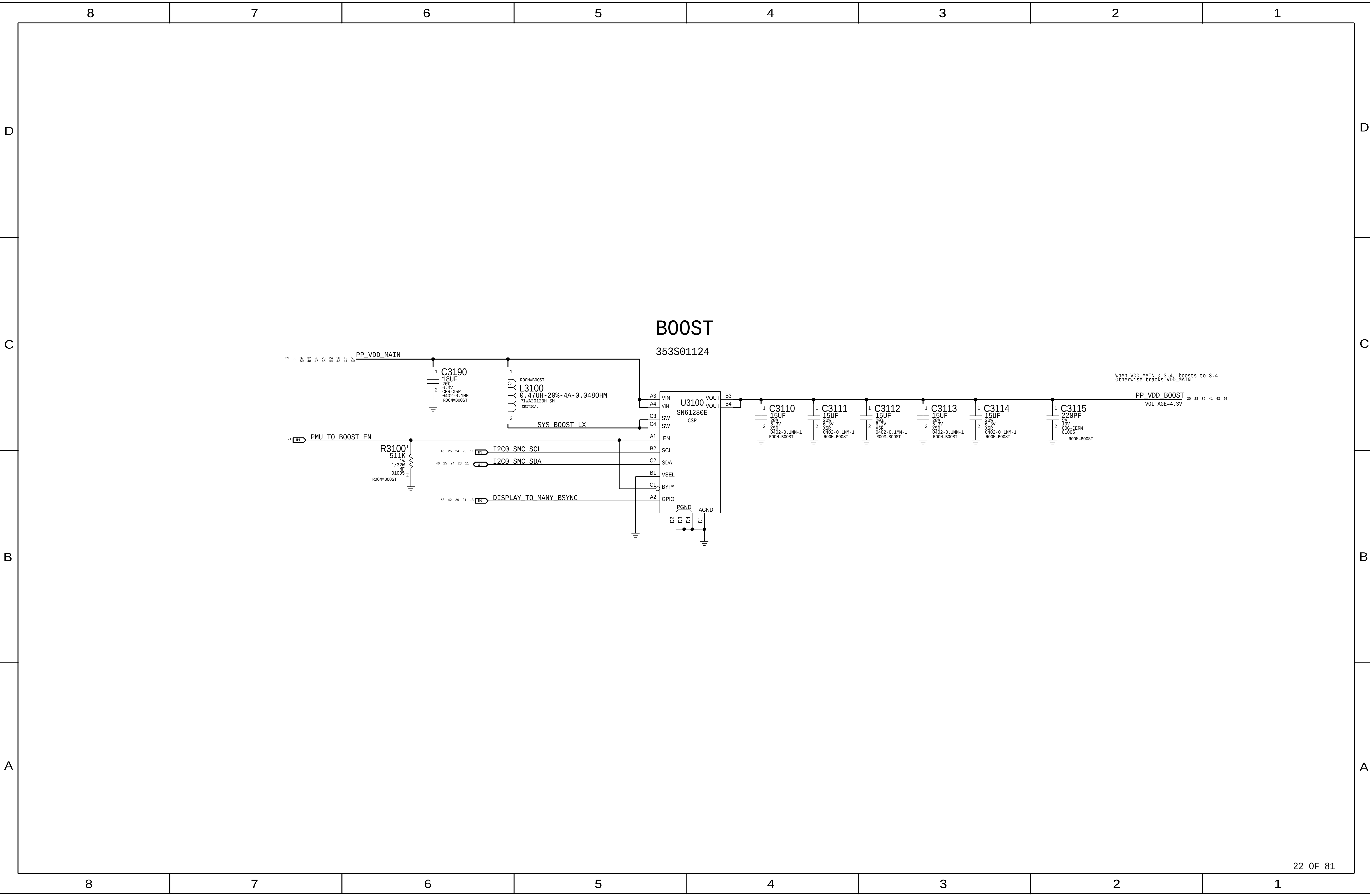
#25244799:200K pull-down

#25244799:CHESTNUT AMUX AND PWR_EN ON SAME CHANNEL
#28748132:CHESTNUT AMUX AND PWR_EN SWAP BETWEEN A6/B6

#25244799:0TP-AJ Connects XTAL1 to GND

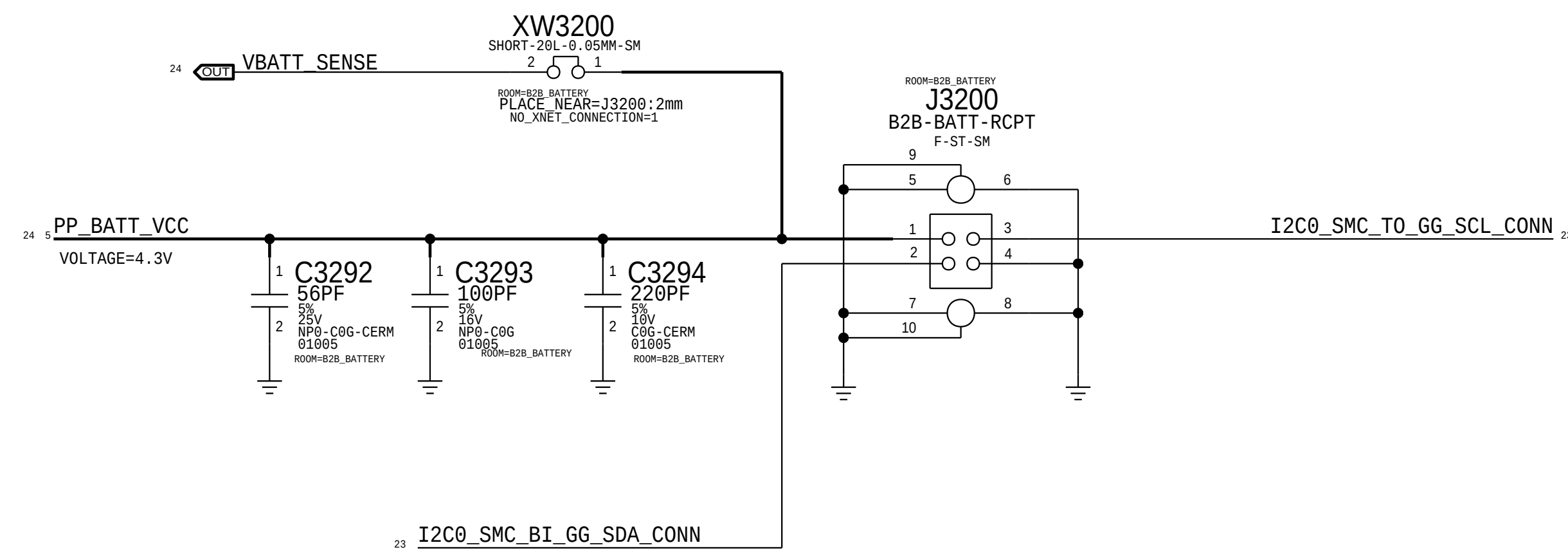
#29312542:REPLACE PMU XTAL WITH XO



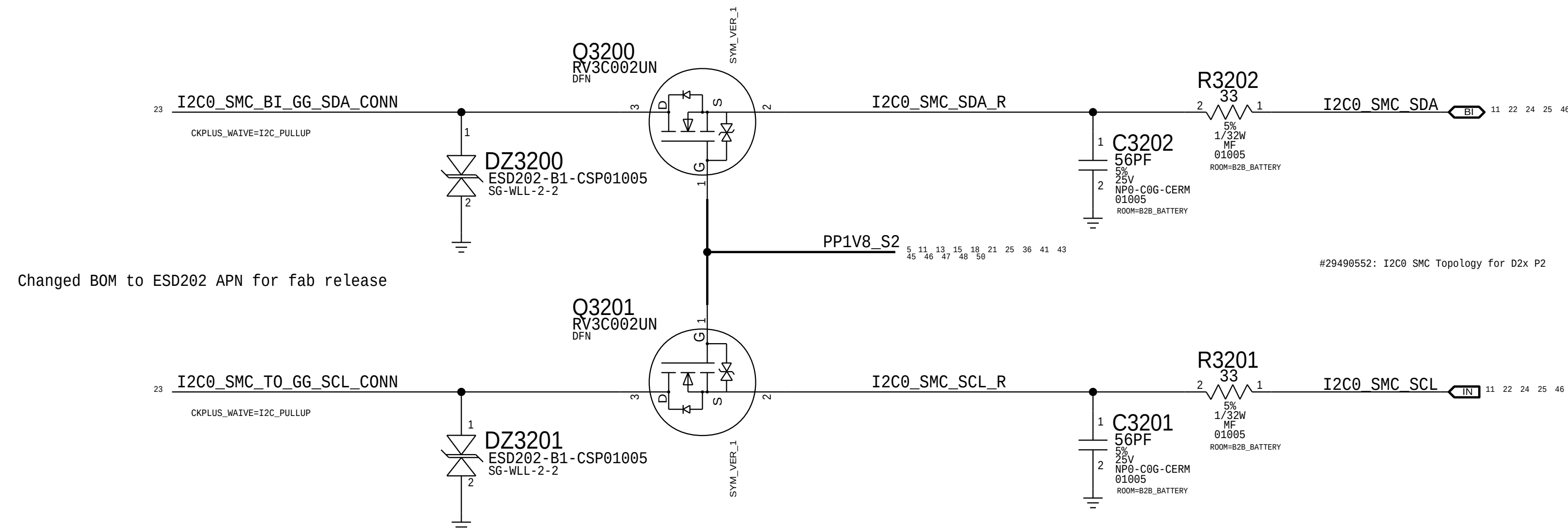


Battery Connector

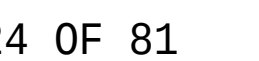
THIS ONE ON MLB ---> 516S00232



```
#32515053:Add DZ3200/DZ3201 on I2C CONN with 377S00001
#31734657:Replace battery I2C FETs with 376S00134
    Gas gauge I2C level translator
```

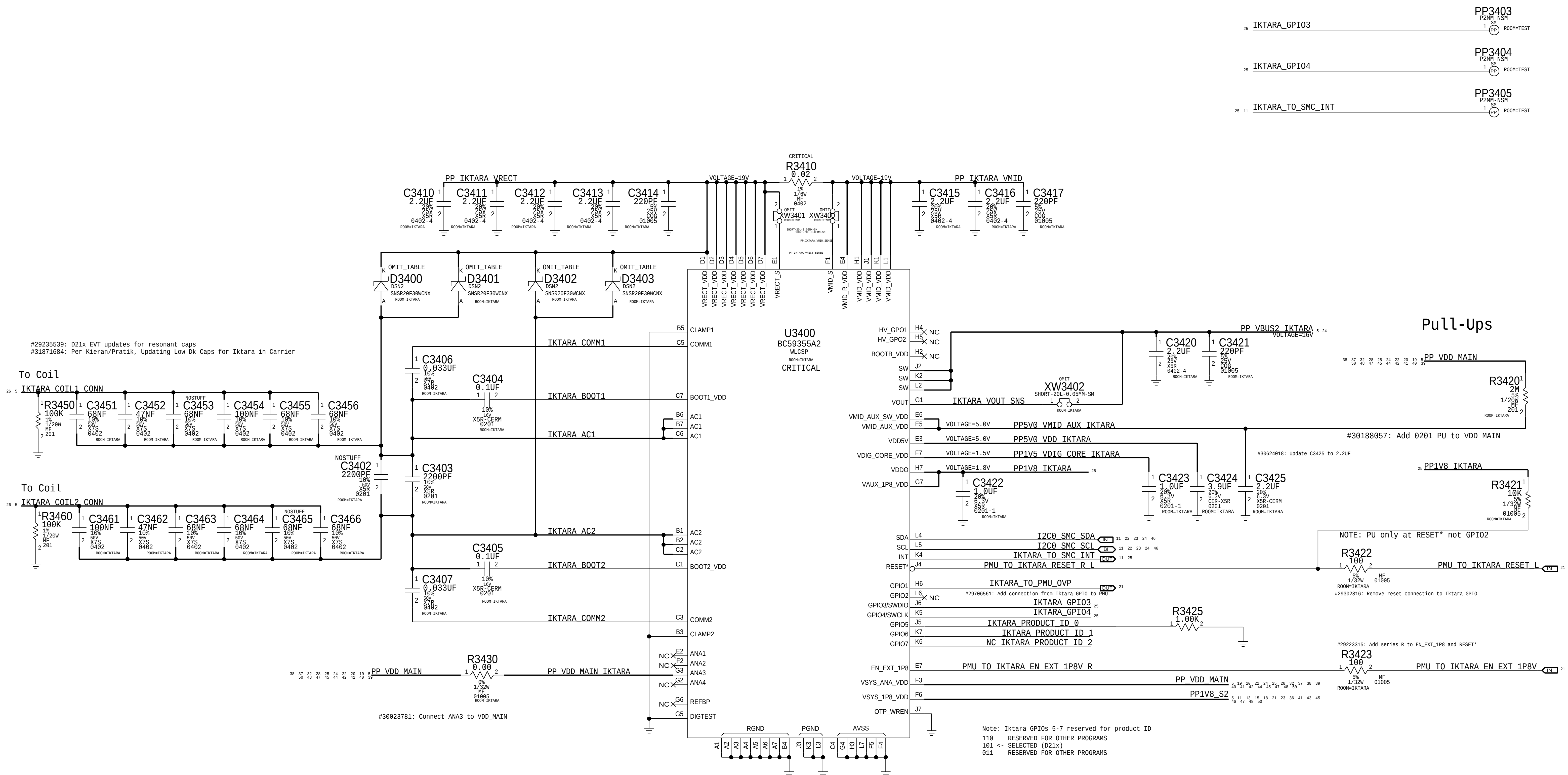


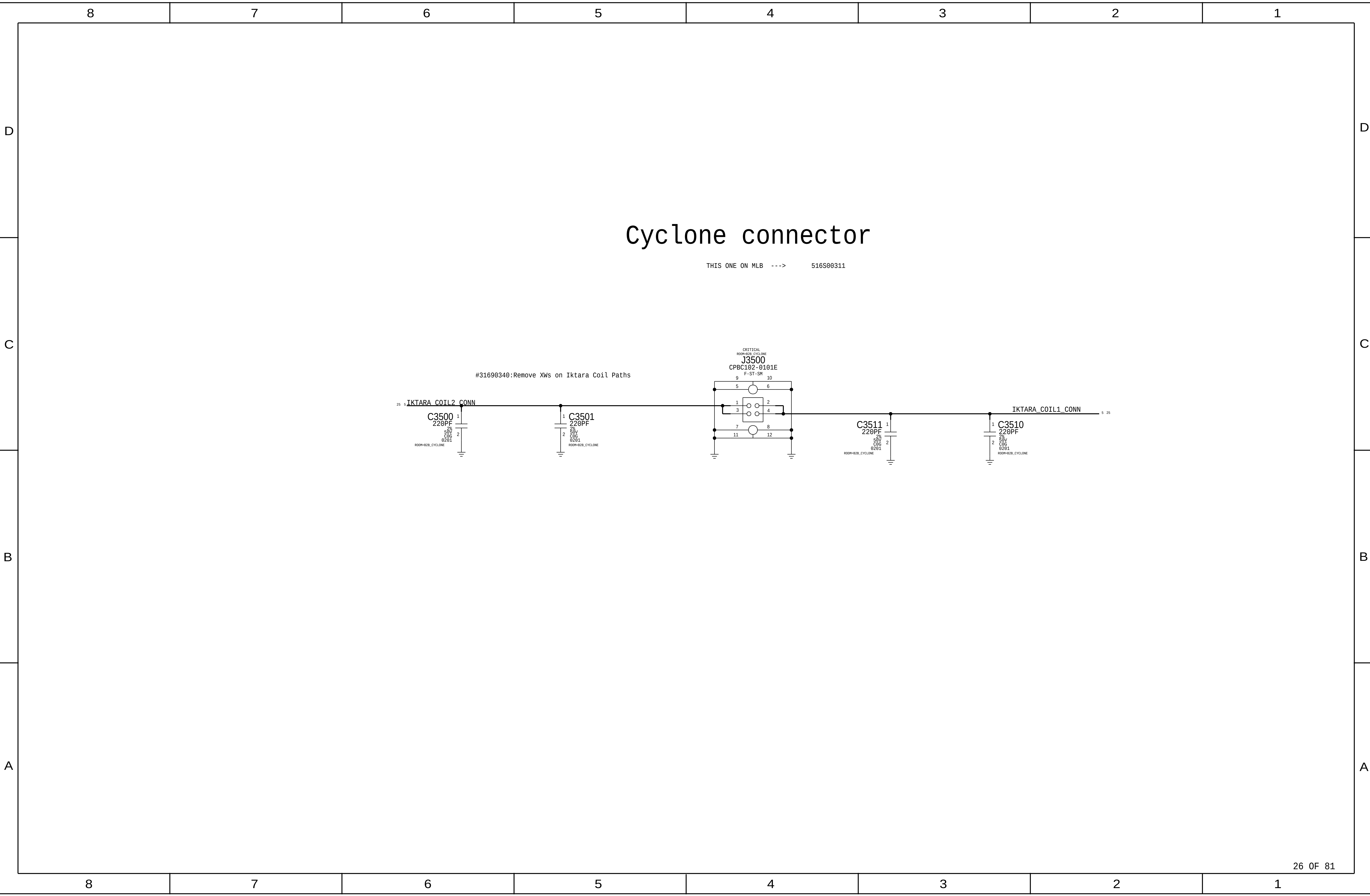
APN:343S00170



Iktara

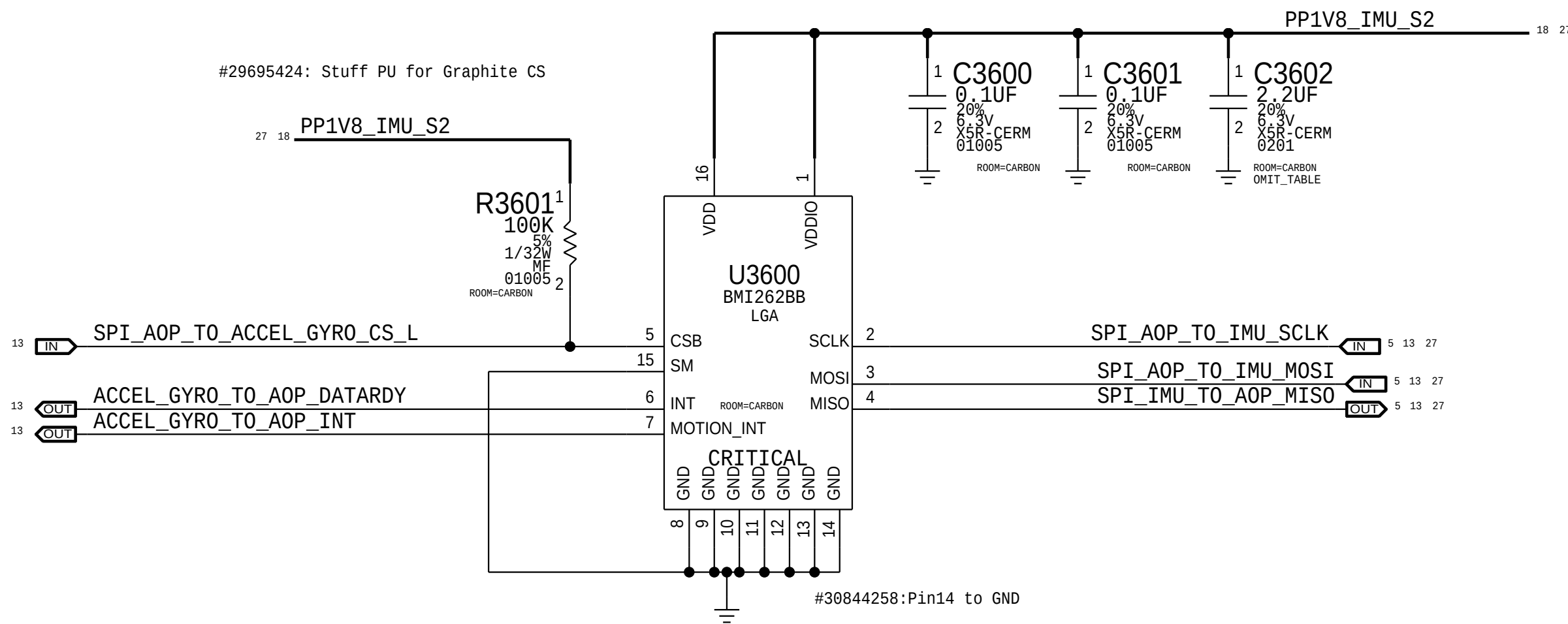
Iktara Debug





Graphite - Accel & Gyro

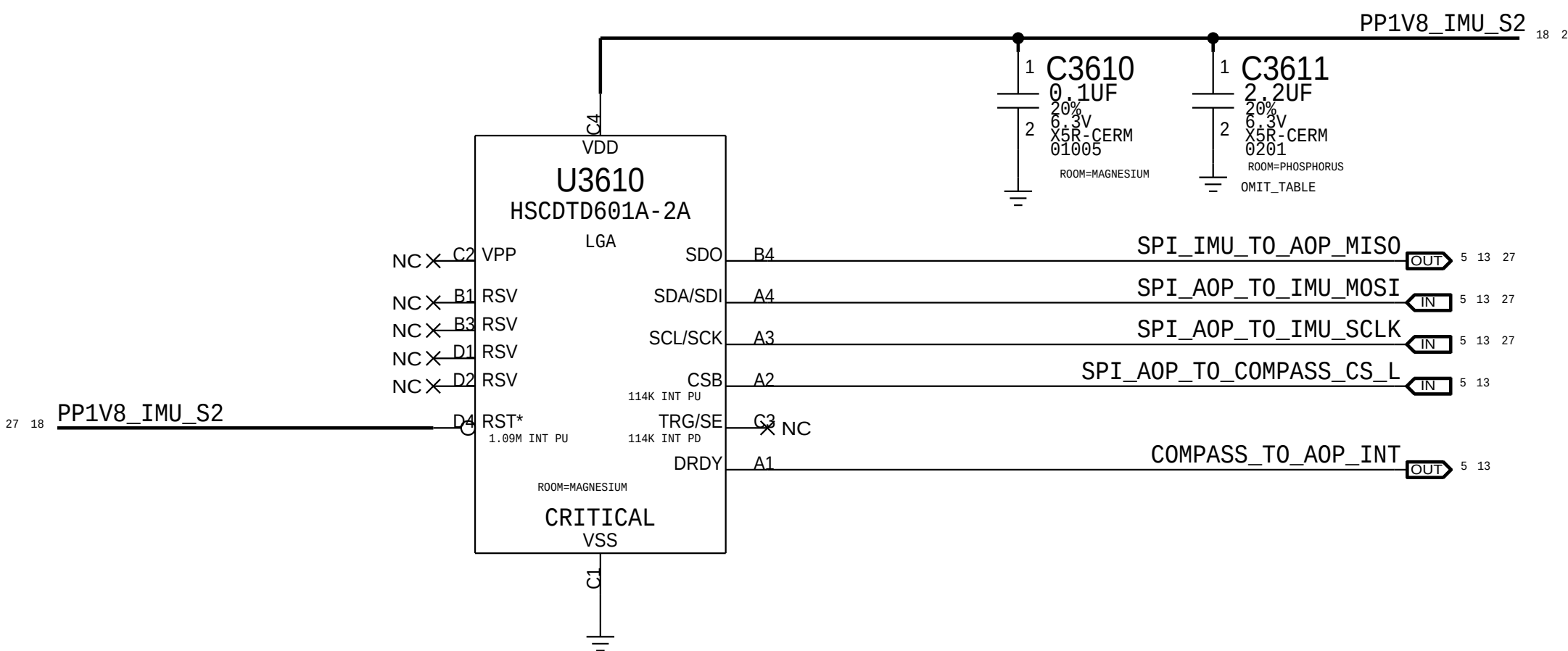
APN: 338S00304 (A1)



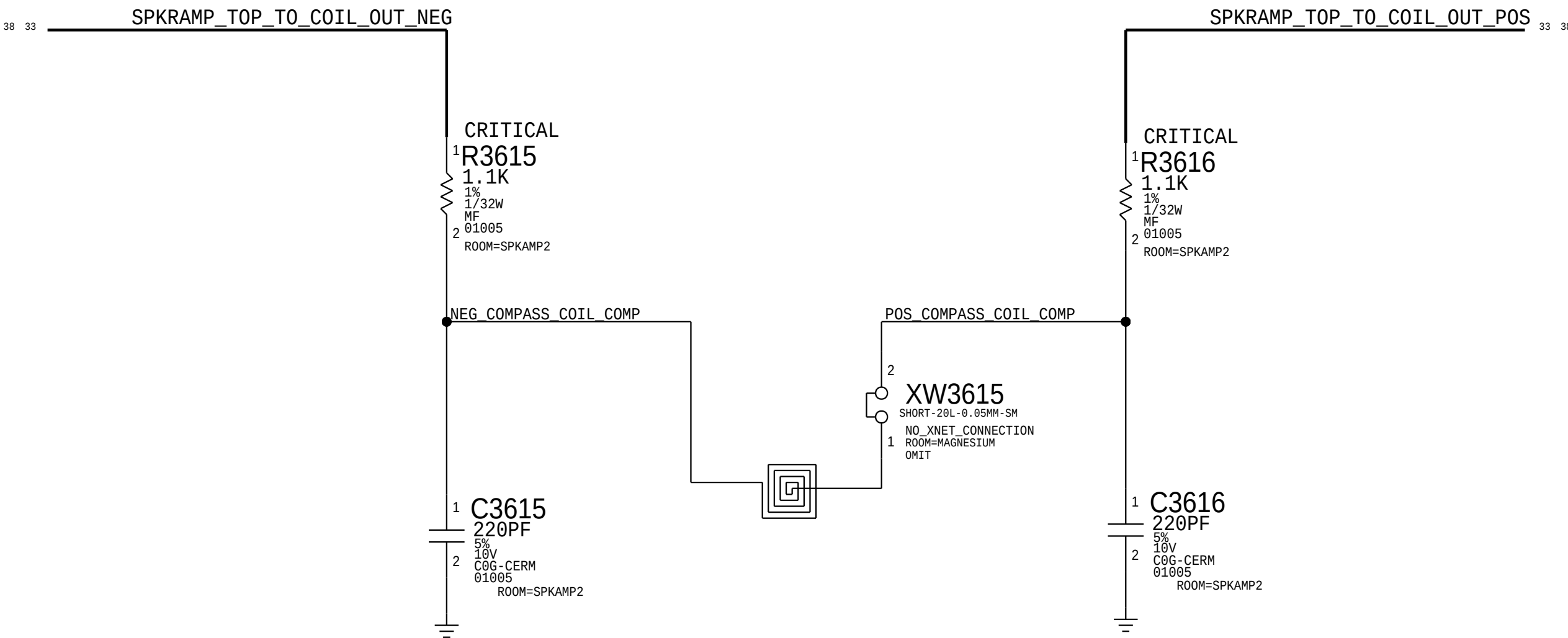
Magnesium - Compass

APN:338S00084

Shared decoupling with Phosphorus

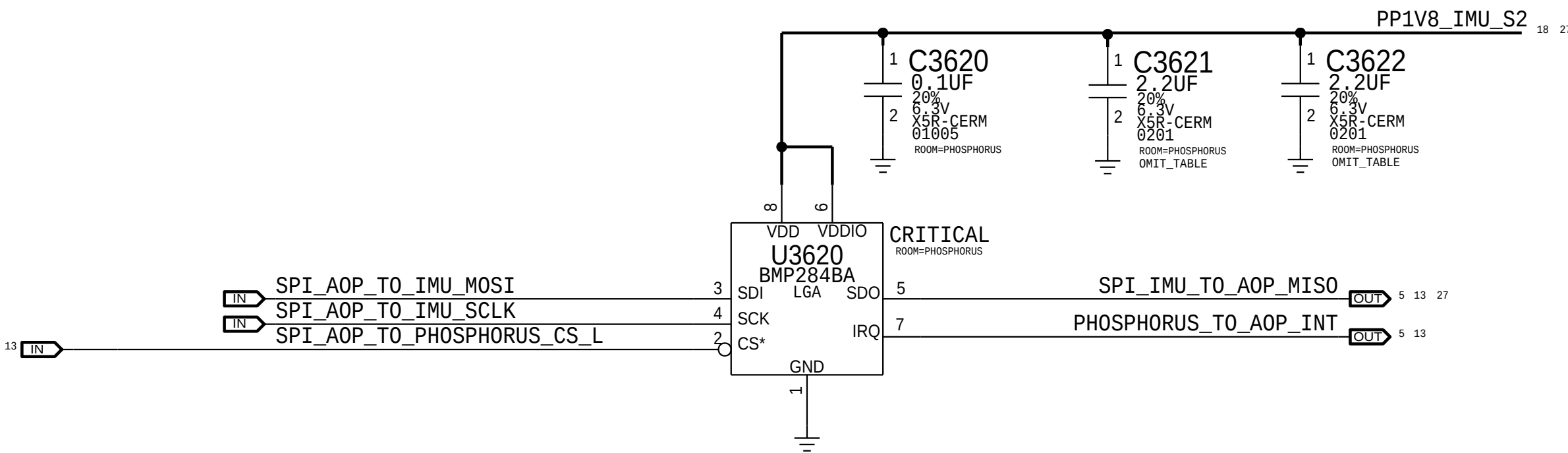


North Speaker Compass Compensation Coil

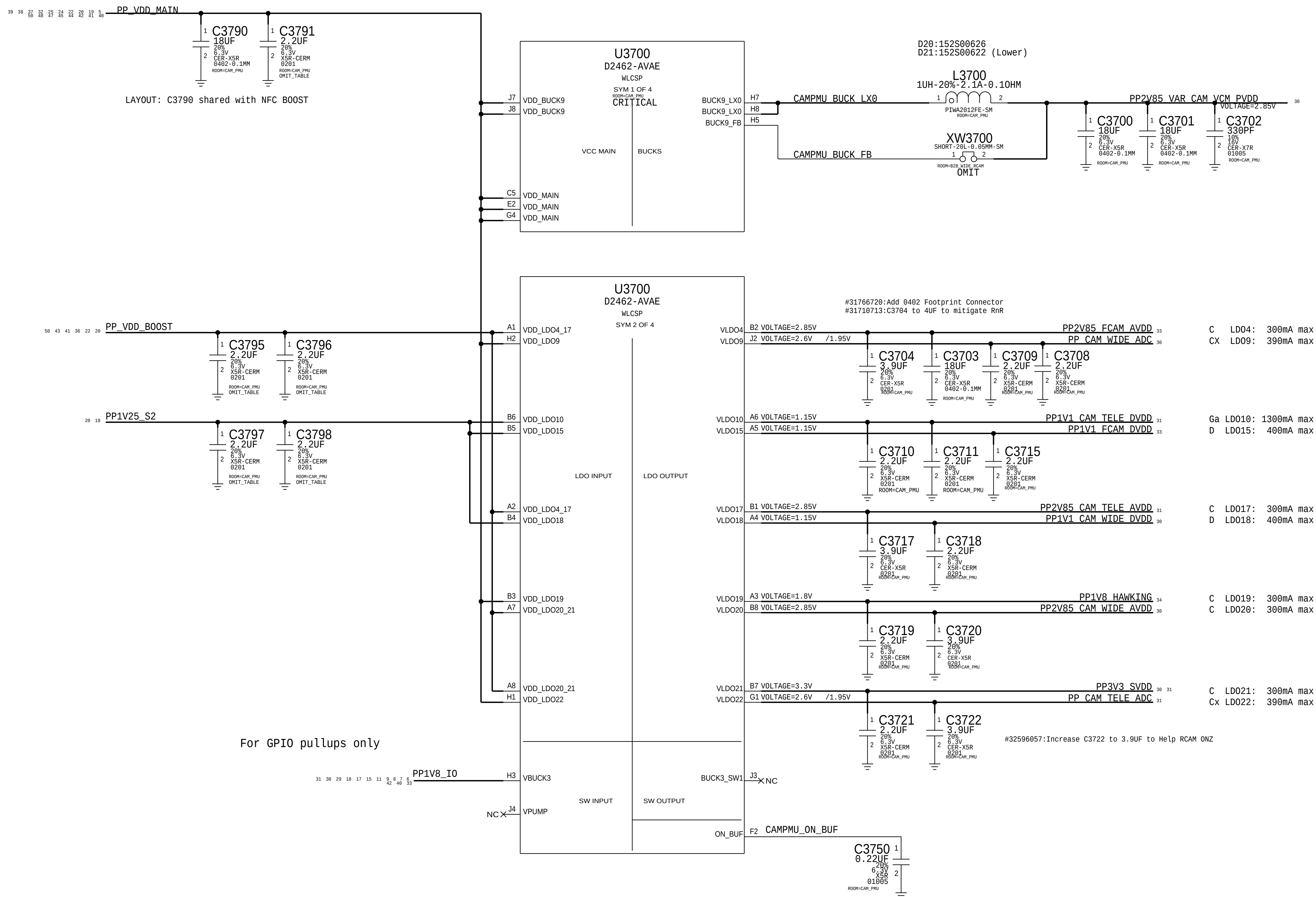


Phosphorus

APN: 338S00334



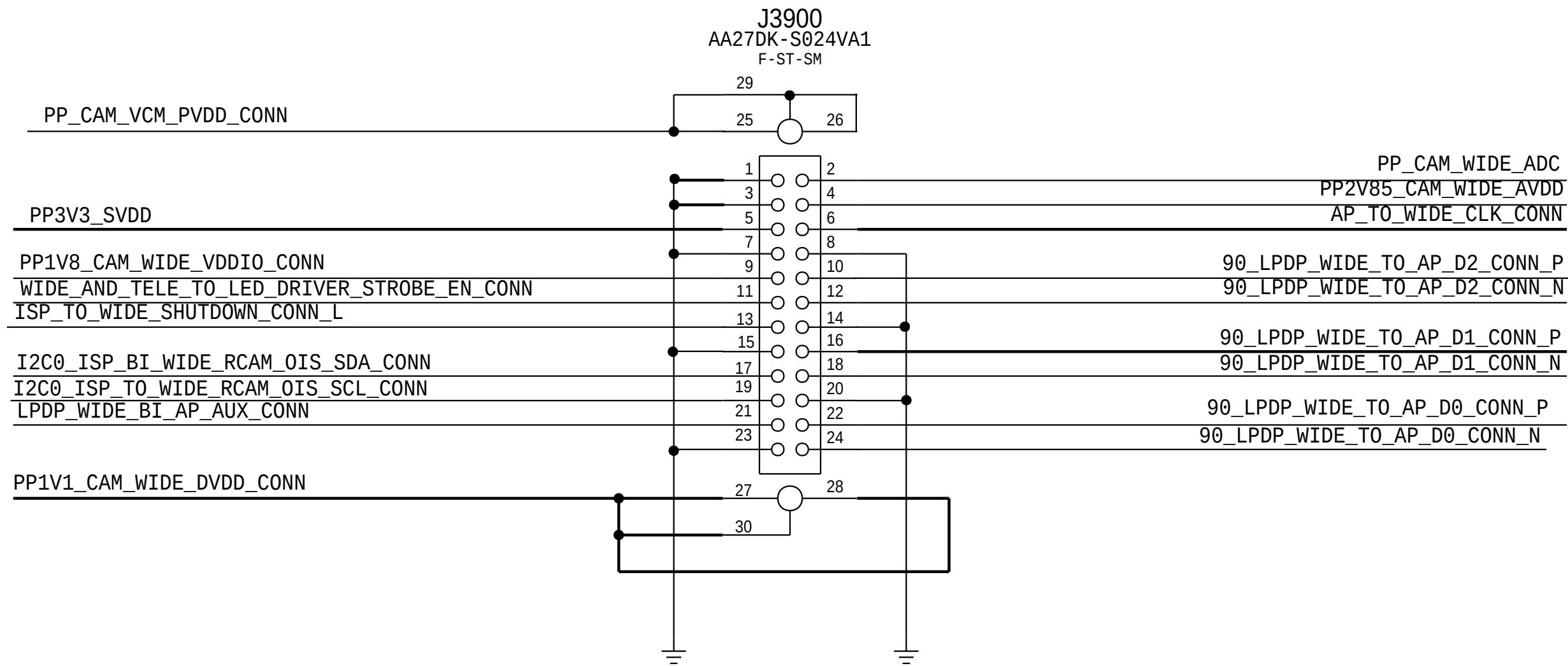
Camera PMU



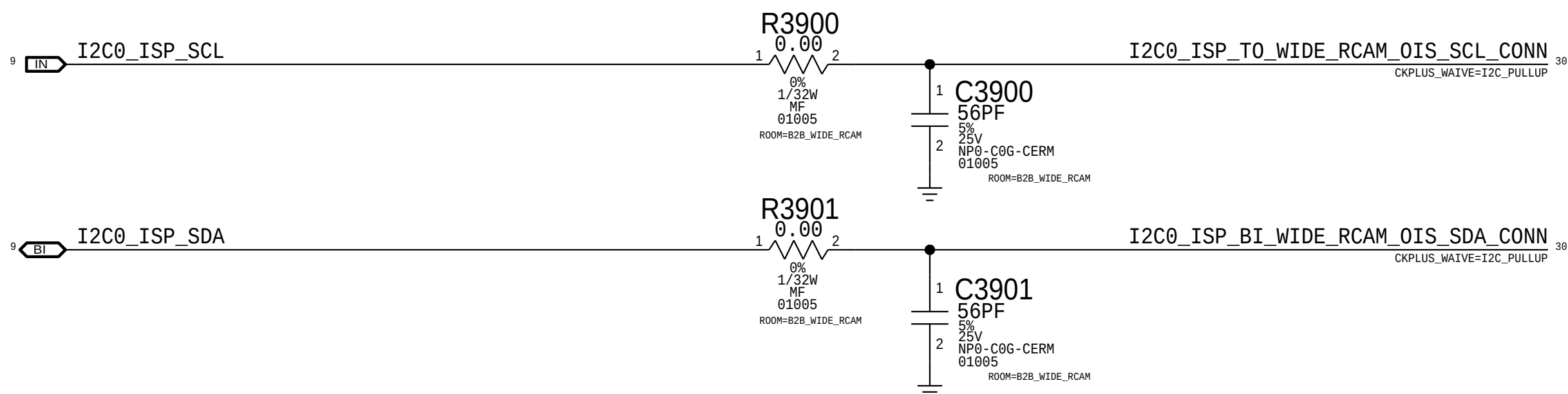


Wide Camera Connector

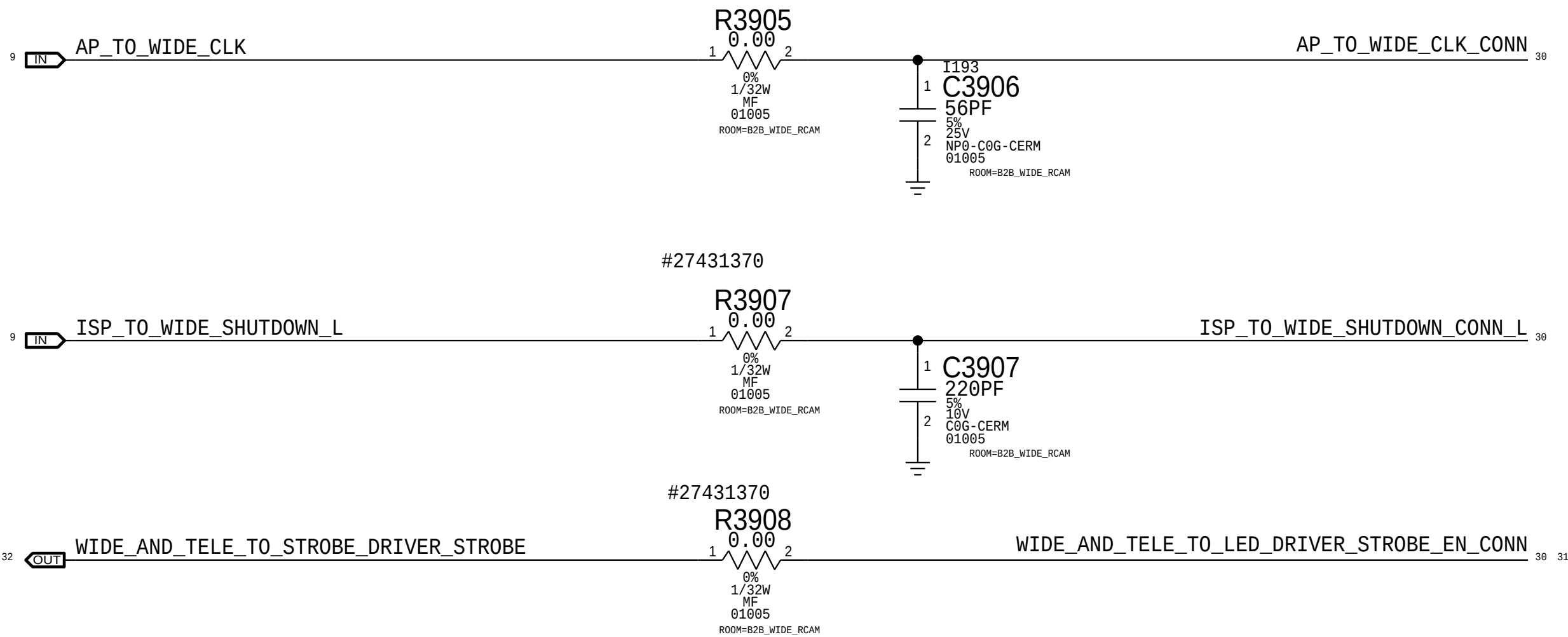
THIS ONE ----> 516S00310/P2 RCPT (USED ON MLB)
516S00309/P2 PLUG (USED ON FLEX)



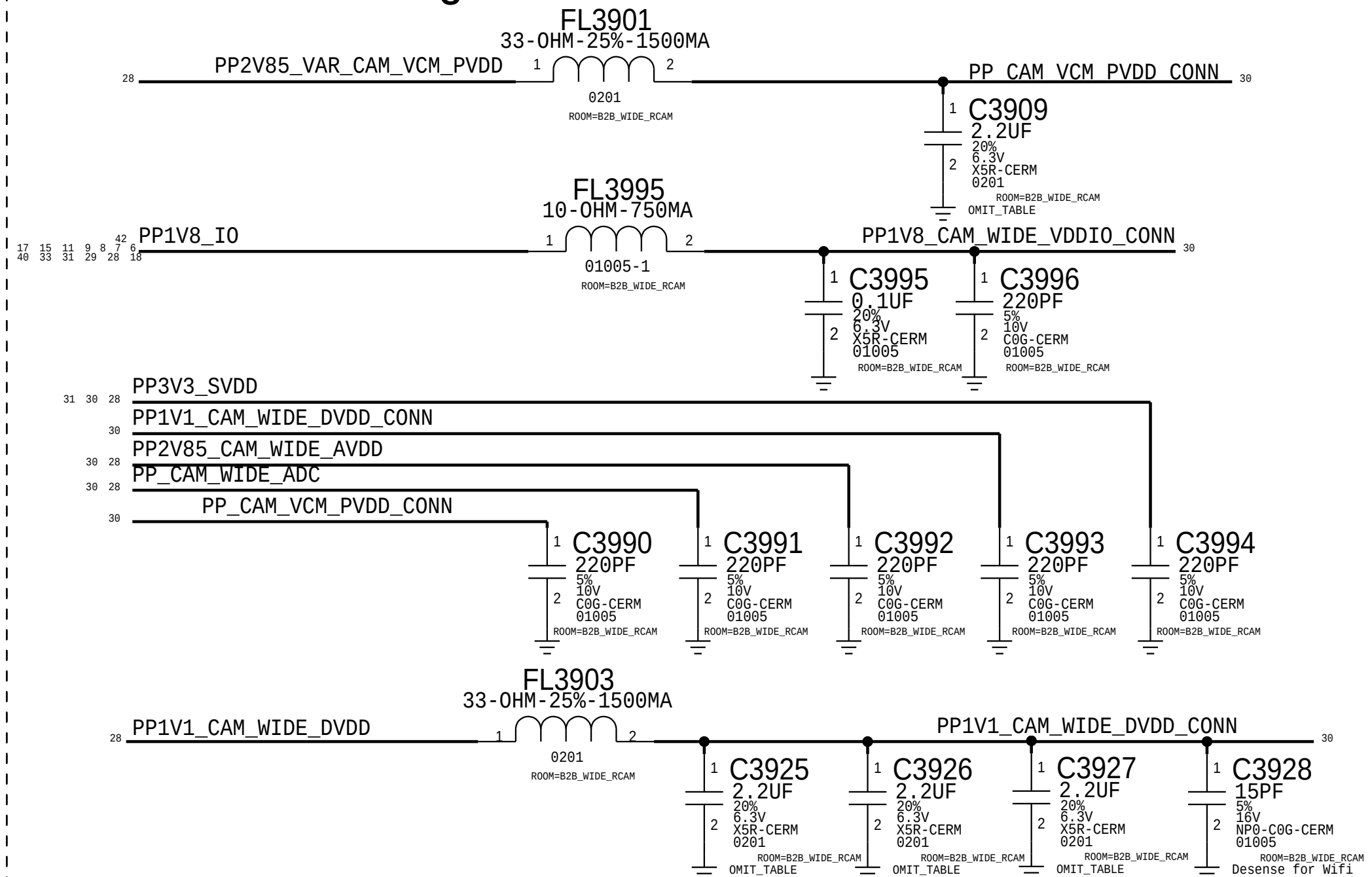
ISP I2C



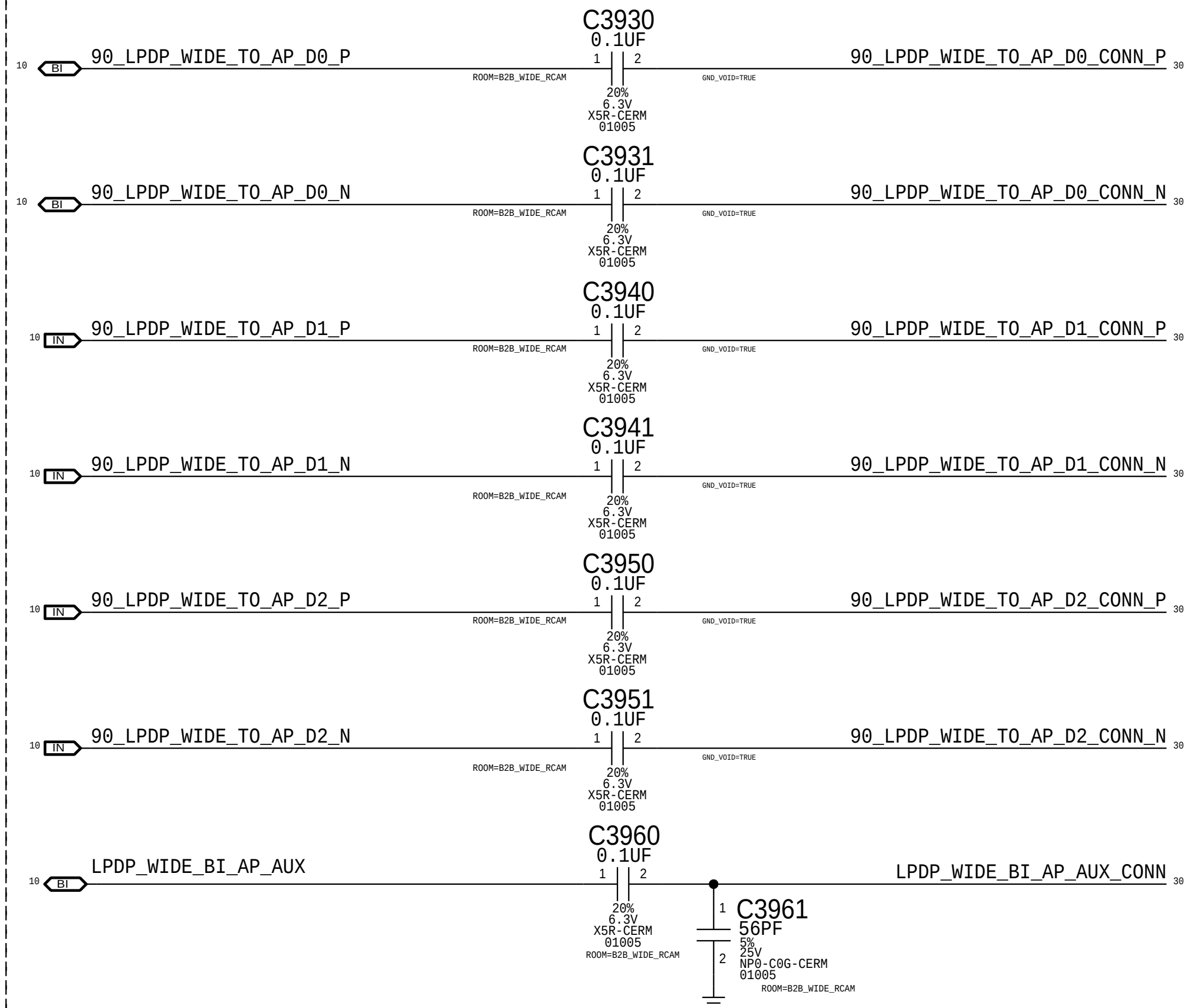
IO Filters



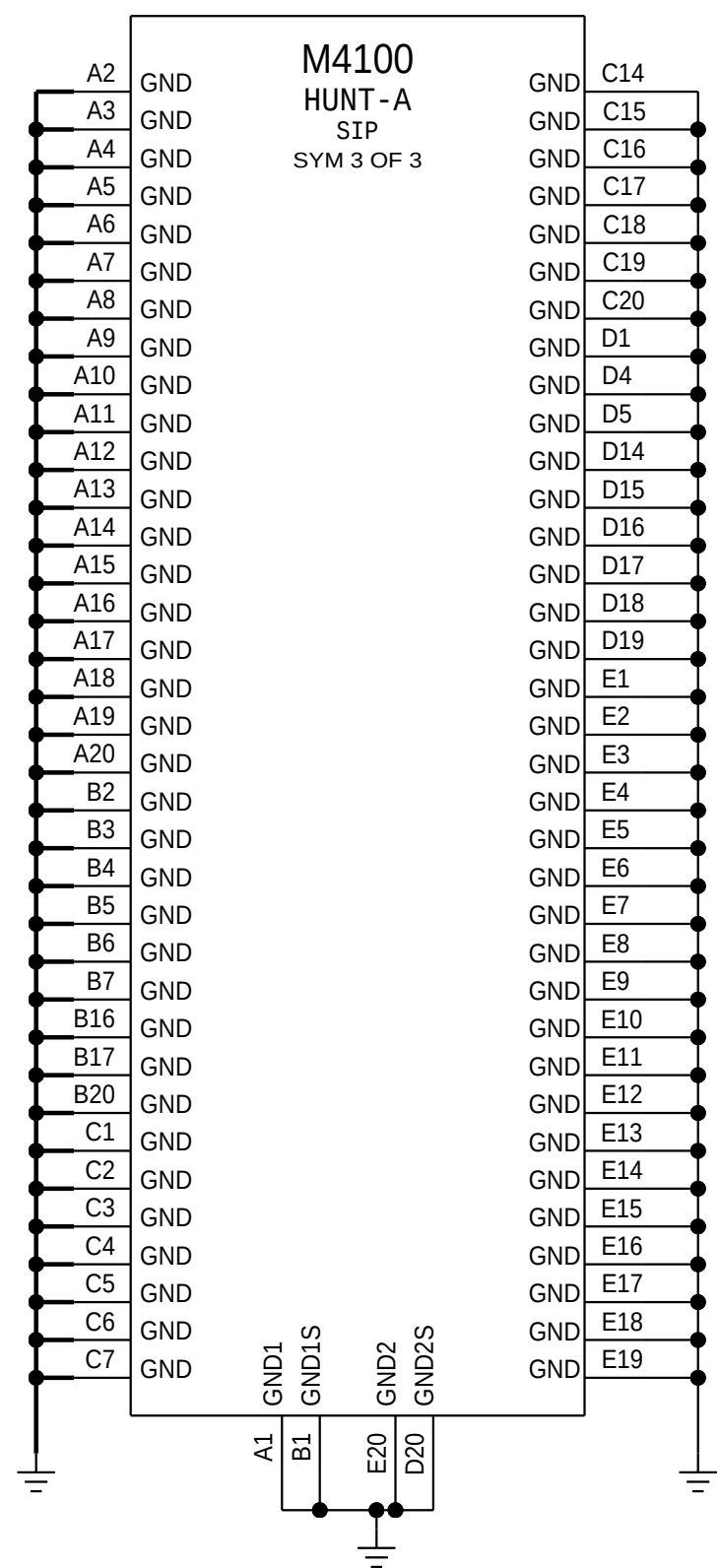
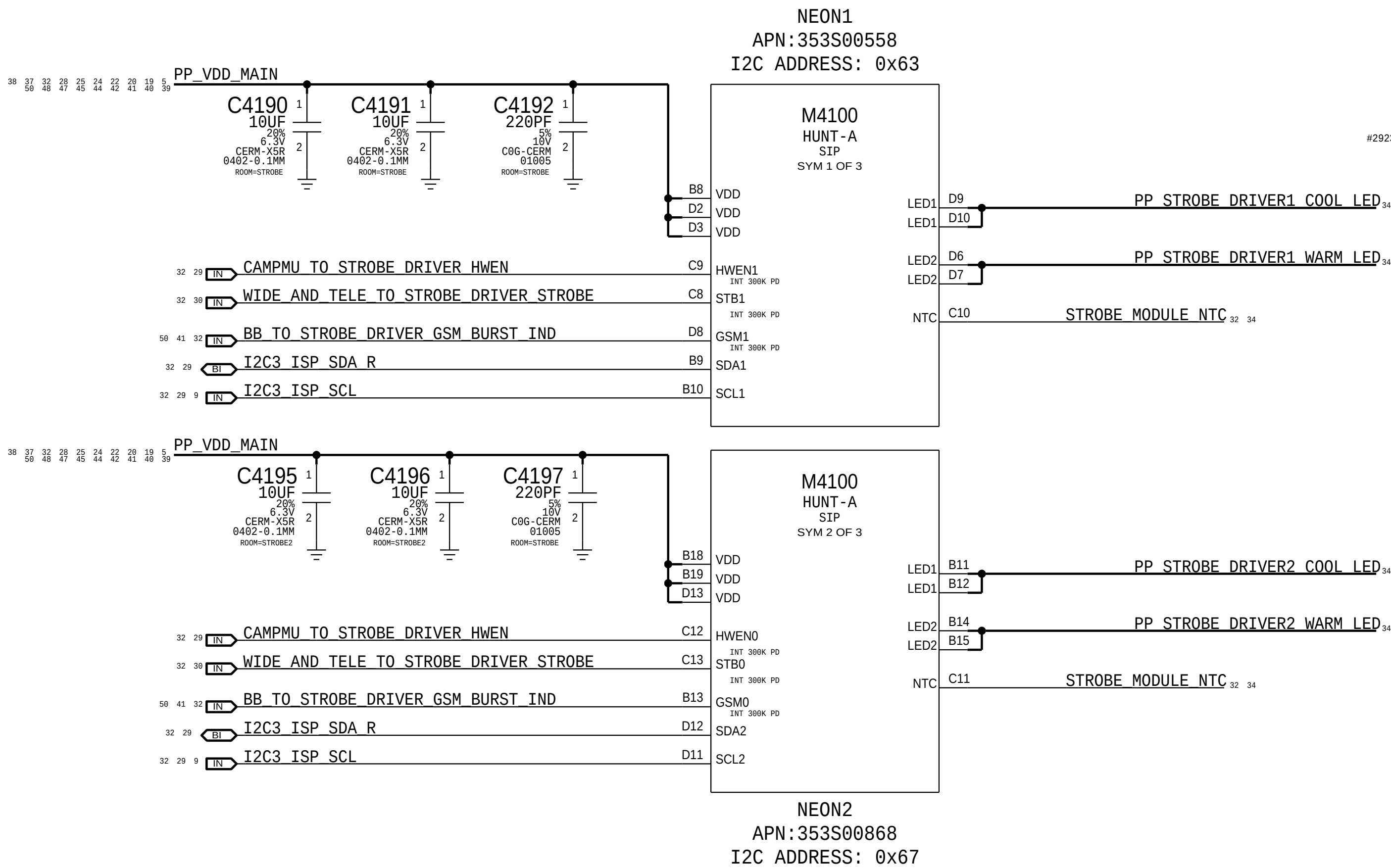
Power Filtering



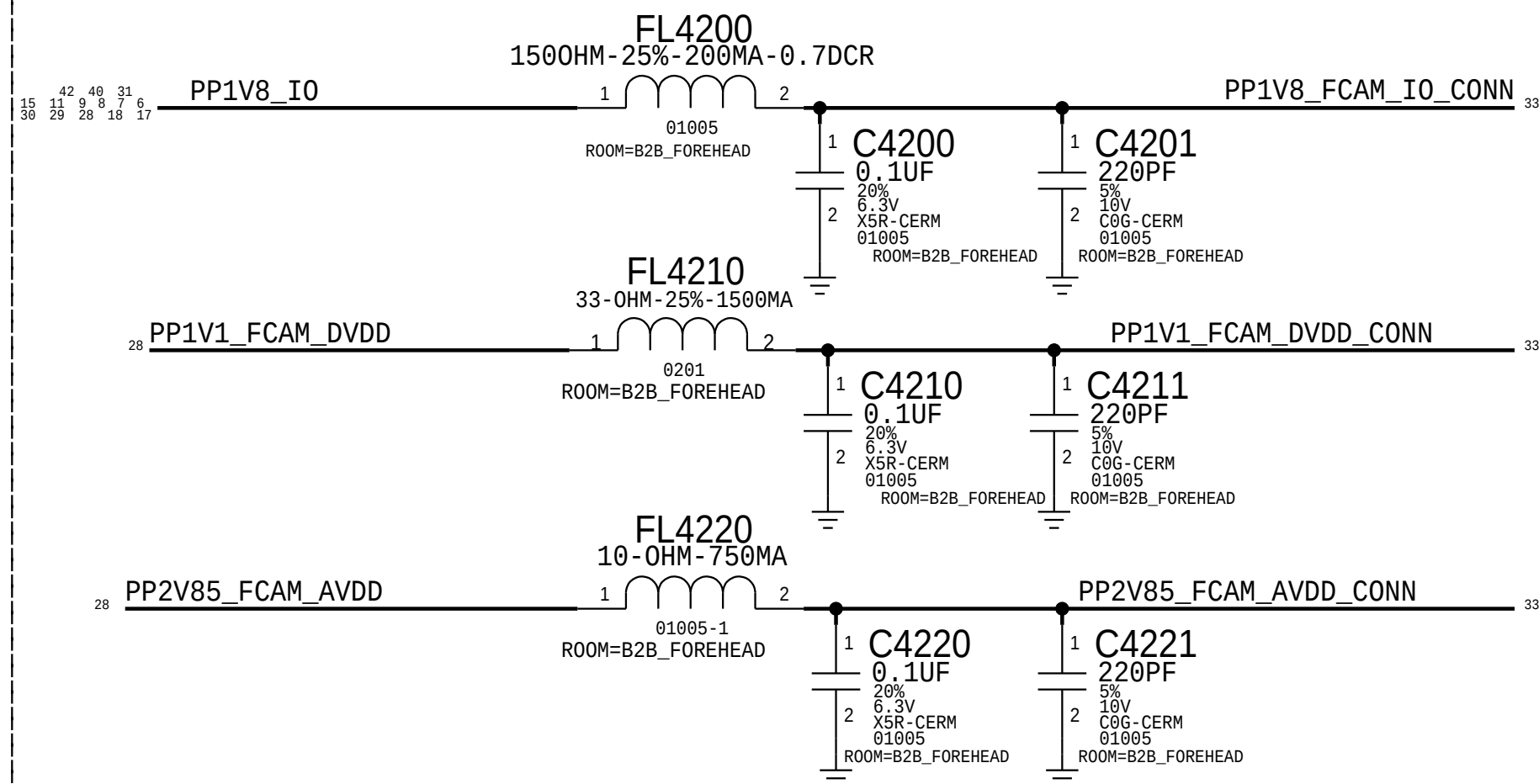
LPDP Filters



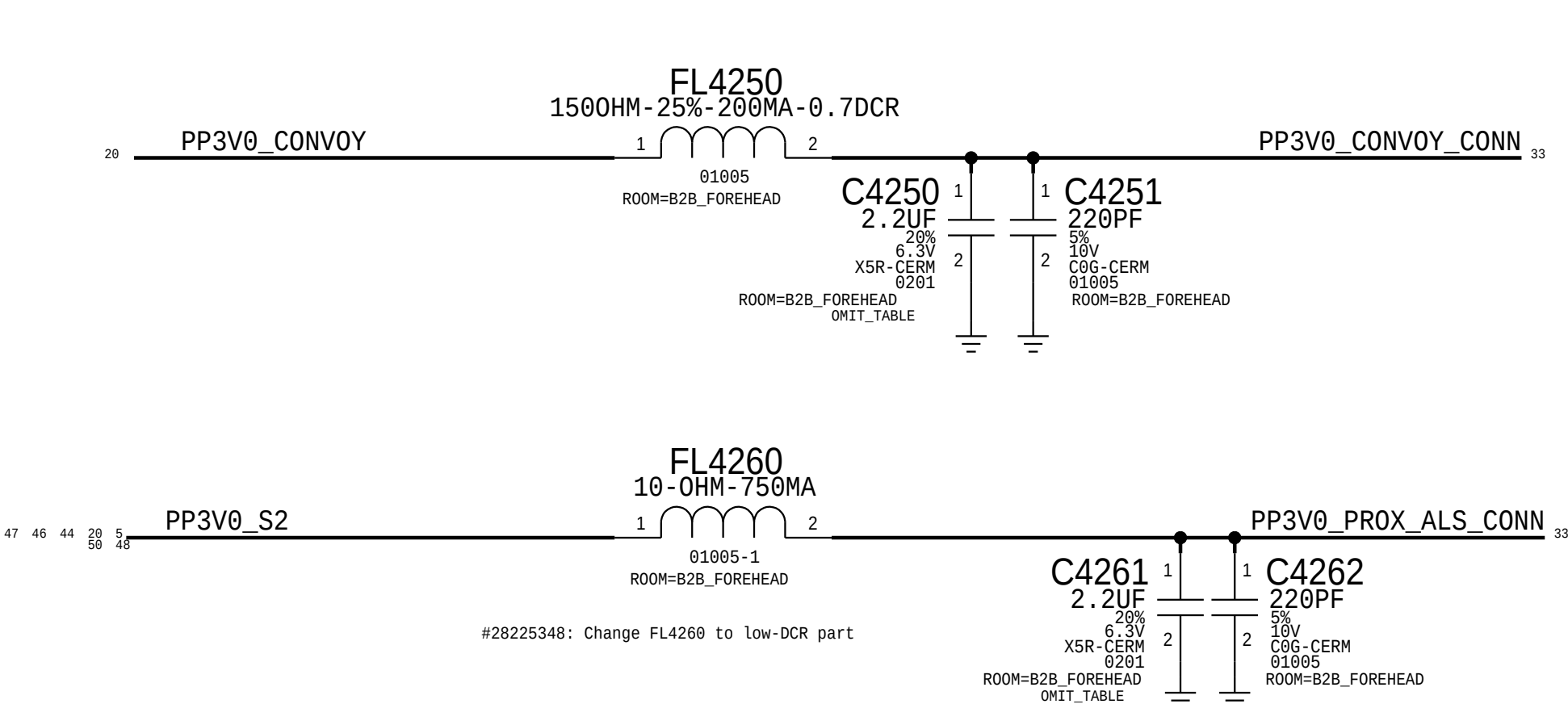
LED STROBE DRIVERS (NEON)



FCAM POWER

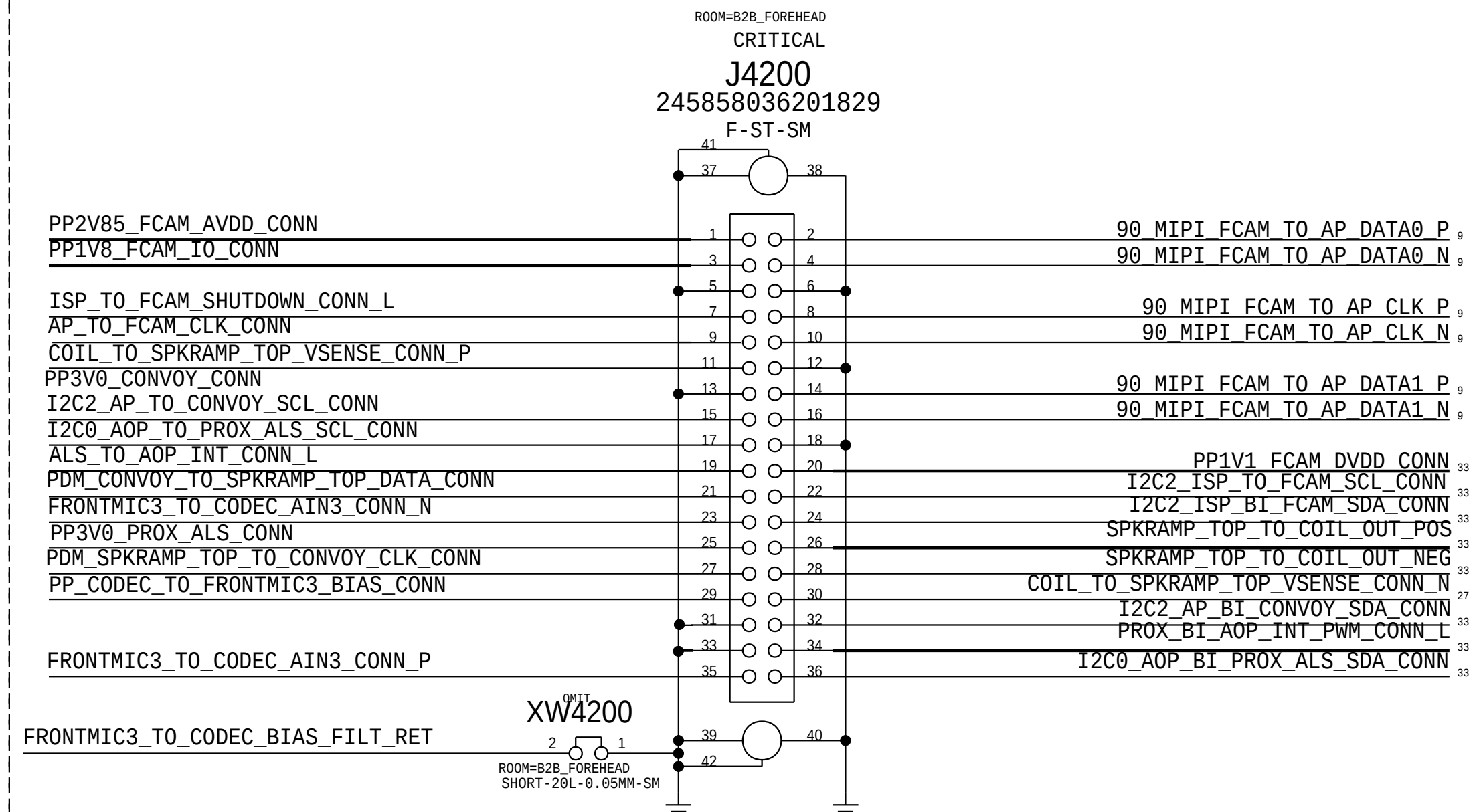


PROX + ALS + CONVOY POWER

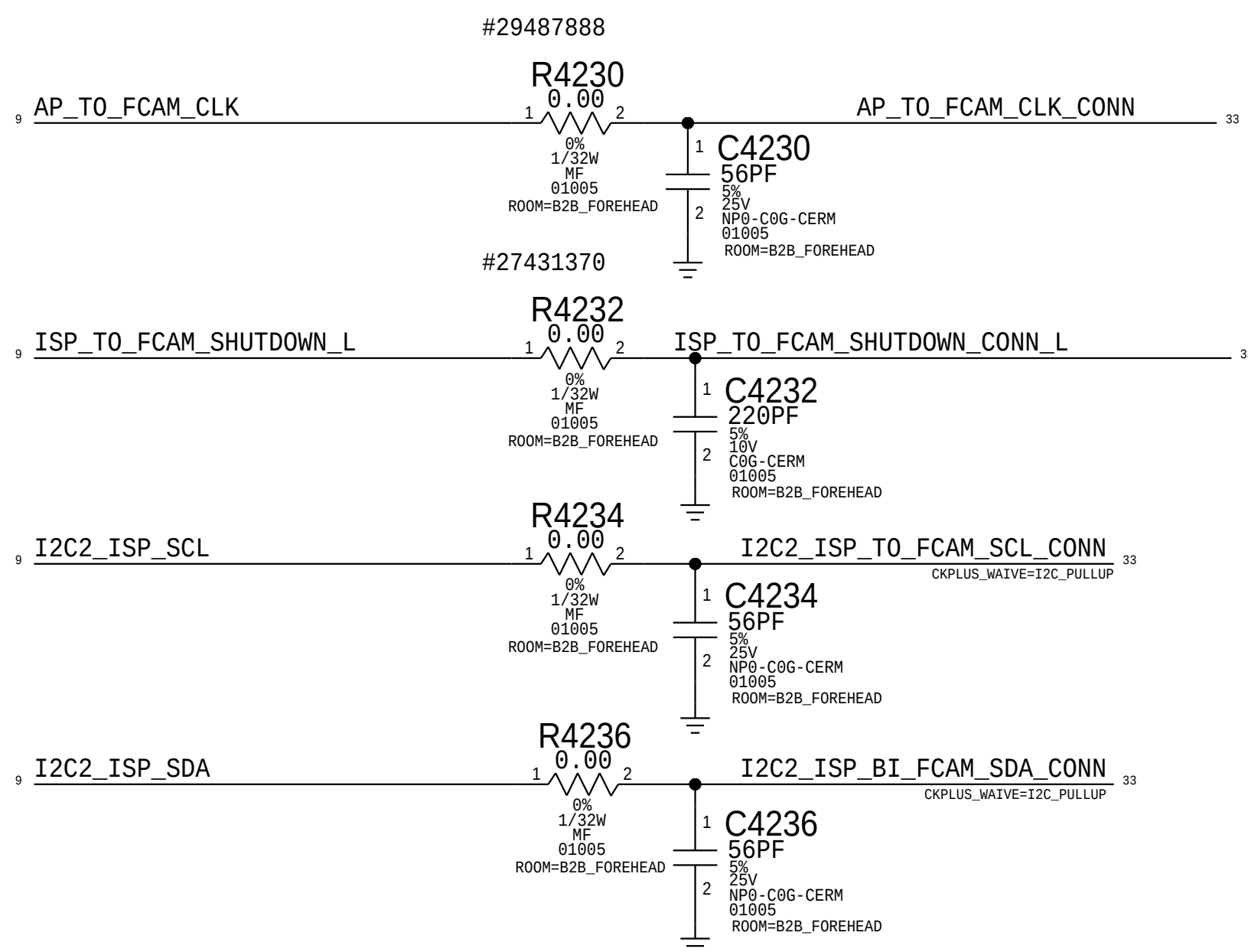


FOREHEAD CONNECTOR

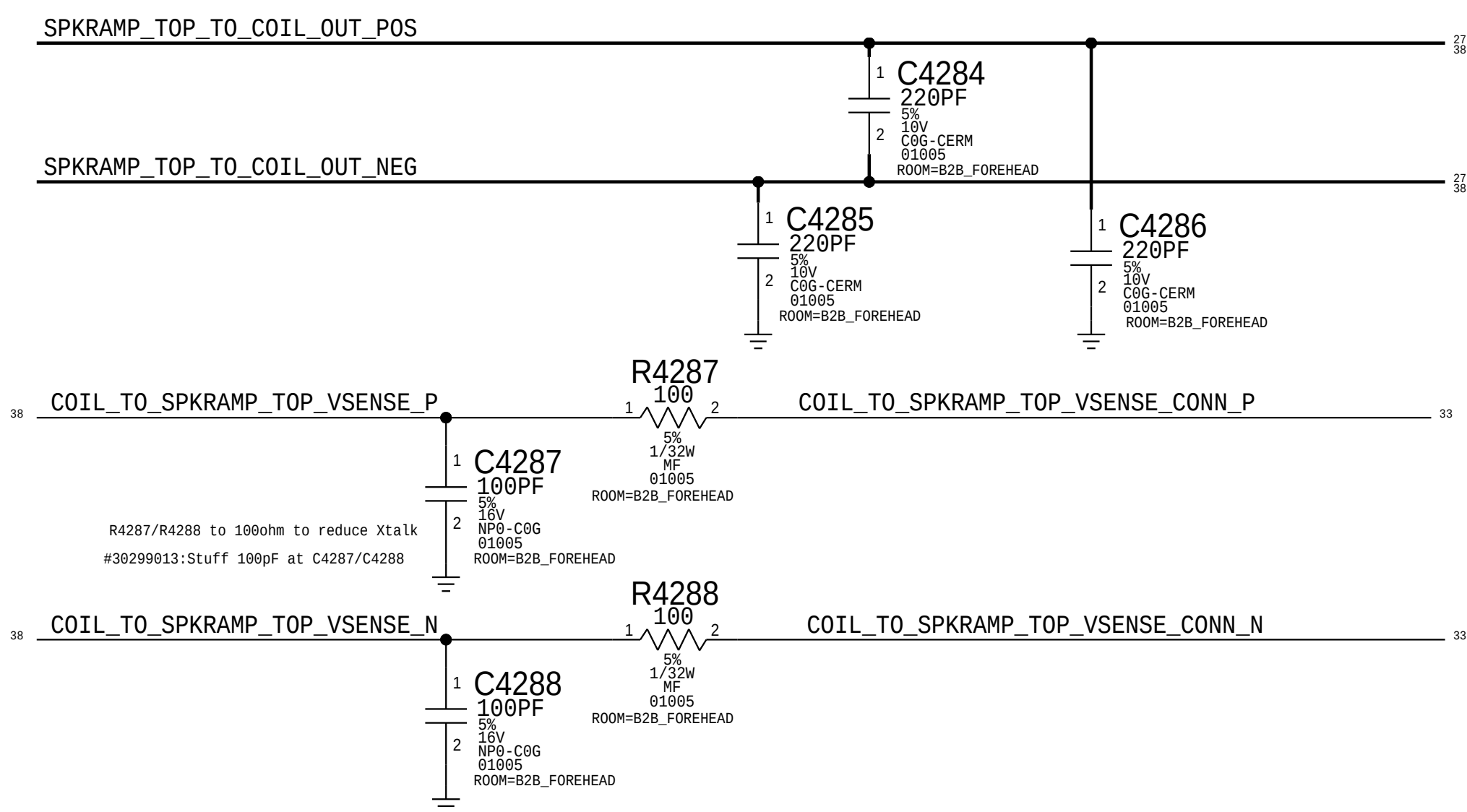
APN: 516S00146



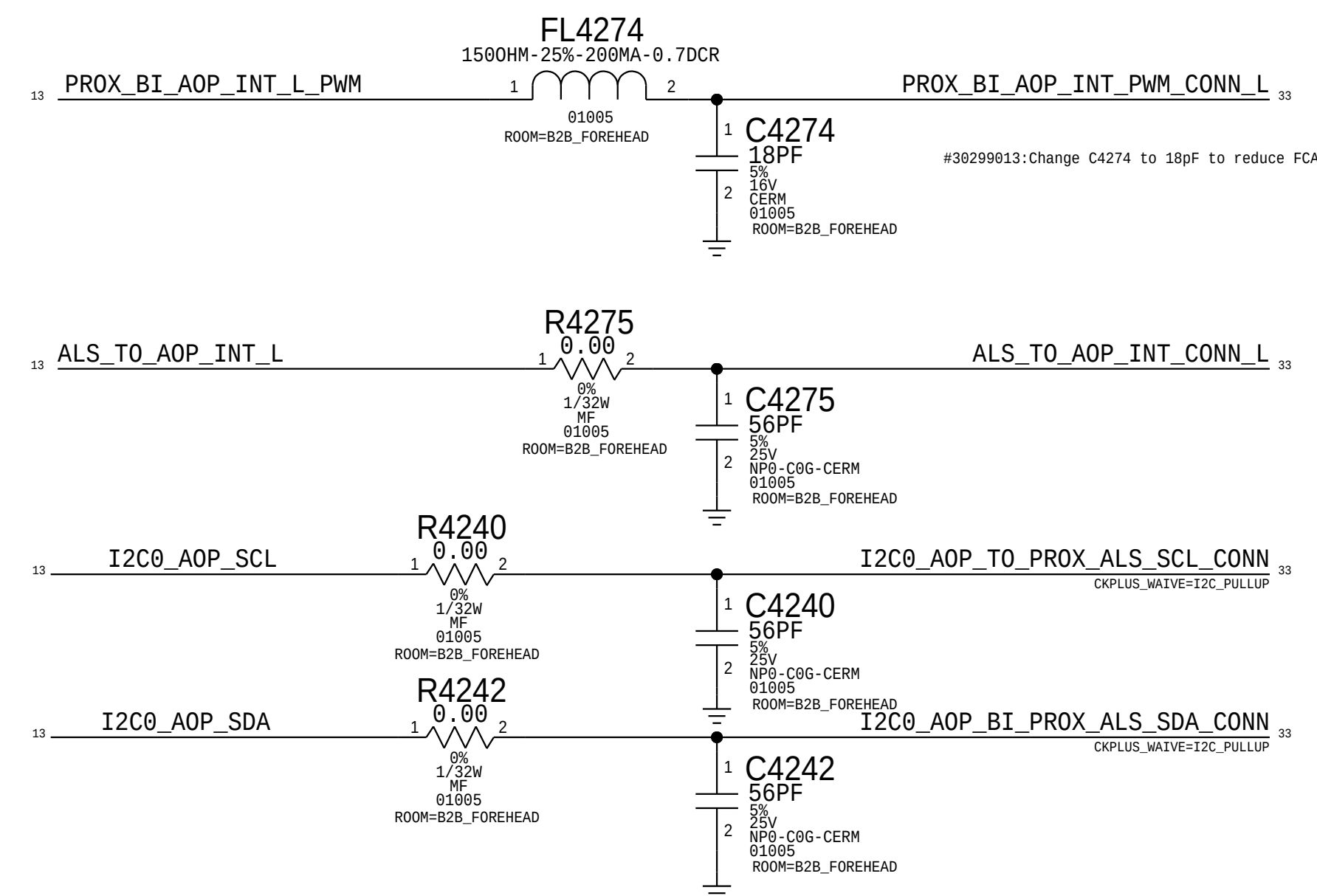
FCAM I/O



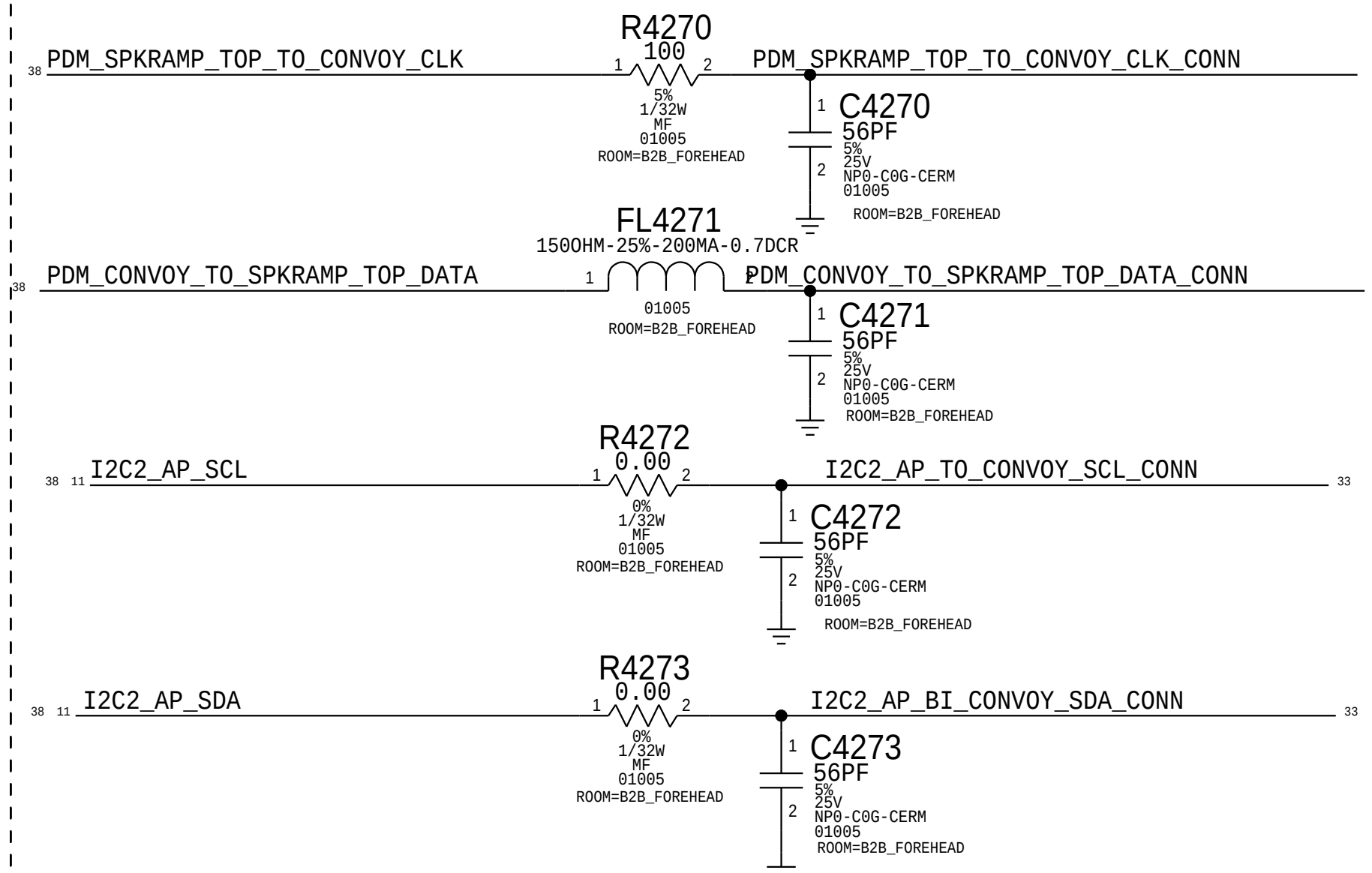
NORTH SPEAKER



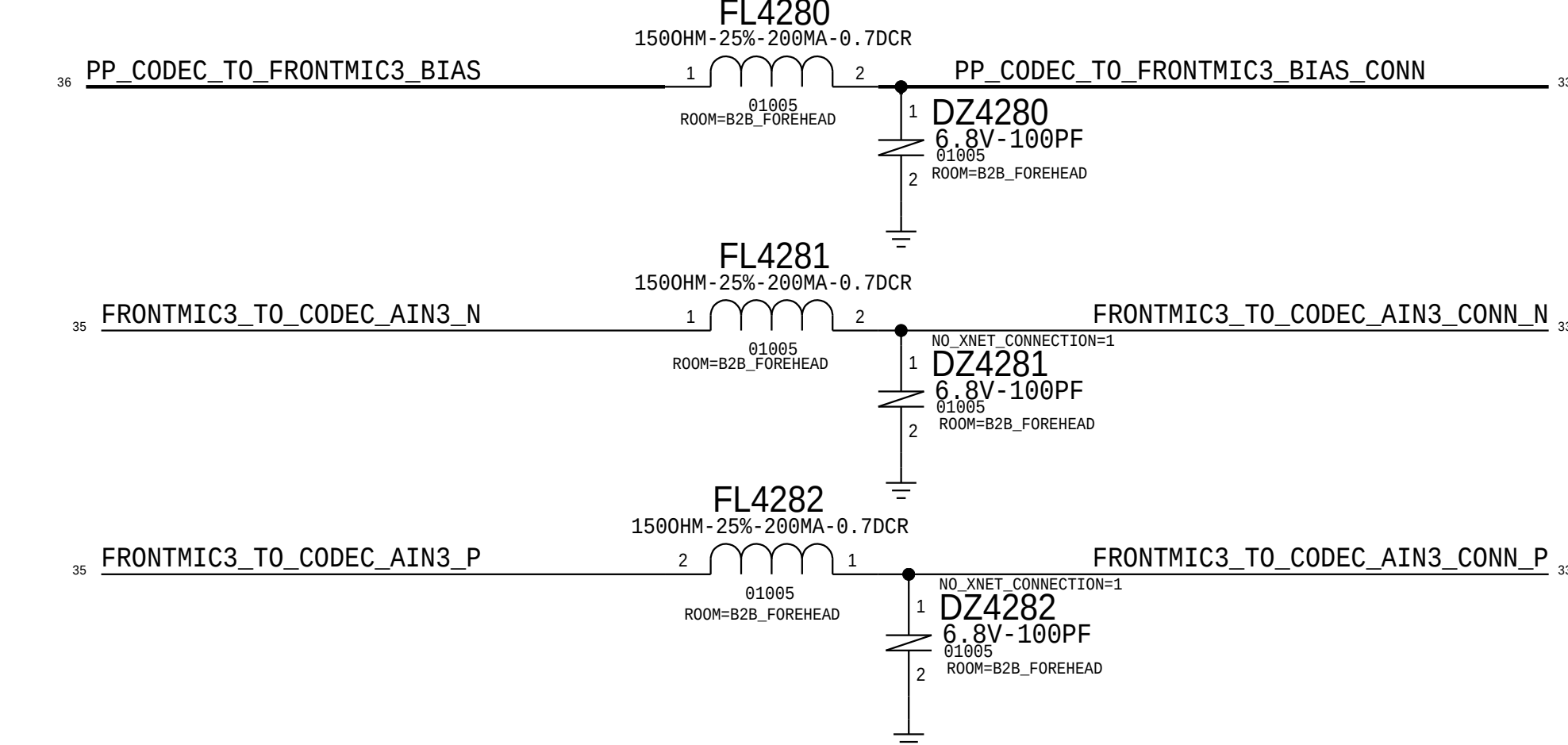
PROX + ALS I/O



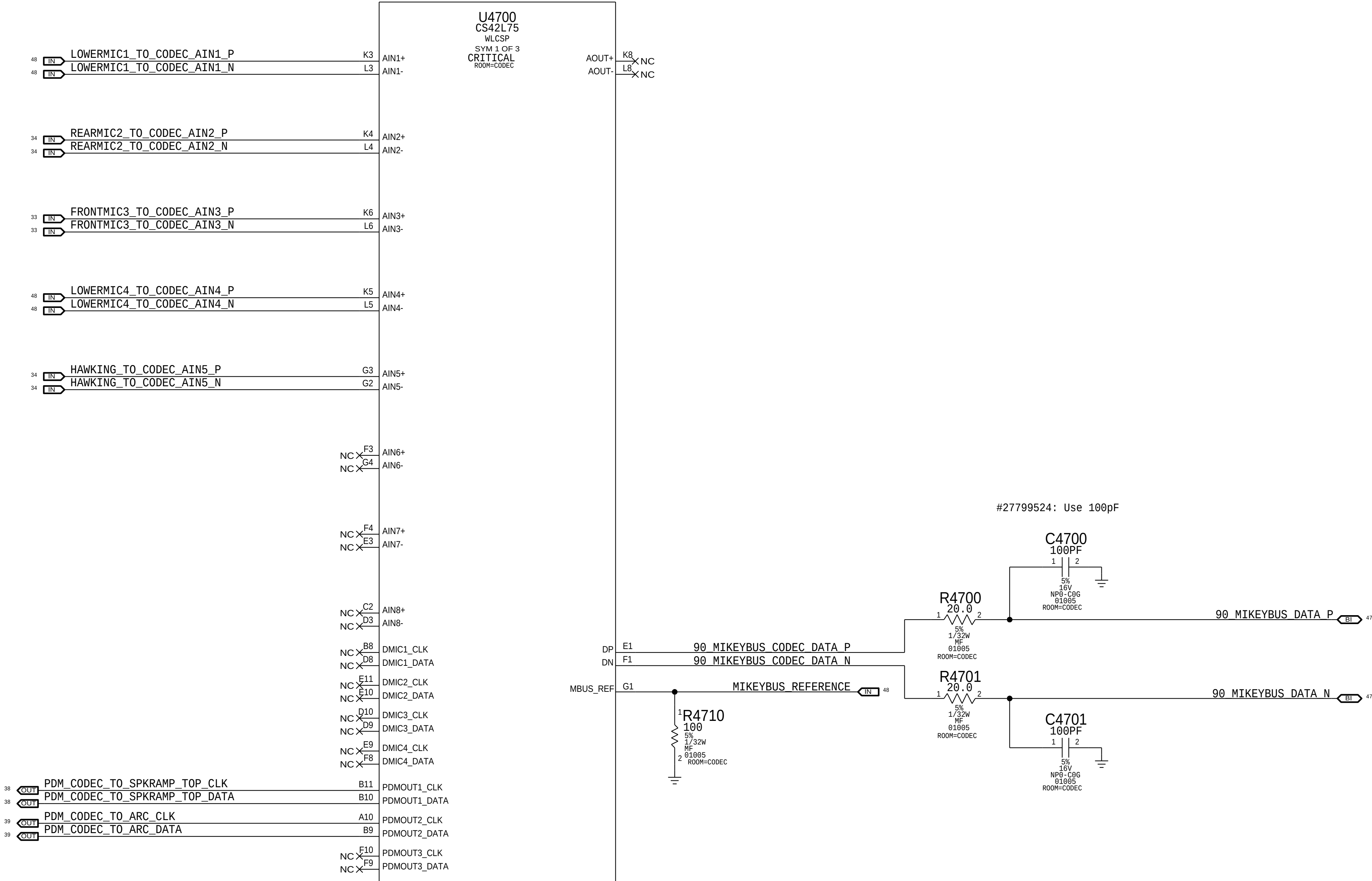
CONVOY I/O



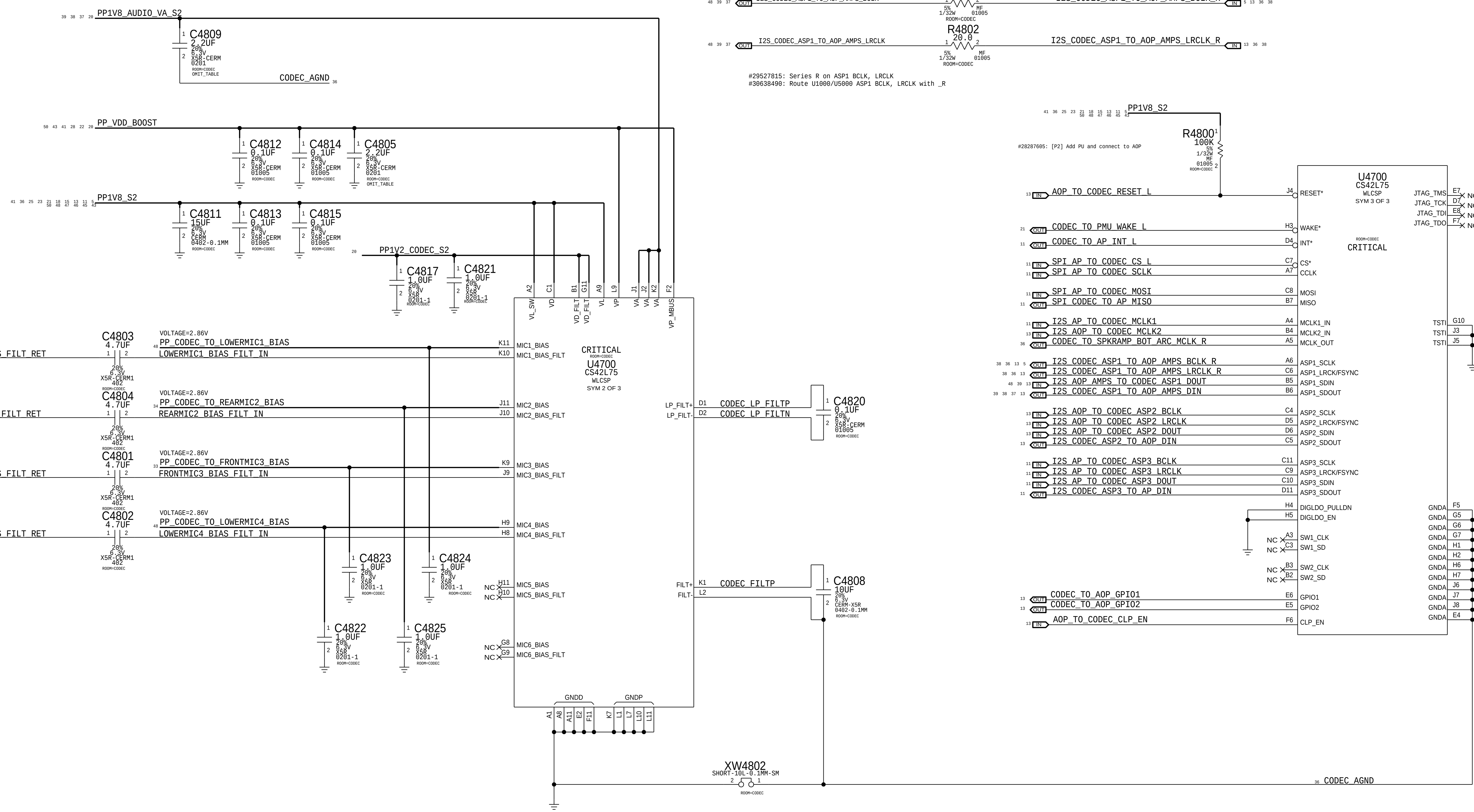
MIC3



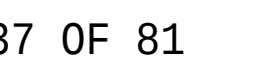
CALLAN AUDIO CODEC (ANALOG INPUTS & OUTPUTS)



CALLAN AUDIO CODEC (POWER & I/O)



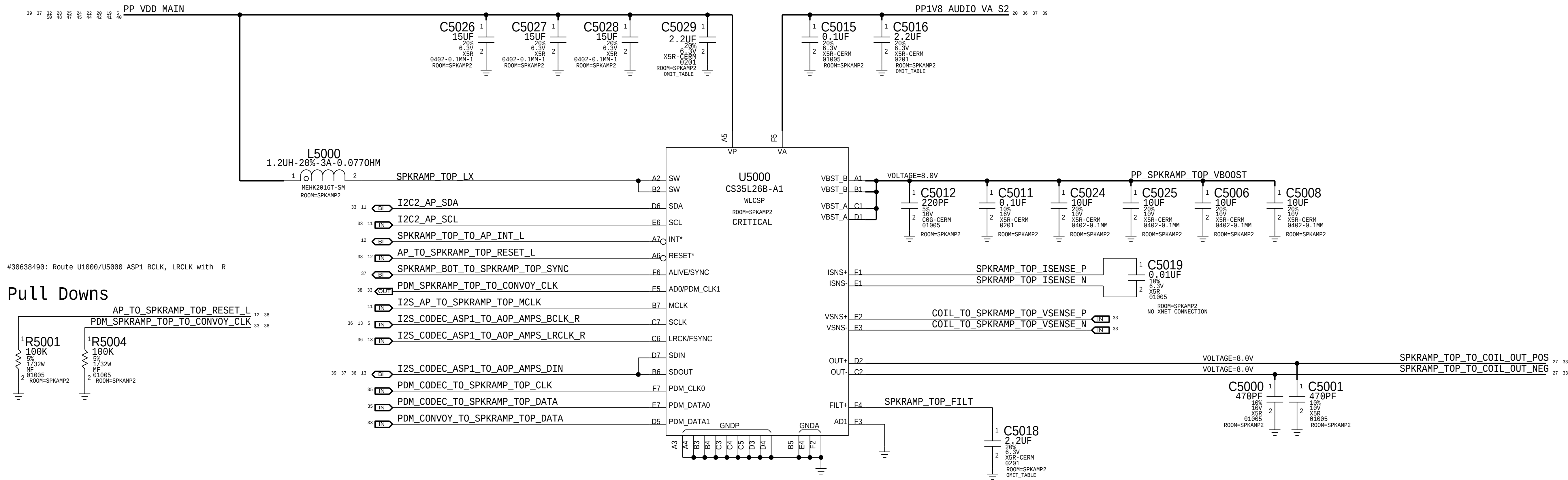

```
APN: 338S00143
I2C ADDRESS: 1000 000x
                0x80
```



North Speaker Amplifier

APN: 338S00295

I2C ADDRESS: 1000 000x (0x80)

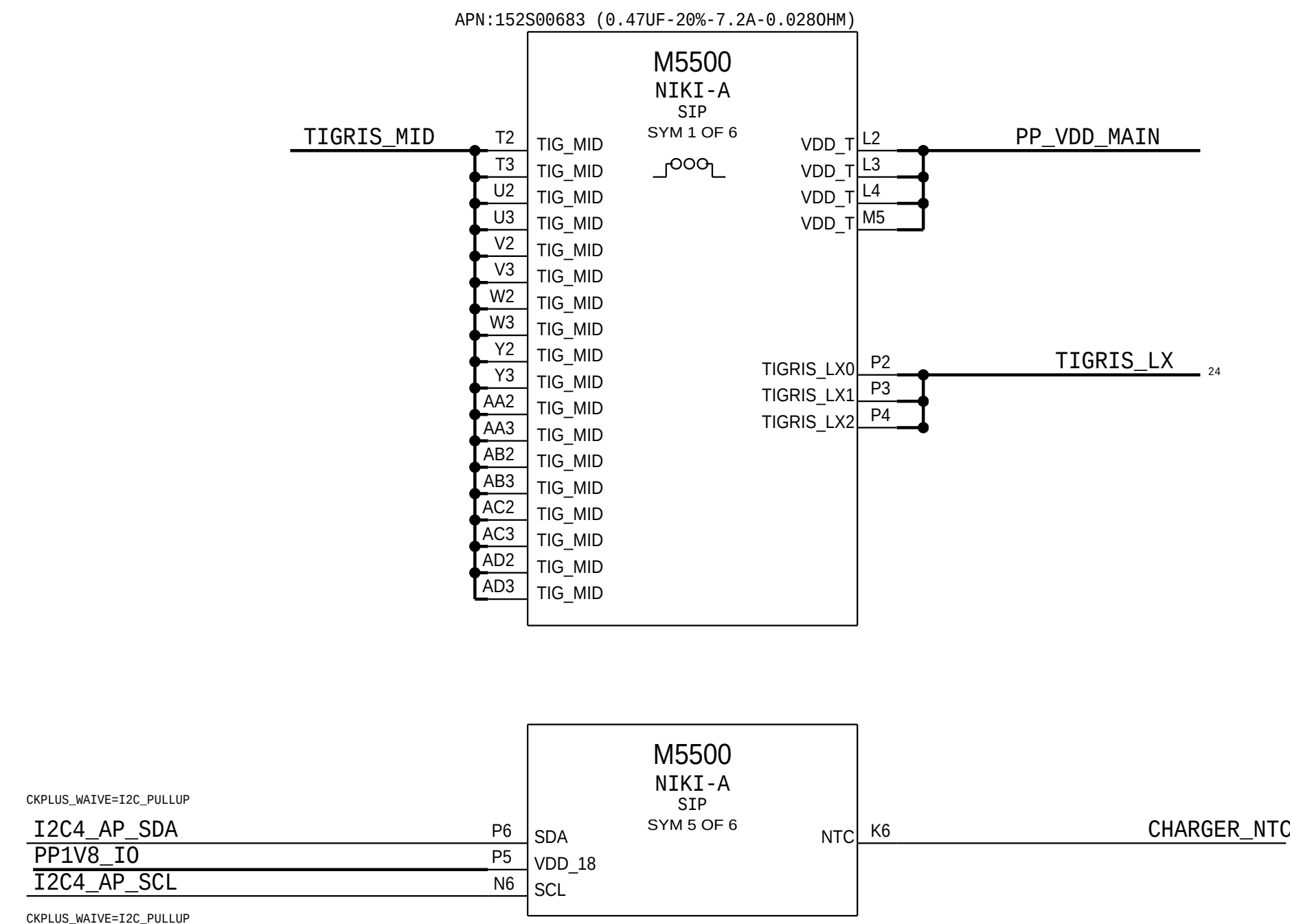


ARC D
APN: 338S00295
I2C ADDRESS: 1

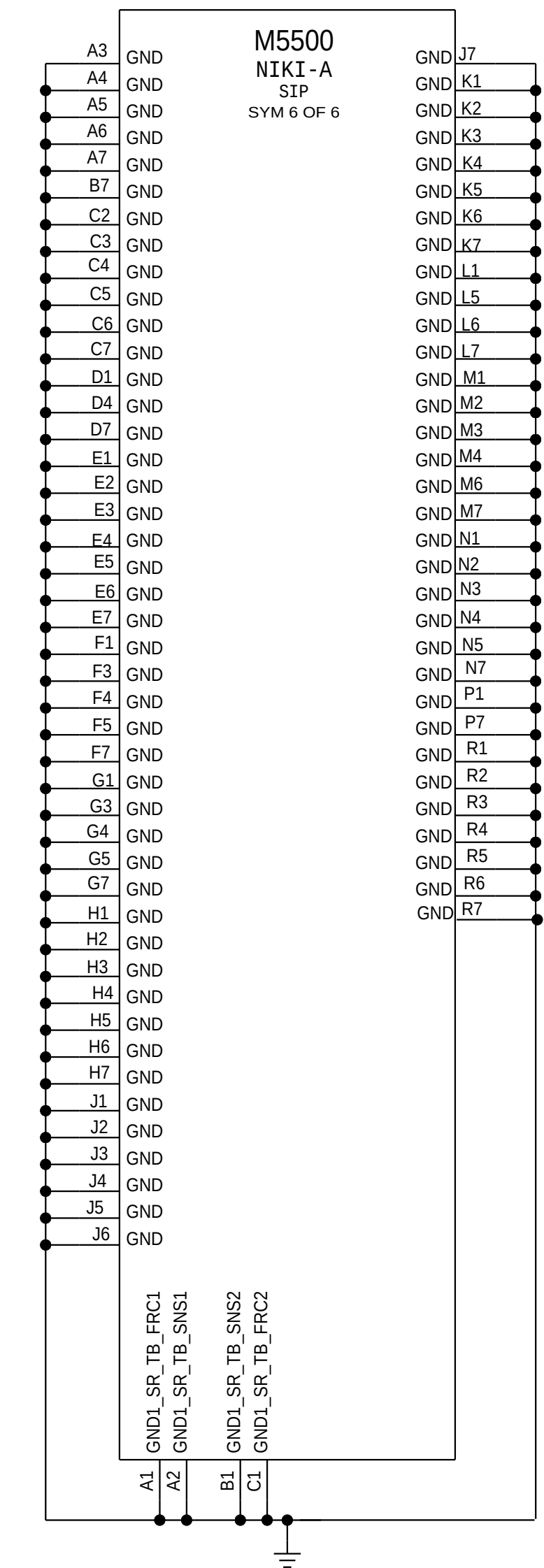
I2C ADDRESS: 1000 001x (0x82)



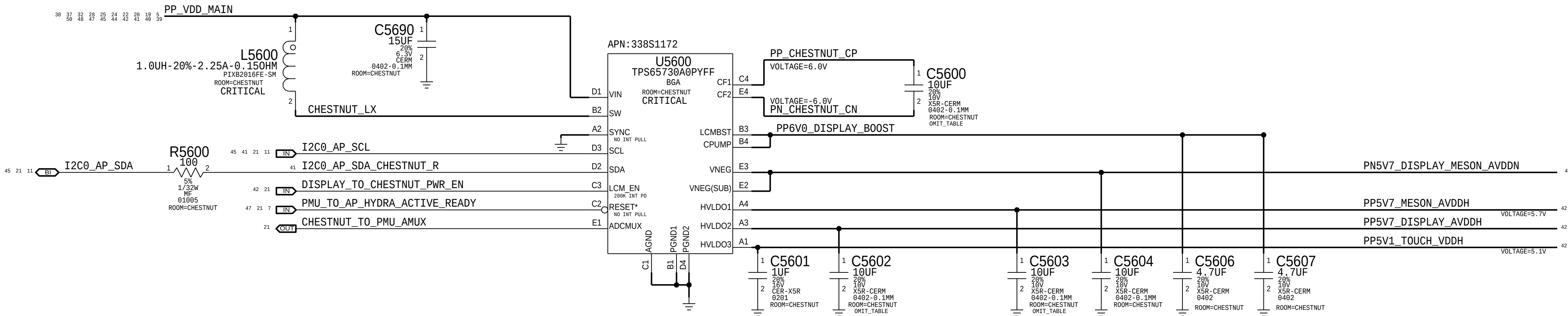
Niki SIP



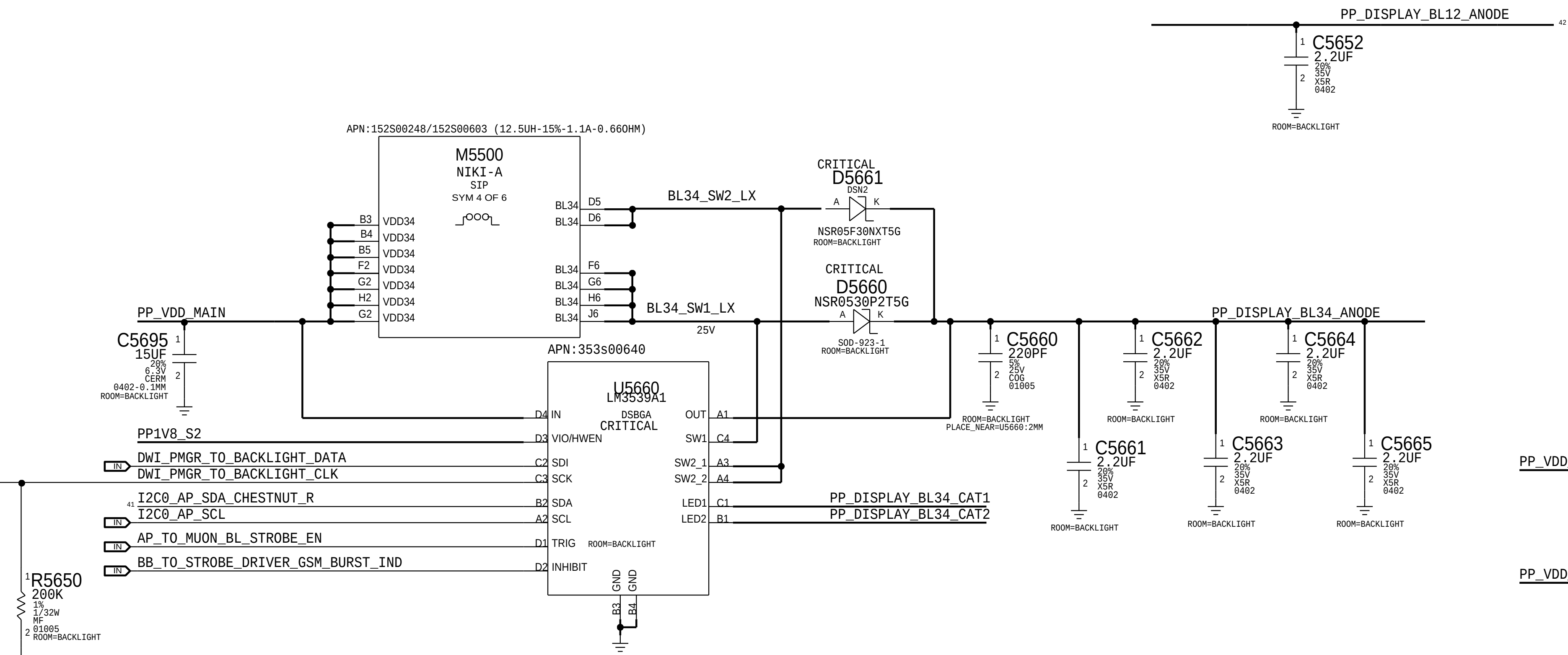
Note: inductors in SIP used for bottom speaker amplifier and backlight
See pages 49, and 56



CHESTNUT DISPLAY PMU

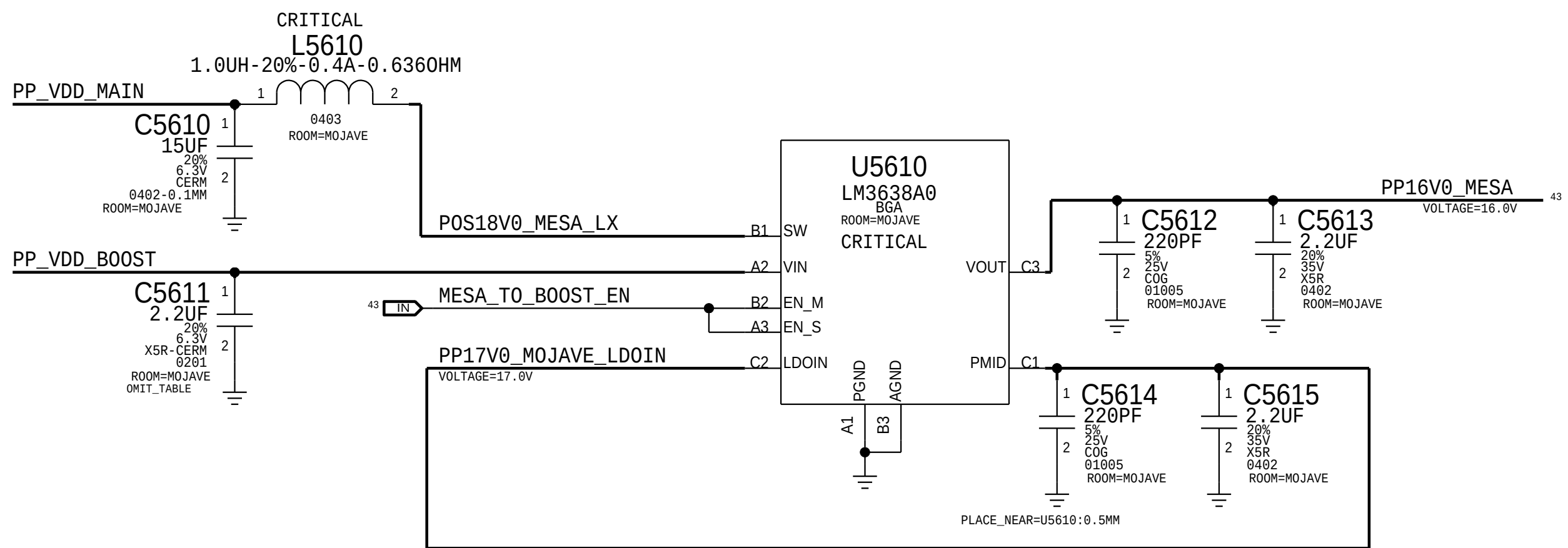


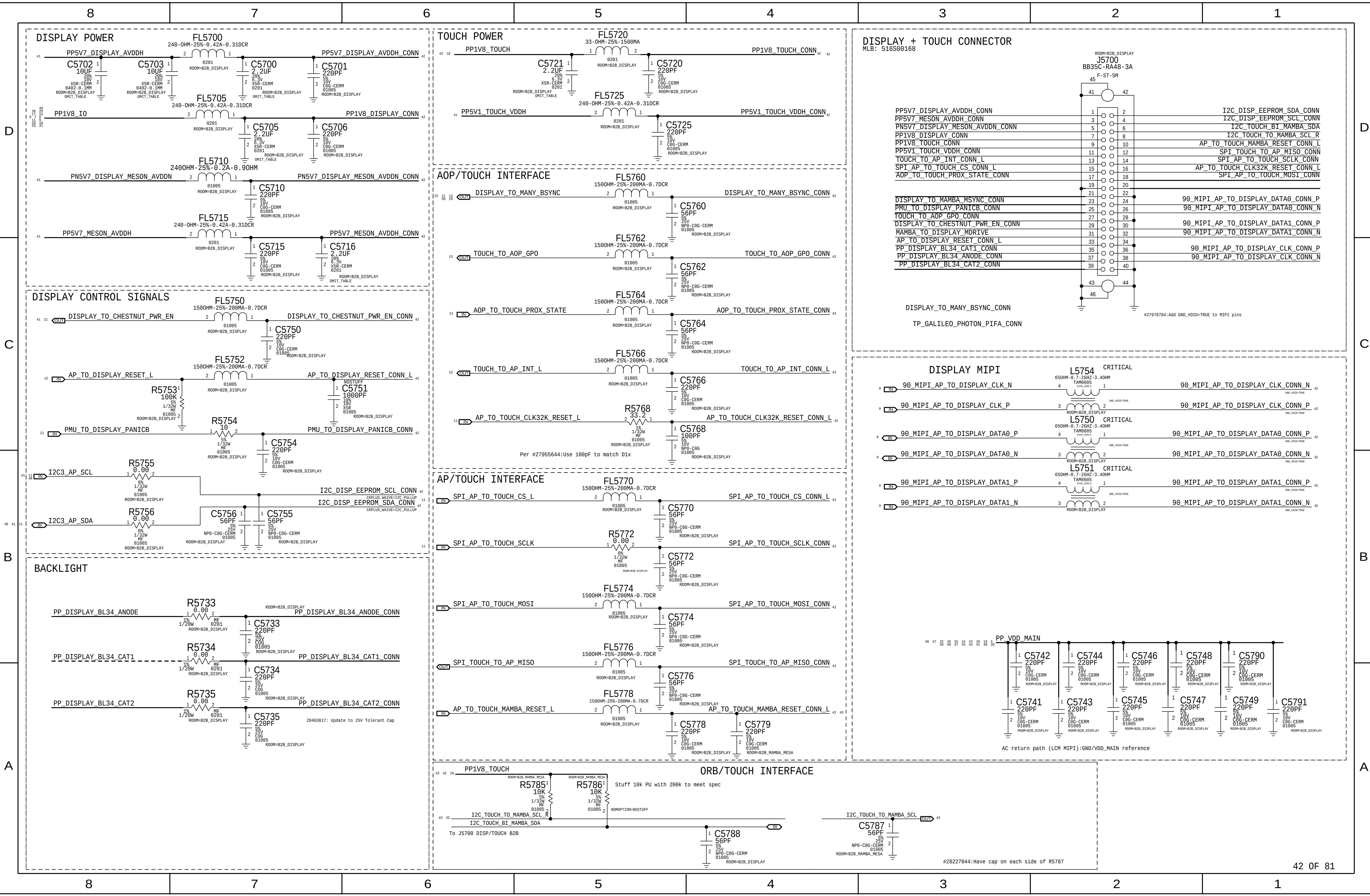
#28282573: Update C5606/C5607 to match D1x



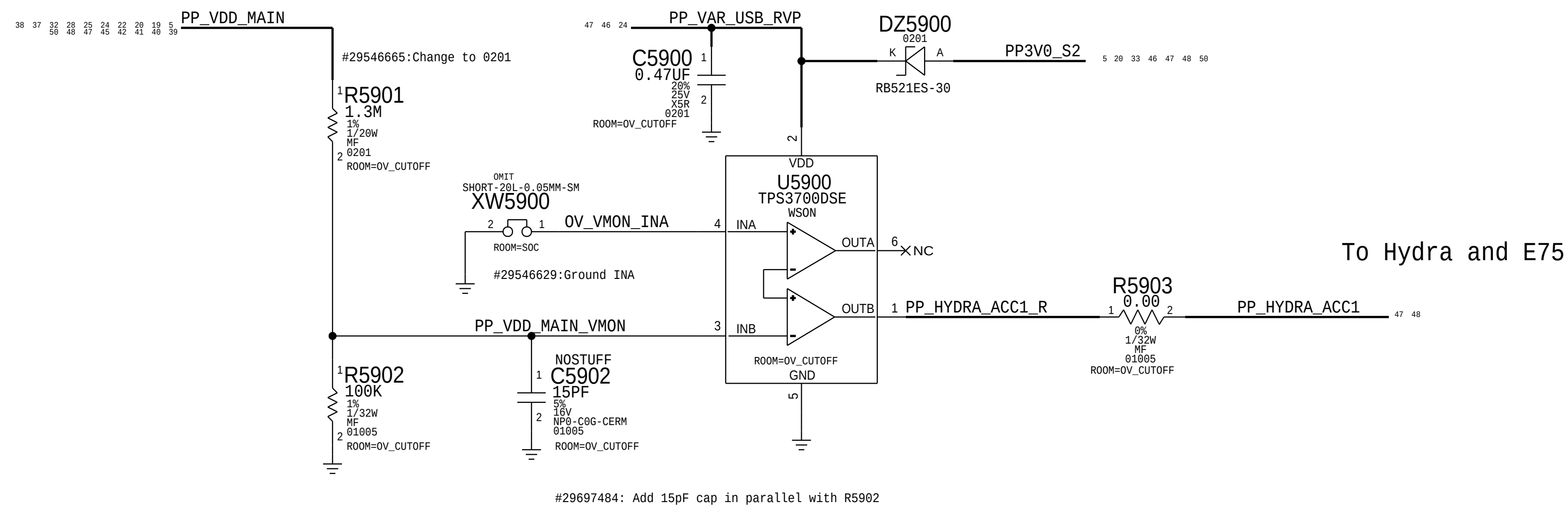
MOJAVE MESA BOOST

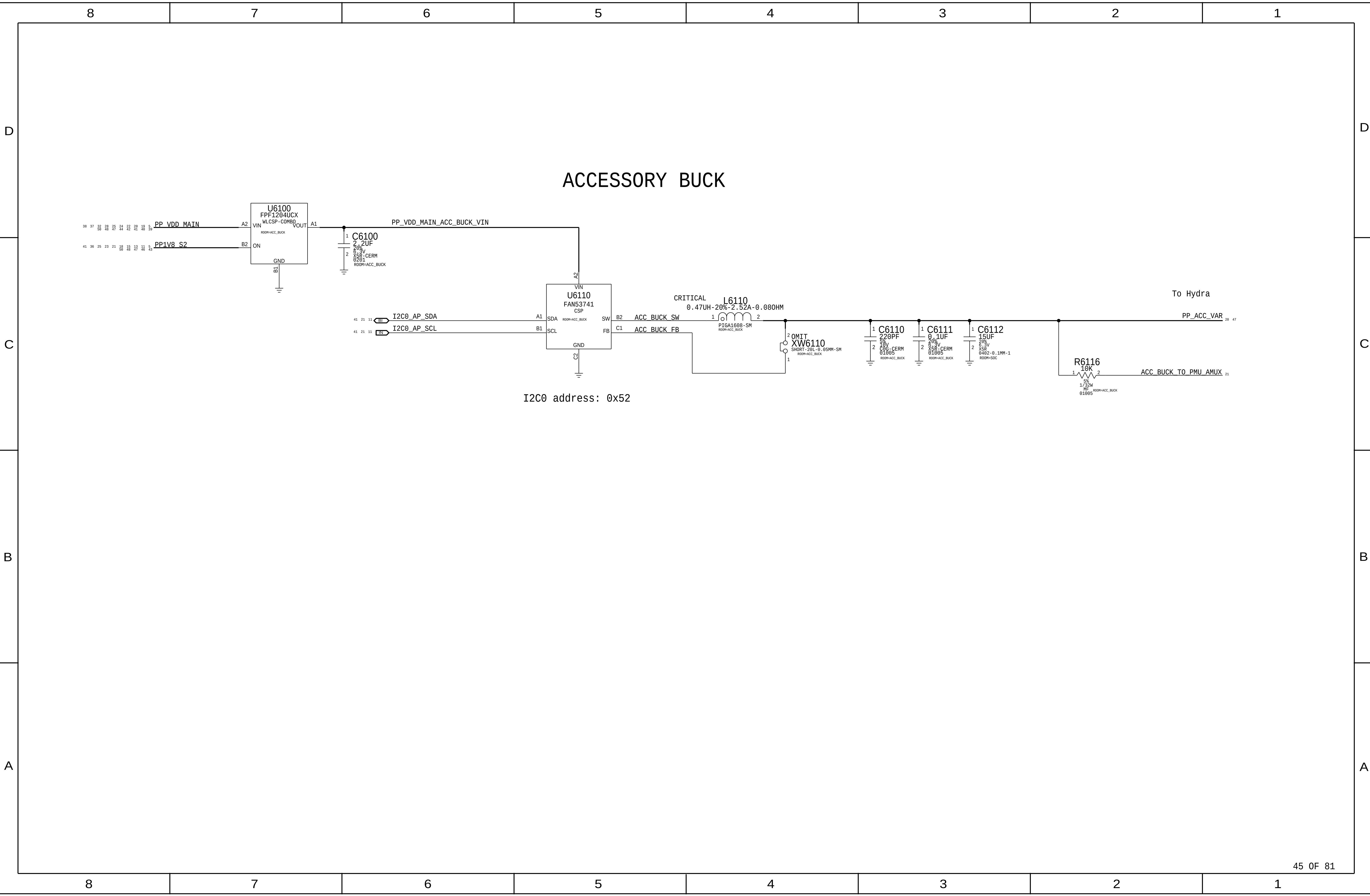
APN:353S00671

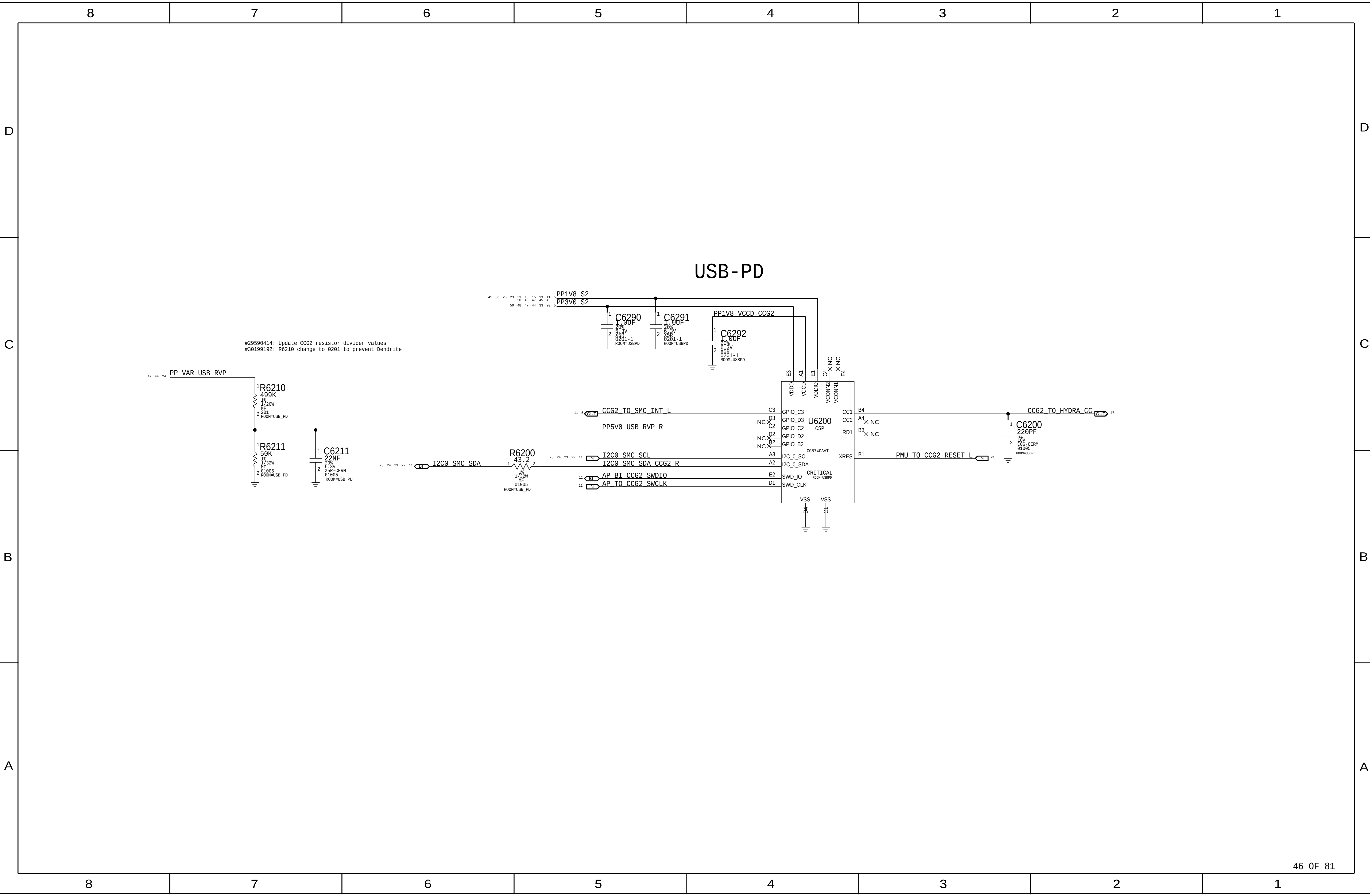




VDD_MAIN OV CUT-OFF CIRCUIT

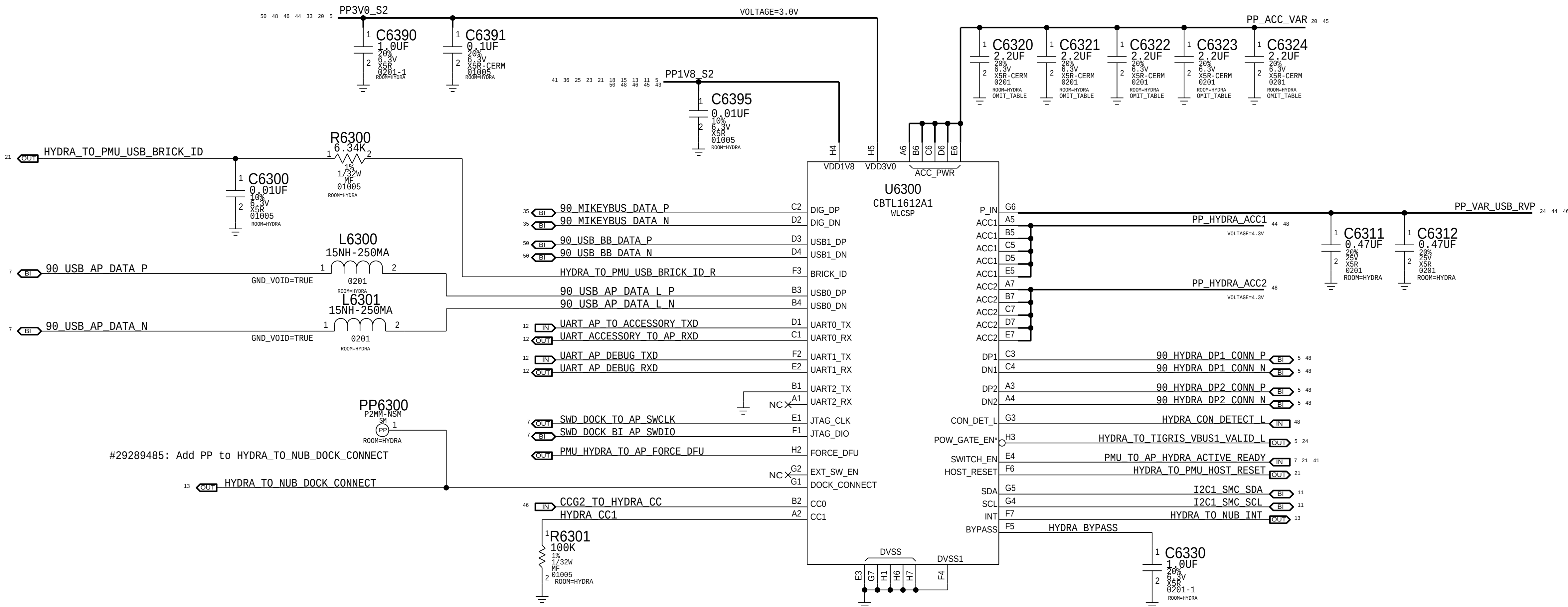




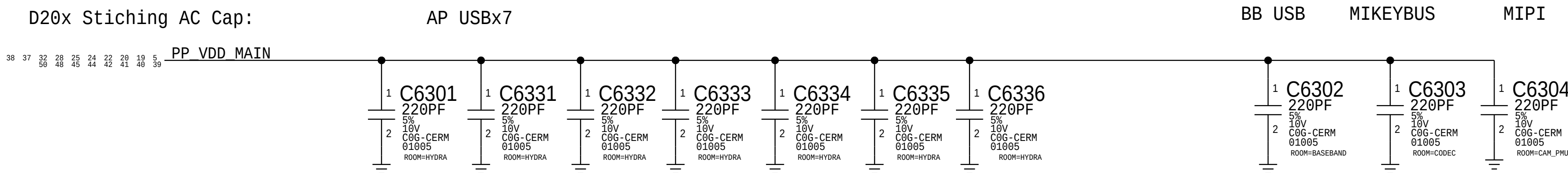


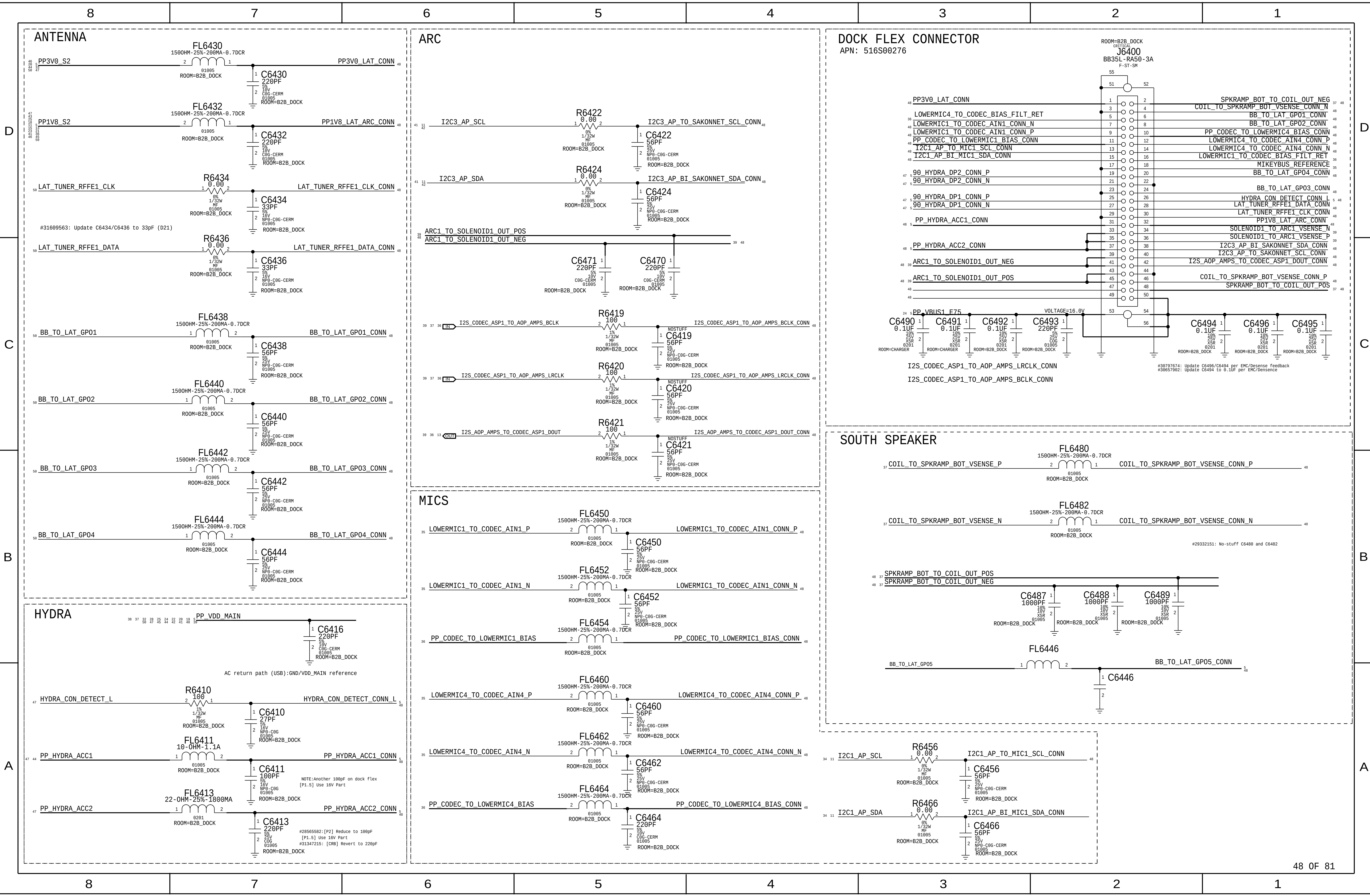
HYDRA

I2C ADDRESS: 0011010X



UART TX/RX pin names on Hydra match signal names (i.e. TX pin is input and connects to TX signal)





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I2C MAP

(see #28857723 updates)

Diags ID	Bus	Device	7-Bit Address	8-Bit Address	Device Max Speed	Operating Speed	Net Name
0	AP I2C0	Agnes (System PMU)	0x74	0xE8	400 Kbps	400 Kbps	I2C0_AP
		Chestnut (Display PMU)	0x27	0x4E	400 Kbps		
		AccessoryBuck	0x52	0xA4	400 Kbps		
		Muon (Backlight Controller 2)	0x62	0xC4	1 Mbps		
1	AP I2C1	MIC1 Temp Sensor (South)	0x54	0xA8	1 Mbps	100 Kbps	I2C1_AP
		MIC2 Temp Sensor (North Rear)	0x56	0xAC	1 Mbps		
2	AP I2C2	Adare (Top Speaker Amp)	0x40	0x80	1 Mbps	1 Mbps	I2C2_AP
		Convoy (Top Speaker Cap Sense)	0x21	0x42	1 Mbps		
3	AP I2C3	EEPROM	0x51	0xA2	400 Kbps	400 Kbps	I2C3_AP
		Muon (Backlight Controller 1)	0x62	0xC4	400 Kbps		
		Sakonnet (Arc: Hall Effect Sensor)	0x08	0x10	1 Mbps		
		EEPROM	0x50	0xA0	400 Kbps		
	ISP I2C0	Casper (Rear Wide Camera)	0x10	0x20	1 Mbps	1 Mbps	I2C0_ISP
		Grunberg+ (AF/OIS Wide Camera)	0x3C	0x78	1 Mbps		
	ISP I2C1	Billings (Rear Tele Camera)	0x20	0x40	1 Mbps	1 Mbps	I2C1_ISP
		Edwin (AF Tele Camera)	0x0D	0x1A	1 Mbps		
	ISP I2C2	Concord (Front Camera)	0x10	0x20	1 Mbps	400 Kbps	I2C2_ISP
	ISP I2C3	Neon (Strobe - CW : LM3566)	0x63	0xC6	1 Mbps	400 Kbps	I2C3_ISP
		Ansel (Camera PMU)	0x40	0x80	1 Mbps		
		Neon (Strobe - AM : LM35662)	0x67	0xCE	1 Mbps		
	SEP I2C	EEPROM	0x51	0xA2	400 Kbps	400 Kbps	I2C4_AP
4	AOP I2C0	Doppler (Prox Sensor)	0x58	0xB0	1 Mbps	400 Kbps	I2C0_AOP
		CT709 (ALS : Firefish)	0x29	0x52	400 Kbps		
5	AOP I2C1	Adare (Bottom Speaker Amp)	0x40	0x80	1 Mbps	1 Mbps	I2C1_AOP
		Adare (Arc : Driver)	0x41	0x82	1 Mbps		
		Apache3 (Mesa Power Supply)	0x61	0xC2	1 Mbps		I2C1_AOP_ISO
		Turtle	0x2C	0x58	1 Mbps		
6	SMC I2C0	Tigris2 (Charger)	0x71	0xE2	400 Kbps	400 Kbps	I2C0_SMC
		TPS61280 (VDD Boost)	0x75	0xEA	400 Kbps		
		CCG2 (USB PD)	0x12 (0x08)	0x24 (0x10)	1 Mbps		
		Iktara	0x39	0x72	400 Kbps		
		SN45546 (Gas Gauge)	0x55	0xAA	400 Kbps		
7	SMC I2C1	Hydra	0x1A	0x34	400 Kbps	400 Kbps	I2C1_SMC

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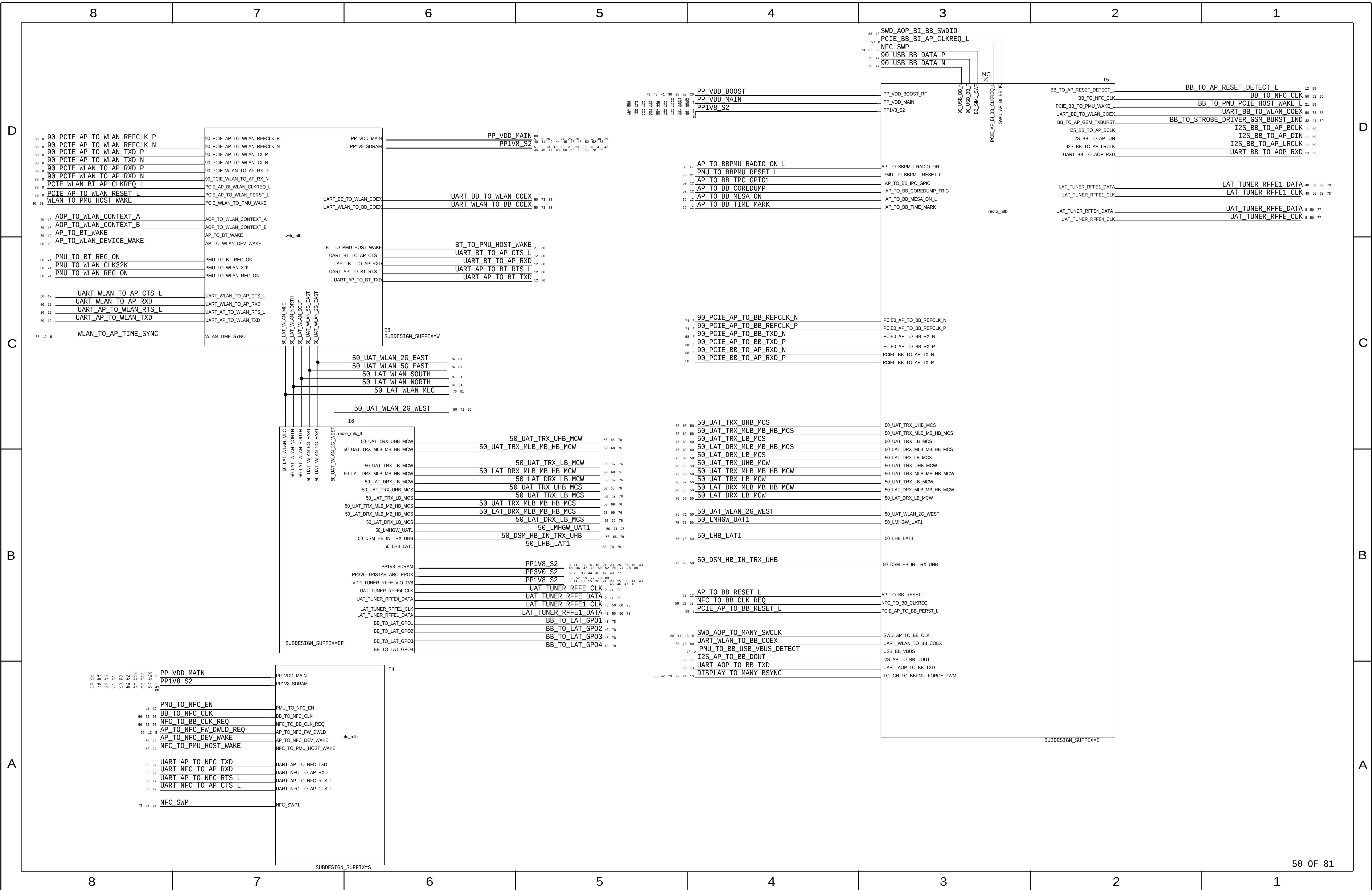
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3

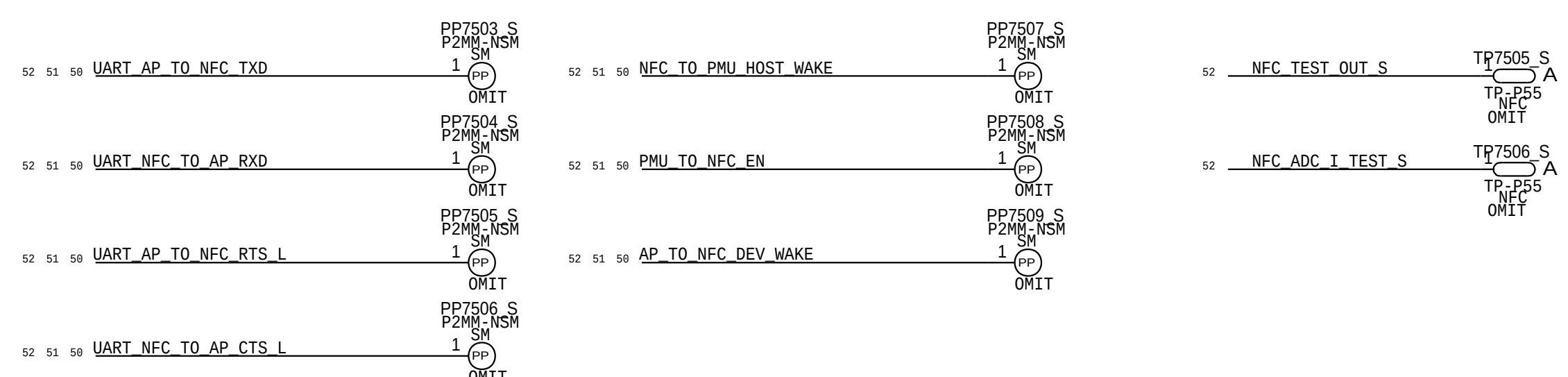
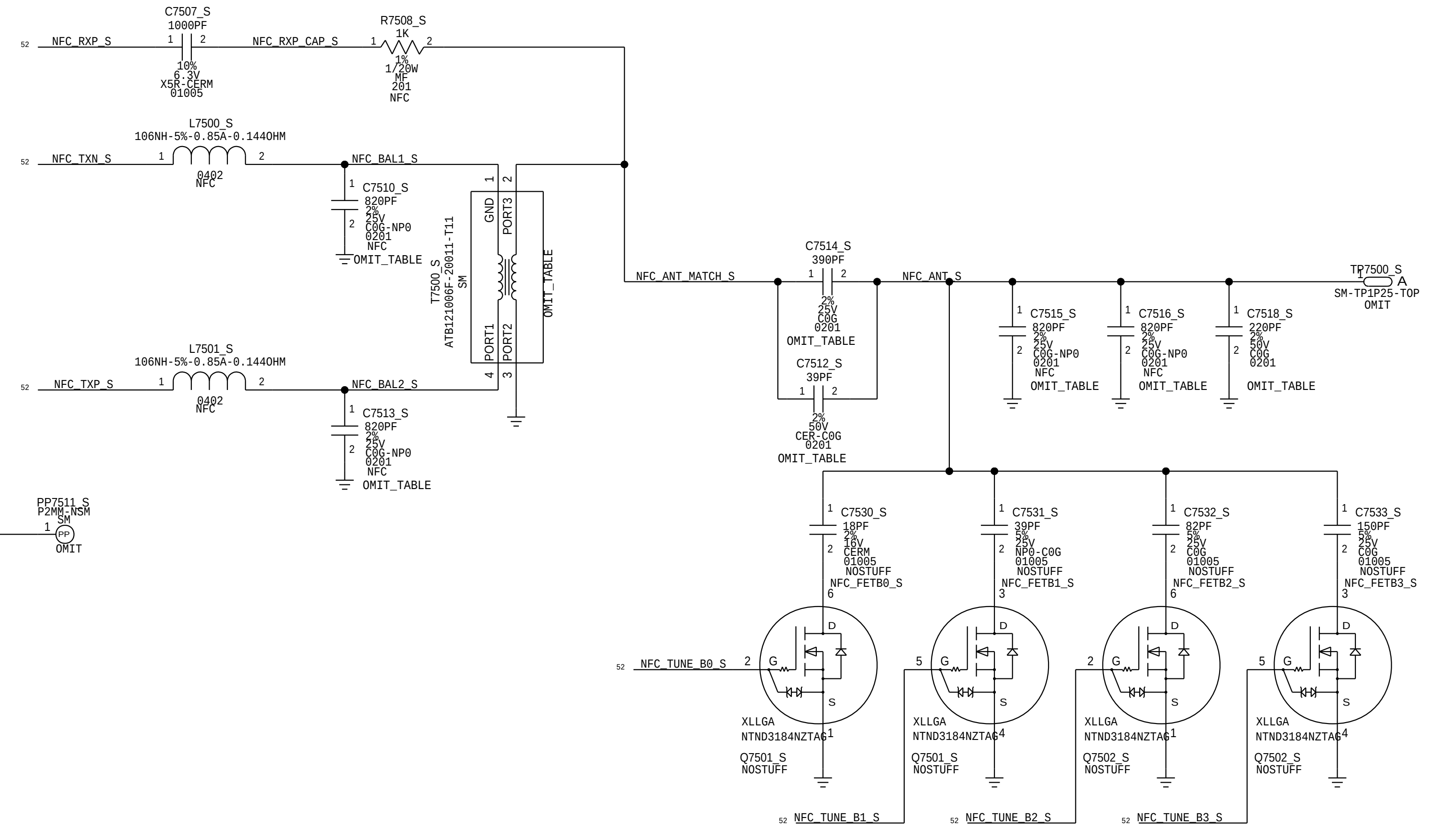
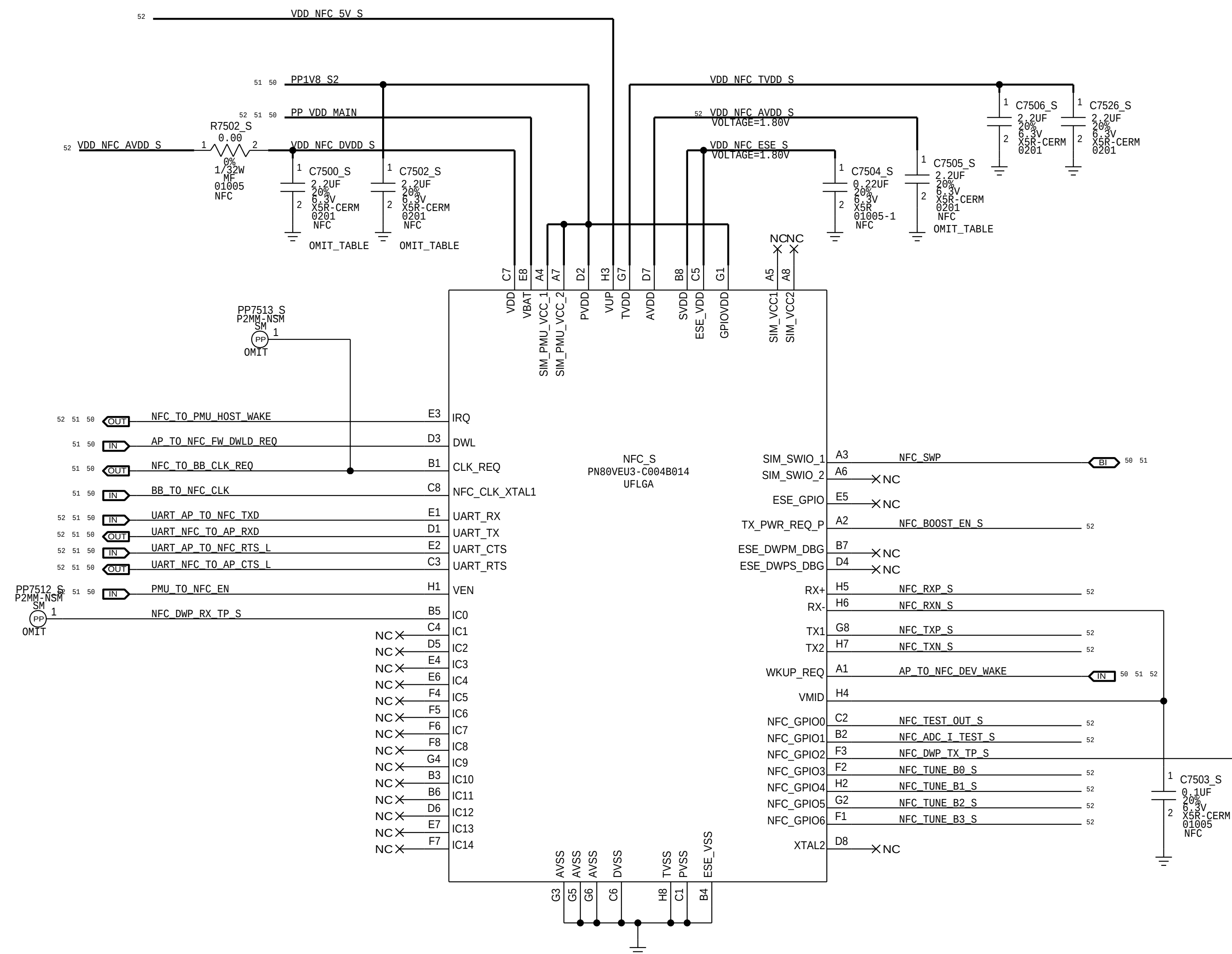
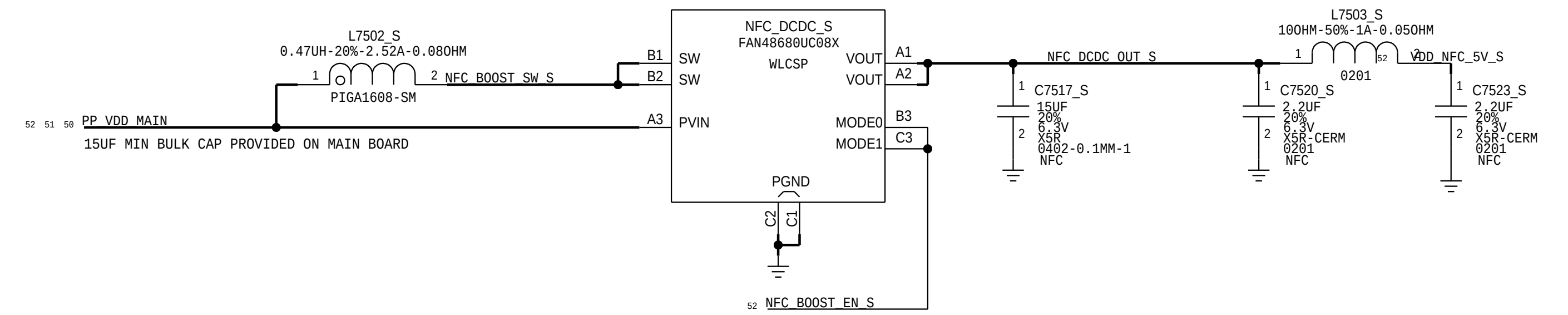
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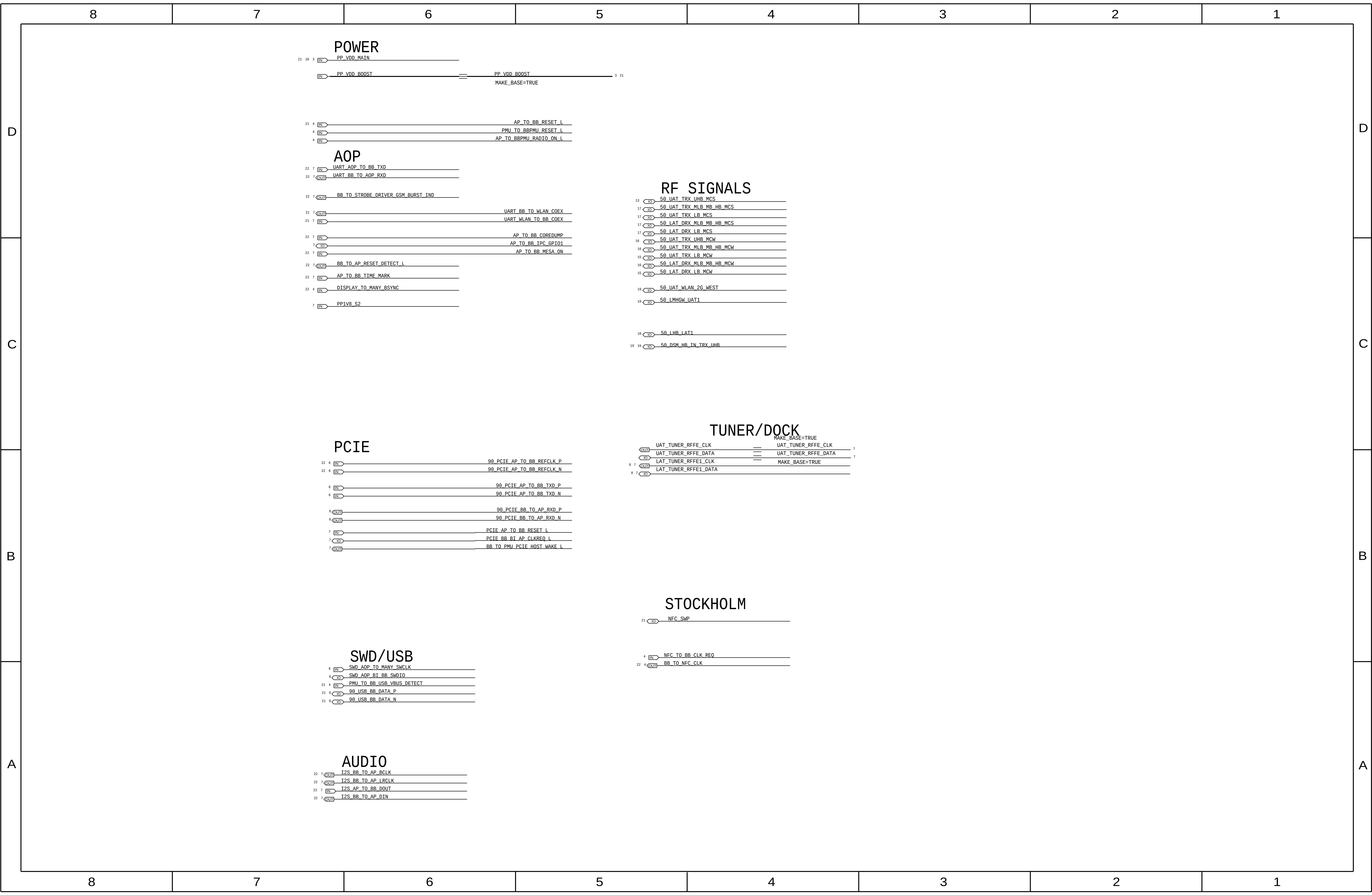
1

STOCKHOLM

5V BOOSTER

NFC CONTROLLER





RF PMIC: SWITCHERS & LDOS

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D

D

C

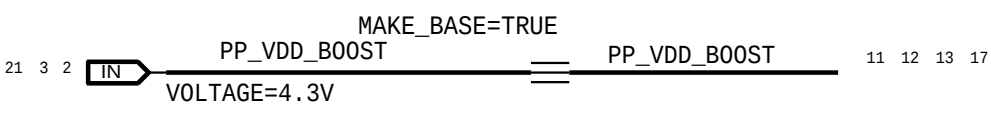
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B

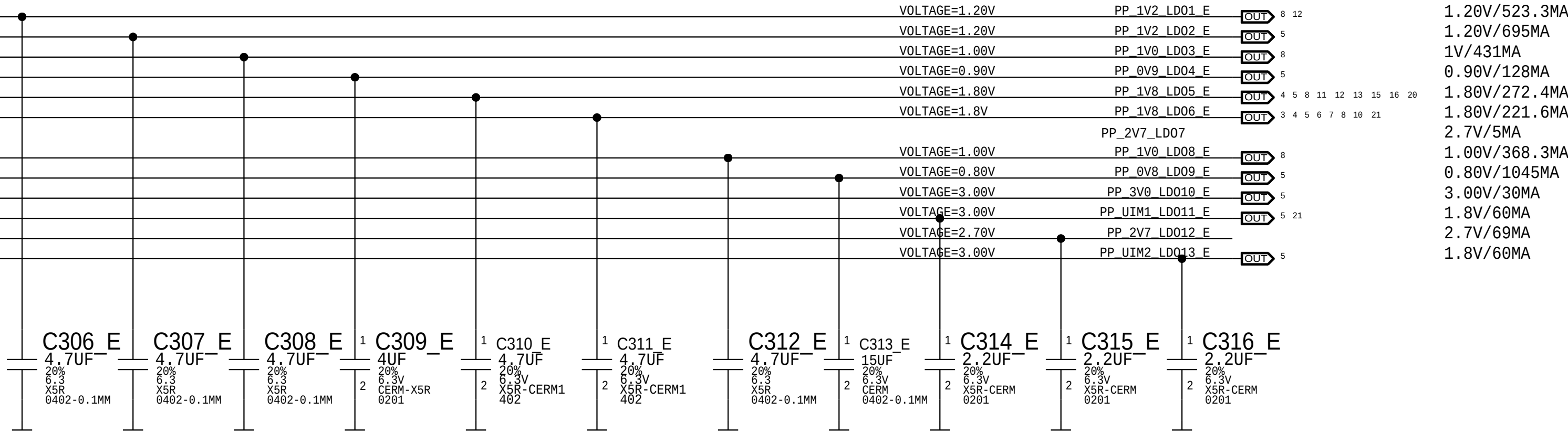
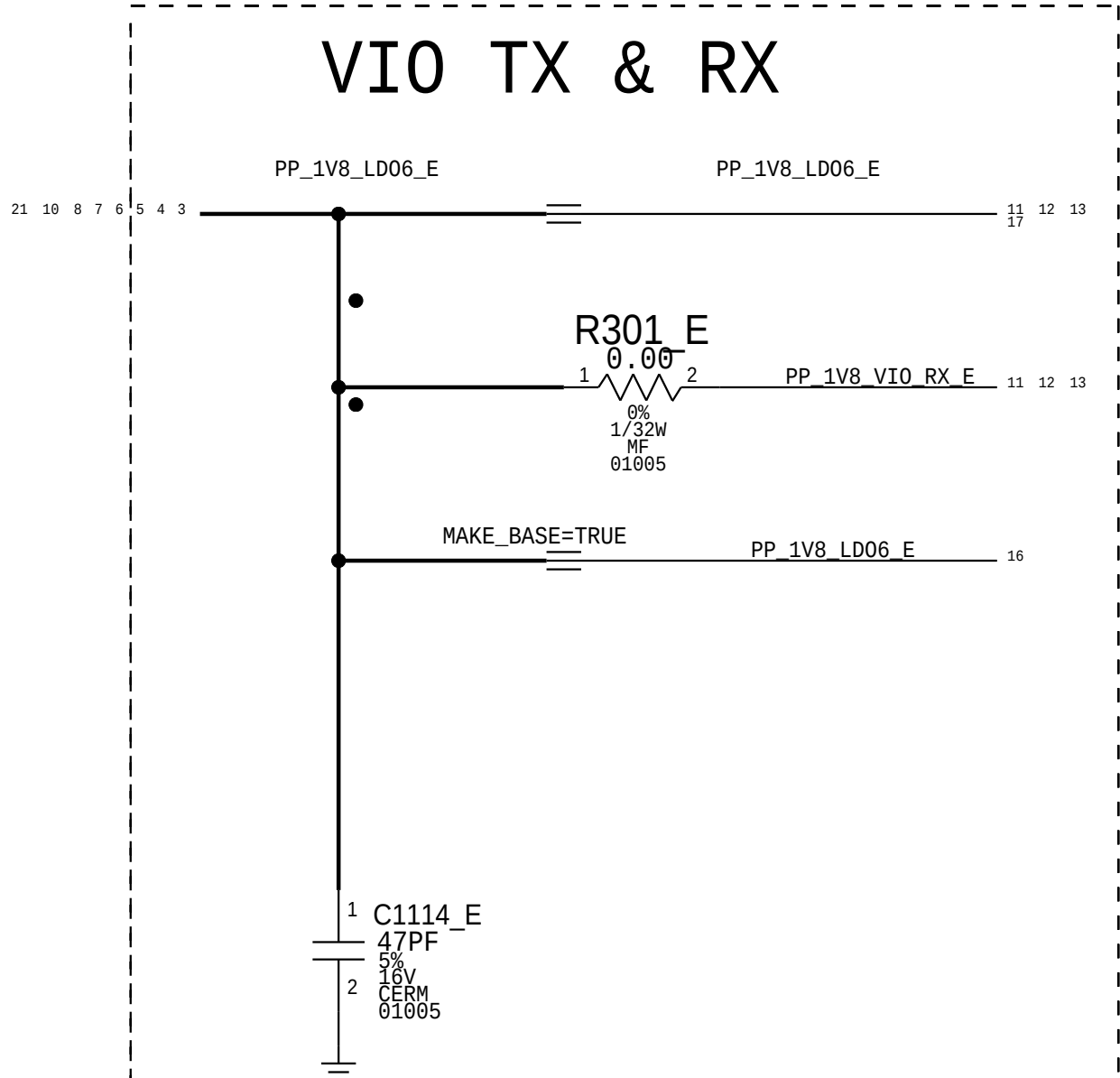
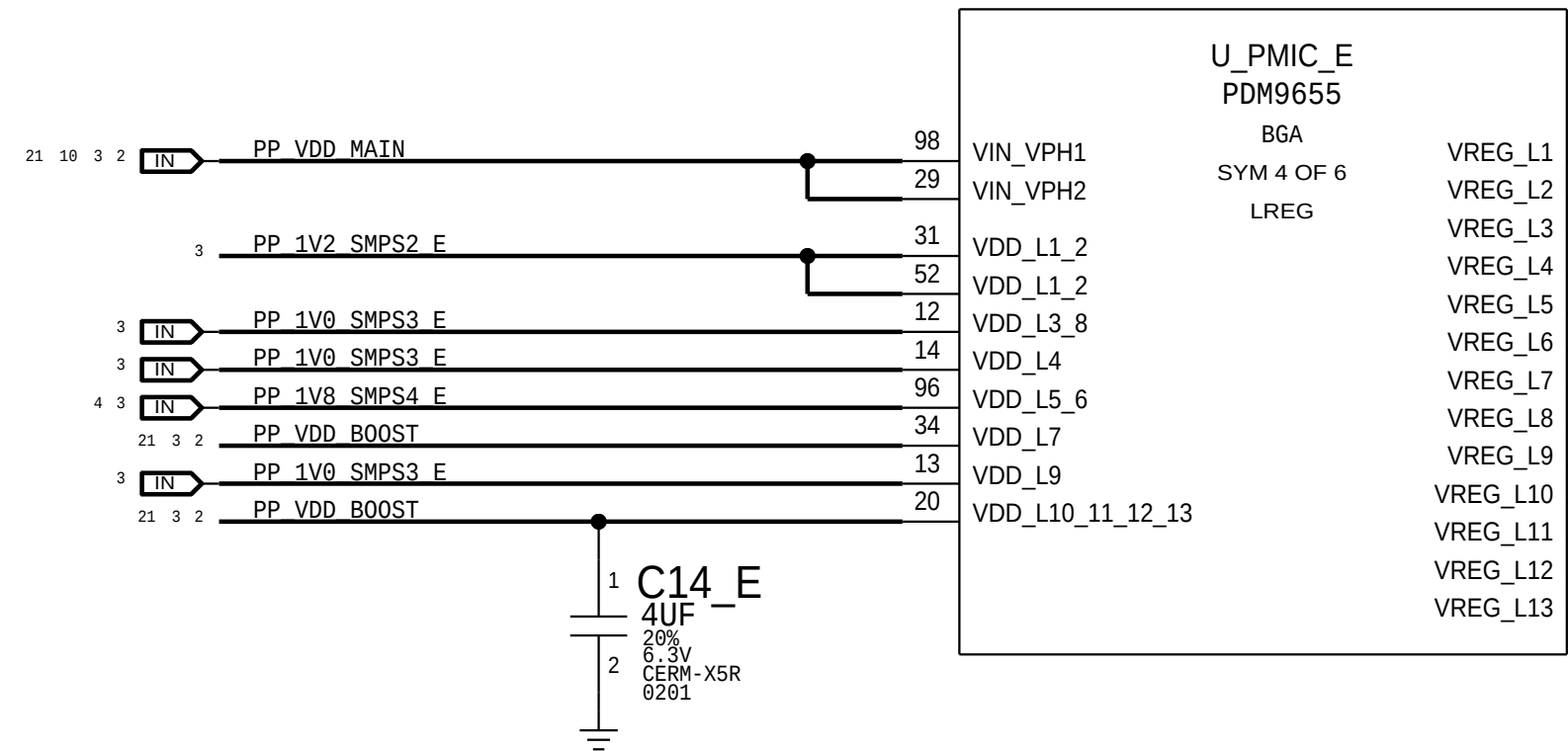
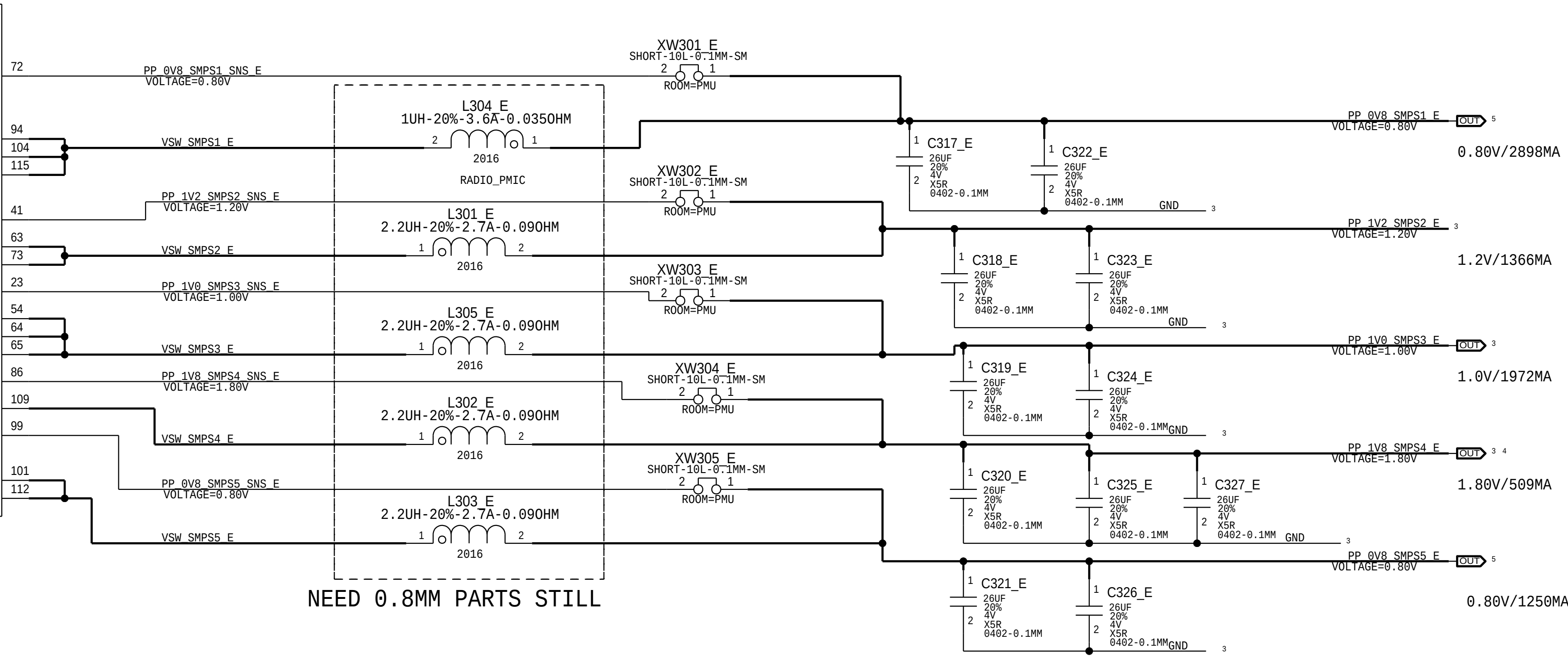
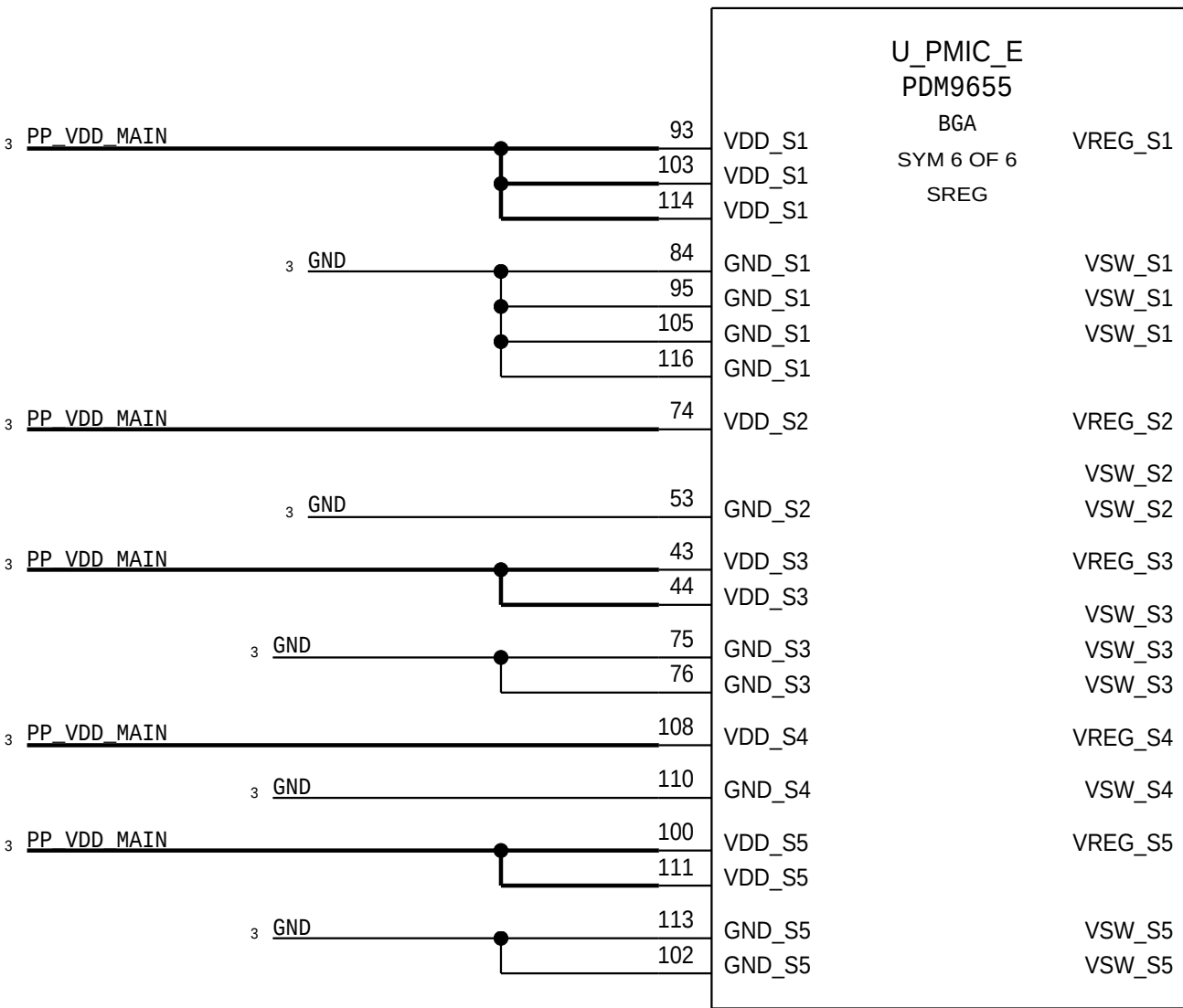
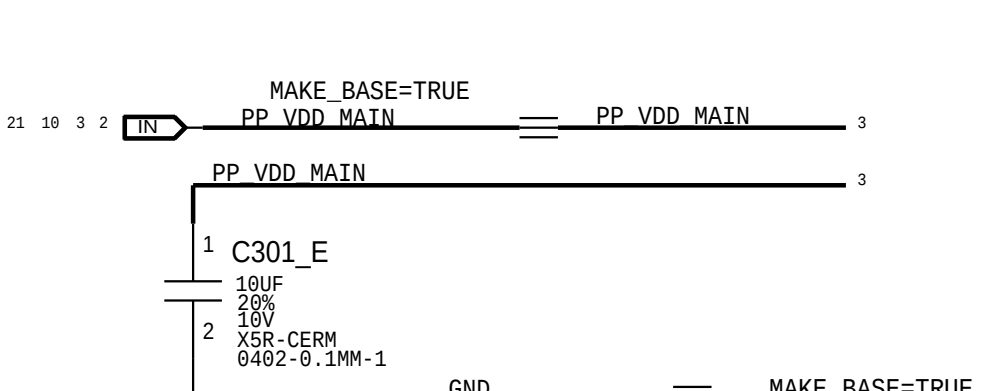
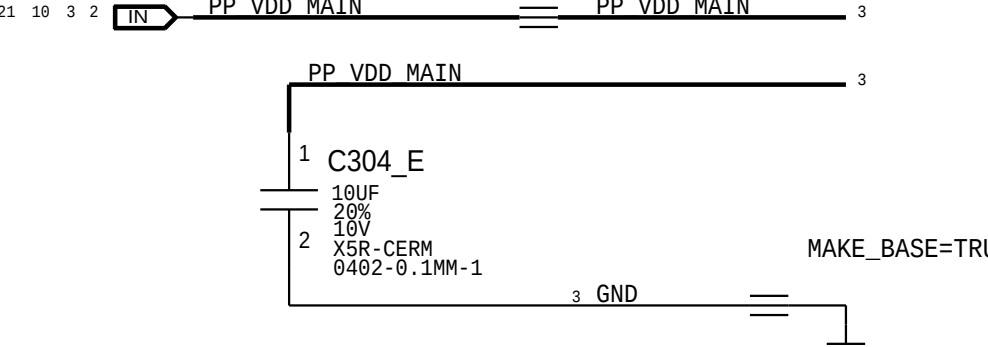
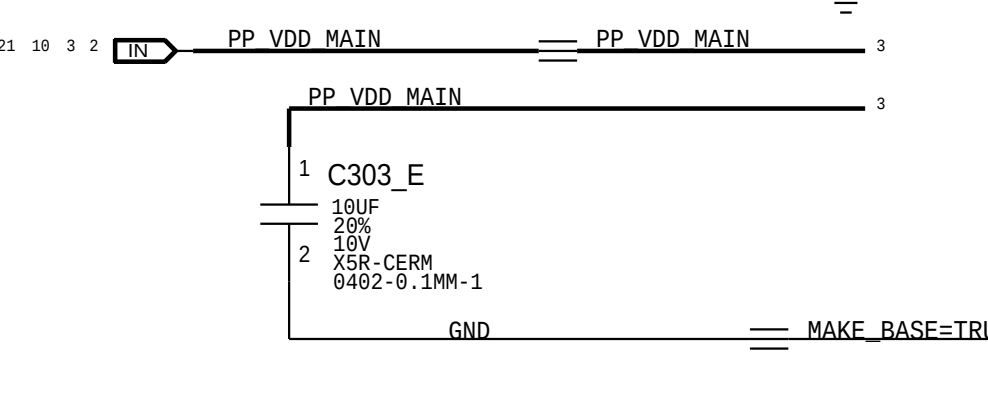
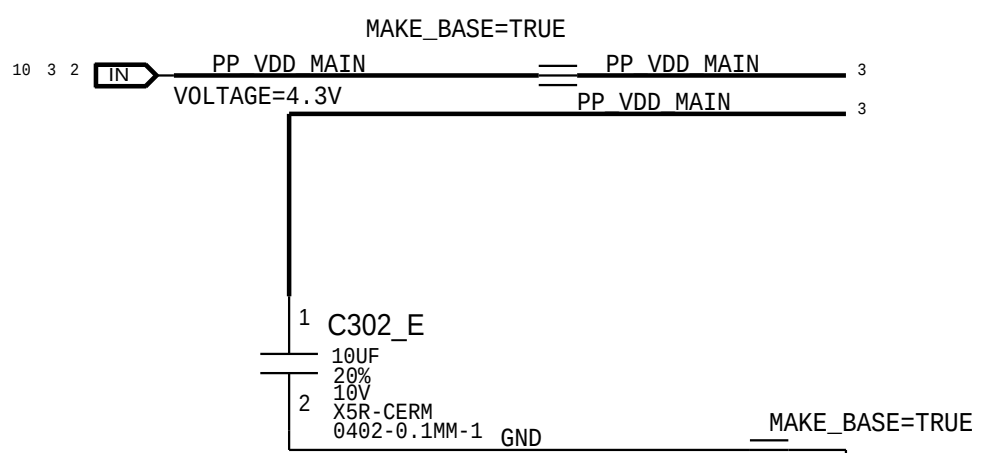
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A

A



SWITCHERS BULK CAPS



RF PMIC: XTAL, CLOCKS AND MISCELLANEOUS

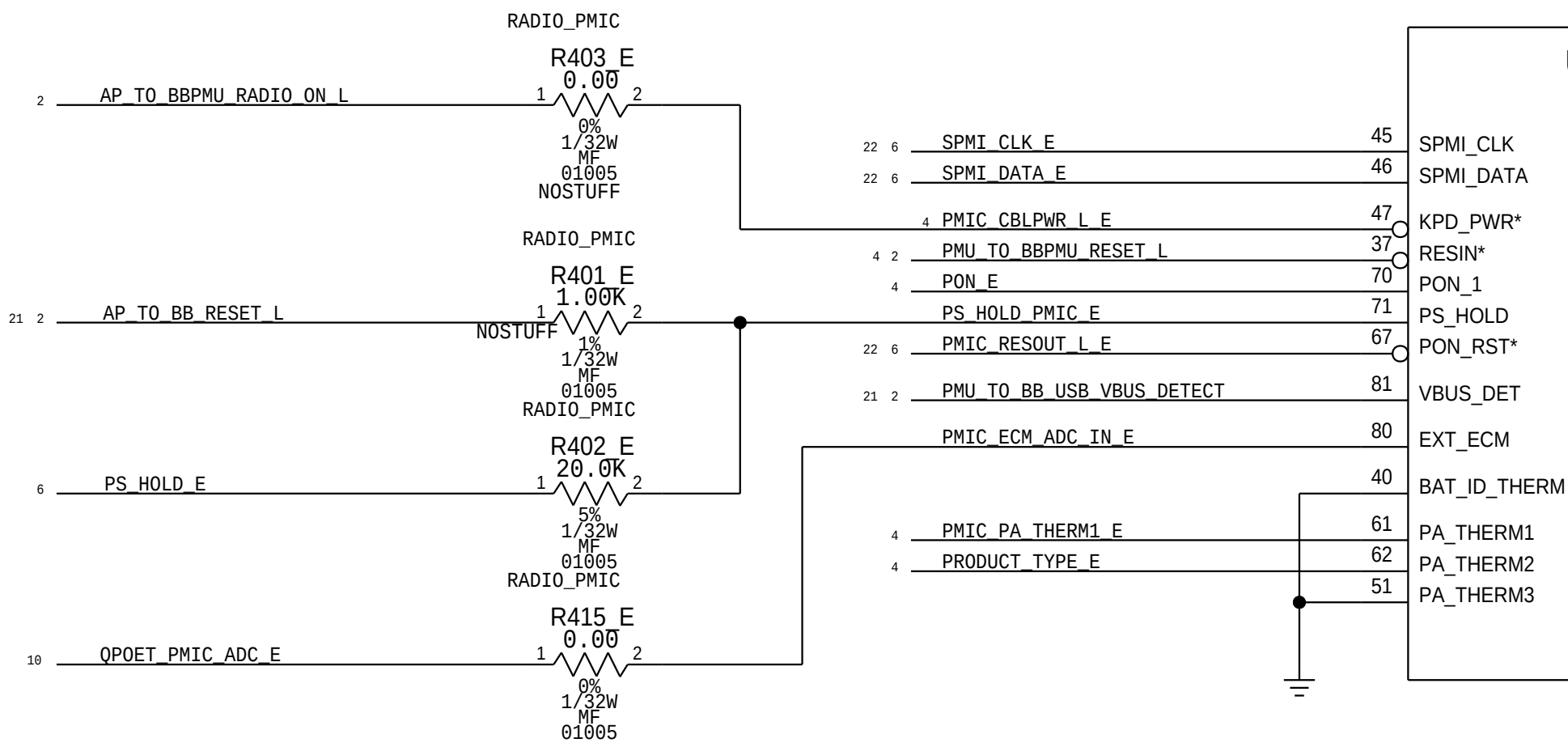
HW_REV_ID	R407	R408	MLB/RADIO_DEV
<0.10V	NOSTUFF	51.1K	T500/DEV1
0.15V-0.25V	422K	51.1K	PROTO1.5
0.35V-0.45V	180K	51.1K	PROTO2.5/DEV2
0.55V-0.65V	105K	51.1K	PROTO1/DEV3
0.75V-0.85V	66.5K	51.1K	PROTO2/DEV4
0.95V-1.05V	44.2K	51.1K	EVT
1.15V-1.25V	28K	51.1K	CARRIER/DEV5
1.35V-1.45V	16.5K	51.1K	DVT/DEV6
1.55V-1.65V	7.87K	51.1K	PVT/DEV7
1.75V-1.85V	10K	NOSTUFF	SPARE

NOTE XTAL SYMBOL MIRRORED COMPARED TO PREVIOUS SCH

PA_THERM2	R409	R410	PRODUCT TYPE	Y
0.00V-0.10V	NOSTUFF	51.1K	RADIO DEV	1
0.15V-0.25V	422K	51.1K	3RD TYPE	2
0.35V-0.45V	180K	51.1K	POR MLB (FF)	3
0.55V-0.65V	105K	51.1K	SPARE	4
0.75V-0.85V	66.5K	51.1K	SPARE	5
0.95V-1.05V	44.2K	51.1K	SPARE	6
1.15V-1.25V	28K	51.1K	SPARE	7
1.35V-1.45V	16.5K	51.1K	SPARE	8
1.55V-1.65V	7.87K	51.1K	SPARE	9
1.75V-1.85V	10K	NOSTUFF	SPARE	A

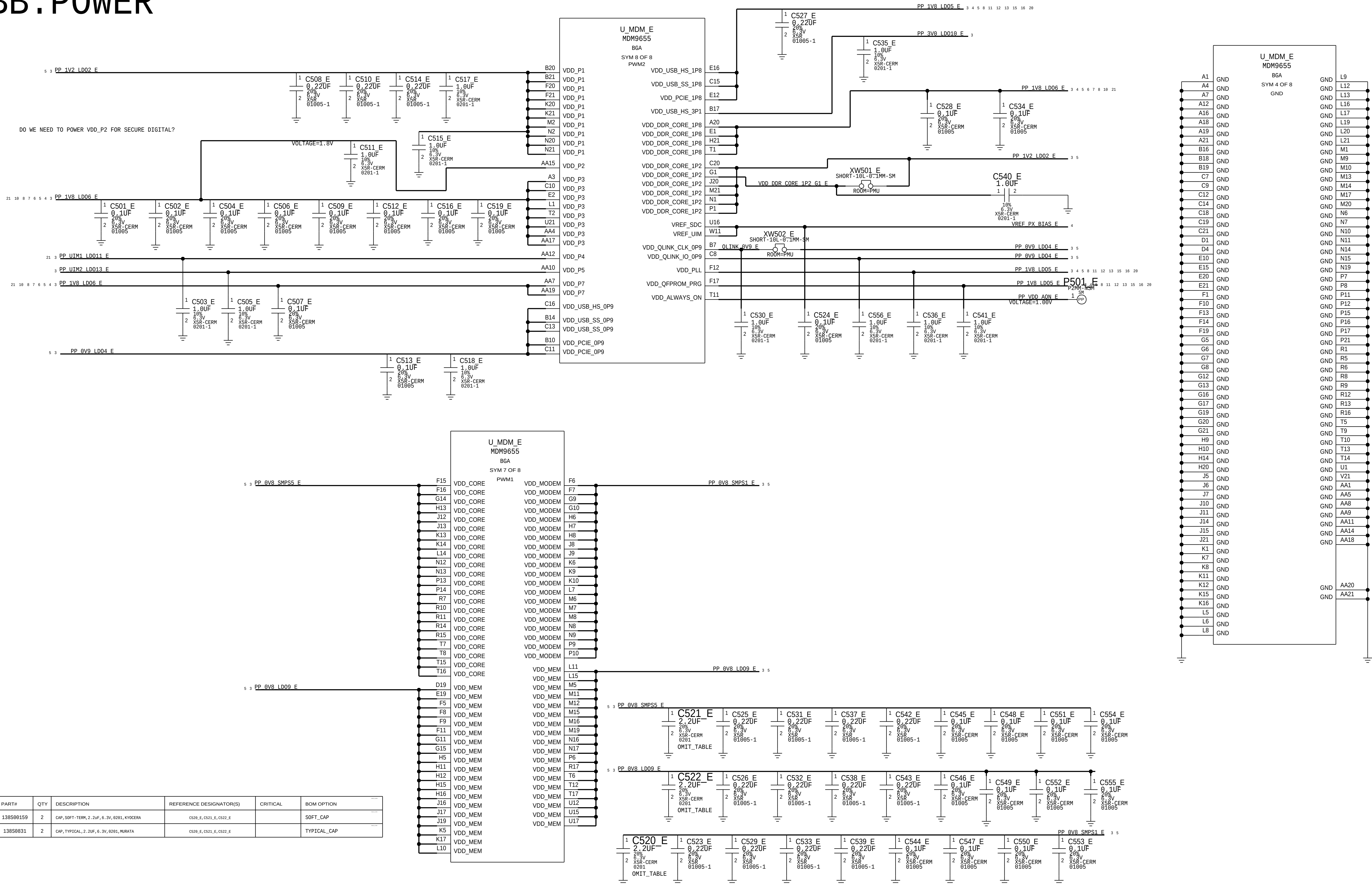
JPN_ROW_SEL	R430	R431	MAV PLATFORM	Y	SKU
<0.10V	NOSTUFF	51.1K	MAV17.0	0	ROW
0.15V-0.25V	422K	51.1K	MAV17.1	1	JPN
0.35V-0.45V	180K	51.1K	MAV17.2	2	ROW
0.55V-0.65V	105K	51.1K	MAV17.3	3	JPN
0.75V-0.85V	66.5K	51.1K	MAV17.4	4	ROW
0.95V-1.05V	44.2K	51.1K	MAV17.5	5	JPN
1.15V-1.25V	28K	51.1K	MAV17.6	6	ROW
1.35V-1.45V	16.5K	51.1K	MAV17.7	7	JPN
1.55V-1.65V	7.87K	51.1K	MAV17.8	8	ROW
1.75V-1.85V	10K	NOSTUFF	MAV17.9	9	JPN

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
118S0606	1	RESISTOR, 01005, 105K OHMS	R430_E	CRITICAL	JPN
117S0197	1	RESISTOR, 01005, 100K OHMS	R430_E	CRITICAL	ROW

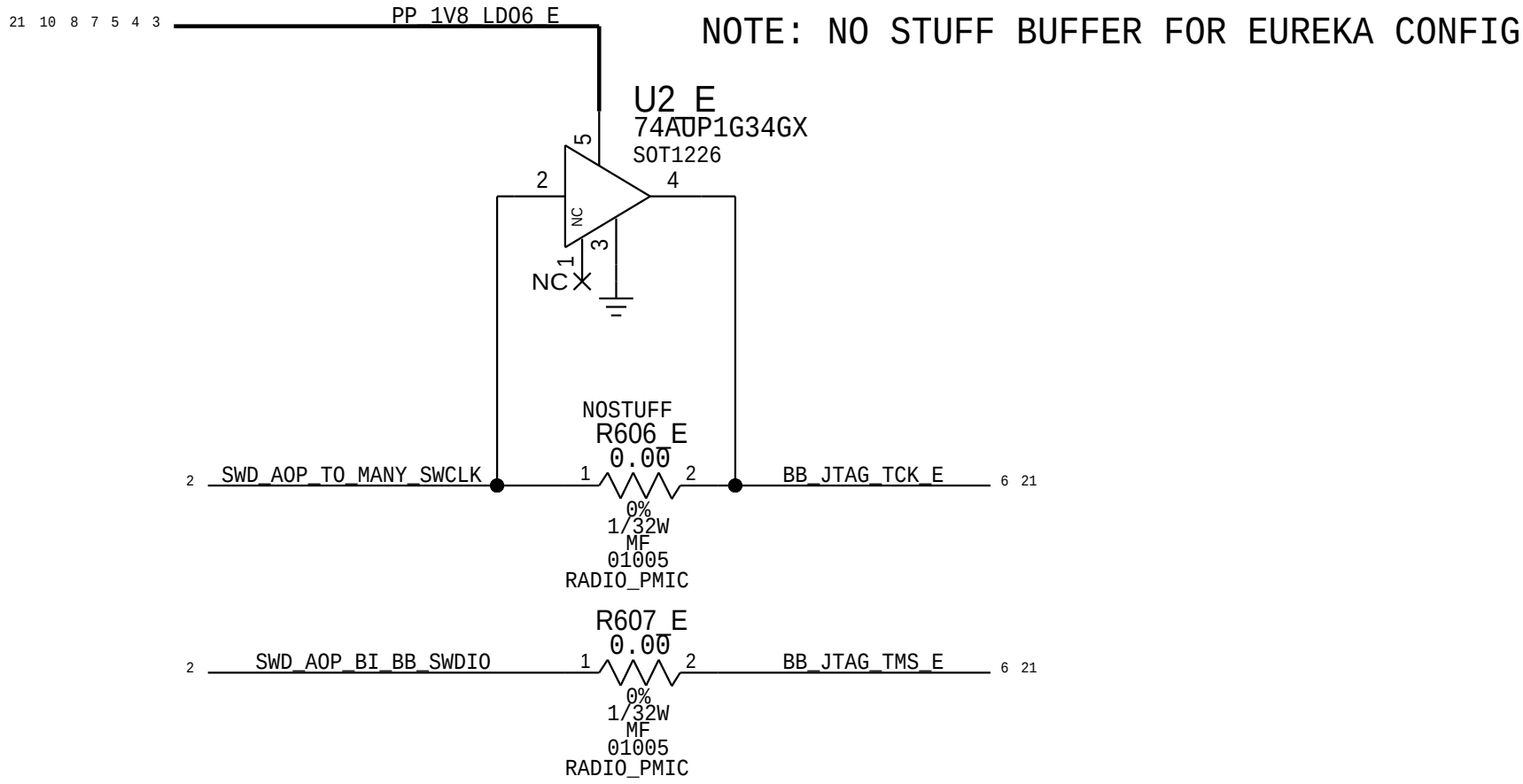
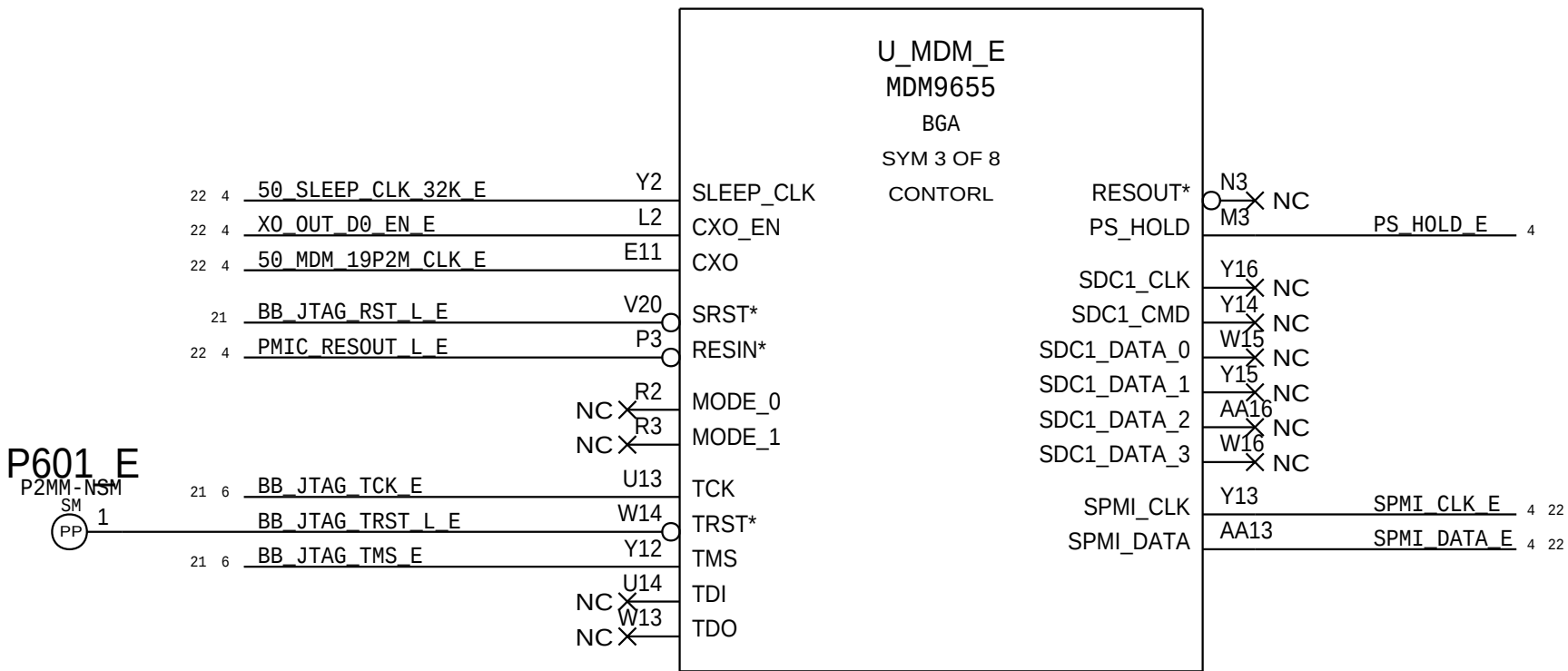
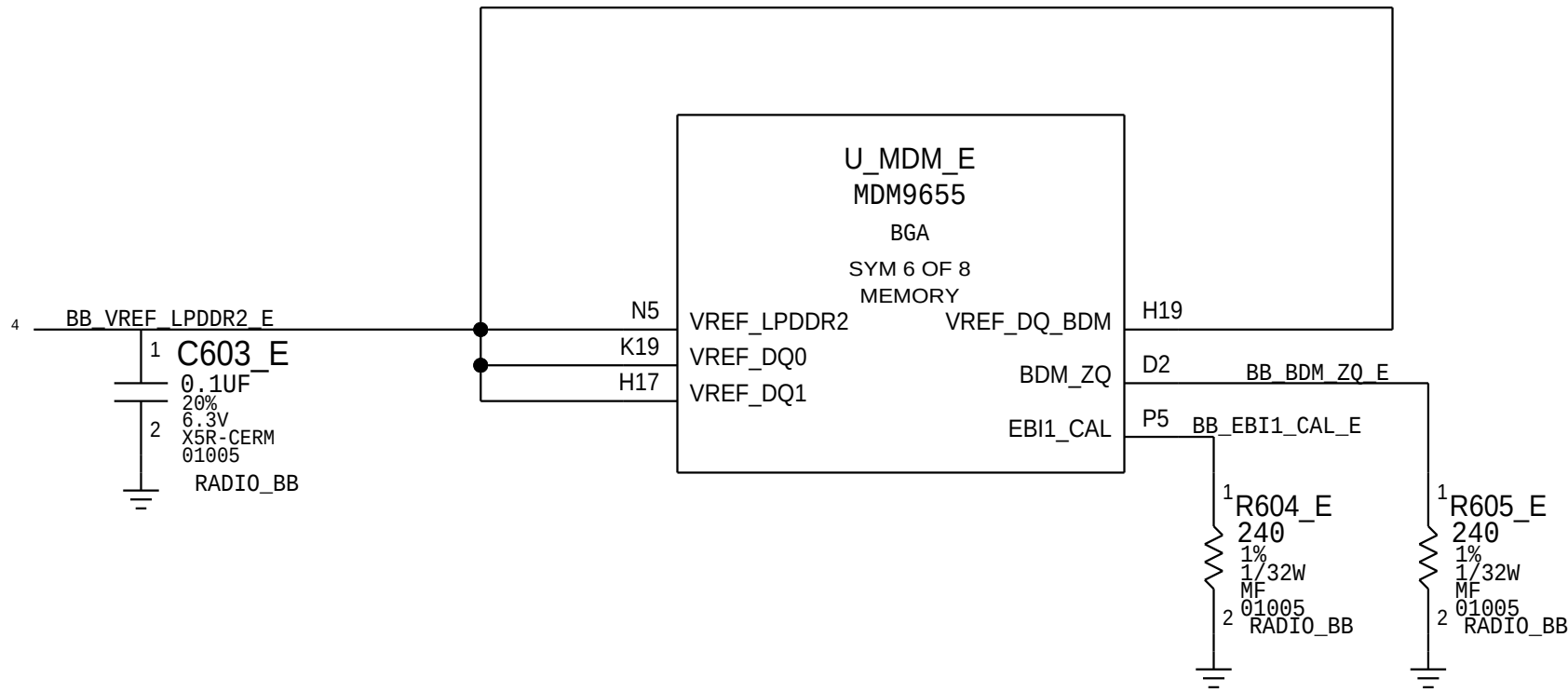
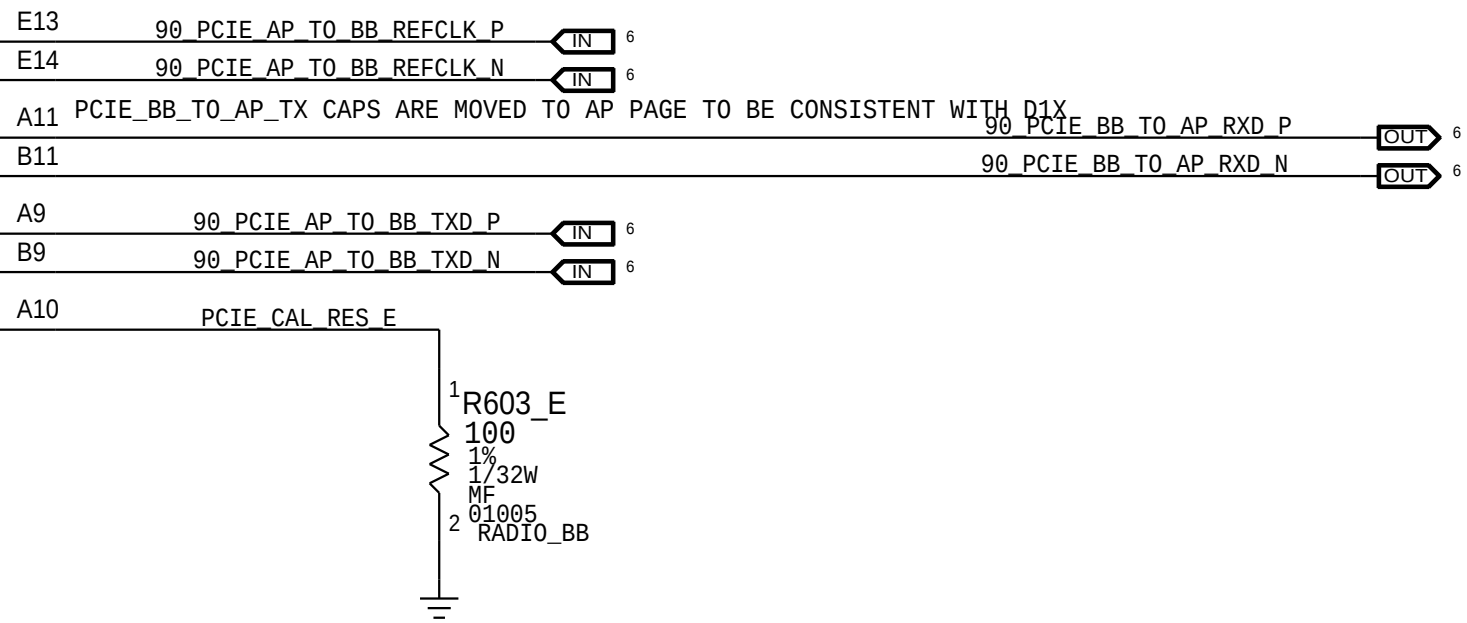
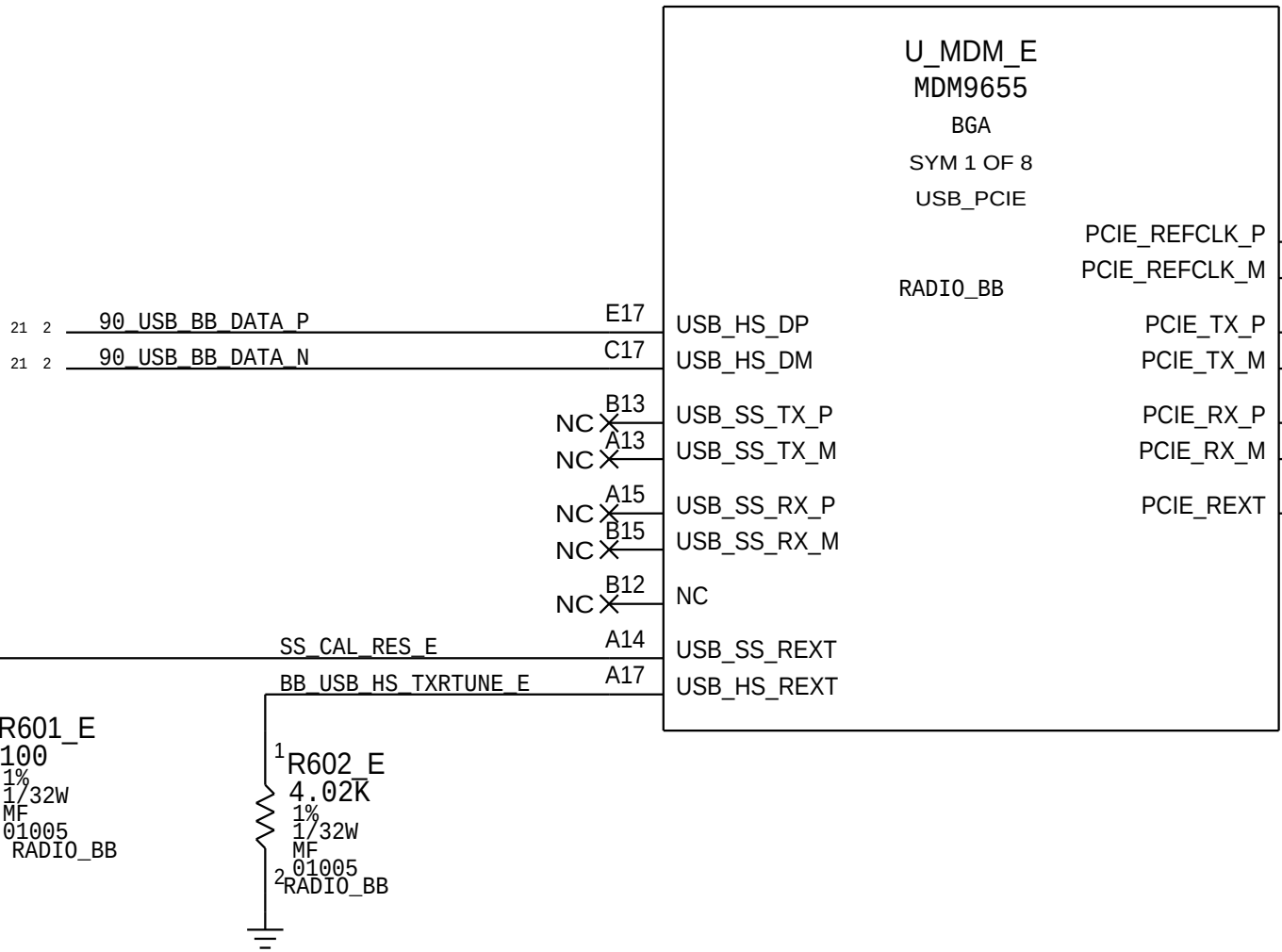
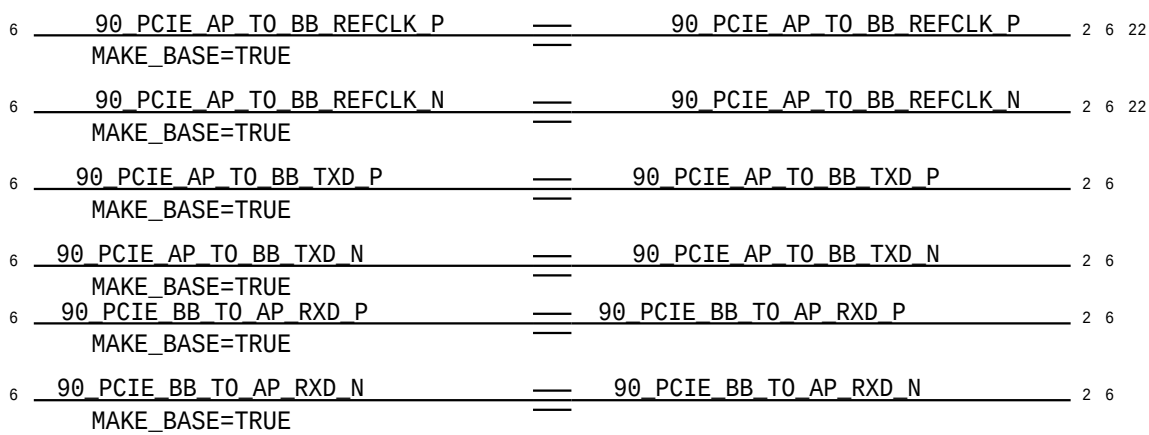
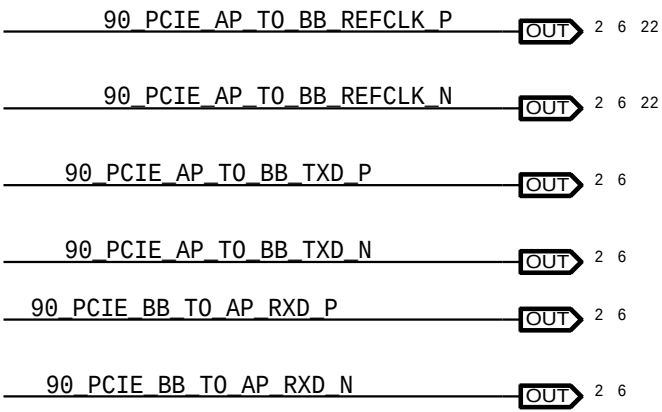


BB : POWER

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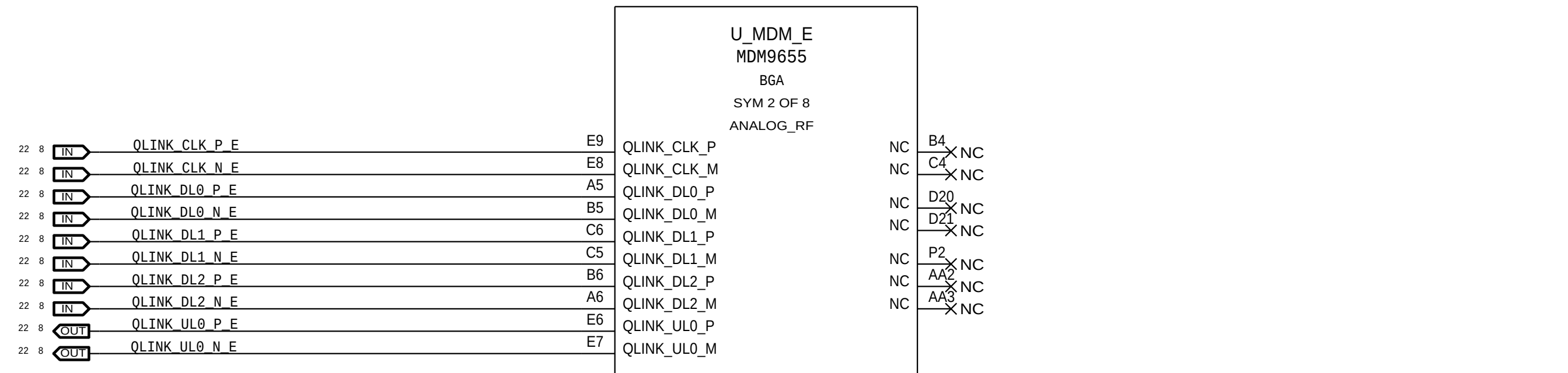
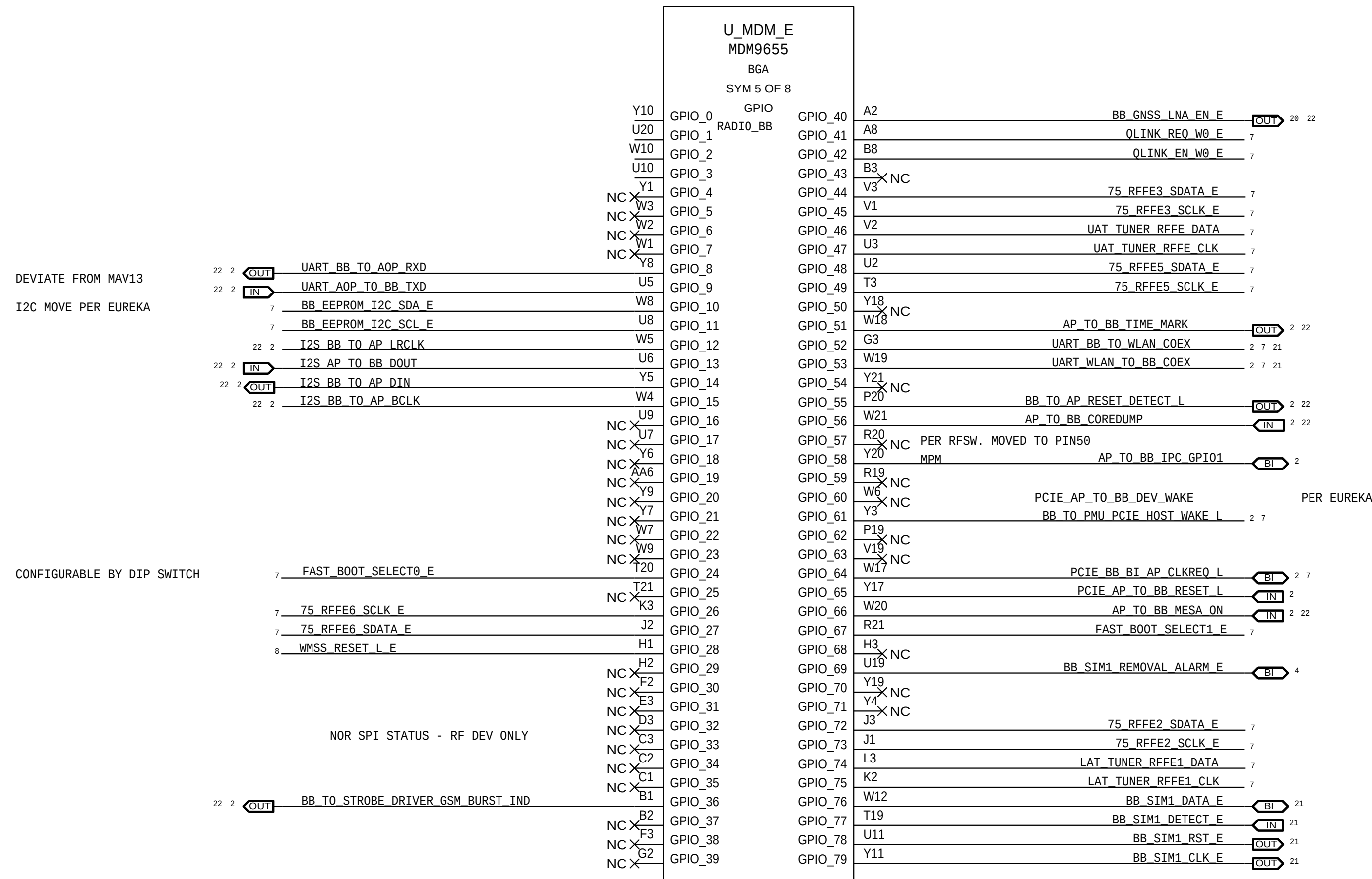


BB: CONTROL & HS PERIPHERALS

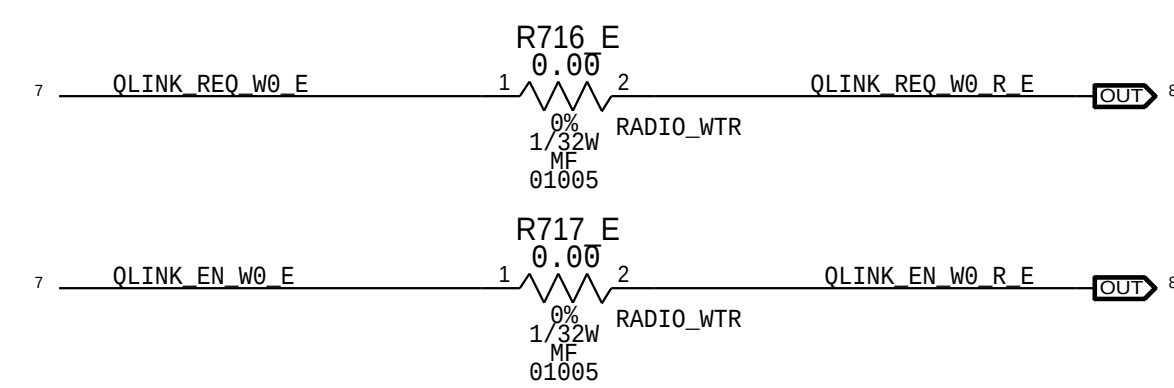
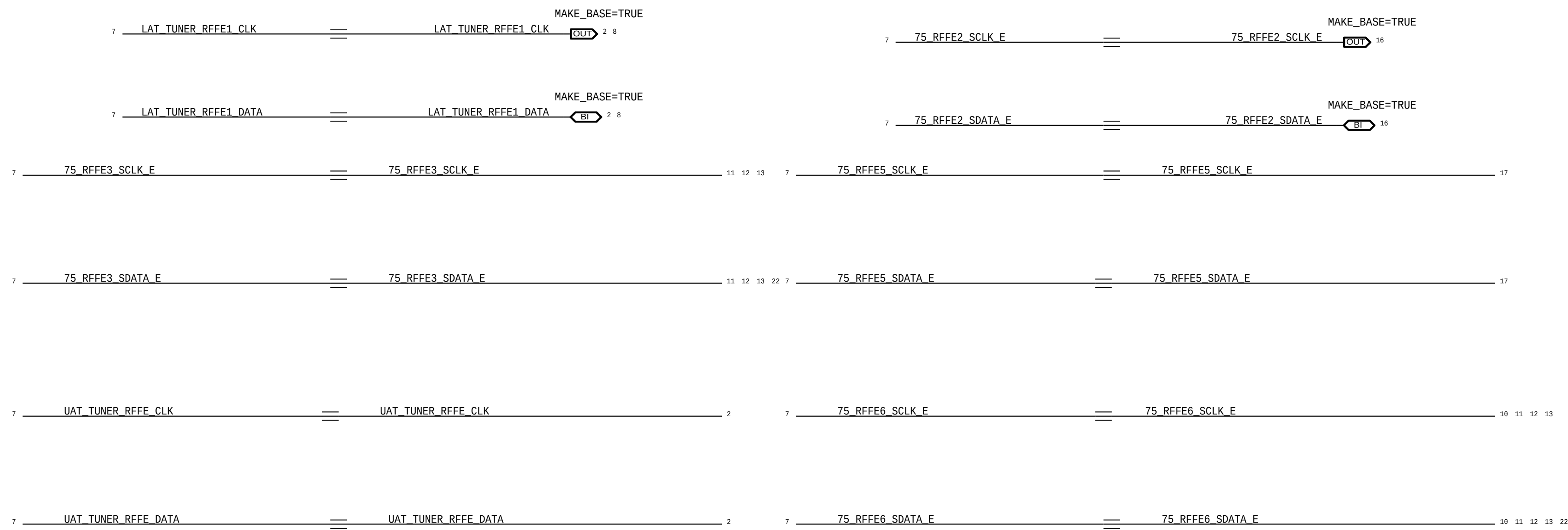
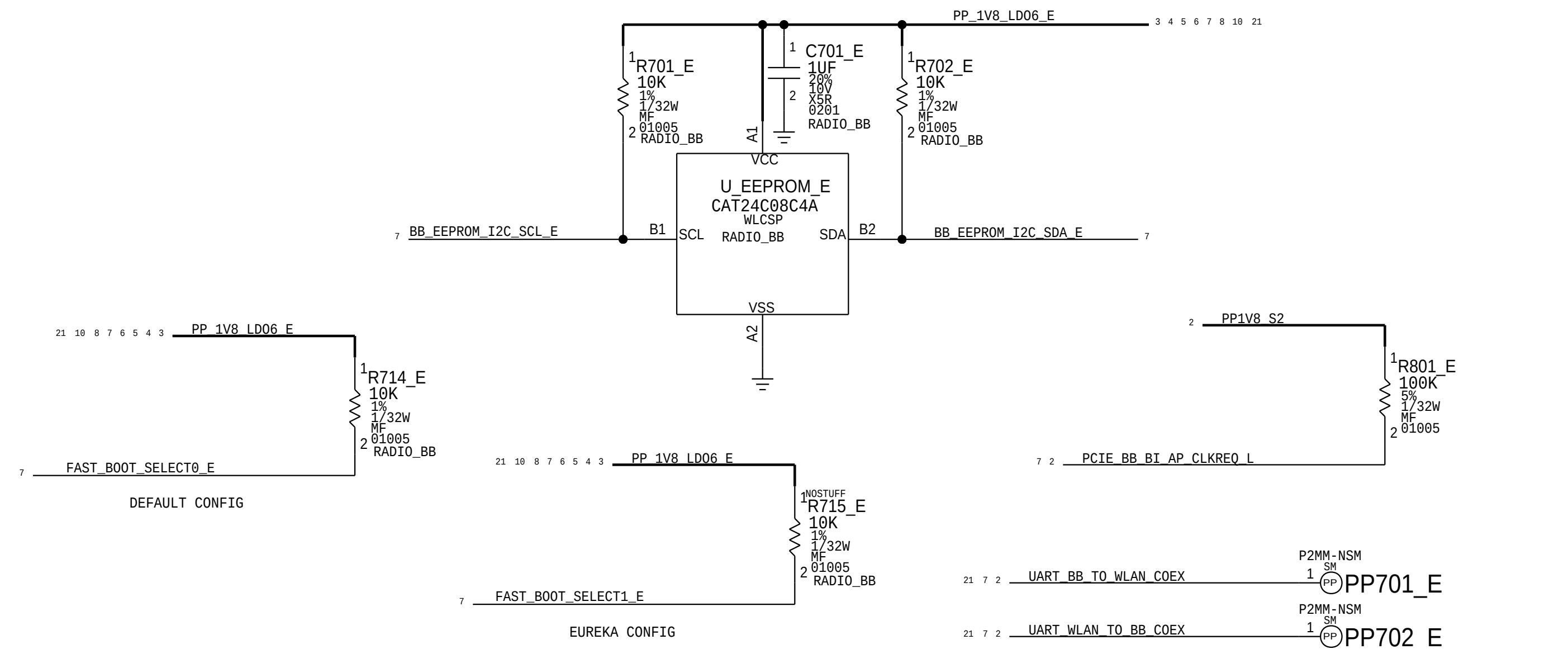


BB: GPIOs & QLINK

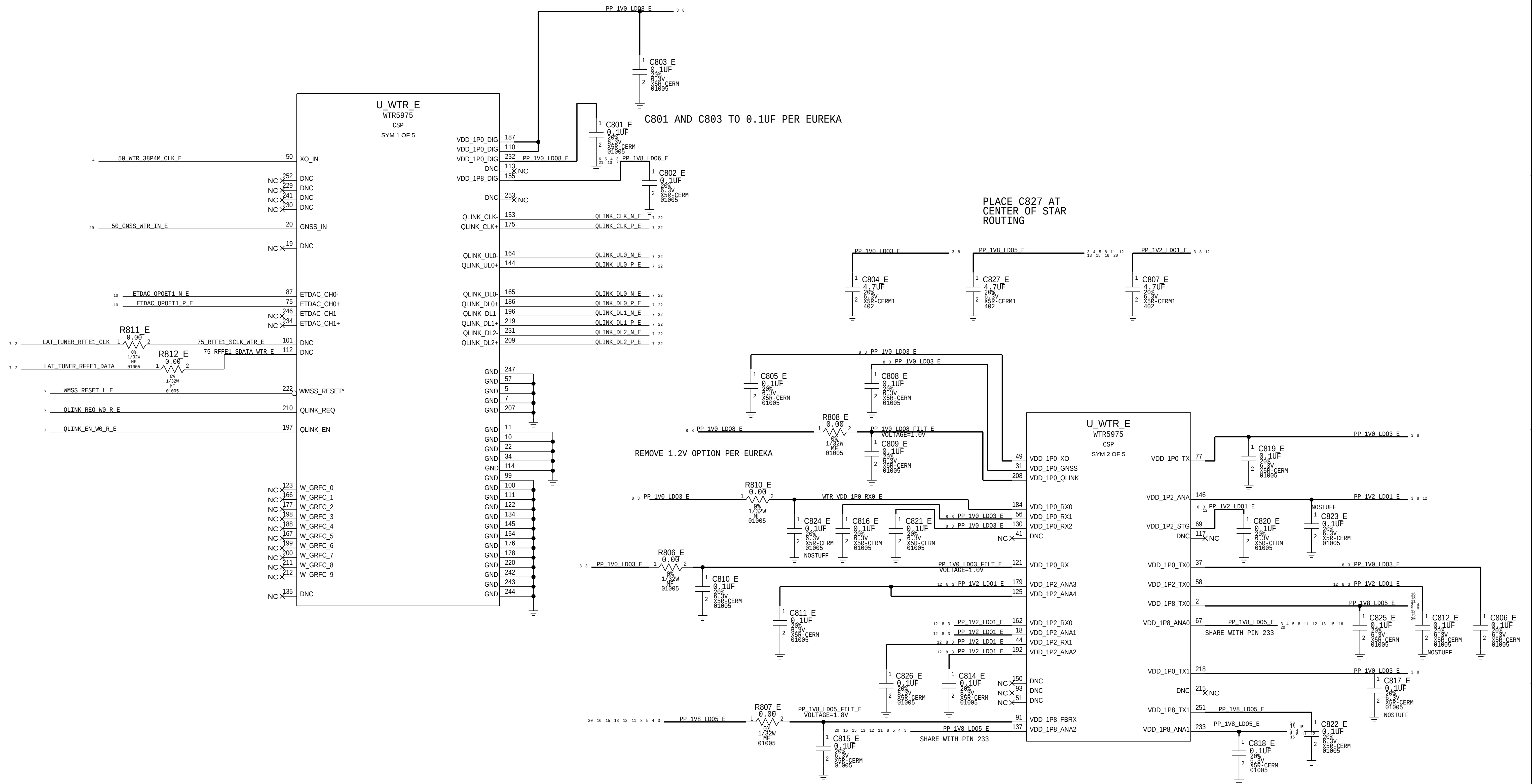
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BB EEPROM



TRANSCEIVER: QLINK & POWER

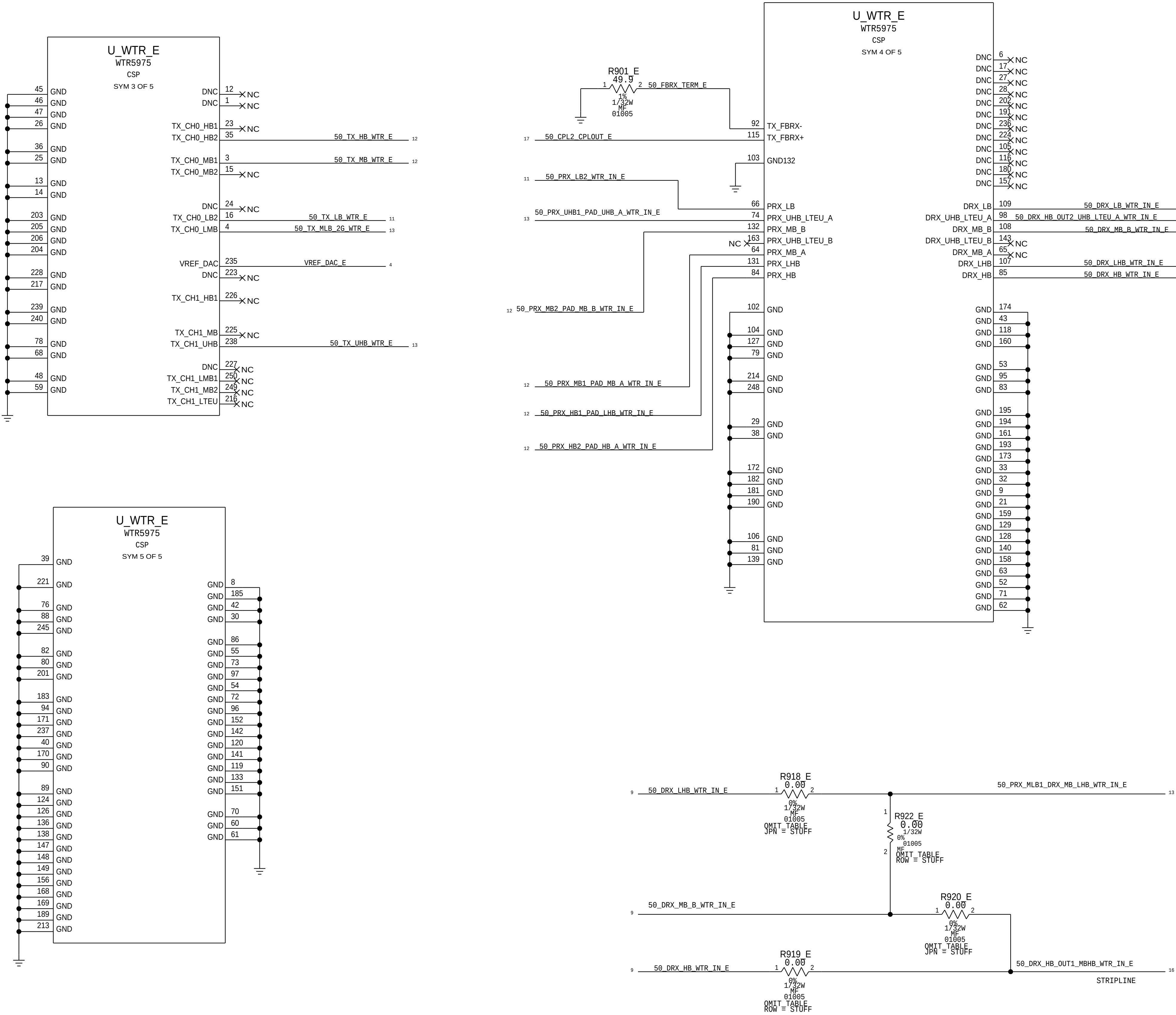


TRANSCEIVER: TX & RX

CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE ONLY - NOT A CHANGE REQUEST

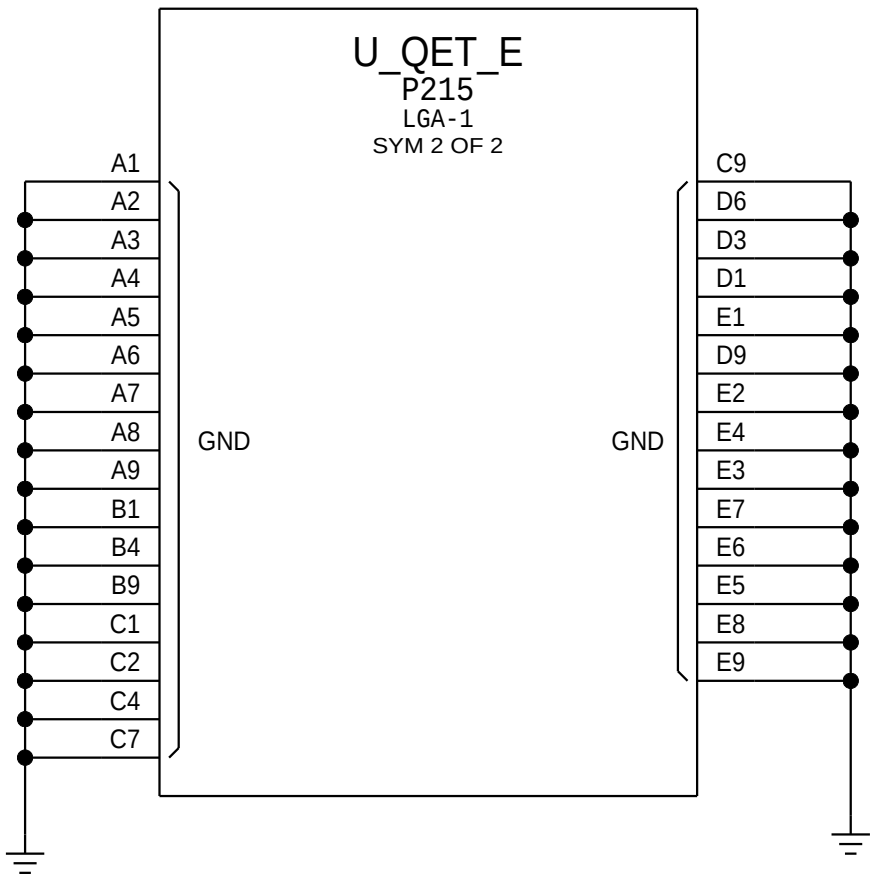
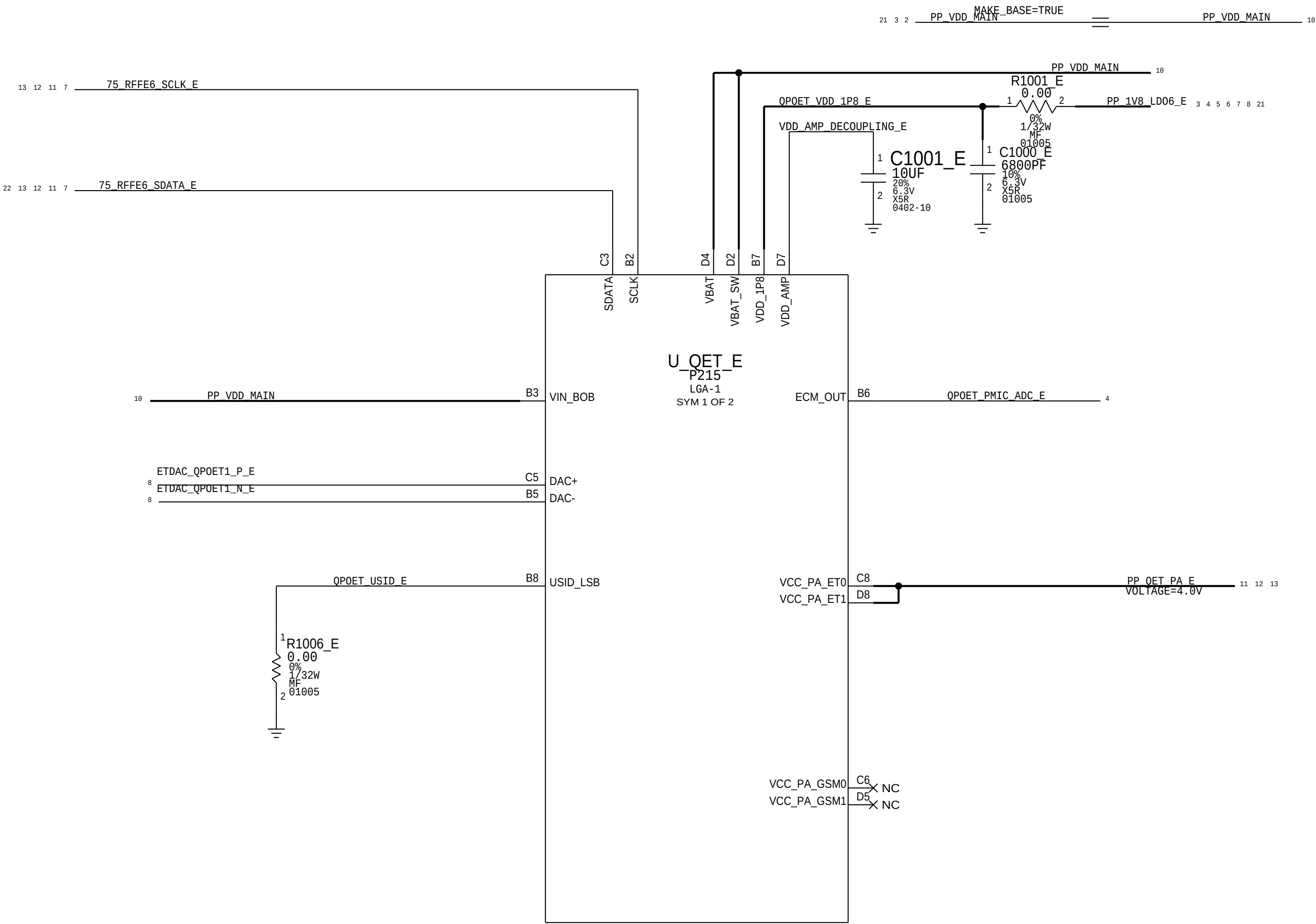
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
117S0161	1	01005 OR	R919_E	CRITICAL	ROW
117S0161	1	01005 OR	R922_E	CRITICAL	ROW

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
117S0161	1	01005 OR	R918_E	CRITICAL	JPN
117S0161	1	01005 OR	R920_E	CRITICAL	JPN



QET:MODULE

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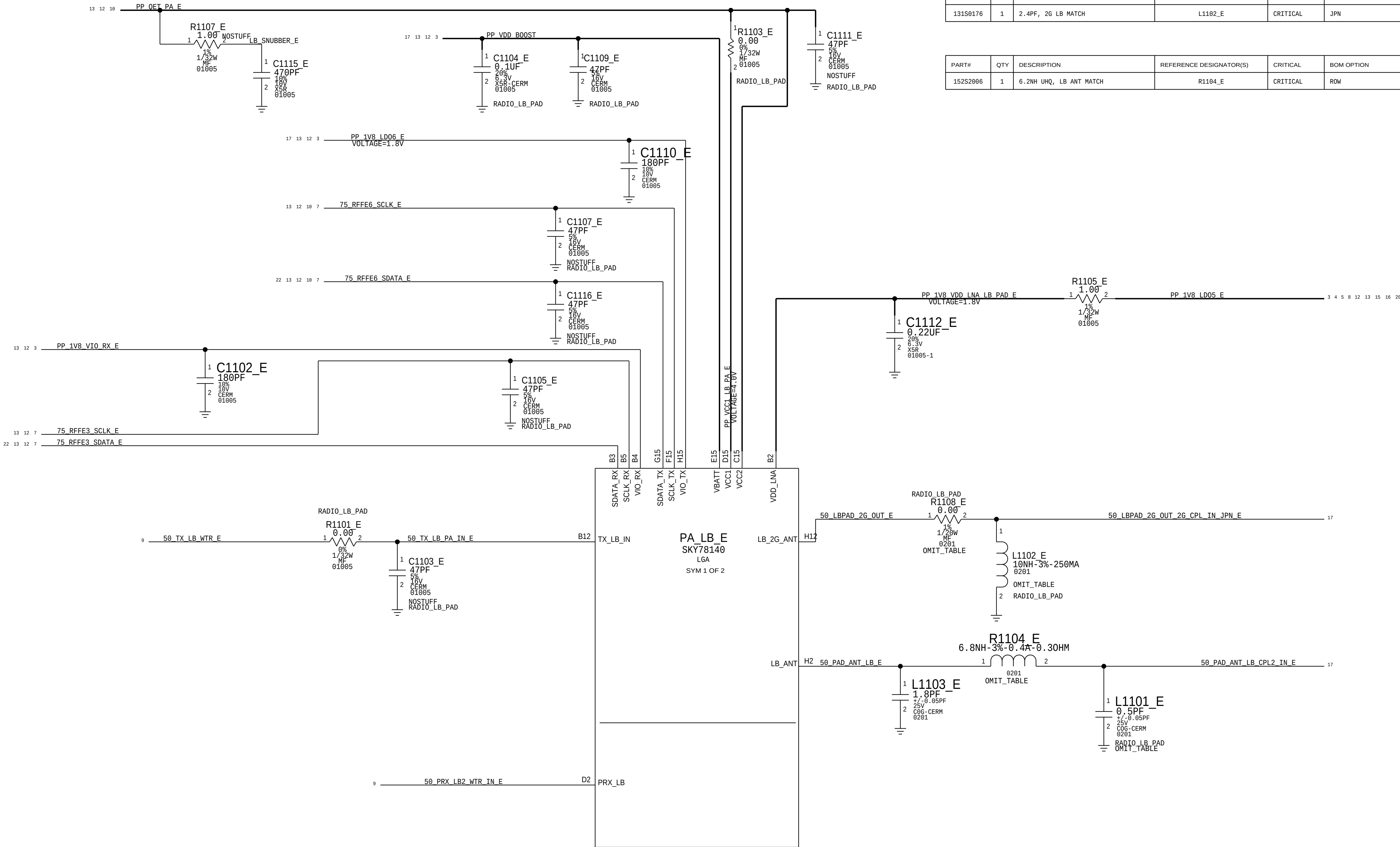


LB PAD

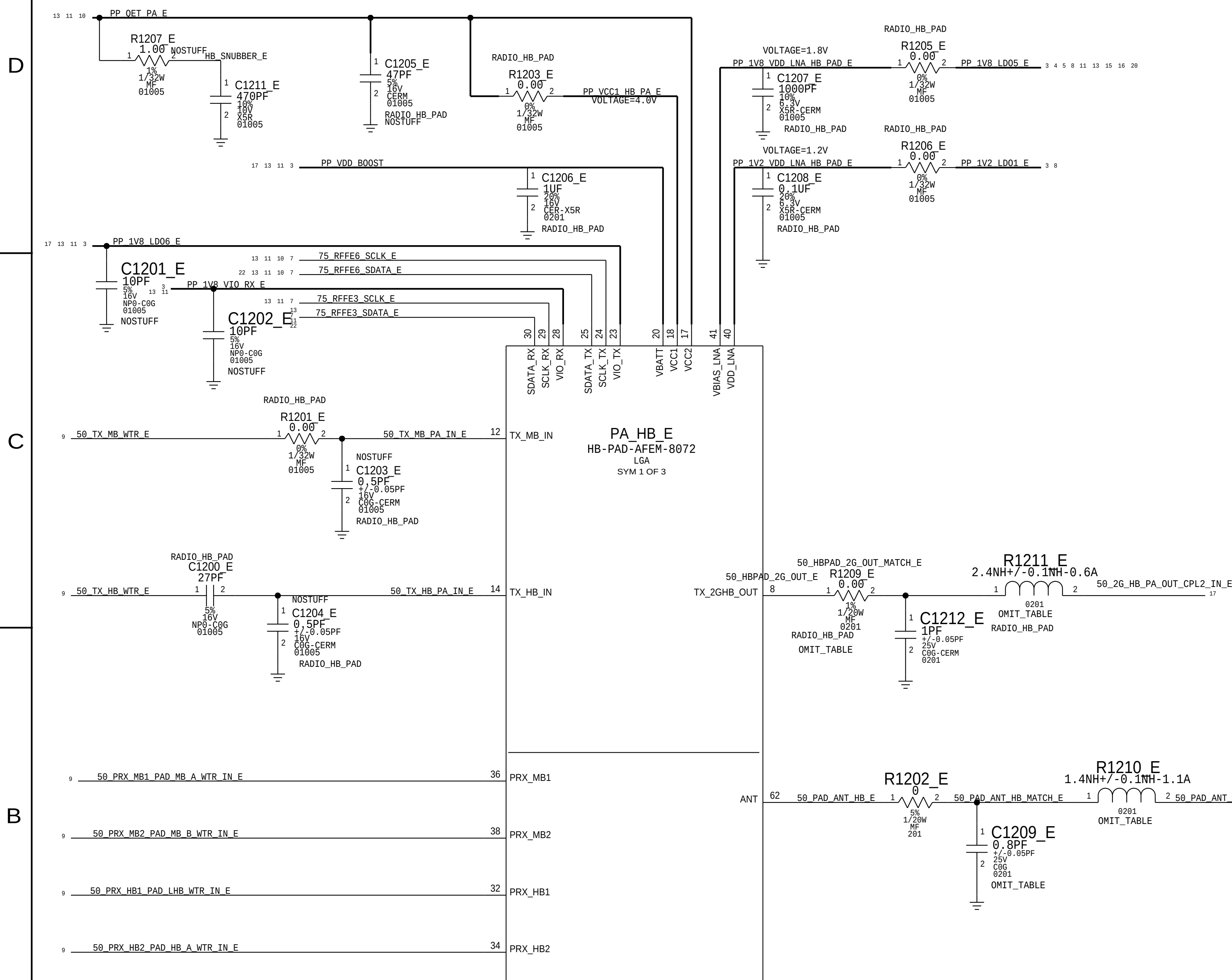
CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE ONLY - NOT A CHANGE REQUEST

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
118S0724	1	2G LB MATCH, 0 OHMS	R1108_E	CRITICAL	JPN
152S00021	1	6.8NH UHQ, LB ANT MATCH	R1104_E	CRITICAL	JPN
131S0555	1	1PF, LB ANT MATCH	L1101_E	CRITICAL	JPN
131S0176	1	2.4PF, 2G LB MATCH	L1102_E	CRITICAL	JPN

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
152S2086	1	6.2NH UHQ, LB ANT MATCH	R1104_E	CRITICAL	ROW

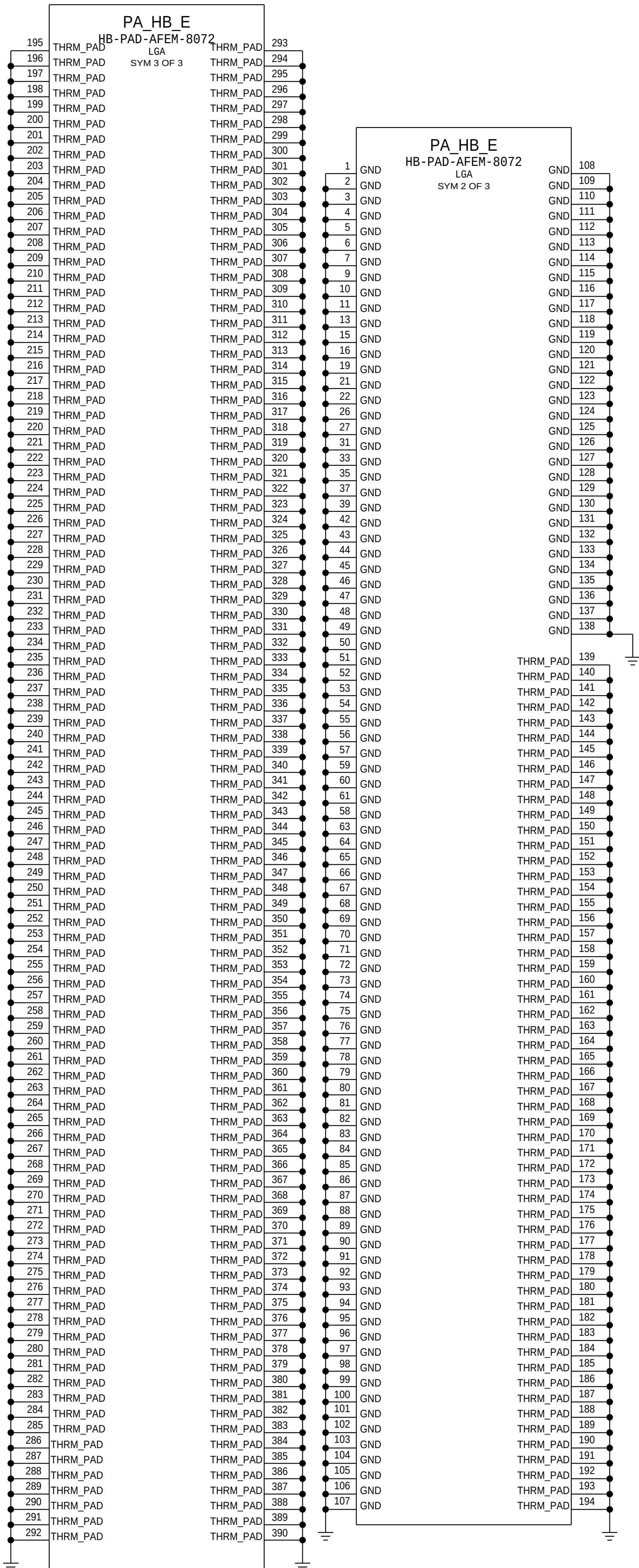


HB PAD



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
152S2023	1	3.2NH UHQ, 2G HB PAD MATCH	R1209_E	CRITICAL	JPN
152S00153	1	1NH UHQ, 2G HB PAD MATCH	R1211_E	CRITICAL	JPN
131S0425	1	0.5PF, HB ANT MATCH	C1209_E	CRITICAL	JPN
152S2042	1	1.8NH UHQ, HB ANT MATCH	R1210_E	CRITICAL	JPN

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
13159329	1	0.4PF, HB ANT MATCH	C1269_E	CRITICAL	ROW
15252080	1	2.0NH UHQ, HB ANT MATCH	R1210_E	CRITICAL	ROW



D



D

CB

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2G PA - ROW

D

D

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
353S01247	1	2G PA	PA_UHB_E	CRITICAL	ROW
117S0161	1	MLB/2GTX IN MCH2, 0 OHMS	R1302_E	CRITICAL	ROW
117S0161	1	UHB 1NPUT MCH1, 0 OHMS	R1303_E	CRITICAL	ROW
118S0724	1	MLB/2G MB OUT MCH1, 0 OHMS	R1305_E	CRITICAL	ROW
118S0724	1	UHB2 OUT/DRX BP MCH1, 0 OHMS	R1307_E	CRITICAL	ROW
152S053	1	4.7NH UHQ, 2G LB OUTPUT MCH2	R1309_E	CRITICAL	ROW
131S0214	1	VBAT DECOUPLING, 18PF	C1311_E	CRITICAL	ROW
138S0692	1	VBAT DECOUPLING, 1UF	C1323_E	CRITICAL	ROW
117S0161	1	VCC2/VCC1 RESISTOR OPTION, 0 OHM	R1304_E	CRITICAL	ROW
103S00089	1	50 OHM TERMINATION FOR ROW ONLY	C1319_E	CRITICAL	ROW

C

C

B

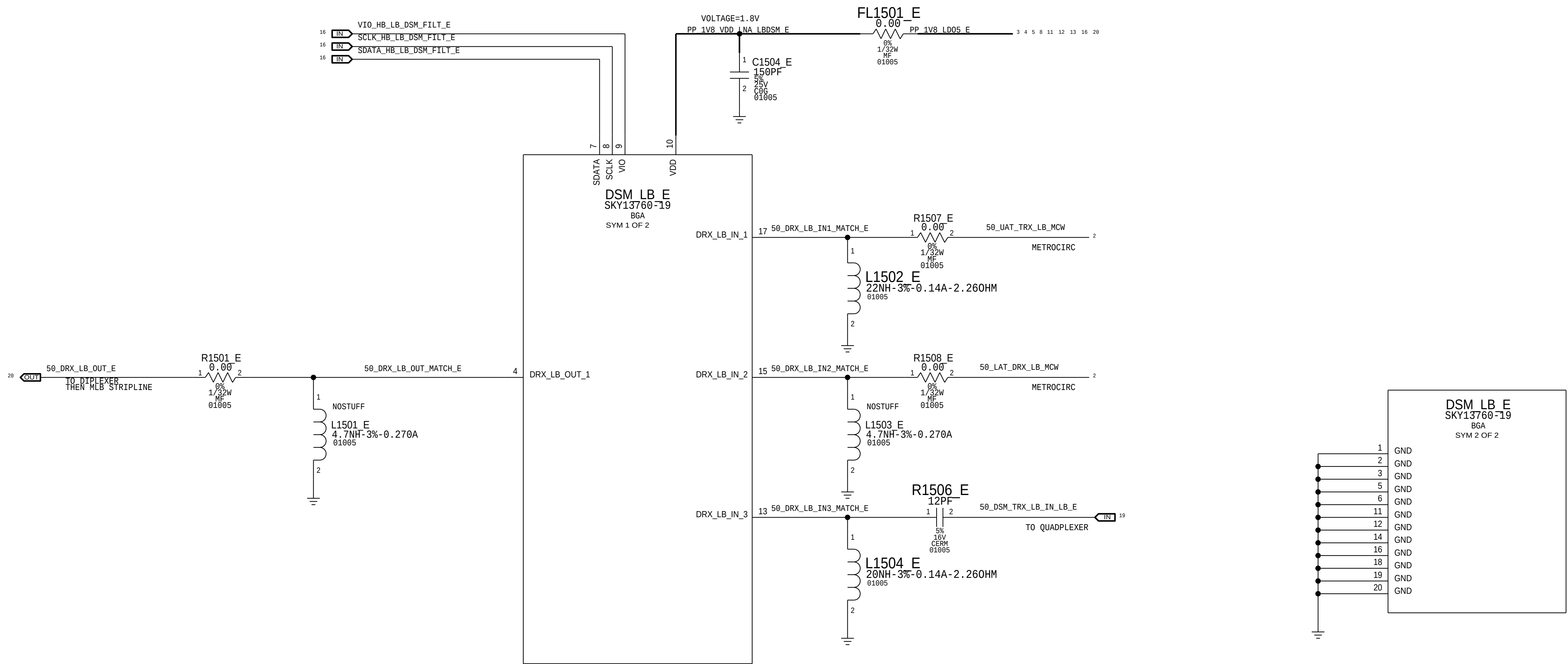
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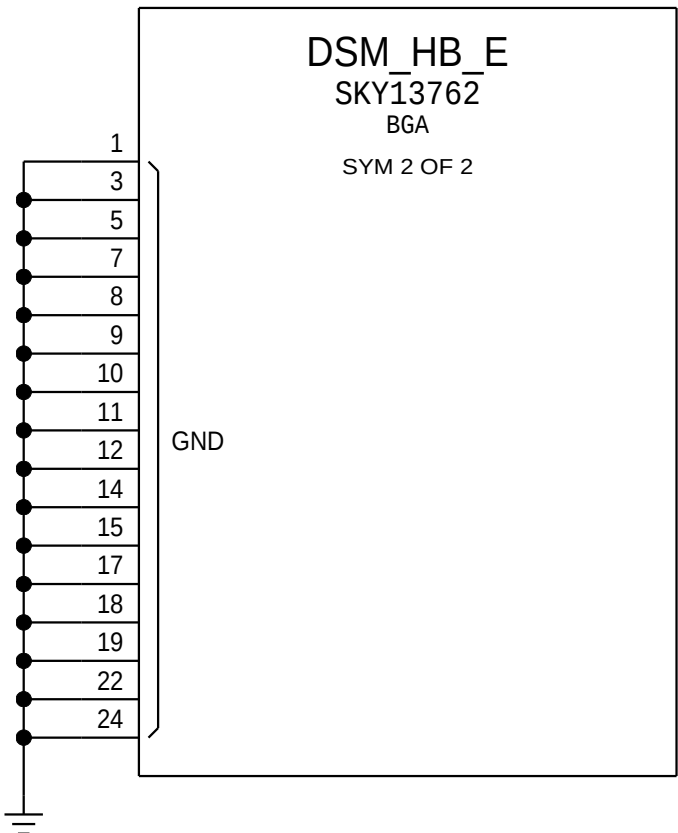
A

A

LB DSM

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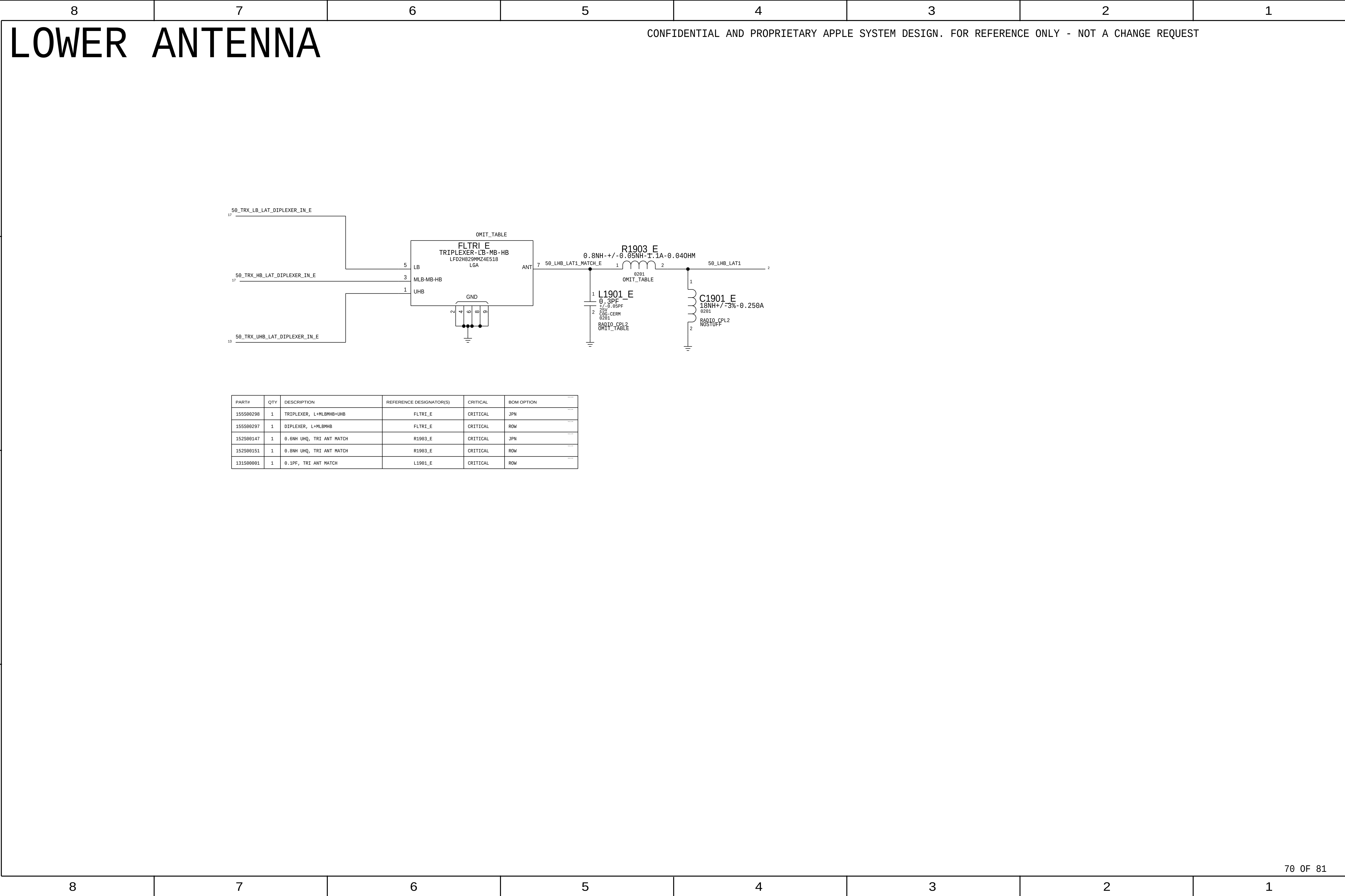


PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
11758161	1	01005 0R	R1603_E	CRITICAL	JPN
15251985	1	01005 1.7NH SHQ	R1612_E	CRITICAL	JPN
13158616	1	01005 0.6PF	L1606_E	CRITICAL	JPN

D

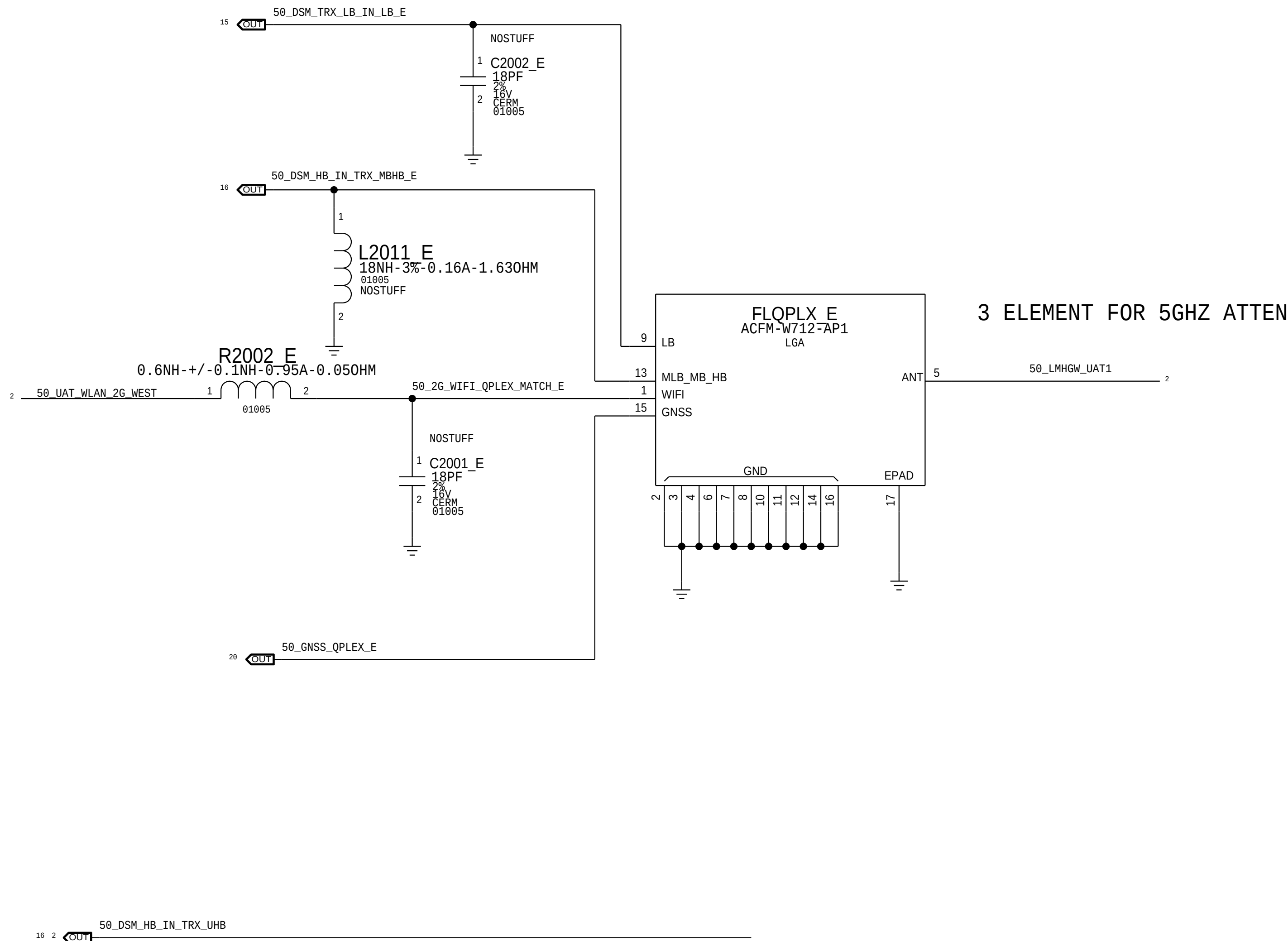


PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
118S9724	1	0201 0R	R1809_E	CRITICAL	JPN
152S2807	1	8.2NH 2G LB MATCH	R1812_E	CRITICAL	JPN
131S9431	1	0.2PF, CPLR HB LAT MATCH	L1801_E	CRITICAL	JPN
152S08055	1	2.6NH, CPLR HB LAT MATCH	R1807_E	CRITICAL	JPN
131S0249	1	2.2PF, 2G LB MATCH	C1810_E	CRITICAL	JPN

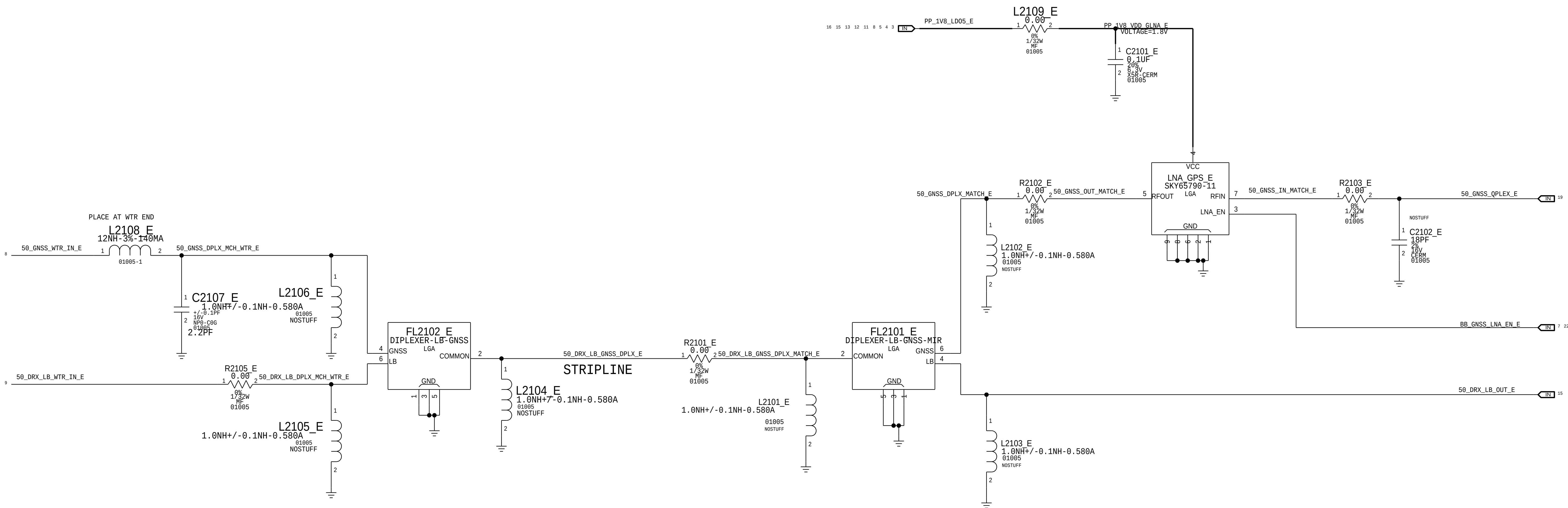


UAT & LB/UHB DIPLEXER

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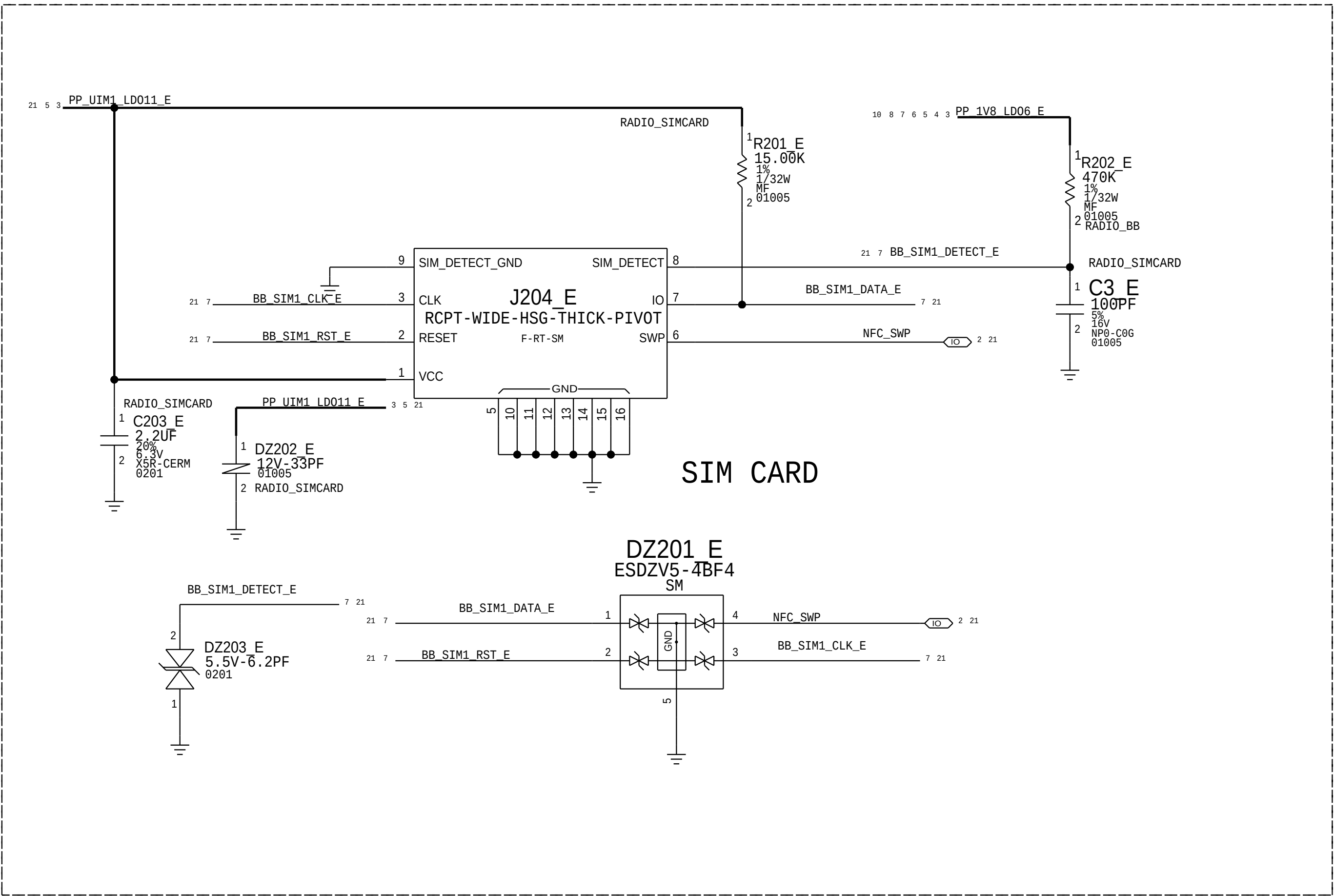


GPS & GPS/LB DIPLEXER



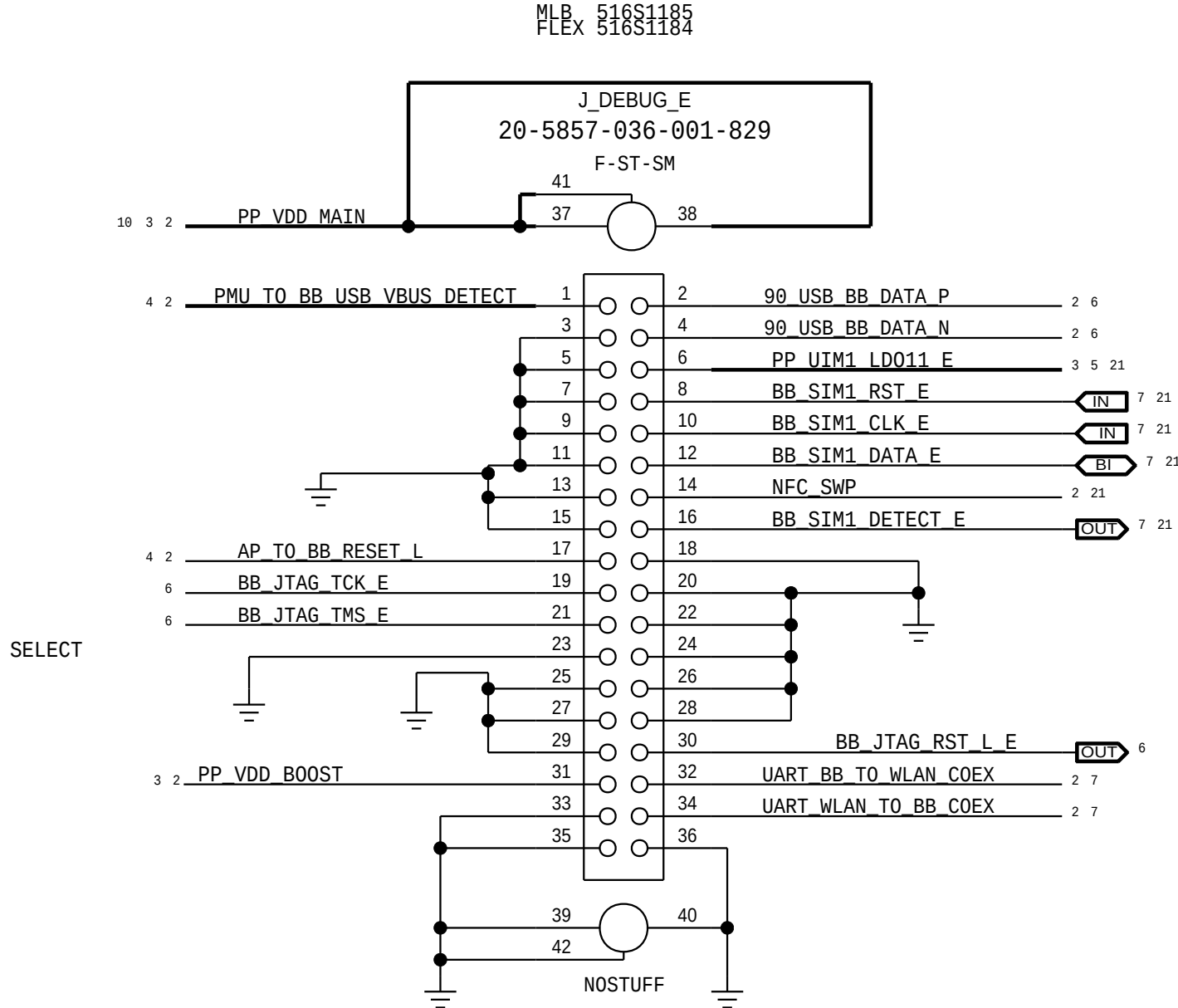
SIM CARD: CONNECTOR

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PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
S12S00026	S12S00017	BOM_TABLE_ALTS	3204_E	SIMCRO, KYOCERA

DEBUG CONNECTOR



BB PROBE POINTS

QLINK

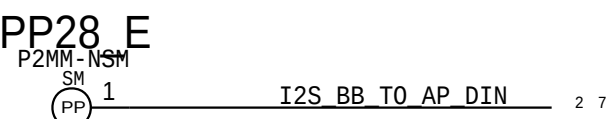
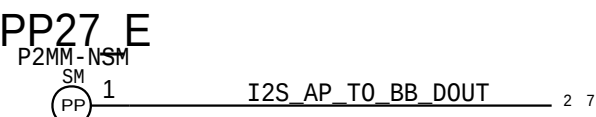
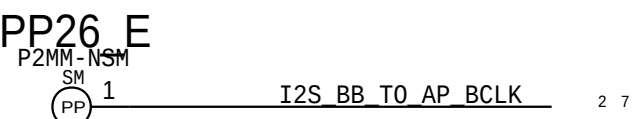
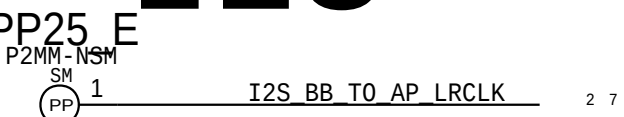
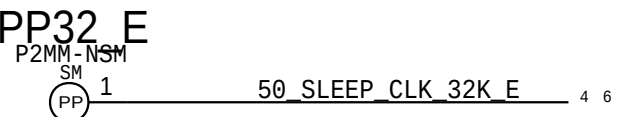
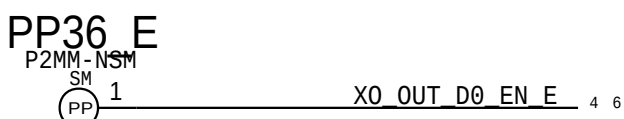
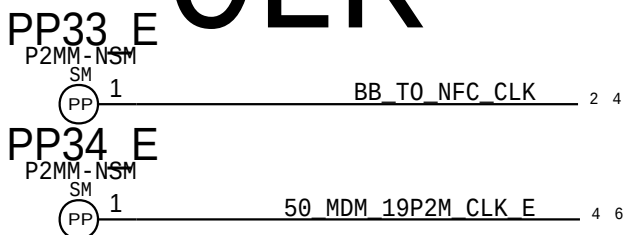
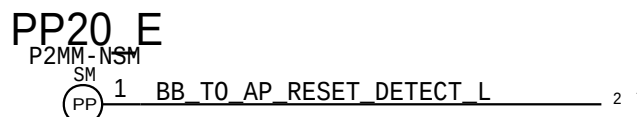
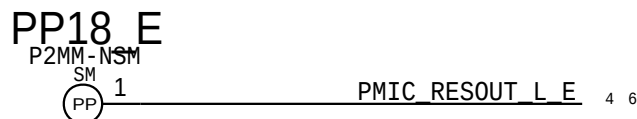
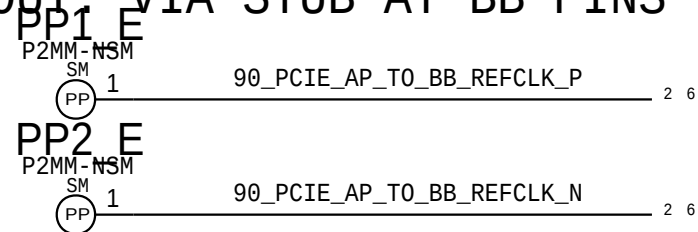
BASEBAND

CLK

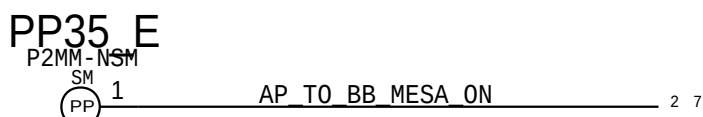
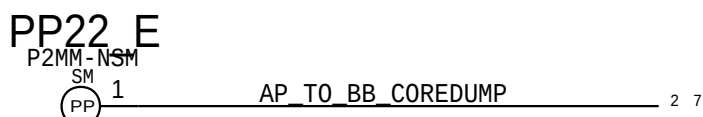
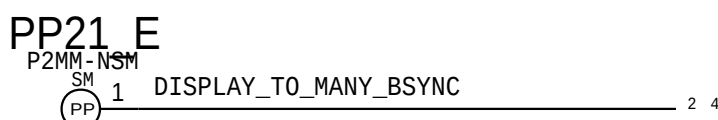
PWR SEQ

PCIE

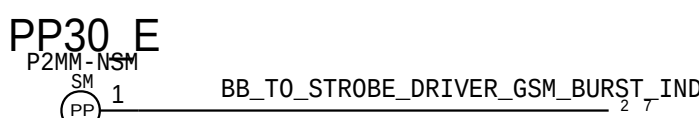
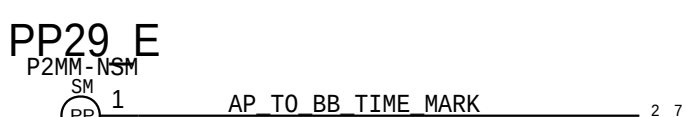
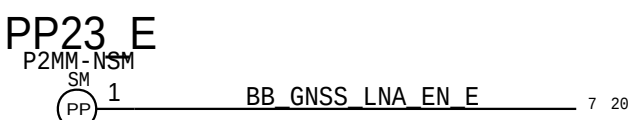
LAYOUT: VIA STUB AT BB PINS



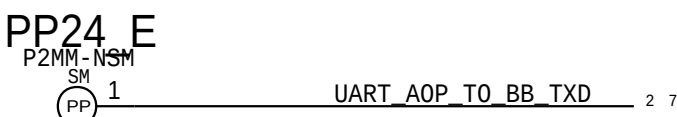
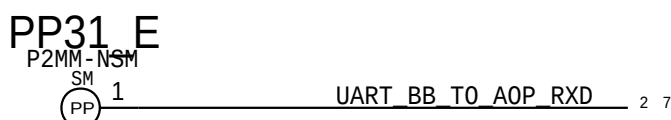
OTHERS



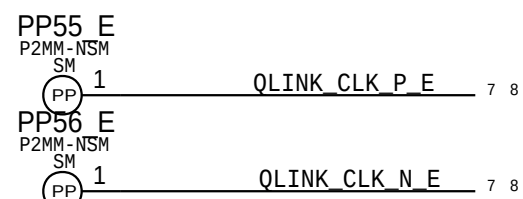
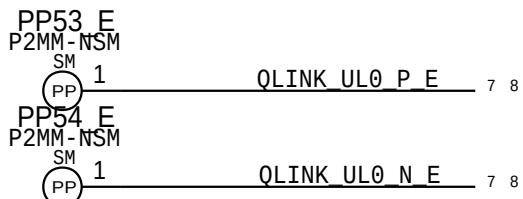
CTRL



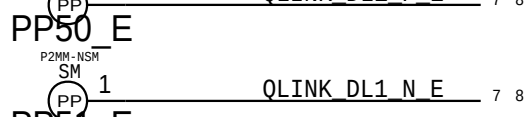
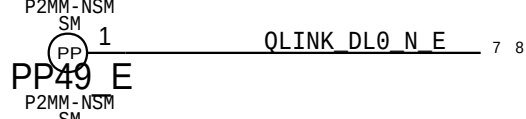
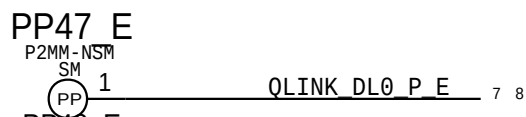
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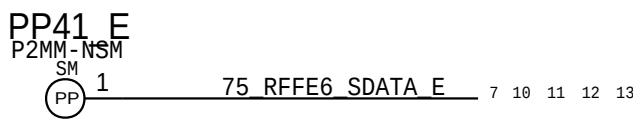
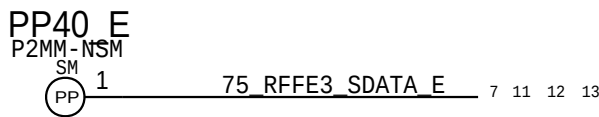
LAYOUT: VIA STUB AT WTR PINS



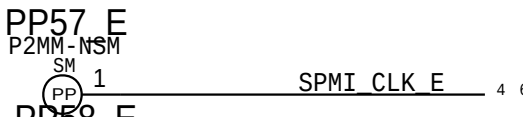
LAYOUT: VIA STUB AT BB PINS



RFFE

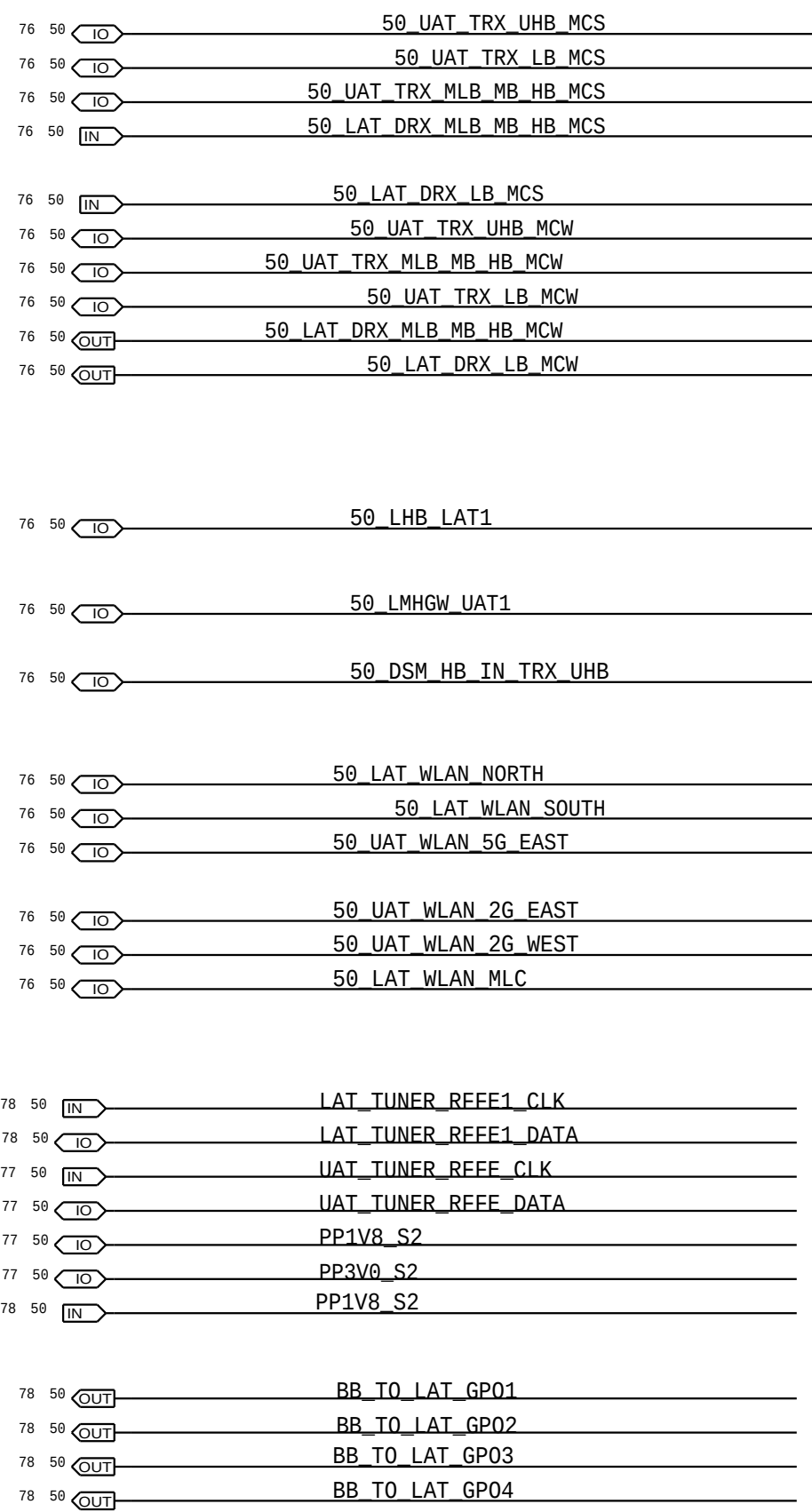


SPMI



D21 RADIO_MLB_FF

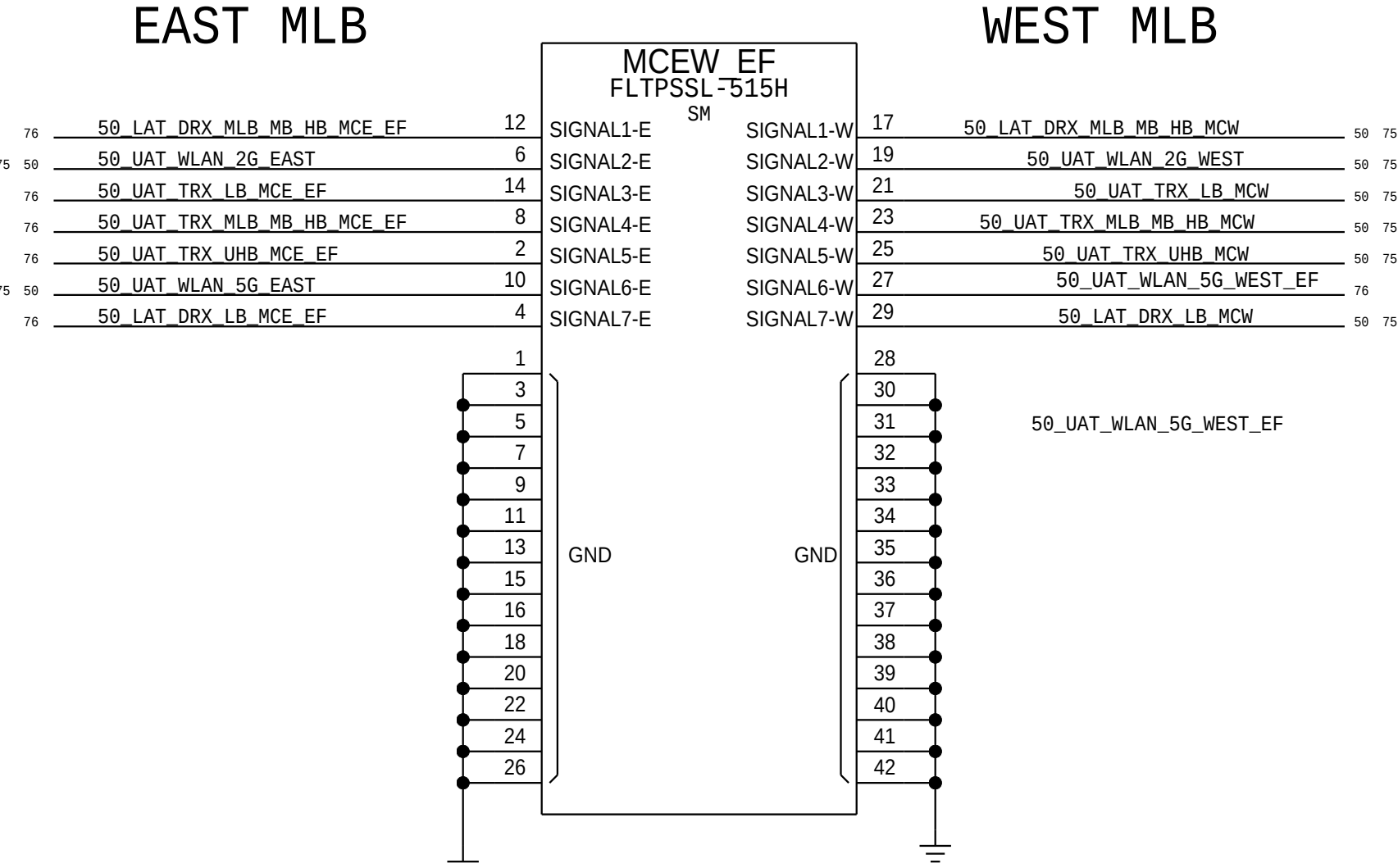
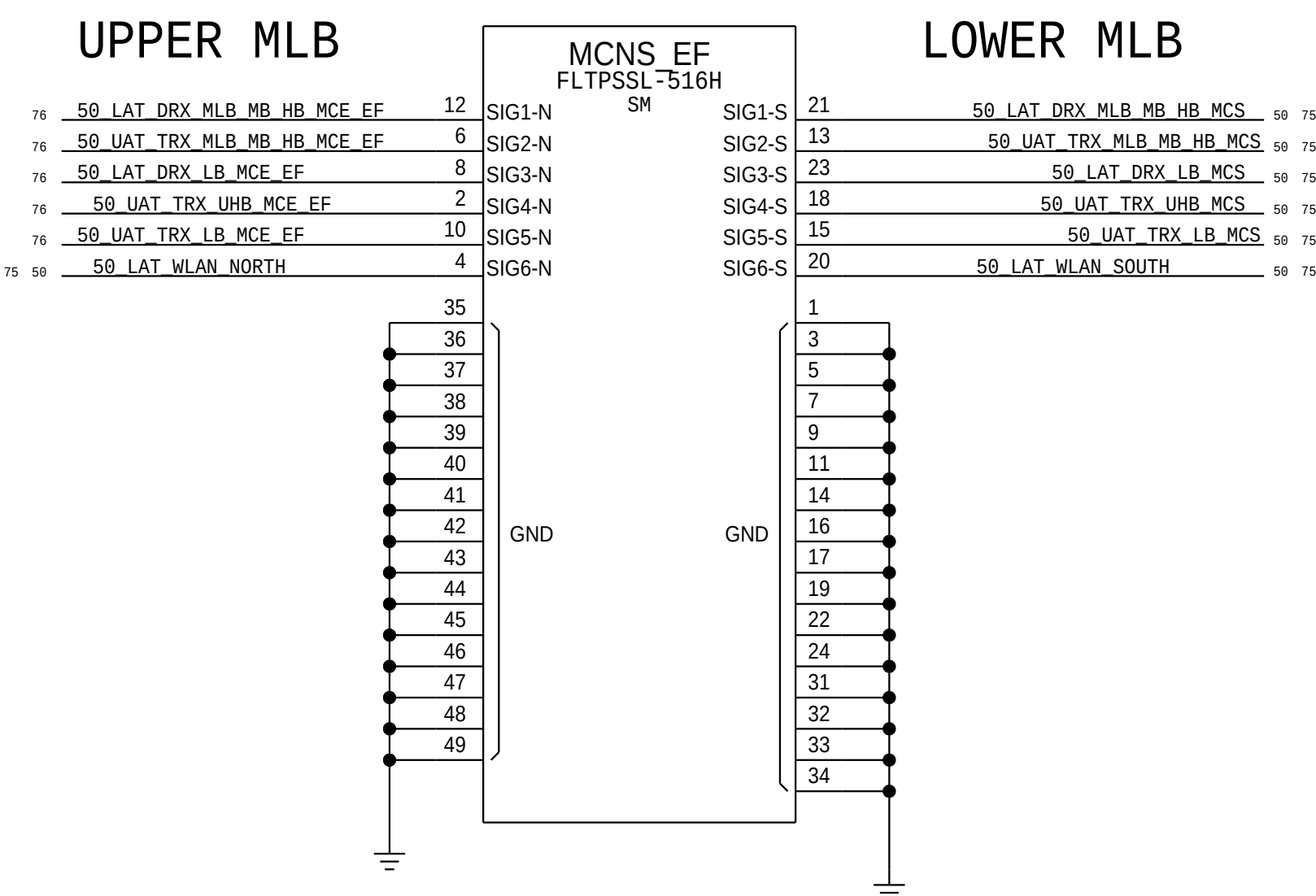
5/22/2017



METROCIRC

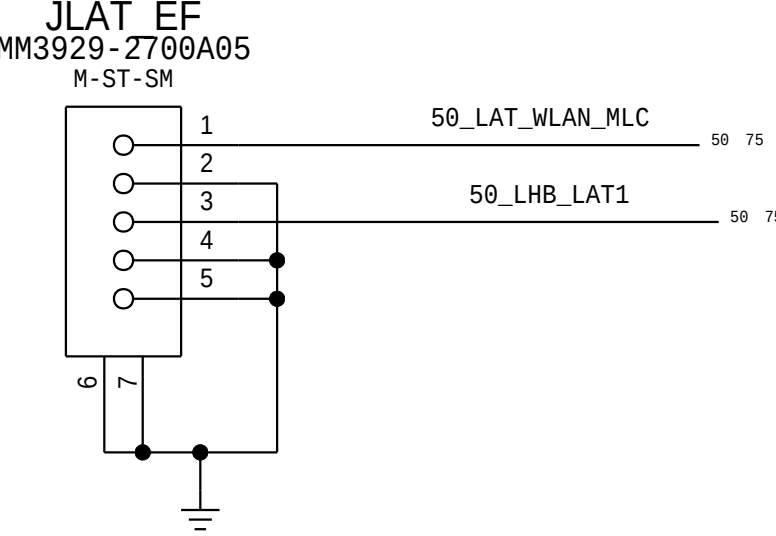
NORTH-SOUTH METROCIRC 339S00297

EAST-WEST METROCIRC 339S00296

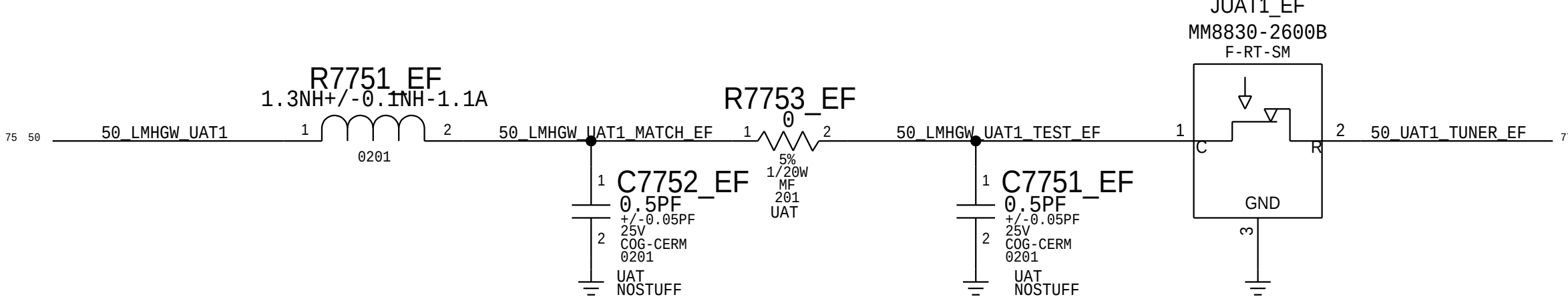


UHB/WIFI 5GHZ TEST

LAT MLC



LB/MLB/MB/HB WIFI 2.4GHZ TEST



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
155S00274	1	B42/5G DIPLEXER	FL2002_EF	CRITICAL	JPN
155S00275	1	5G PASSTHROUGH	FL2002_EF	CRITICAL	ROW
131S00598	1	B42/5G DIPLEXER 5G_IN MATCH	L7751_EF	CRITICAL	JPN
131S00385	1	5G PASSTHROUGH INPUT MATCH	L7751_EF	CRITICAL	ROW

UAT TUNER FLEX

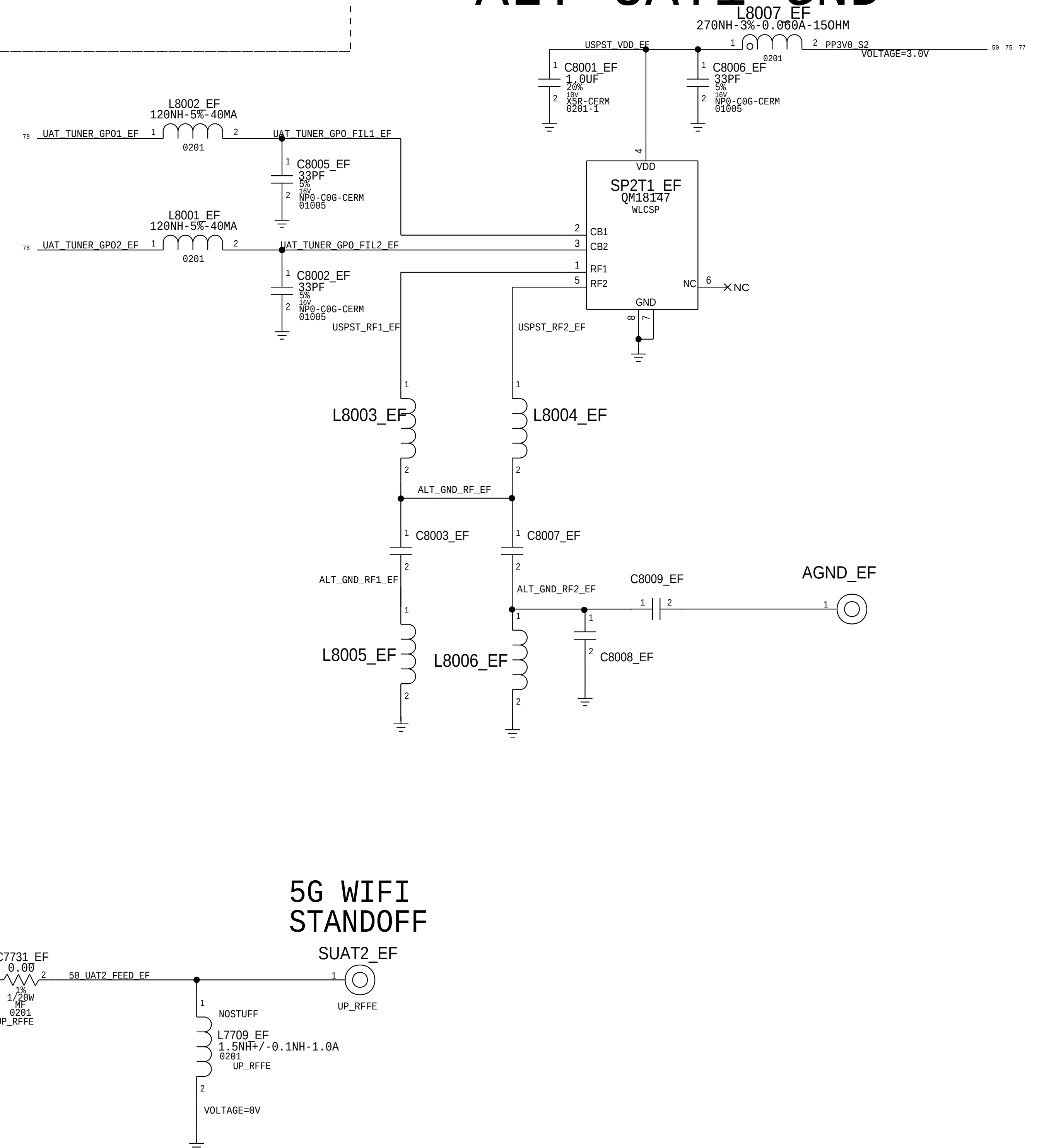
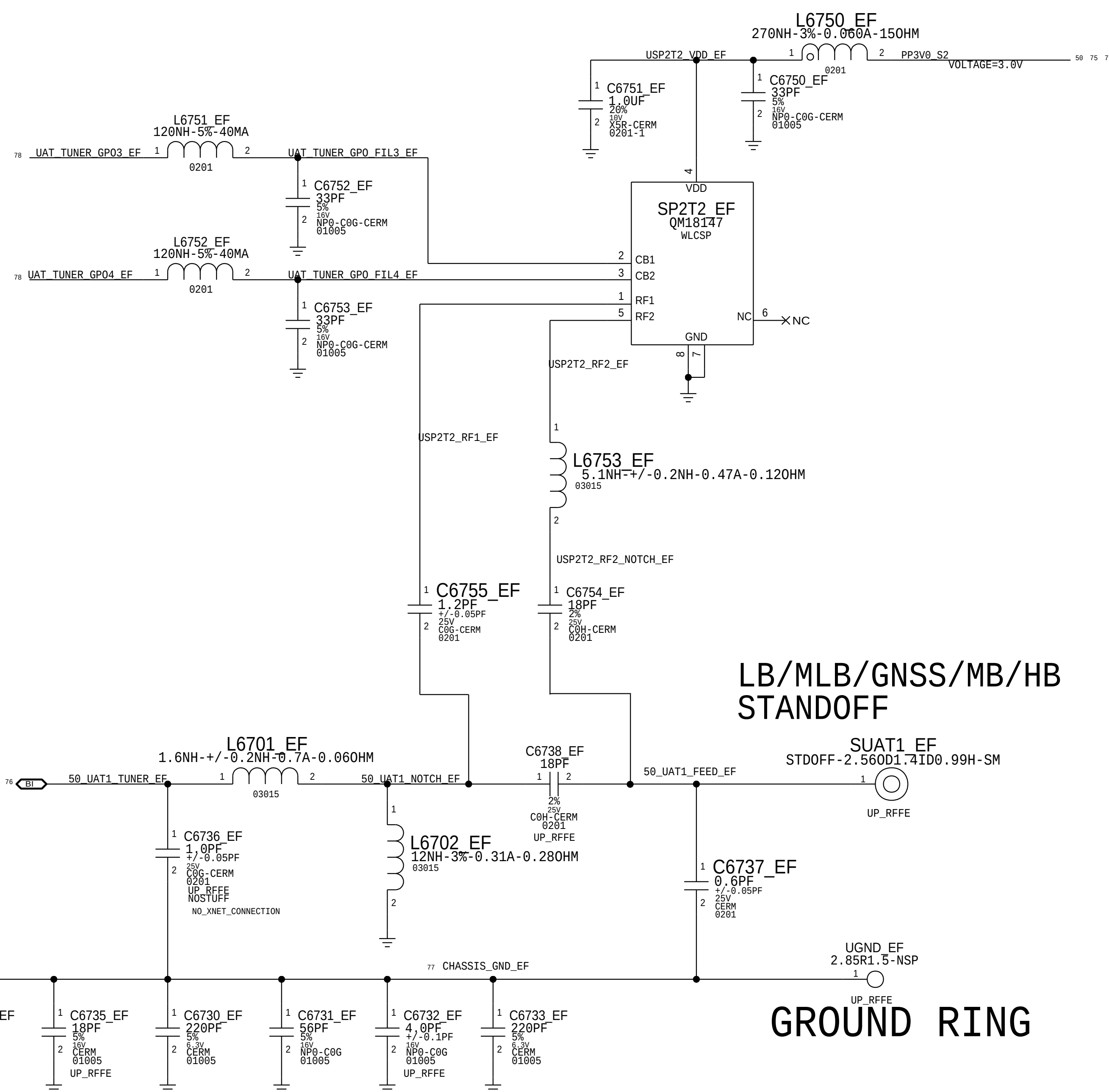
PLEASE CONTACT ANTENNA (MATT MOW)
FOR ANY COMPONENT CHANGE.

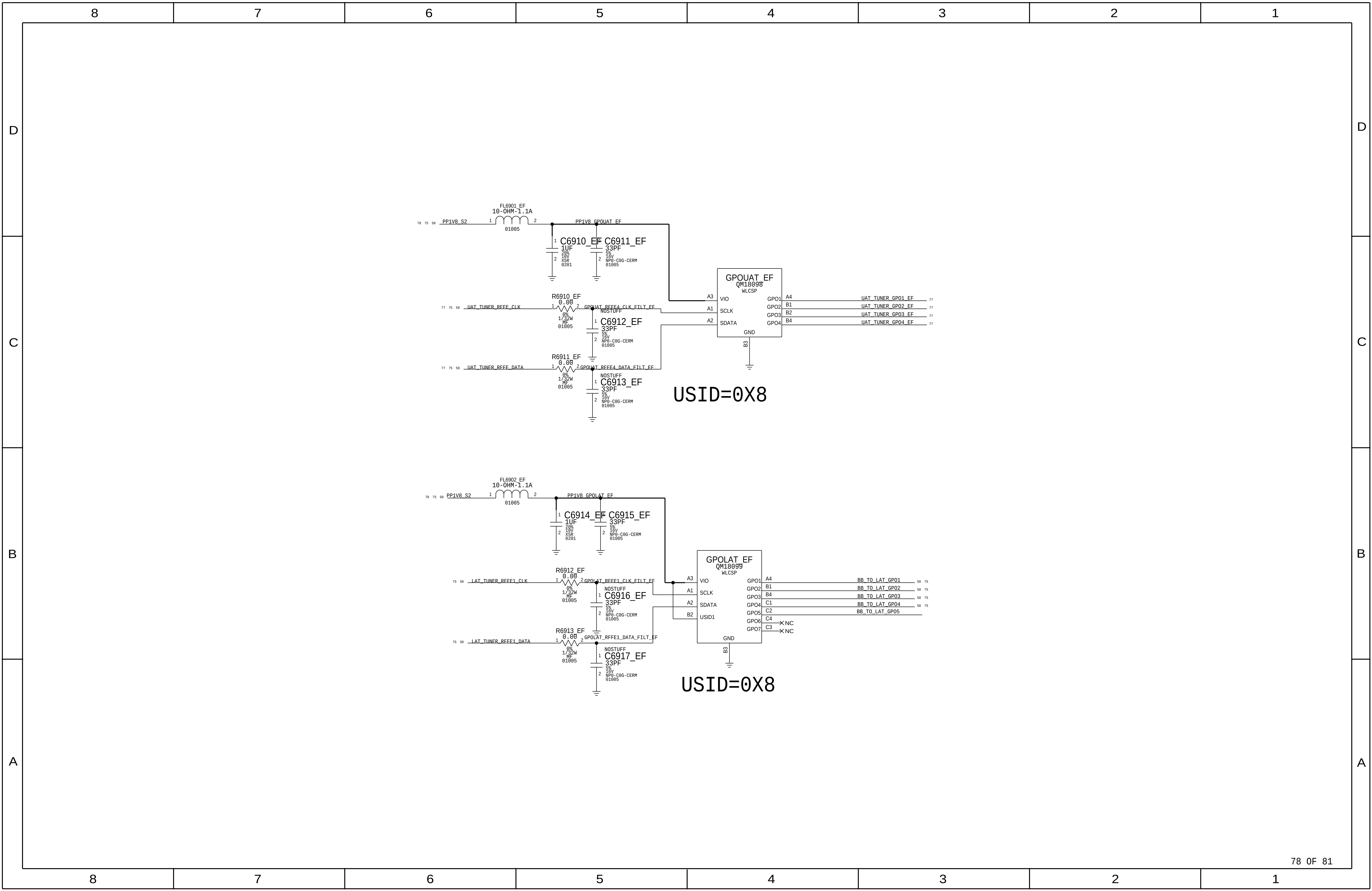
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
152S1248	1	4.3NH 03015 WW INDUCTOR	L8005_EF	CRITICAL	JPN
152S06749	1	3.6NH 03015 WW INDUCTOR	L8005_EF	CRITICAL	ROW

PINOUT IS ROTATED 180
COMPARED TO D20 & D201

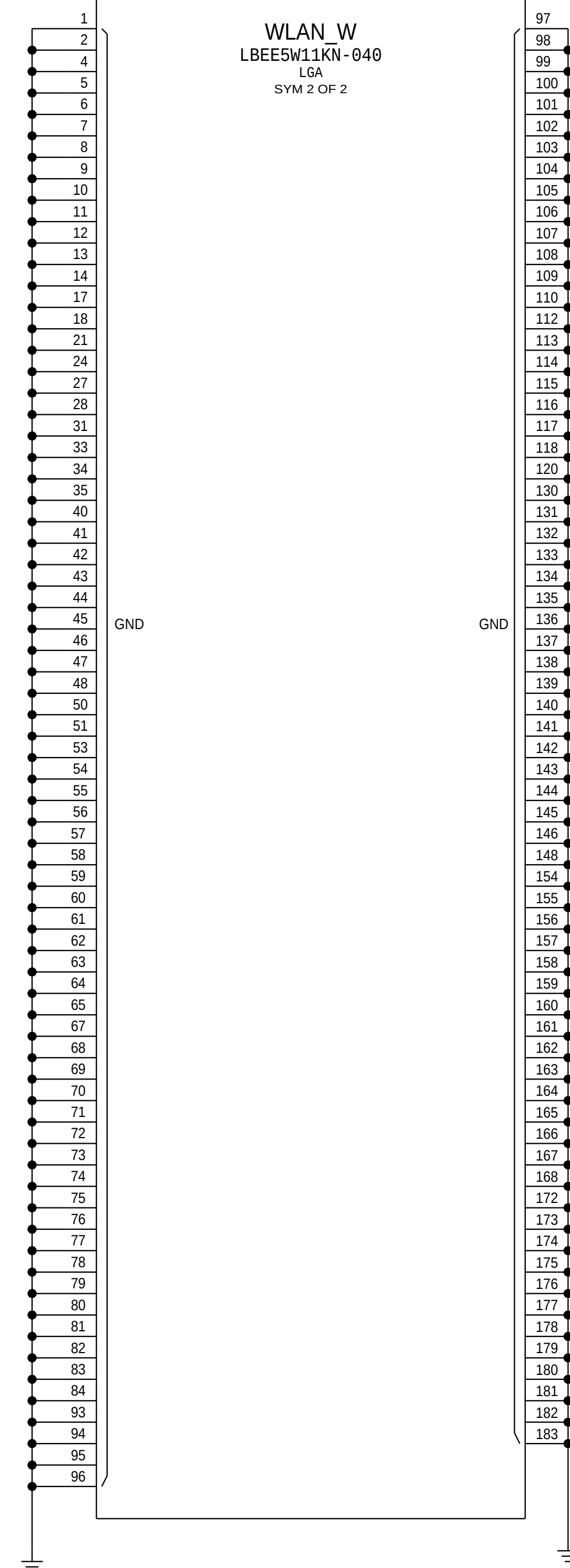
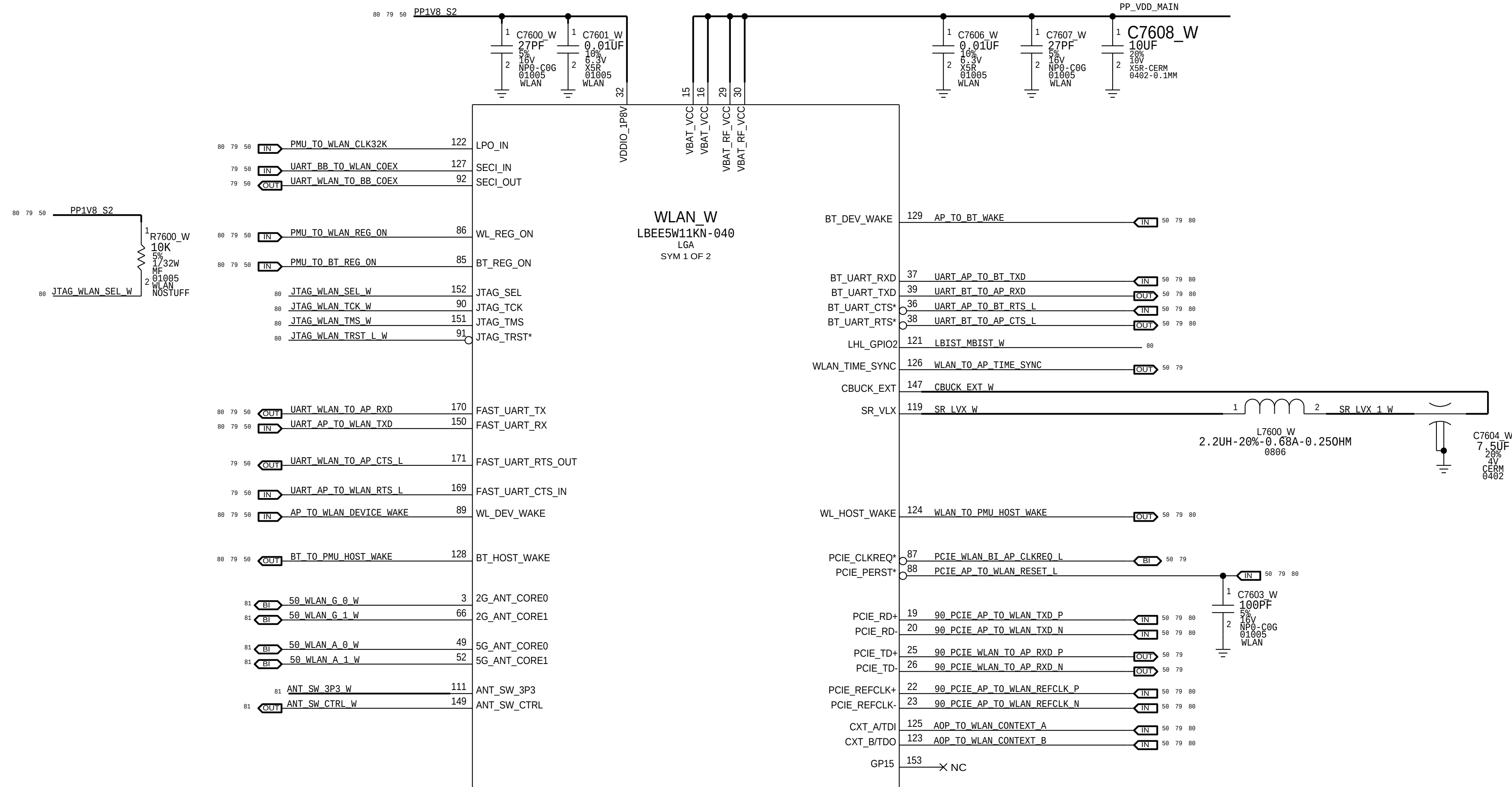
UAT TUNER

ALT UAT1 GND



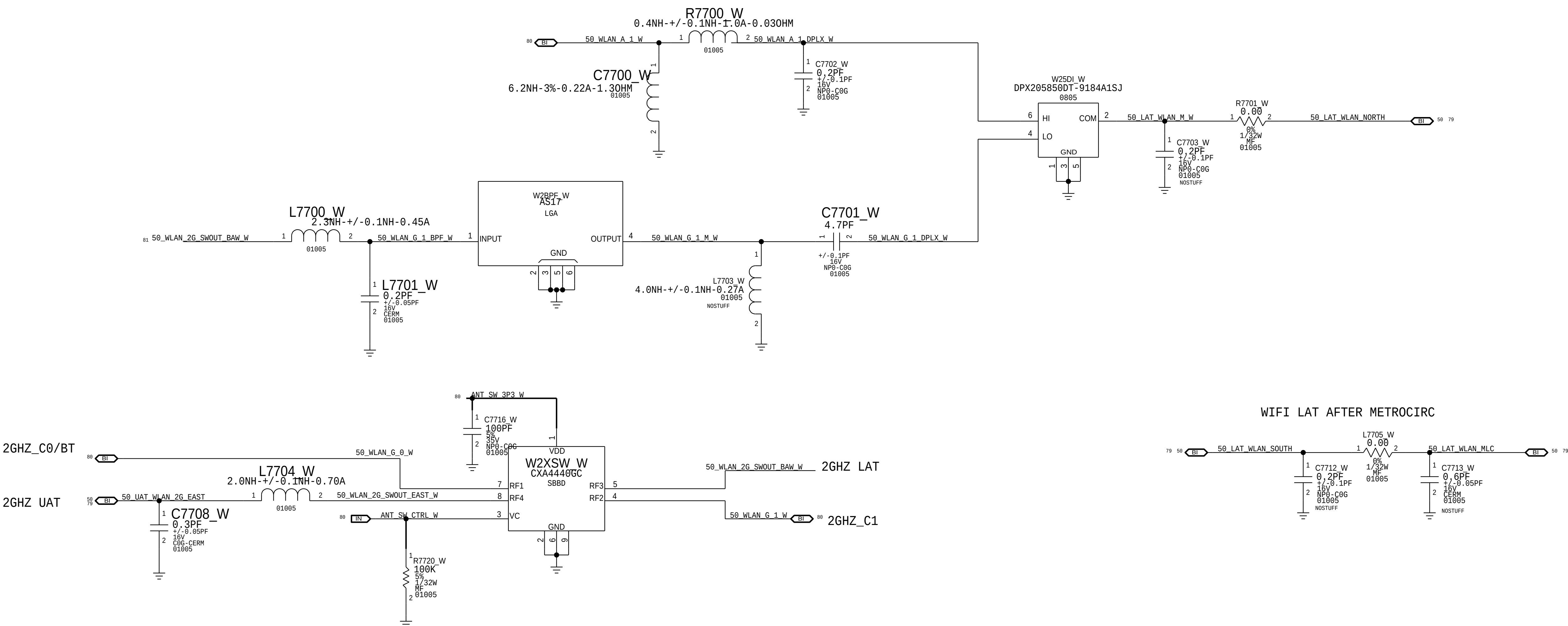


WIFI/BT

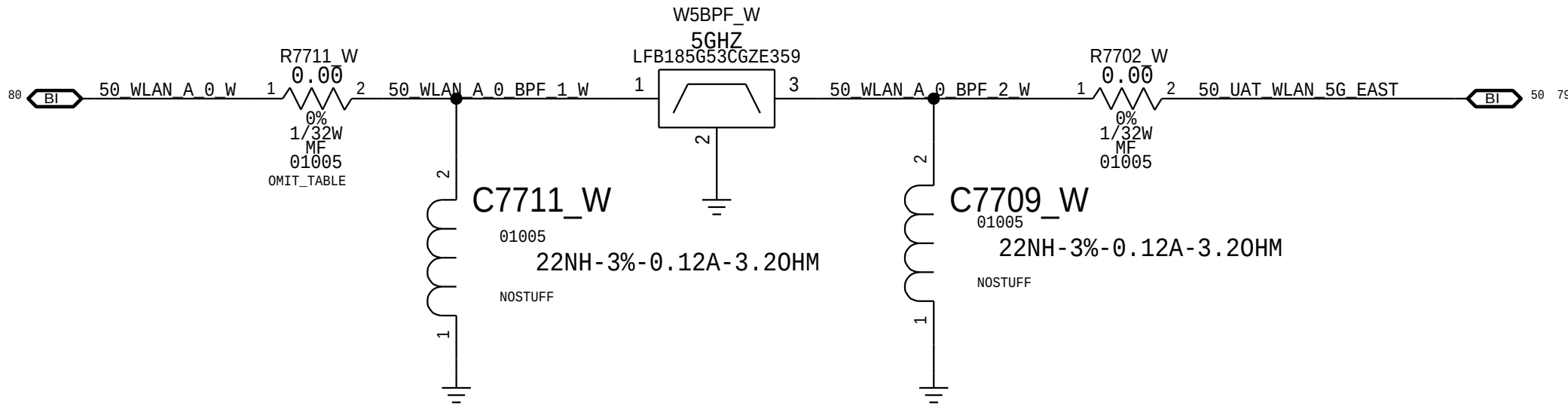


PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
339S00397	339S00399	BOM_TABLE_ALTS	WLAN_W	ALT WIFI/BT MODULE

WIFI LOWER ANTENNA FEED



WIFI UPPER ANTENNA FEED



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
152S00498	1	5G CORE0 BPF MATCH	R7711_W	CRITICAL	D211
117S0161	1	5G CORE0 BPF MATCH	R7711_W	CRITICAL	JPN
117S0161	1	5G CORE0 BPF MATCH	R7711_W	CRITICAL	ROW